

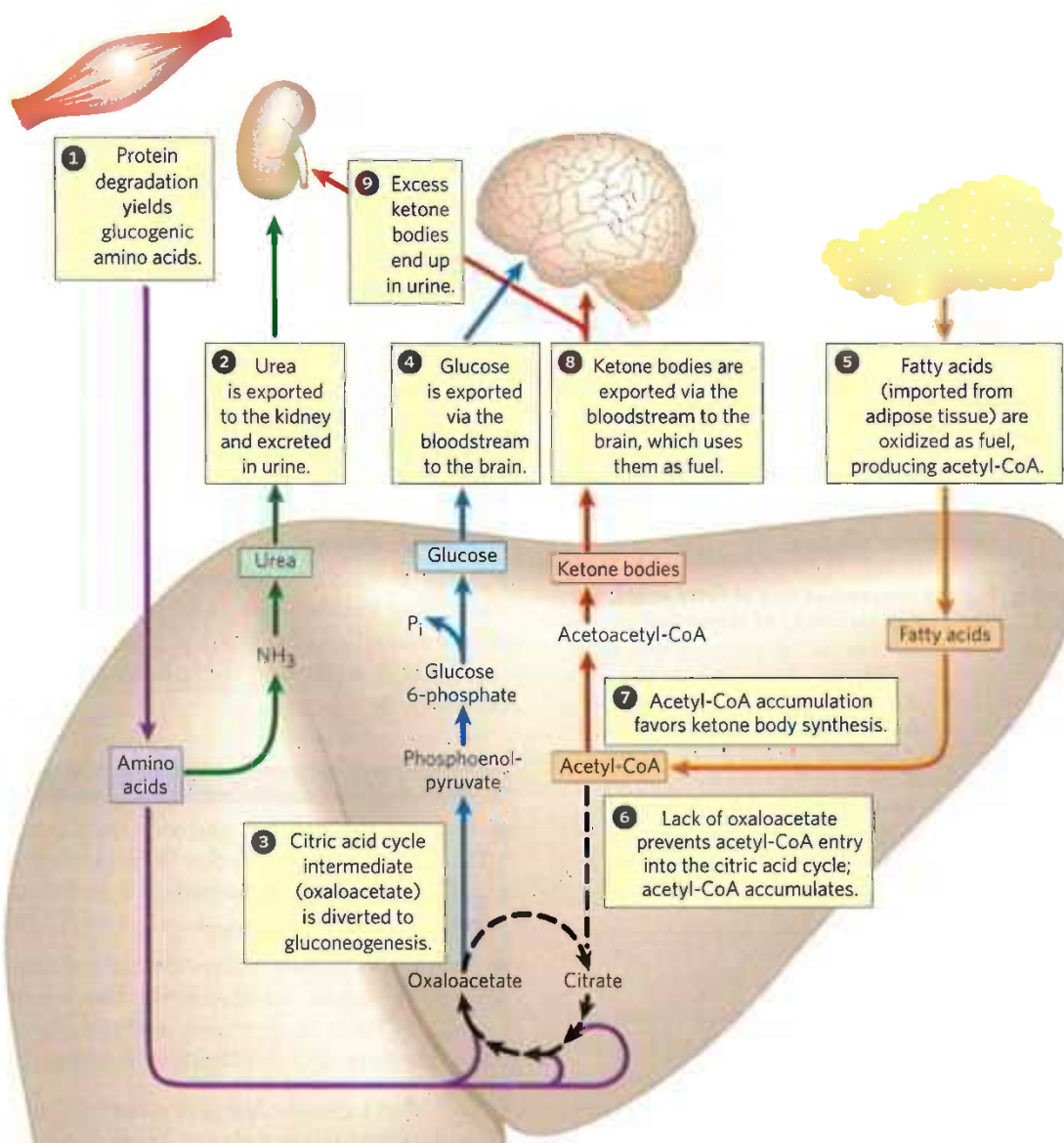


FIGURE 23-27 Fuel metabolism in the liver during prolonged fasting or in uncontrolled diabetes mellitus. The numbered steps are described in the text. After depletion of stored carbohydrates (glycogen), gluconeogenesis in the liver becomes the main source of glucose for the brain (blue arrows). NH_3 from amino acid deamination is converted into urea and excreted (green arrows). Glucogenic amino acids from protein breakdown

(purple arrows) provide substrates for gluconeogenesis, and glucose is exported to the brain. Fatty acids from adipose tissue are imported into the liver and oxidized to acetyl-CoA (orange arrows), and acetyl-CoA is the starting material for ketone body formation in the liver and for export to the brain to serve as an energy source (red arrows). Excess ketone bodies are excreted in the urine.

conversion of *liver* glycogen to blood glucose, the fuel for anaerobic muscular work. Epinephrine also promotes the anaerobic breakdown of *muscle* glycogen by lactic acid fermentation, stimulating glycolytic ATP formation. The stimulation of glycolysis is accomplished by raising the concentration of fructose 2,6-bisphosphate, a potent allosteric activator of the key glycolytic enzyme phosphofructokinase-1. Epinephrine also stimulates fat mobilization in adipose tissue, by activating hormone-sensitive lipase and moving aside perilipin (see Fig. 17-2). Finally, epinephrine stimulates glucagon

secretion and inhibits insulin secretion, reinforcing its effect of mobilizing fuels and inhibiting fuel storage.

Cortisol Signals Stress, Including Low Blood Glucose

A variety of stressors (anxiety, fear, pain, hemorrhage, infection, low blood glucose, starvation) stimulate release of the glucocorticoid **cortisol** from the adrenal cortex (see Fig. 23-6). **P3** Cortisol acts on muscle, liver, and adipose tissue to supply the organism with fuel to withstand the stress. Cortisol is a relatively slow-acting hormone that

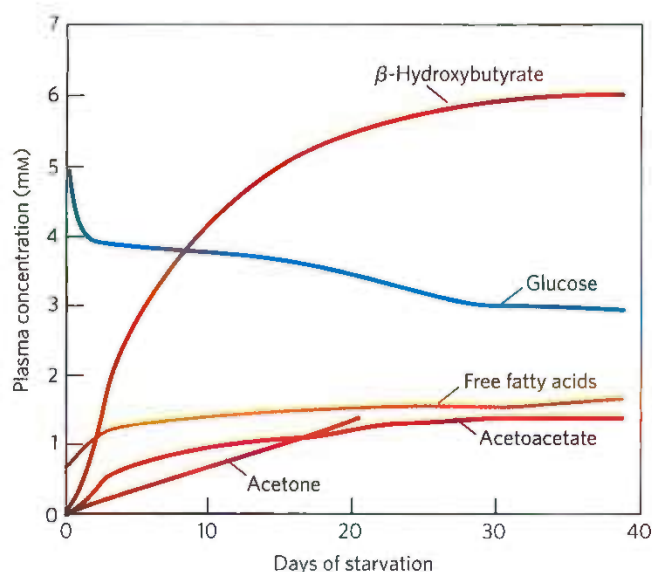


FIGURE 23-28 Plasma concentrations of fatty acids, glucose, and ketone bodies during six weeks of starvation. Despite the hormonal mechanisms for maintaining the glucose level in blood, glucose begins to diminish within 2 days of beginning a fast. The levels of ketone bodies, almost unmeasurable before the fast, rise dramatically after 2 to 4 days of fasting, with β -hydroxybutyrate as the major contributor. These water-soluble ketones, acetoacetate and β -hydroxybutyrate, supplement glucose as an energy source for the brain during a long fast. Acetone, a minor ketone body, is not metabolized but is eliminated in the breath. A much smaller rise in blood fatty acids also occurs, but this does not contribute to energy metabolism in the brain, as fatty acids do not cross the blood-brain barrier. [Data from G. F. Cahill, Jr., *Annu. Rev. Nutr.* 26:1, 2006, Fig. 2.]

alters metabolism by changing the kinds and amounts of certain enzymes synthesized in its target cells, rather than by regulating the activity of existing enzyme molecules.

In adipose tissue, cortisol leads to an increased release of fatty acids from stored TAGs. The exported fatty acids serve as fuel for other tissues, and the glycerol is used for gluconeogenesis in the liver. Cortisol stimulates the breakdown of nonessential muscle proteins and the export of

amino acids to the liver, where they serve as precursors for gluconeogenesis. In the liver, cortisol promotes gluconeogenesis by stimulating synthesis of PEP carboxykinase; glucagon has the same effect, whereas insulin has the opposite effect. Glucose produced in this way is stored in the liver as glycogen or exported immediately to tissues that need glucose for fuel. The net effect of these metabolic changes is to restore blood glucose to its normal level and to increase glycogen stores, ready to support the fight-or-flight response commonly associated with stress. The effects of cortisol therefore counterbalance those of insulin. During extended periods of stress, the continued release of cortisol loses its positive adaptive value and begins to cause damage to muscle and bone and to impair endocrine and immune function.



Cushing disease is a medical condition in which a tumor on the pituitary gland causes the adrenal glands to overproduce cortisol. It is treated by surgery to remove the tumor, followed by chemotherapy to kill remaining tumor cells. Addison disease results from underproduction of cortisol, and is treated by administering hydrocortisone (the pharmaceutical name for cortisol). ■

SUMMARY 23.3 Hormonal Regulation of Fuel Metabolism

- Fluctuations in blood glucose (normally 70 to 100 mg/100 mL, or about 4.5 mM) due to dietary intake or vigorous exercise are counterbalanced by a variety of hormonally triggered changes in metabolism in several organs.
- High blood glucose elicits the release of insulin, which speeds the uptake of glucose by tissues and favors the storage of fuels as glycogen and triacylglycerols while inhibiting fatty acid mobilization in adipose tissue.
- Low blood glucose triggers release of glucagon, which stimulates glucose release from liver glycogen and shifts fuel metabolism in liver and muscle to fatty acid oxidation, sparing glucose for use by the brain. In prolonged fasting, TAGs become the principal fuel; the liver

TABLE 23-6 Physiological and Metabolic Effects of Epinephrine: Preparation for Action

Immediate effect	Overall effect
Physiological ↑ Heart rate ↑ Blood pressure ↑ Dilation of respiratory passages	Increase delivery of O ₂ to tissues (muscle)
Metabolic ↑ Glycogen breakdown (muscle, liver) ↓ Glycogen synthesis (muscle, liver) ↑ Gluconeogenesis (liver)	
↑ Glycolysis (muscle)	
↑ Fatty acid mobilization (adipose tissue)	Increases ATP production in muscle Increases availability of fatty acids as fuel
↑ Glucagon secretion ↓ Insulin secretion	Reinforce metabolic effects of epinephrine

converts the fatty acids to ketone bodies for export to other tissues, including the brain.

■ Epinephrine prepares the body for increased activity by mobilizing glucose from glycogen and other precursors, releasing the glucose into the blood.

■ Cortisol, released in response to a variety of stressors (including low blood glucose), stimulates gluconeogenesis from amino acids and glycerol in the liver, thus raising blood glucose and counterbalancing the effects of insulin.

23.4 Obesity and the Regulation of Body Mass

 In the U.S. population, more than 40% of adults are obese, including more than 10% who are severely obese, as defined in terms of **body mass index (BMI)**, calculated as (weight in kg)/(height in m)². A BMI below 25 is considered normal; an individual with a BMI of 25 to 30 is overweight; a BMI greater than 30 indicates **obesity**; a BMI greater than 40 indicates severe obesity. Obesity is life-threatening. It significantly increases the likelihood of developing type 2 diabetes, as well as heart attack, stroke, and cancers of the colon, breast, prostate, and endometrium. Consequently, there is great interest in understanding how body mass and the storage of fats in adipose tissue are regulated. ■

To a first approximation, obesity is the result of taking in more calories in the diet than are expended by the body's fuel-consuming activities. The body can deal with an excess of dietary calories in three ways: (1) convert excess fuel to fat and store it in adipose tissue, (2) burn excess fuel by extra exercise, and (3) "waste" fuel by diverting it to heat production (thermogenesis) by uncoupled mitochondria. **P4** In mammals, a complex set of hormonal and neuronal signals acts to keep fuel intake and energy expenditure in balance so as to hold the amount of adipose tissue at a suitable level. Dealing effectively with obesity requires understanding how the various checks and balances work under normal conditions and how these homeostatic mechanisms can fail.

Adipose Tissue Has Important Endocrine Functions

One early hypothesis to explain body-mass homeostasis, the "adiposity negative-feedback" model, postulated a mechanism that inhibits eating behavior and increases energy expenditure whenever body weight exceeds a certain value, called the set point; the inhibition is relieved when body weight drops below the set point (**Fig. 23-29**). This model predicts that a feedback signal originating in adipose tissue influences the brain centers that control eating behavior and metabolic and motor activity. The first such signal to be discovered was leptin, in 1994. Subsequent research revealed that **P4** adipose

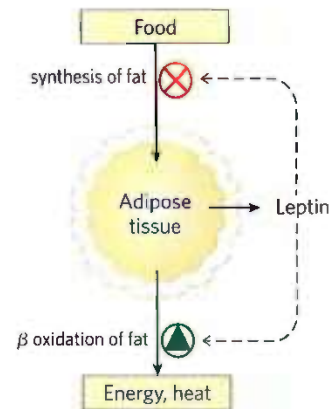


FIGURE 23-29 Set-point model for maintaining constant mass. When the mass of adipose tissue increases (dashed outline), released leptin inhibits feeding and fat synthesis and stimulates oxidation of fatty acids. When the mass of adipose tissue decreases (solid outline), lowered leptin production favors greater food intake and less fatty acid oxidation.

tissue is an important endocrine organ that produces peptide hormones, known as **adipokines**. Adipokines may act locally (autocrine and paracrine action) or systemically (endocrine action), carrying information about the adequacy of the energy reserves (TAGs) stored in adipose tissue to other tissues and to the brain. Normally, adipokines produce changes in fuel metabolism and feeding behavior that reestablish adequate fuel reserves and maintain body mass. When adipokines are over- or underproduced, the resulting dysregulation may result in life-threatening disease.

Leptin (Greek *leptos*, "thin") is an adipokine (167 amino acid residues) that, on reaching the brain, acts on receptors in the hypothalamus to curtail appetite. Leptin was first identified as the product of a gene designated *OB* (obese) in laboratory mice. Mice with two defective copies of this gene (*ob/ob* genotype; lowercase letters signify a mutant form of the gene) show the behavior and physiology of animals in a constant state of starvation: their plasma cortisol levels are elevated; they exhibit unrestrained appetite, are unable to stay warm, grow abnormally large, and do not reproduce. As a consequence of unrestrained appetite, they become severely obese, weighing as much as three times more than normal mice (**Fig. 23-30**). They also have metabolic disturbances similar to those seen in diabetes, and they are insulin-resistant. When leptin is injected into *ob/ob* mice, they eat less, lose weight, and increase their locomotor activity and thermogenesis.

A second mouse gene, designated *DB* (diabetic), also has a role in appetite regulation. Mice with two defective copies (*db/db*) are obese and diabetic. The *DB* gene encodes the **leptin receptor**. When the receptor is defective, the signaling function of leptin is lost.

The leptin receptor is expressed primarily in regions of the brain known to regulate feeding behavior—neurons of



FIGURE 23-30 Obesity caused by defective leptin production.

Both of these mice, which are the same age, have defects in the *OB* gene. The mouse on the right was injected daily with purified leptin and weighs 35 g. The mouse on the left got no leptin and consequently ate more food and was less active; it weighs 67 g. [The Rockefeller University/AP Photo.]

the **arcuate nucleus** of the hypothalamus (Fig. 23-31a). Leptin carries the message that fat reserves are sufficient, and it promotes reduction of fuel intake and increase in expenditure of energy. Leptin-receptor interaction in the hypothalamus alters the release of neuronal signals to the region of the brain that affects appetite. Leptin also stimulates the sympathetic nervous system, increasing blood pressure, heart rate, and thermogenesis by uncoupling the mitochondria of brown adipocytes (Fig. 23-31b). Recall that the uncoupling protein UCP1 forms a channel in the inner mitochondrial membrane that allows protons to reenter the mitochondrial matrix without passing through the ATP synthase complex. This permits constant oxidation of fuel (fatty acids in a brown or beige adipocyte) without ATP synthesis, dissipating energy as heat and consuming dietary calories or stored fats in potentially large amounts.

Leptin Stimulates Production of Anorexigenic Peptide Hormones

Two types of neurons in the arcuate nucleus control fuel intake and metabolism (Fig. 23-32). The **orexigenic** (appetite-stimulating) neurons stimulate eating by producing and releasing **neuropeptide Y (NPY)**, which causes the next neuron in the circuit to send the signal to the brain: Eat! The blood level of NPY rises during starvation and is elevated in mice with either two defective copies of the leptin gene (*ob/ob*) or two defective copies of the leptin receptor gene (*db/db*). The high NPY concentration presumably contributes to the obesity of these mice, which eat voraciously.

The **anorexigenic** (appetite-suppressing) neurons in the arcuate nucleus produce **α -melanocyte-stimulating hormone (α -MSH)**; also known as melanocortin), formed from its polypeptide precursor pro-opiomelanocortin (POMC; Fig. 23-5). Release of α -MSH causes the next neuron in the circuit to send the signal to the brain: Stop eating!

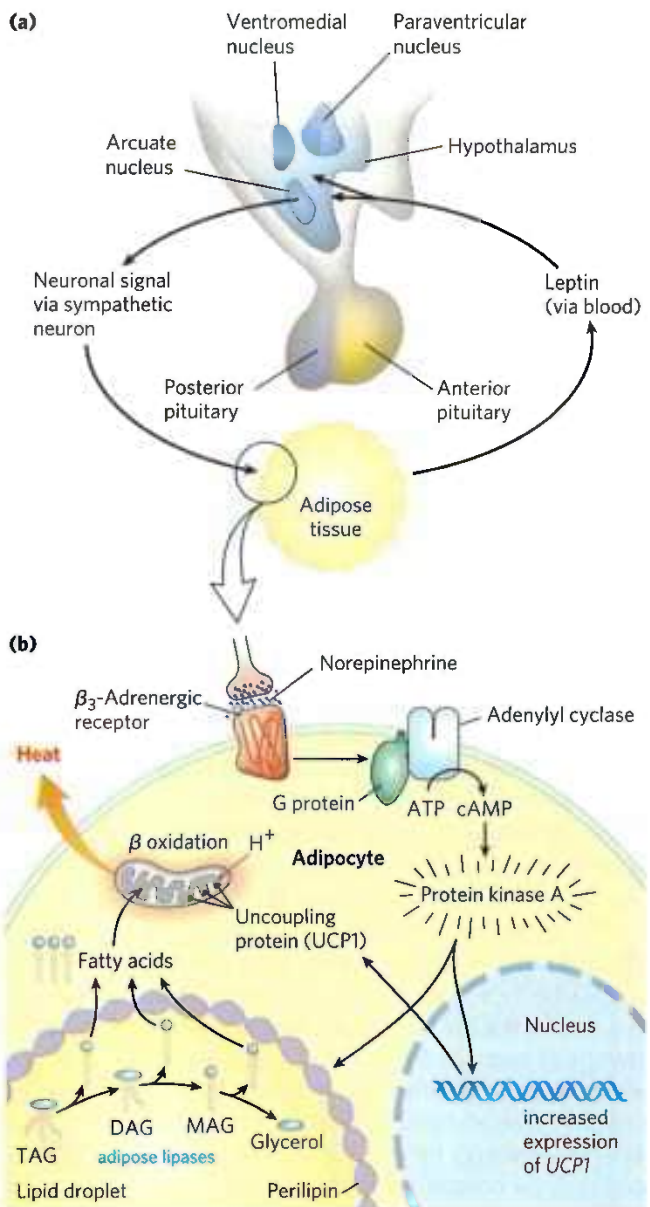


FIGURE 23-31 Hypothalamic regulation of food intake and energy expenditure.

(a) Role of the hypothalamus in its interaction with adipose tissue. The hypothalamus receives input (leptin) from adipose tissue and responds with neuronal signals to adipocytes. (b) This signal (norepinephrine) activates protein kinase A, which triggers mobilization of fatty acids from TAG and their uncoupled oxidation in mitochondria, generating heat but not ATP. DAG, diacylglycerol; MAG, monoacylglycerol.

The amount of leptin released by adipose tissue depends on both the number and the size of adipocytes. When weight loss decreases the mass of lipid tissue, leptin levels in the blood decrease, the production of NPY increases, and the processes in adipose tissue shown in Figure 23-31 are reversed. Uncoupling is diminished, slowing thermogenesis and saving fuel, and fat mobilization slows in response to reduced signaling by cAMP. Consumption of more food, combined with more efficient utilization of fuel, results in replenishment of the fat reserve in adipose tissue, bringing the system back into balance.

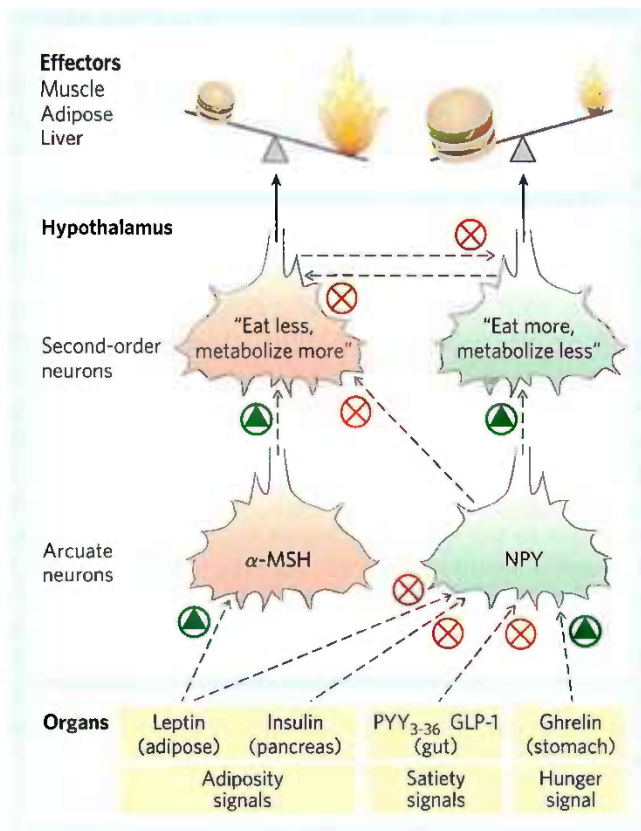


FIGURE 23-32 Hormones that control eating. In the arcuate nucleus of the hypothalamus, two sets of neurosecretory cells receive hormonal input and relay neuronal signals to the cells of muscle, adipose tissue, and liver. Leptin and insulin are released from adipose tissue and the pancreas, respectively, in proportion to the mass of body fat. The two hormones act on anorexigenic neurosecretory cells to trigger release of α -MSH (melanocyte-stimulating hormone); α -MSH carries the signal to second-order neurons in the hypothalamus, which puts out the signals to eat less and metabolize more fuel. Leptin and insulin also act on orexigenic neurosecretory cells to inhibit the release of NPY, reducing the “eat more” signal sent to the tissues. As described later in the text, the gastric hormone ghrelin stimulates appetite by activating the NPY-expressing cells; PYY₃₋₃₆, released from the colon, inhibits these neurons and decreases appetite. Each of the two types of neurosecretory cells inhibits hormone production by the other, so any stimulus that activates orexigenic cells inactivates anorexigenic cells, and vice versa. This strengthens the effect of stimulatory inputs.

Leptin Triggers a Signaling Cascade That Regulates Gene Expression

The leptin signal is transduced by a mechanism also used by receptors for some growth factors. The leptin receptor has a single transmembrane segment. When leptin binds to the extracellular domains of two monomers, they dimerize and undergo phosphorylation on several Tyr residues. This initiates a chain of events that ends in the nucleus with the increased synthesis of target genes including the gene for POMC, from which α -MSH is produced.

The increased catabolism and thermogenesis triggered by leptin are due in part to increased synthesis of the mitochondria in brown and beige adipocytes. Leptin

stimulates UCP1 synthesis by altering synaptic transmissions from neurons of the arcuate nucleus to adipose and other tissues via the sympathetic nervous system. The consequent increased release of norepinephrine in these tissues acts through β_3 -adrenergic receptors to stimulate transcription of the *UCP1* gene. The resulting uncoupling of electron transfer from oxidative phosphorylation consumes fat and is thermogenic (Fig. 23-31).

Might human obesity be the result of insufficient leptin production and therefore be treatable by the injection of leptin? Blood levels of leptin are, in fact, usually much *higher* in obese animals (including humans) than in animals of normal body mass (except, of course, in *ob/ob* mutants, which cannot make leptin). Some downstream element in the leptin response system must be defective in obese individuals, and the elevation in leptin is the result of an (unsuccessful) attempt to overcome the leptin resistance. In those very rare humans with extreme obesity who have a defective leptin gene (*OB*), leptin injection does result in dramatic weight loss. In the vast majority of obese individuals, however, the *OB* gene is intact. In clinical trials, the injection of leptin did not have the weight-reducing effect observed in obese *ob/ob* mice. Clearly, most cases of human obesity involve one or more factors in addition to leptin.

Adiponectin Acts through AMPK to Increase Insulin Sensitivity

Adiponectin is a peptide hormone produced almost exclusively in adipose tissue, an adipokine that sensitizes other organs to the effects of insulin. Adiponectin circulates in the blood and powerfully affects the metabolism of fatty acids and carbohydrates in liver and muscle. It increases the uptake of fatty acids from the blood by myocytes and the rate at which fatty acids undergo β oxidation in muscle. It also blocks fatty acid synthesis and gluconeogenesis in hepatocytes, and stimulates glucose uptake and catabolism in muscle and liver.

These effects of adiponectin are indirect and not fully understood, but the **AMP-activated protein kinase (AMPK)** mediates many of them. Acting through its GPCR, adiponectin triggers phosphorylation and activation of AMPK. Recall that AMPK is activated by factors that signal the need to shift metabolism toward energy generation and away from energy-requiring biosynthesis (Fig. 23-33; see also p. 503). When activated, AMPK profoundly affects the metabolism of individual cells and, through its actions in the brain, the metabolism of the whole animal.

AMPK Coordinates Catabolism and Anabolism in Response to Metabolic Stress

AMPK has emerged as a central player in the coordination of metabolic pathways, organism activity, and feeding behavior (Fig. 23-34). This heterotrimeric, AMP-activated protein kinase monitors the energy and

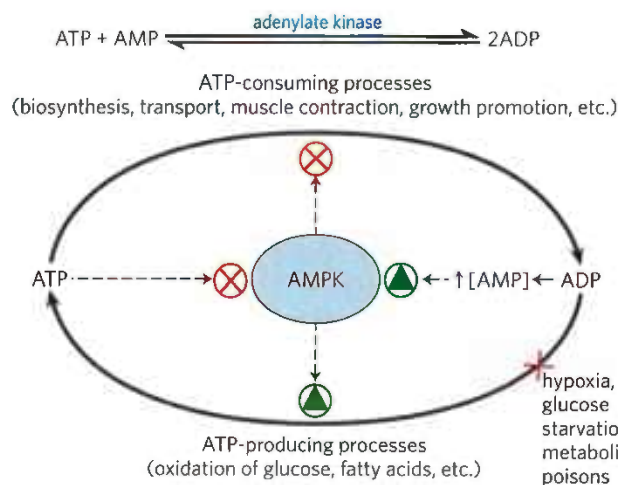


FIGURE 23-33 The role of AMP-activated protein kinase (AMPK) in maintaining energy homeostasis. ADP produced by synthetic reactions is converted to ATP and AMP by adenylate kinase. AMP activates AMPK, which reciprocally regulates ATP-consuming and ATP-generating pathways by phosphorylating key enzymes (see Fig. 23-34). Conditions or agents that inhibit ATP production by catabolic reactions (such as hypoxia, lack of glucose, or metabolic poisons) raise [AMP], activate AMPK, and stimulate catabolism. Cellular or organismal activities that consume ATP (muscle contraction, growth) increase [AMP] and stimulate catabolic reactions to replenish ATP. When [ATP] is high, ATP prevents AMP binding to AMPK, thus lowering AMPK activity and slowing catabolism.

nutrient status in individual cells and shifts metabolism toward energy generation when necessary to maintain metabolic homeostasis. Furthermore, by responding to a variety of hormone signals, AMPK in the hypothalamus acts to keep the whole organism in energetic balance (Fig. 23-8).

AMPK monitors the energy status of a cell through its response to increased [AMP]/[ATP]. Many of the energy-consuming reactions in cells convert ATP to ADP or AMP. Adenylate kinase catalyzes the reaction $2\text{ADP} \rightarrow \text{AMP} + \text{ATP}$, so [AMP] is a sensitive measure of the cell's energy status. AMPK is allosterically activated by AMP binding, and ATP prevents AMP binding, so the enzyme is activated when the cell is energetically depleted (high [AMP]) and inactivated when energy is plentiful (high [ATP], and high [ATP]/[AMP]). AMPK responds to the energetic needs of the whole organism through a second mode of regulation. The enzyme is activated 100-fold by phosphorylation of Thr¹⁷² by liver kinase B1 (LKB1), which is itself subject to regulation by upstream components, including adiponectin. When activated by phosphorylation and AMP binding, AMPK phosphorylates specific enzymes in metabolic pathways that are crucial to energy homeostasis (Fig. 23-34).

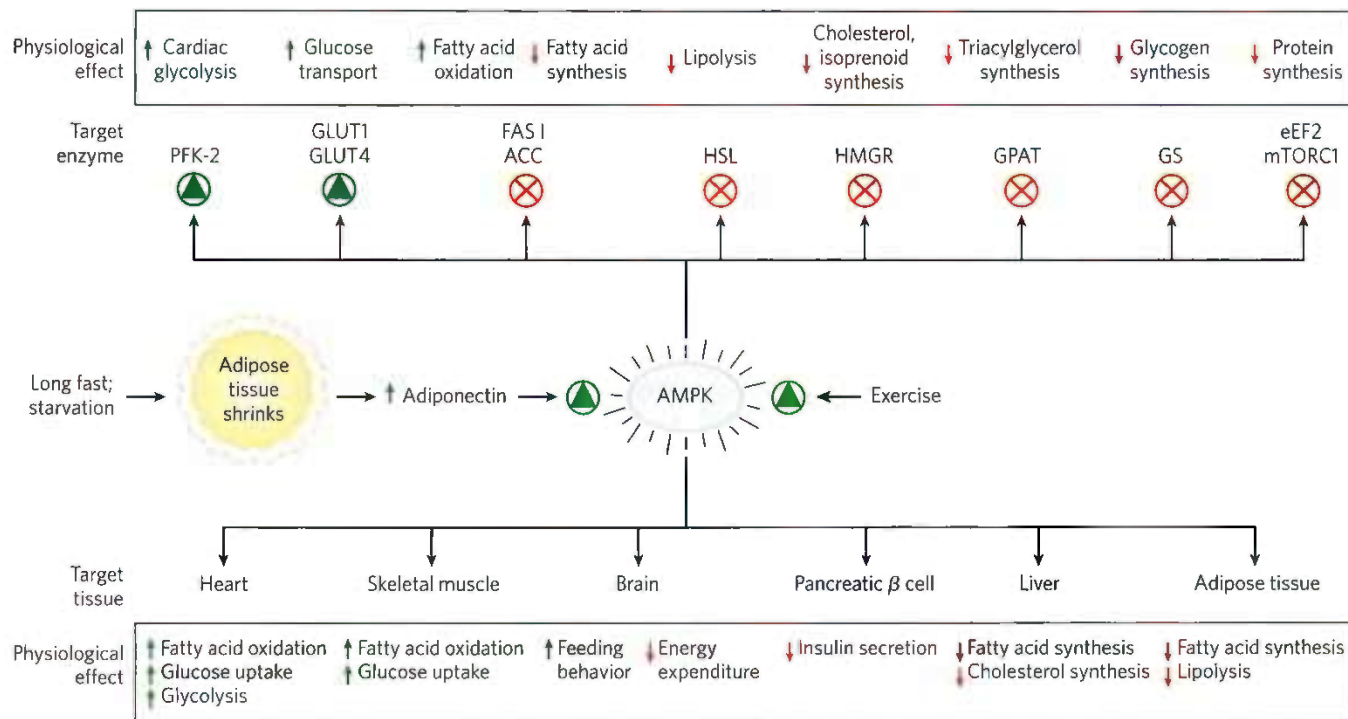


FIGURE 23-34 Formation of adiponectin and its actions through AMPK. Extended fasting or starvation decreases triacylglycerol reserves in adipose tissue, which triggers adiponectin production and release from adipocytes. Adiponectin acts through its plasma membrane receptors in various cell types and organs to inhibit energy-consuming processes and stimulate energy-producing processes. It acts in the brain to stimulate feeding behavior and inhibit energy-consuming physical activity, and in brown fat to inhibit thermogenesis. Adiponectin exerts some of its metabolic effects by activating AMPK, which regulates (by

phosphorylation) specific enzymes in key metabolic processes. PFK-2, phosphofructokinase-2; GLUT1 and GLUT4, glucose transporters; FAS I, fatty acid synthase I; ACC, acetyl-CoA carboxylase; HSL, hormone-sensitive lipase; HMGR, HMG-CoA reductase; GPAT, an acyl transferase; GS, glycogen synthase; eEF2, eukaryotic elongation factor 2 (required for protein synthesis; see Chapter 27); and mTORC1, mammalian target of rapamycin complex 1 (a protein kinase complex that regulates protein synthesis on the basis of nutrient availability). Exercise, through conversion of ATP to ADP and AMP, also stimulates AMPK.

When AMPK senses depletion of ATP in an individual cell, lipid synthesis is inhibited and use of lipid as fuel is stimulated. One enzyme regulated by AMPK in the liver and in white adipose tissue is acetyl-CoA carboxylase, which produces malonyl-CoA, the first intermediate committed to fatty acid synthesis. Malonyl-CoA is a powerful inhibitor of the enzyme carnitine acyltransferase 1, which starts the process of β oxidation by transporting fatty acids into the mitochondrion (see Fig. 17-6). By phosphorylating and inactivating acetyl-CoA carboxylase, AMPK inhibits fatty acid synthesis while relieving inhibition (by malonyl-CoA) of β oxidation (see Fig. 17-13). Cholesterol synthesis, a heavy energy consumer, is also inhibited by AMPK, which phosphorylates and inactivates HMG-CoA reductase, an enzyme central to sterol synthesis (see Fig. 21-34). Similarly, AMPK inhibits fatty acid synthase and acyl transferase, effectively blocking the synthesis of TAGs. In addition to its effects on lipid metabolism, AMPK inhibits the synthesis of glycogen and of protein (see Fig. 23-35). Inadequate supplies of oxygen (hypoxia) or blood glucose (hypoglycemia) are among the stressors that trigger AMPK activation.

In the hypothalamus, AMPK is positioned to receive a variety of signals from throughout the body (Fig. 23-8). Ghrelin and adiponectin signal “empty stomach” and “fat tissue depleted,” and, like low blood glucose, they elicit hypothalamic signals, mediated by AMPK, that stimulate feeding and inhibit energy-requiring biosynthetic processes. Leptin, as we have seen, brings to the brain the signal “adipose tissue is full,” slowing catabolic processes and favoring growth and biosynthesis.

At the cytoplasmic surface of lysosomes, AMPK interacts with a second central regulator of cellular activity, the protein kinase mTOR. This enzyme, at the center of an enormous complex, mTORC1, gauges whether sufficient nutrients and low molecular weight substrates are available to support cell growth and proliferation. Together, the two protein kinases, AMPK and mTOR, control major aspects of a cell's activity and fate.

The mTORC1 Pathway Coordinates Cell Growth with the Supply of Nutrients and Energy

The highly conserved Ser/Thr kinase **mTOR** forms a complex, **mTORC1**, with a scaffold protein, raptor, and other regulatory proteins. mTORC1 is recruited to the cytosolic surface of the lysosome through raptor by another complex, Ragulator, and a number of associated Rag G proteins that sense amino acid sufficiency in the cell. The Ragulator-Rag complex is tethered to the lysosomal membrane by covalently linked lipids. When mTORC1 is docked to the Ragulator-Rag complex, it is in contact with another G protein, **Rheb** (Fig. 23-35). This massive complex integrates signals from inside and outside of the lysosome about the energy status of the cell, the availability of critical amino acids needed for protein synthesis, and the presence of growth factors. When these signals

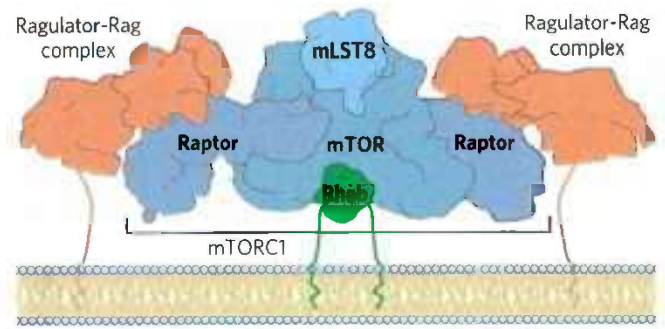


FIGURE 23-35 The mTORC1-Ragulator-Rag complex on the lysosomal surface. This model integrates the structures of mTORC1 and Ragulator-Rag, separately determined by cryo-EM. Raptor acts as a scaffold protein organizing the assembly, which puts the mTOR protein kinase in contact with the G protein Rheb for activation. [Information from J. H. Park et al., *Trends Biochem. Sci.* 45:367, 2019, Fig. 1; K. B. Rogala et al., *Science* 366:468, 2019, Fig. 5A.]

indicate that the cell has what it needs to grow, GTP-bound Rheb activates the protein kinase activity of mTOR, which then phosphorylates many different proteins required for transcription, increased ribosome synthesis, and expression of genes encoding the enzymes of lipid synthesis and mitochondrial proliferation (Fig. 23-36). Because the lysosome is where the cell recycles defective or unneeded components, or extracellular substances brought into the cell through phagocytosis, it is, in effect, the warehouse of parts for cellular construction. The mTORC1-Ragulator-Rag complex is the supply chain manager determining whether the assembly line can operate.

Fasting results in inactivation of mTORC1 by AMPK, leading to increased breakdown of protein and glycogen in the liver and muscle and mobilization of TAGs in adipose tissue. Chronic activation of mTORC1 by overeating results in excess deposition of TAGs in adipose tissue, as well as in liver and muscle, which may contribute to insulin insensitivity and type 2 diabetes. Mutations that produce constantly activated mTORC1 are commonly associated with human cancers.

Diet Regulates the Expression of Genes Central to Maintaining Body Mass

Proteins in a family of ligand-activated transcription factors known as **peroxisome proliferator-activated receptors (PPARs)** respond to changes in dietary lipid by altering the expression of genes involved in fat and carbohydrate metabolism. (These transcription factors were first recognized for their roles in peroxisome synthesis—thus their names.) Their normal ligands are fatty acids or fatty acid derivatives. PPAR γ , PPAR α , and PPAR δ act in the nucleus by forming heterodimers with another nuclear receptor, RXR, then binding to regulatory regions of DNA near the genes under their control and changing the rate of transcription of those genes (Fig. 23-37).

PPAR γ turns on genes that act in the differentiation of fibroblasts into adipocytes and genes that encode

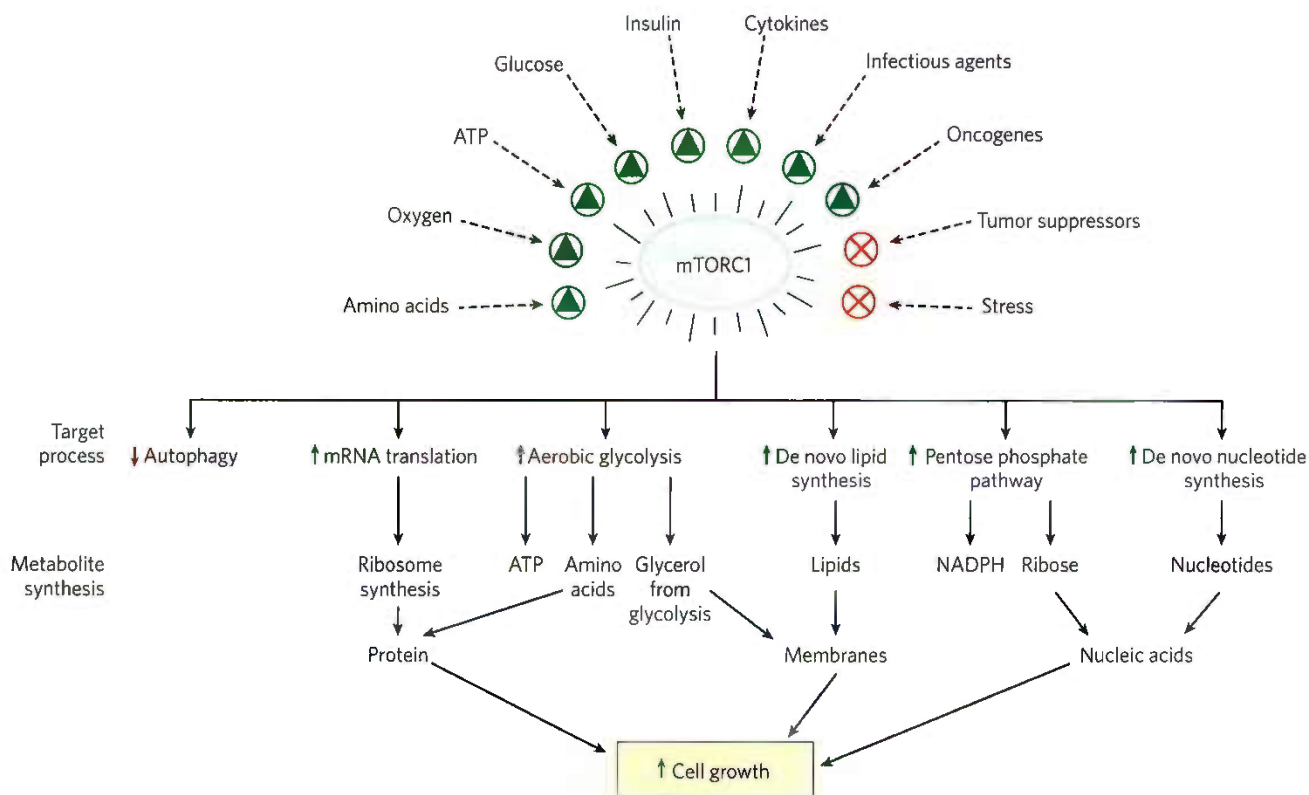


FIGURE 23-36 A summary of mTORC1 activation signals and the cellular processes that active mTORC1 stimulates. The Ser/Thr protein kinase of mTORC1 is activated by the G protein Rheb, reflecting the integration of many signals that indicate that the cell is

prepared for growth. By phosphorylating key target proteins, mTORC1 activates energy (ATP and NADPH) production for biosynthesis and stimulates the synthesis of proteins and lipids, allowing cell growth and proliferation.

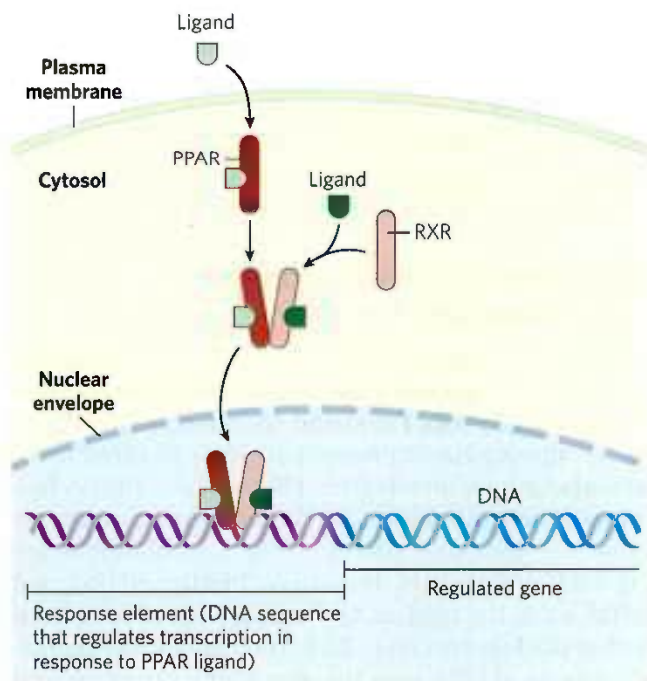


FIGURE 23-37 Mode of action of PPARs. PPARs are transcription factors that, when bound to their cognate ligand, form heterodimers with the nuclear receptor RXR. The dimer binds specific regions of DNA known as response elements, stimulating transcription of genes in those regions. [Information from R. M. Evans et al., *Nat. Med.* 10:355, 2004, Fig. 3.]

proteins required for lipid synthesis and storage in adipocytes (**Fig. 23-38**). PPAR γ is activated by the thiazolidinedione drugs that are used to treat type 2 diabetes (discussed below).

PPAR α is expressed in liver, kidney, heart, skeletal muscle, and brown adipose tissue. The ligands that activate this transcription factor include eicosanoids and free fatty acids. In hepatocytes, PPAR α turns on the genes necessary for the uptake and β oxidation of fatty acids and for the formation of ketone bodies during fasting.

PPAR δ is a key regulator of fat oxidation, which responds to changes in dietary lipid. PPAR δ acts in liver and muscle, stimulating the transcription of at least nine genes encoding proteins for β oxidation and for energy dissipation through uncoupling of mitochondria. By stimulating fatty acid breakdown in uncoupled mitochondria, PPAR δ causes fat depletion, weight loss, and thermogenesis.

Short-Term Eating Behavior Is Influenced by Ghrelin, PYY₃₋₃₆, and Cannabinoids

The peptide hormone **ghrelin** is produced in cells lining the stomach. It is a powerful appetite stimulant that works on a shorter time scale (between meals) than leptin and insulin. Ghrelin receptors are located in the hypothalamus, affecting appetite, as well as in heart muscle and adipose tissue. Ghrelin acts through a GPCR

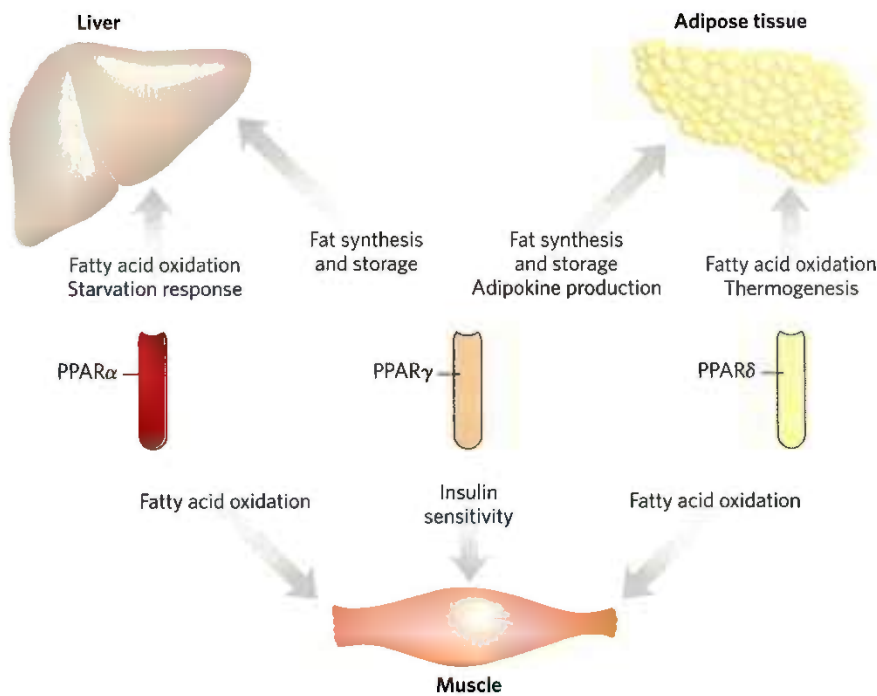


FIGURE 23-38 Metabolic integration by PPARs. The three PPAR isoforms regulate lipid and glucose homeostasis through their coordinated effects on gene expression in liver, muscle, and adipose tissue. PPAR α and PPAR δ regulate lipid utilization; PPAR γ regulates lipid storage and the insulin sensitivity of various tissues.

to generate the second messenger IP₃, which mediates the hormone's action. The concentration of ghrelin in the blood fluctuates strikingly throughout the day, peaking just before a meal and dropping sharply just after a meal (Fig. 23-39). Injection of ghrelin into humans produces immediate sensations of intense hunger. Individuals with Prader-Willi syndrome, whose blood levels of ghrelin are exceptionally high, have an uncontrollable appetite, leading to extreme obesity that often results in death before the age of 30.

PYY₃₋₃₆ is a peptide hormone (34 amino acid residues) secreted by endocrine cells in the lining of the small intestine and colon in response to food entering from the stomach. The level of PYY₃₋₃₆ in the blood rises after a meal and remains high for some hours. The hormone is carried in the blood to the arcuate nucleus, where it acts on orexigenic neurons, inhibiting NPY release and reducing hunger (Fig. 23-32). Humans injected with PYY₃₋₃₆ feel little hunger and eat less than normal amounts for about 12 hours.

Endocannabinoids (Fig. 23-40) are eicosanoid lipid messengers that act through specific GPCRs in the brain and peripheral nervous system to increase

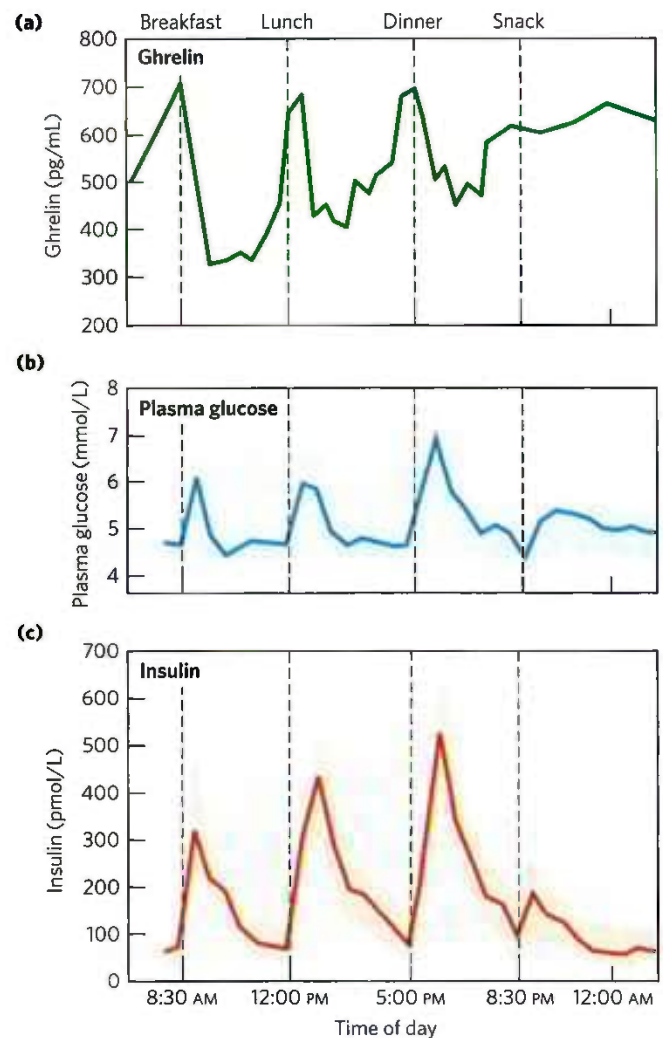


FIGURE 23-39 Variations in blood concentrations of glucose, ghrelin, and insulin relative to meal times. (a) Plasma levels of ghrelin rise sharply just before the usual time for meals (7 AM breakfast, 12 noon lunch, 5:30 PM dinner) and drop precipitously just after meals, paralleling subjective feelings of hunger. (b) Plasma glucose rises sharply after a meal, (c) followed immediately by a rise in insulin level in response to the increased blood glucose. [(a, c) Data from D. E. Cummings et al., *Diabetes* 50:1714, 2001, Fig. 1. (b) Data from M. D. Feher and C. J. Bailey, *Br. J. Diabet. Vasc. Dis.* 4:39, 2004.]

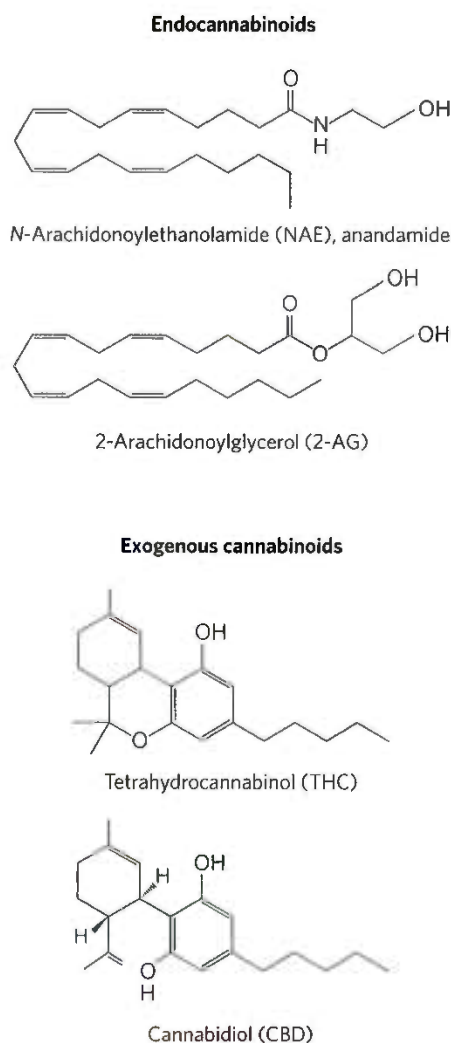


FIGURE 23-40 Cannabinoids. Two endocannabinoids produced by animals, and two products from the cannabis plant. Endocannabinoids carry retrograde signals: secreted by a postsynaptic cell, they diffuse across the synaptic cleft and activate GPCR receptors in the presynaptic cell. The receptors are of two types: CB₁ in the central nervous system, and CB₂ in the peripheral nervous system. Δ^9 -Tetrahydrocannabinol (THC), the psychoactive compound in cannabis, is an agonist of both types of cannabinoid receptors in animals; cannabidiol (CBD) does not bind either.

appetite, heighten the sensory response to food (especially sweet and fatty foods), and elevate mood. When food enters the mouth, neuronal signals travel to the brain, and from there through the vagus nerve to the intestine, which then produces and releases endocannabinoids. The receptors for endocannabinoids control ion channels of sensory neurons, changing their membrane potentials and sending signals to the brain. Palatable food sensed in this way motivates further consumption of that food. The taste of fats (which are particularly high in caloric content) causes cannabinoid release, which effectively triggers further consumption. Well-conserved across vertebrate species, this system probably evolved to maximize the intake of food and to guard against starvation. In mammals, cannabinoid

action stimulates an increase in fat mass and inhibits energy loss by motor activity or thermogenesis. Cannabinoid receptors also mediate the psychoactive effects of Δ^9 -tetrahydrocannabinol (Fig. 23-40), the main active ingredient in marijuana, long known for its stimulating effect on appetite.

Microbial Symbionts in the Gut Influence Energy Metabolism and Adipogenesis

An adult human is host to about 10^{14} microbial cells that inhabit the gut. These microbes function as a major endocrine organ, producing a variety of metabolites with profound effects on host metabolism, feeding behavior, and body mass. Lean and obese individuals harbor different combinations of microbial symbionts in the gut. Investigation of this observation led to the discovery that gut microbes release fermentation products—the short-chain fatty acids acetate, propionate, butyrate, and lactate—that enter the bloodstream and trigger metabolic changes in adipose tissue (Fig. 23-41). Propionate, for example, drives the expansion of white adipose tissue by acting on GPCRs in the plasma membranes of several cell types, including adipocytes. These receptors trigger differentiation of precursor cells (preadipocytes) into adipocytes and inhibit lipolysis in existing adipocytes, leading to an increase in white adipose tissue mass—that is, obesity. Gut microbes also convert primary bile acids, synthesized in the liver, into the secondary bile acids deoxycholate and lithocholate, which enter the bloodstream and act through GPCRs and steroid receptors to activate beige adipocytes to produce UCP1 and increase energy expenditure.

These findings raise the possibility of preventing obesity by altering the makeup of the microbial community in the gut. Weight reduction might be accomplished either by adding, directly to the gut, microbial species (**probiotics**) that disfavor adipogenesis, or by adding to the diet nutrients (**prebiotics**) that favor the dominance of probiotic microbes. For example, experiments in mice have shown that fructans, polymers of fructose that are indigestible by animals, favor a specific microbial community. When this combination of microorganisms is present, fat storage in white adipose tissue and liver decreases, and there is none of the decrease in insulin sensitivity that is associated with obesity and lipid deposition in the liver (see below). Investigators have transplanted fecal material from lean mice to fat ones, and found that a new collection of microbes became established in the gut of the recipient animals, and the animals lost weight.

P4 Endocrine cells in the lining of the intestinal tract secrete peptides—the anorexigenic PYY_{3–36} and GLP-1 and the orexigenic ghrelin—that modulate food intake and energy expenditure. Interaction with specific microbes in the gut, or with their fermentation products, may trigger release of these peptides. Understanding how

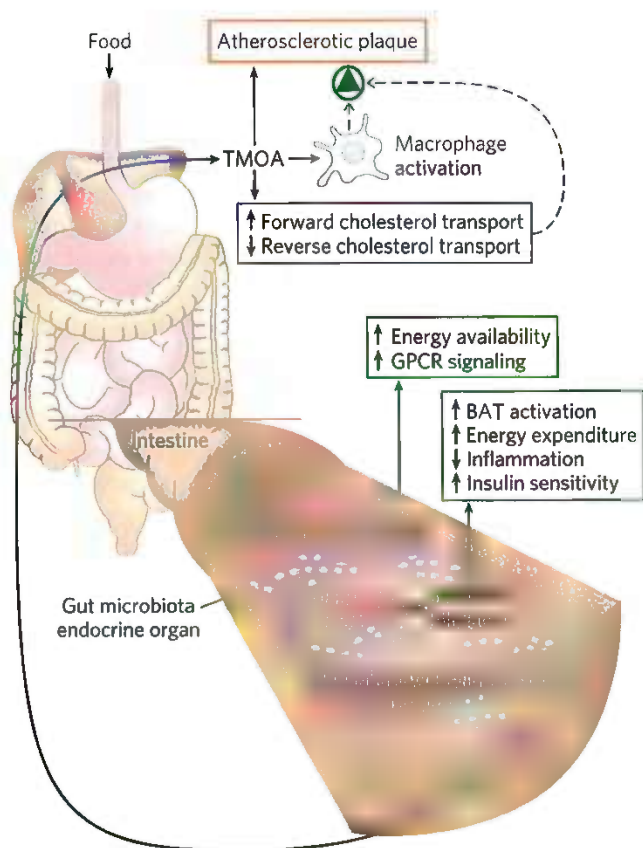


FIGURE 23-41 Effects of gut microbe metabolism on health.

The enormous number and diversity of microorganisms (the microbiota) in the colon generate metabolic products that may have significant effects on health, both positive and negative. For example, the metabolism of primary bile acids by microbiota produces secondary products that act through nuclear receptors to stimulate thermogenesis in host brown adipose tissue (BAT) and increase both energy consumption and insulin sensitivity, while reducing inflammation. Metabolism of undigested carbohydrates by microbiota produces short-chain fatty acids (SCFAs) that signal expansion of the host's white adipose tissue (WAT), promoting obesity. SCFAs produced by microbiota are also a readily metabolizable energy source for the host. Microbial production of the SCFA propionate prevents lipogenesis in the liver and lowers blood cholesterol, both favorable to health. On the other hand, metabolic conversion of phosphatidylcholine and L-carnitine to trimethylamine (TMA), and its further conversion in liver to trimethylamine-N-oxide (TMAO), results in receptor-mediated changes in cholesterol transport and macrophage activity. The combination of altered sterol transport and increased macrophage activity can lead to formation of atherosclerotic plaque (see Fig. 21-46).

diet and microbial symbionts interact to affect energy metabolism and adipogenesis is an important key to understanding the development of obesity, metabolic syndrome, and type 2 diabetes and is a major challenge for the future.

The exquisitely interconnected system of neuroendocrine controls of food intake and metabolism presumably evolved to protect against starvation and to eliminate counterproductive accumulation of fat (extreme obesity). The difficulty most people face in trying to lose weight testifies to the remarkable effectiveness of these controls.

SUMMARY 23.4 Obesity and the Regulation of Body Mass

- Adipose tissue produces leptin, a hormone that regulates feeding behavior and energy expenditure so as to maintain adequate reserves of fat. Leptin production and release increase with the number and size of adipocytes.
- Leptin acts on receptors in the brain, causing the release of anorexigenic (appetite-suppressing) peptides, including α -MSH, that act in the brain to inhibit eating.
- Adiponectin stimulates fatty acid uptake and oxidation and inhibits fatty acid synthesis. It also sensitizes muscle and liver to insulin. At least some of the actions of adiponectin are mediated by AMPK, which is also activated by metabolic stress (low [AMP]), and by exercise.
- The protein kinase complex mTORC1 gauges the supply of essential amino acids and other metabolites, triggering cell growth if all required nutrients are available. It thus complements AMPK in determining the energetic status of a cell.
- Expression of the enzymes of lipid synthesis is under tight and complex regulation. PPARs are transcription factors that determine the rate of synthesis of many enzymes involved in lipid metabolism and adipocyte differentiation.
- Ghrelin, a hormone produced in the stomach, acts on orexigenic (appetite-stimulating) neurons in the arcuate nucleus to produce hunger before a meal. PYY₃₋₃₆, a peptide hormone of the intestine, acts at the same site to lessen hunger after a meal. Endocannabinoids signal the availability of sweet or fatty food and stimulate its consumption.
- Microbial symbionts in the gut produce fermentation products and secondary bile acids, which influence release of gut hormones that regulate body mass.

23.5 Diabetes Mellitus



Diabetes mellitus is a relatively common disease: about 9% of the U.S. population, and nearly 25% of the U.S. population over the age of 65, show some degree of abnormality in glucose metabolism that is indicative of diabetes or a tendency toward the condition. There are two major clinical classes of diabetes mellitus: **type 1 diabetes**, sometimes referred to as insulin-dependent diabetes mellitus (IDDM), and **type 2 diabetes**, or non-insulin-dependent diabetes mellitus (NIDDM), also called insulin-resistant diabetes. The discovery of insulin and its role in diabetes led to its development as a pharmaceutical, saving millions of lives (**Box 23-2**). ■

Diabetes Mellitus Arises from Defects in Insulin Production or Action



Type 1 diabetes usually begins early in life, and symptoms quickly become severe. This disease responds to insulin injection, because the metabolic defect



The Arduous Path to Purified Insulin

Millions of people with type 1 diabetes mellitus inject themselves daily with pure insulin to compensate for the lack of production of this critical hormone by their own pancreatic β cells. Insulin injection is not a cure for diabetes, but it allows people who otherwise would have died young to lead long and productive lives. The discovery of insulin, which began with an accidental observation, illustrates the combination of serendipity and careful experimentation that led to the discovery of many hormones.

In 1889, Oskar Minkowski, a young assistant at the Medical College of Strasbourg, and Josef von Mering, at the Hoppe-Seyler Institute in Strasbourg, had a friendly disagreement about whether the pancreas, known to contain lipases, was important in fat digestion in dogs. To resolve the issue, they began an experiment on the digestion of fats. They surgically removed the pancreas from a dog, but before their experiment got any farther, Minkowski noticed that the dog was now producing far more urine than normal (a common symptom of untreated diabetes). Also, the dog's urine had glucose levels far above normal (another symptom of diabetes). These findings suggested that lack of some pancreatic product caused diabetes.

Minkowski tried unsuccessfully to prepare an extract of dog pancreas that would reverse the effect of removing the pancreas—that is, would lower the urinary or blood glucose levels. Despite considerable effort, no significant progress was made in the isolation or characterization of the “antidiabetic factor” until the summer of 1921, when Frederick G. Banting, a young scientist working in the laboratory of J. J. R. MacLeod at the University of Toronto, and a student assistant, Charles Best, took up the problem. By that time, several lines of evidence pointed to a group of specialized cells in the pancreas (the islets of Langerhans; see Fig. 23-23) as the source of the antidiabetic factor, which came to be called insulin (from Latin *insula*, “island”).

Taking precautions to prevent proteolysis by the pancreatic proteases trypsin and chymotrypsin, Banting and Best (later aided by biochemist J. B. Collip) succeeded in December 1921 in preparing a purified pancreatic extract that cured the symptoms of experimentally induced diabetes in dogs. On January 25, 1922 (just one month later!), their insulin preparation was injected into Leonard Thompson, a 14-year-old boy severely ill with diabetes mellitus. Within days, the levels of ketone bodies and glucose in Thompson's urine dropped dramatically; the extract saved his life and the lives of other seriously ill children who also received these early preparations (Fig. 1). In 1923, Banting and MacLeod won the Nobel Prize for their isolation of

insulin. Banting immediately announced that he would share his prize with Best; MacLeod shared his with Collip.

By 1923, pharmaceutical companies were supplying thousands of patients throughout the world with insulin extracted from porcine pancreas. With the development of genetic engineering techniques in the 1980s, it became possible to produce unlimited quantities of human insulin by inserting the cloned human gene for insulin into a microorganism, which was then cultured on an industrial scale. Many people with diabetes are now fitted with implanted insulin pumps, which release adjustable amounts of insulin on demand to meet changing needs at meal times and during exercise. There is a reasonable prospect that, in the future, transplanted pancreatic tissue will provide a source of insulin that responds as well as a normal pancreas, releasing insulin into the bloodstream only when blood glucose rises.



FIGURE 1 A child with type 1 diabetes, before (left) and after (right) three months of treatment with an early preparation of insulin.

[H. R. Geyelin et al., *J. Metabol. Res.* 2:767, 1922.]

stems from an autoimmune destruction of pancreatic β cells and a consequent inability to produce sufficient insulin. Type 1 diabetes requires both insulin therapy and careful, lifelong control of the balance between dietary intake, activity, and insulin dose. Characteristic symptoms of untreated type 1 (and type 2) diabetes are excessive

thirst and frequent urination, leading to the intake of large volumes of water. These symptoms are due to excretion of large amounts of glucose in the urine. (“Diabetes mellitus” means “excessive excretion of sweet urine.”)

Type 2 diabetes is slower to develop (typically but not always in obese adults), and the symptoms are milder

and often go unrecognized at first. This is really a group of diseases in which the regulatory activity of insulin is disordered: insulin is produced, but some feature of the insulin-response system is defective. Many individuals with this disorder are insulin-resistant. The connection between type 2 diabetes and obesity (discussed below) is an active area of research.

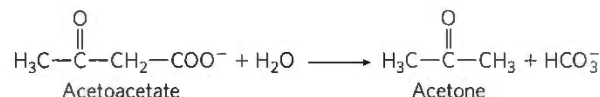
The pathology of diabetes includes cardiovascular disease, renal failure, blindness, and neuropathy. In 2019, the global mortality from diabetes was estimated at 4.2 million and rising. It is essential to understand diabetes and its relationship to obesity and to find countermeasures that prevent or reverse the damage done by this disease.

P5 Individuals with either type of diabetes are unable to take up glucose efficiently from the blood. Recall that insulin triggers the movement of GLUT4 glucose transporters to the plasma membrane in muscle and adipose tissue (see Fig. 12-23 and Box 11-1). Biochemical measurements on blood and urine samples are essential in the diagnosis and treatment of diabetes. A sensitive diagnostic criterion is the level of HbA1c, a glucose derivative of hemoglobin, which forms in the blood and reflects the average blood glucose level (see Box 7-2). Another measurement to confirm the diagnosis of diabetes is the **glucose-tolerance test**. The individual fasts overnight, then drinks a test dose of 100 g of glucose dissolved in a glass of water. The blood glucose concentration is measured before the test dose and at 30 min intervals for several hours thereafter. A healthy individual assimilates the glucose readily, the blood glucose rising to no more than about 9 or 10 mM; little or no glucose appears in the urine. In diabetes, individuals assimilate the test dose of glucose poorly; their blood glucose level rises dramatically and returns to the fasting level very slowly. Because the blood glucose levels exceed the kidney threshold (about 10 mM), glucose also appears in the urine. ■

Carboxylic Acids (Ketone Bodies) Accumulate in the Blood of Those with Untreated Diabetes

 With glucose unavailable to cells, fatty acids become the principal fuel, which leads to another characteristic metabolic change in diabetes: excessive but incomplete oxidation of fatty acids in the liver. The acetyl-CoA produced by β oxidation cannot be completely oxidized by the citric acid cycle, because the high [NADH]/[NAD⁺] ratio produced by β oxidation inhibits the cycle (recall that three steps of the cycle convert NAD⁺ to NADH). Accumulation of acetyl-CoA leads to overproduction of the ketone bodies β -hydroxybutyrate and acetoacetate, which cannot be used by extrahepatic tissues as fast as they are made in the liver (see Figs. 17-15, 17-16). In addition to β -hydroxybutyrate and acetoacetate, the blood of individuals with diabetes contains small

amounts of acetone, which results from the spontaneous decarboxylation of acetoacetate:



The overproduction of ketone bodies, called **ketosis**, results in greatly increased concentrations of ketone bodies in the blood (ketonemia) and urine (ketonuria). The ketone bodies are carboxylic acids, which ionize, releasing protons. In uncontrolled diabetes, this acid production can overwhelm the capacity of the blood's bicarbonate buffering system and produce a lowering of blood pH called **acidosis** or, in combination with ketosis, **ketoacidosis**, a potentially life-threatening condition. ■

In Type 2 Diabetes the Tissues Become Insensitive to Insulin

 In the industrialized world, where the food supply is more than adequate, there is a growing epidemic of obesity and the type 2 diabetes associated with it.

P5 The hallmark of **type 2 diabetes** is the development of insulin resistance, a state in which more insulin is needed to bring about the biological effects produced by a lower concentration of insulin in the normal, healthy state. In the early stages of the disease, pancreatic β cells secrete enough insulin to overcome the lower insulin sensitivity of muscle and liver. But the β cells eventually fail, and the lack of insulin becomes apparent in the body's inability to regulate blood glucose. The intermediate stage, preceding type 2 diabetes mellitus, is sometimes called **metabolic syndrome**. This is typified by obesity, especially in the abdomen; hypertension (high blood pressure); abnormal blood lipids (high TAG and LDL, low HDL); slightly high fasting blood glucose; and a reduced ability to clear glucose in the glucose-tolerance test. Individuals with metabolic syndrome often also show changes in blood proteins associated with abnormal clotting (high fibrinogen concentration) or inflammation (high concentration of the C-reactive peptide, not to be confused with the C peptide generated during the proteolytic maturation of insulin). About 30% of the adult population in the United States has these indicators of metabolic syndrome.

What predisposes individuals with metabolic syndrome to develop type 2 diabetes? According to the "lipid toxicity" hypothesis (**Fig. 23-42**), the action of PPAR δ on adipocytes normally keeps the cells ready to synthesize and store triacylglycerols—the adipocytes are insulin-sensitive and produce leptin, which leads to their continued intracellular deposition of TAGs. However, excess caloric intake in obese individuals causes adipocytes to become filled with TAGs, leaving adipose tissue unable to meet any further demand for TAG storage. Lipid-filled adipose tissue releases protein factors, including MCP-1

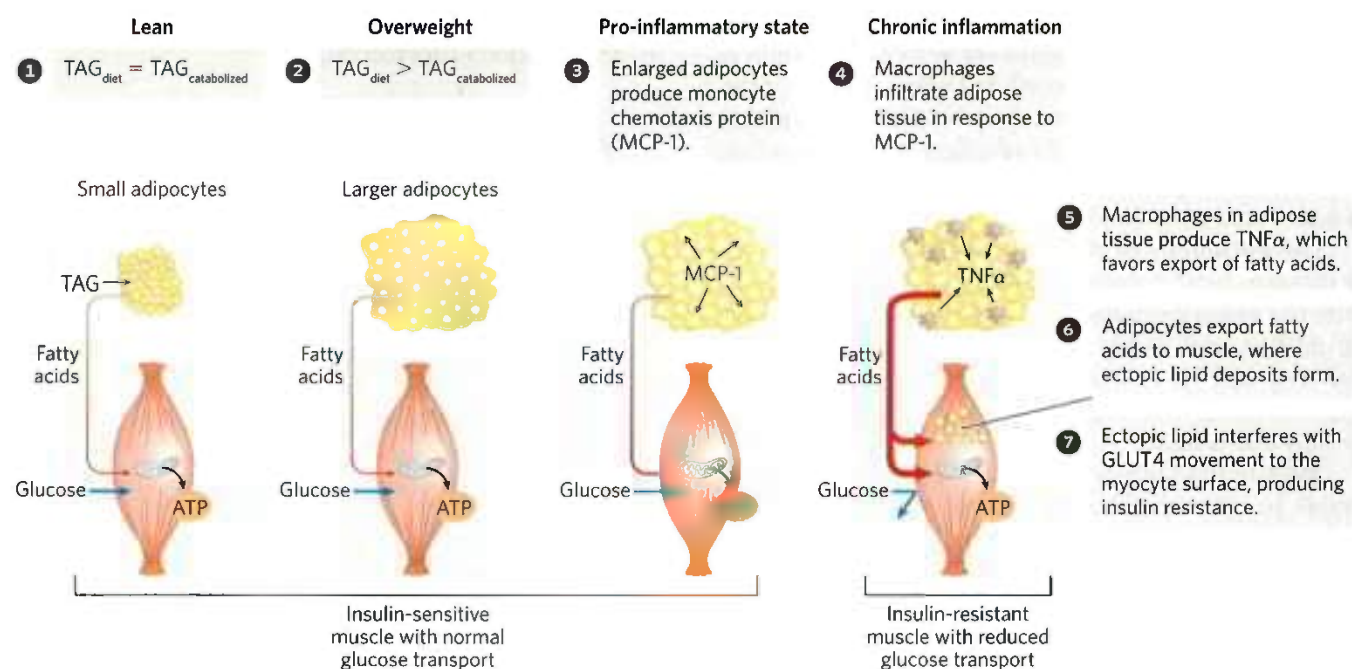


FIGURE 23-42 Overloading adipocytes with triacylglycerols triggers inflammation in fat tissue and ectopic lipid deposition and insulin resistance in muscle. In an individual of healthy body mass, dietary TAG uptake equals TAG oxidation for energy. In overweight individuals, excess caloric intake results in enlarged adipocytes, engorged with TAG and unable to store more. Enlarged adipocytes secrete MCP-1 (monocyte chemotaxis protein-1), attracting macrophages. Macrophages infiltrate the adipose tissue and produce TNF α (tumor necrosis factor α), which triggers lipid breakdown and release of fatty acids into the blood. The fatty acids enter myocytes, where they

accumulate in small lipid droplets. This ectopic lipid storage in muscle somehow causes insulin resistance, perhaps by triggering lipid-activated protein kinases that inactivate some element in the insulin-signaling pathway. GLUT4 glucose transporters leave the myocyte surface, preventing glucose entry into muscle; the myocyte has now become insulin-resistant. It cannot use blood glucose for its fuel, so fatty acids are mobilized from adipose tissue and become the primary fuel. The increased influx of fatty acids into muscle leads to further deposition of ectopic lipids. In some individuals, insulin resistance develops into type 2 diabetes. [Information from A. Guilherme et al., *Mol. Cell Biol.* 9:367, 2008, Fig. 1.]

(monocyte chemotaxis protein-1), that attract macrophages, which infiltrate the tissue and may eventually represent as much as 50% of the adipose tissue by mass. Macrophages trigger the inflammatory response mediated by TNF α release, which impairs TAG deposition in adipocytes and favors release of free fatty acids into the blood. These excess fatty acids enter liver and muscle cells, where they are converted to TAGs that accumulate as lipid droplets. This ectopic (Greek *ektos*, "out of place") deposition of TAGs inhibits GLUT4 transporter movement to the plasma membrane, which leads to insulin insensitivity in liver and muscle, the hallmark of type 2 diabetes.

According to this hypothesis, excess stored fatty acids and TAGs are toxic to liver and muscle. Some individuals are less well equipped genetically to handle this burden of ectopic lipids and are more susceptible to the cellular damage that leads to development of type 2 diabetes. Insulin resistance probably involves impairment of several of the mechanisms by which insulin acts on metabolism, which include changes in protein levels and changes in the activities of signaling enzymes and transcription factors. For example, both adiponectin synthesis in adipocytes and adiponectin levels in the blood decrease with obesity and increase with weight loss.

There are genetic factors that predispose individuals to type 2 diabetes. Although 80% of people with type 2 diabetes are obese, most obese individuals do not develop type 2 diabetes. Given the complexity of the regulatory mechanisms we have discussed in this chapter, it is not surprising that the genetics of diabetes is complex, involving interactions among variant genes and environmental factors, including diet and lifestyle. A growing number of genetic loci have been reliably linked to type 2 diabetes; variation in any of these "diabetogenes" alone would cause a relatively small increase in the likelihood of developing type 2 diabetes. ■

Type 2 Diabetes Is Managed with Diet, Exercise, Medication, and Surgery

Studies show that at least four factors improve the health of individuals with type 2 diabetes: dietary restriction, regular exercise, drugs that increase insulin sensitivity or insulin production, and surgery that reroutes food passage through the gastrointestinal tract. Dietary restriction (and accompanying weight loss) reduces the overall burden of handling fatty acids. The lipid composition of the diet influences, through

TABLE 23-7 Treatments for Type 2 Diabetes Mellitus

Intervention/treatment	Direct target	Effect of treatment
Weight loss	Adipose tissue; reduction in TAG content	Reduces lipid burden; increases capacity for lipid storage in adipose tissue; restores insulin sensitivity
Exercise	AMPK, activated by increasing [AMP]/[ATP]	Aids weight loss (see Fig. 23-34)
Bariatric surgery	Unknown	Leads to weight loss, better control of blood glucose
Sulfonylureas: glipizide (Glucotrol), glyburide (Micronase), glimepiride (Amaryl)	Pancreatic β cells; K^+ channels blocked	Stimulates insulin secretion by pancreas (see Fig. 23-24)
Biguanides: metformin (Glucophage)	AMPK, activated	Increases glucose uptake by muscle; decreases glucose production in liver
Thiazolidinediones: rosiglitazone (Avandia), pioglitazone (Actos)	PPAR γ	Stimulates expression of genes potentiating the action of insulin in liver, muscle, adipose tissue; increases glucose uptake; decreases glucose synthesis in liver
GLP-1 modulators: exenatide (Byetta), sitagliptin (Januvia), dulaglutide (Trulicity)	Glucagon-like peptide-1, dipeptide protease IV	Enhances insulin secretion by pancreas

PPARs and other transcription factors, the expression of genes that encode proteins involved in burning fat.

Exercise contributes to weight loss directly by consuming calories. Exercise also increases release of **irisin** from muscle into the blood. Irisin increases the expression of *UCP1* genes in white adipose tissue and also stimulates the development of beige adipocytes, so that even after the exercise ends, energy continues to be used in thermogenesis. Exercise activates AMPK, as does adiponectin; AMPK shifts metabolism toward fat oxidation and inhibits fat synthesis.

Several classes of drugs are used in the management of type 2 diabetes (Table 23-7). Their targets include AMPK, the K^+ channels in β cells, PPARs, and GLP receptors.

In cases of extreme obesity, dramatic weight loss can be achieved by bariatric surgery, which reroutes food movement through the stomach and small intestine. In many cases, this procedure also moderates or even reverses type 2 diabetes. In Roux-en-Y gastric bypass (RYGBP, named for César Roux, the Swiss surgeon who developed the procedure), the stomach is reduced to a small pouch attached to the esophagus, and the middle region of the small intestine (the jejunum) is attached directly to the pouch. Food bypasses most of the stomach and the duodenum, and goes primarily to the “Roux limb” of the intestine. Stomach acid and digestive enzymes travel through bypassed portions of the gut to join the food in the common channel. People who undergo RYGBP surgery not only experience dramatic weight loss but also are less hungry. Remarkably, this surgery also reverses type 2 diabetes in many cases. The explanations for these effects are likely to lie in altered communication among the gut, the brain, and other organs. This may result from changes in the kind and amount of peptide hormones (such as GLP-1 and PYY₃₋₃₆)

secreted in the intestine that signal satiety and inhibit feeding behavior. The last word has not been written on this issue. ■

SUMMARY 23.5 Diabetes Mellitus

- Metabolic syndrome, which includes obesity, hypertension, elevated blood lipids, and insulin resistance, is often the prelude to type 2 diabetes. Uncontrolled diabetes is characterized by high glucose levels in the blood and urine and the production and excretion of ketone bodies.
- In diabetes, insulin is either not produced or not recognized by the tissues, and the uptake of blood glucose is defective. Lacking access to glucose, cells rely on fatty acid oxidation, which results in ketone body formation, producing ketoacidosis.
- The insulin resistance that characterizes type 2 diabetes may be a consequence of abnormal lipid storage in muscle and liver, in response to a lipid intake that cannot be accommodated by adipose tissue.
- Effective treatments for type 2 diabetes include exercise, appropriate diet, and drugs that increase insulin sensitivity or insulin production. Surgical alteration of the digestive tract leads to weight loss and often reverses type 2 diabetes.

KEY TERMS

Terms in bold are defined in the glossary.

hormone 842	endocrine 843
neuroendocrine system 842	paracrine 843
metabotropic 843	autocrine 844
ionotropic 843	hypothalamus 845
	hepatocyte 849

white adipose tissue

(WAT) 851

adipocyte 851**brown adipose tissue**

(BAT) 852

uncoupling protein 1

(UCP1) 852

thermogenesis 852**beige adipose****tissue** 852**myocyte** 852**plasma proteins** 858ATP-gated K^+ channels 860**sulfonylurea****drugs** 861

glucagon 862

cortisol 865

body mass index

(BMI) 867

adipokines 867

leptin 867

arcuate nucleus 868**orexigenic** 868**anorexigenic** 868 α -melanocyte-stimulating hormone (α -MSH) 868

adiponectin 869

AMP-activated protein**kinase (AMPK)** 869**mTORC1 (mechanistic****target of rapamycin****complex 1)** 871**PPAR (peroxisome****proliferator-activated****receptor)** 871

ghrelin 872

endocannabinoids 873**probiotics** 874**prebiotics** 874**diabetes mellitus** 875

type 1 diabetes 875

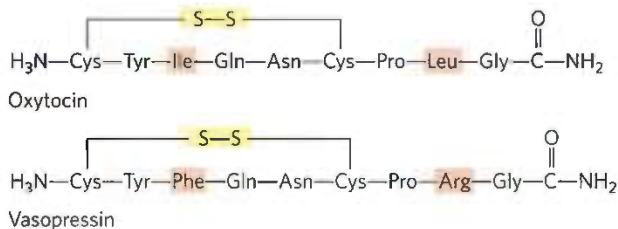
type 2 diabetes 875

glucose-tolerance

test 877

ketosis 877**acidosis** 877**ketoacidosis** 877**metabolic syndrome** 877**PROBLEMS**

1. Peptide Hormone Activity Explain how two peptide hormones as structurally similar as oxytocin and vasopressin can have such different effects (see Fig. 23-8).



2. Metabolism of Glutamate in the Brain Brain tissue takes up glutamate from the blood, transforms it into glutamine, and releases the glutamine into the blood. What does this metabolic conversion accomplish? How does this conversion take place? The amount of glutamine produced in the brain can exceed the amount of glutamate entering from the blood. How does this extra glutamine arise? (Hint: You may want to review amino acid catabolism in Chapter 18; recall that NH_4^+ is very toxic to the brain.)

3. Proteins as Fuel during Fasting When muscle proteins undergo catabolism in skeletal muscle during a fast, what are the fates of the amino acids?

4. Absence of Glycerol Kinase in Adipose Tissue The biosynthesis of triacylglycerols requires glycerol 3-phosphate. Adipocytes, specialized for the synthesis and degradation of TAGs, cannot use glycerol directly because they lack glycerol kinase, which catalyzes the reaction



How does adipose tissue obtain the glycerol 3-phosphate necessary for TAG synthesis?

5. Oxygen Consumption during Exercise A sedentary adult consumes about 0.05 L of O_2 in 10 seconds. A sprinter running a 100 m race consumes about 1 L of O_2 in 10 seconds. After finishing the race, the sprinter continues to breathe at an elevated (but declining) rate for some minutes, consuming an extra 4 L of O_2 above the amount consumed by the sedentary individual.

(a) Why does the need for O_2 increase dramatically during the sprint?

(b) Why does the demand for O_2 remain high after the sprinter finishes the race?

6. Thiamine Deficiency and Brain Function Individuals with thiamine deficiency show some characteristic neurological signs and symptoms, including loss of reflexes, anxiety, and mental confusion. Why might thiamine deficiency manifest as changes in brain function?

7. Potency of Hormones Under normal conditions, the human adrenal medulla secretes epinephrine ($C_9H_{13}NO_3$) at a rate sufficient to maintain a concentration of 10^{-10} M in circulating blood. To appreciate what that concentration means, calculate the volume of water that you would need to dissolve 1.0 g (about 1 teaspoon) of epinephrine to a concentration equal to that in blood.

8. Regulation of Hormone Levels in the Blood The half-life of most hormones in the blood is relatively short. For example, when researchers inject radioactively labeled insulin into an animal, half of the labeled hormone disappears from the blood within 30 min.

(a) What is the importance of the relatively rapid inactivation of circulating hormones?

(b) In what ways can the organism make rapid changes in the level of a circulating peptide hormone?

9. Water-Soluble versus Lipid-Soluble Hormones On the basis of their physical properties, hormones fall into one of two categories: those that are very soluble in water but relatively insoluble in lipids (e.g., epinephrine) and those that are relatively insoluble in water but highly soluble in lipids (e.g., steroid hormones). In their role as regulators of cellular activity, most water-soluble hormones do not enter their target cells. The lipid-soluble hormones, by contrast, do enter their target cells and ultimately act in the nucleus. What is the relationship between solubility, the location of receptors, and the mode of action of these two classes of hormones?

10. Metabolic Differences between Muscle and Liver in a "Fight-or-Flight" Situation When an animal confronts a "fight-or-flight" situation, the release of epinephrine promotes glycogen breakdown in the liver and skeletal muscle. The end product of glycogen breakdown in the liver is glucose; the end product in skeletal muscle is pyruvate.

(a) What is the reason for the different products of glycogen breakdown in the two tissues?

(b) What is the advantage of these specific glycogen breakdown routes to an animal that must fight or flee?



11. Excessive Amounts of Insulin Secretion: Hyperinsulinism Certain malignant tumors of the pancreas cause excessive production of insulin by β cells. Affected

individuals exhibit shaking and trembling, weakness and fatigue, sweating, and hunger.

(a) What is the effect of hyperinsulinism on the metabolism of carbohydrates, amino acids, and lipids by the liver?

(b) What are the causes of the observed symptoms? Suggest why this condition, if prolonged, leads to brain damage.

12. Thermogenesis Caused by Thyroid Hormones Thyroid hormones are intimately involved in regulating the basal metabolic rate. Liver tissue of animals given excess thyroxine shows an increased rate of O_2 consumption and increased heat output (thermogenesis), but the ATP concentration in the tissue is normal. Different explanations have been offered for the thermogenic effect of thyroxine. One is that excess thyroxine causes uncoupling of oxidative phosphorylation in mitochondria. How could such an effect account for the observations? Another explanation suggests that thermogenesis is due to an increased rate of ATP utilization by the thyroxine-stimulated tissue. Is this a reasonable explanation? Why or why not?

13. Function of Prohormones What are the possible advantages of synthesizing hormones as prohormones?

14. Sources of Glucose during Starvation The typical human adult uses about 160 g of glucose per day. Of this, the brain alone uses 120 g. The body's available reserve of glucose (~20 g of circulating glucose and ~190 g of glycogen) is adequate for about one day. After the glucose reserve has been depleted during starvation, how does the body obtain more glucose?

15. Parabiotic *ob/ob* Mice By careful surgery, researchers can connect the circulatory systems of two mice so that the same blood circulates through both animals. In these parabiotic mice, products released into the blood by one animal reach the other animal via the shared circulation. Both animals are free to eat independently. Suppose a researcher parabiotically joins a mutant *ob/ob* mouse (both copies of the *OB* gene are defective) and a normal *OB/OB* mouse (both copies of the *OB* gene are functional). What would happen to the weight of each mouse in the parabiotic pair?

16. Calculation of Body Mass Index A biochemistry professor weighs 260 lb (118 kg) and is 5 feet 8 inches (173 cm) tall. What is his body mass index (BMI)? How much weight would he have to lose to bring his BMI down to 25 (normal)?

17. Insulin Secretion Predict the effects on insulin secretion by pancreatic β cells of exposure to the potassium ionophore valinomycin. Explain your prediction.

18. Effects of a Deleted Insulin Receptor A strain of mice specifically lacking the insulin receptor of liver is found to have mild fasting hyperglycemia (blood glucose = 132 mg/dL, vs. 101 mg/dL in controls) and a more striking hyperglycemia in the fed state (glucose = 363 mg/dL, vs. 135 mg/dL in controls). The mice have higher than normal levels of glucose 6-phosphatase in the liver and elevated levels of insulin in the blood. Explain these observations.



19. Decisions on Drug Safety The drug rosiglitazone (Avandia) is effective in lowering blood glucose levels in patients with type 2 diabetes, but a few years after rosiglitazone came into widespread use, it seemed that using the drug came with an increased risk of heart attack. In response, the U.S. Food and Drug Administration (FDA) severely restricted the

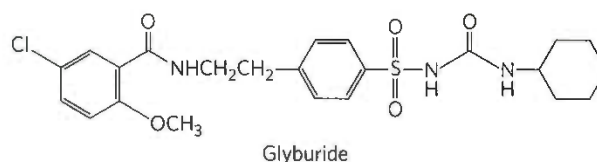
conditions under which it could be prescribed. Two years later, after additional studies had been completed, the FDA lifted the restrictions, and today rosiglitazone is available by prescription in the United States, with no special limitations. Many other countries ban it completely. If it were your responsibility to decide whether this drug should remain on the market (labeled with suitable warnings about its side effects) or should be withdrawn from the market altogether, what factors would you weigh in making your decision?



20. Type 2 Diabetes Medication The drugs acarbose (Precose) and miglitol (Glyset), used in the treatment of type 2 diabetes mellitus, inhibit α -glucosidases in the brush border of the small intestine. These enzymes degrade oligosaccharides derived from glycogen or starch to monosaccharides. Suggest a possible mechanism for the salutary effect of these drugs for individuals with diabetes. What side effects, if any, would you expect from these drugs? Why? (Hint: Review lactose intolerance, p. 523.)

DATA ANALYSIS PROBLEM

21. Cloning the Sulfonyleurea Receptor of the Pancreatic β Cell Glyburide, a member of the sulfonyleurea family of drugs, is used to treat type 2 diabetes. It binds to and closes the ATP-gated K^+ channel shown in Figures 23-26 and 23-27.



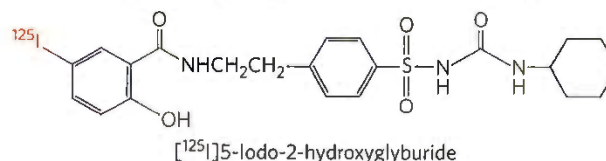
(a) Given the mechanism shown in Figure 23-27, would treatment with glyburide result in increased or decreased insulin secretion by pancreatic β cells? Explain your reasoning.

(b) How does treatment with glyburide help reduce the symptoms of type 2 diabetes?

(c) Would you expect glyburide to be useful for treating type 1 diabetes? Explain your answer.

Aguilar-Bryan and coauthors (1995) cloned the gene for the sulfonyleurea receptor (SUR) portion of the ATP-gated K^+ channel from hamsters. The research team went to great lengths to ensure that the gene they cloned was, in fact, the SUR-encoding gene. Here we explore how it is possible for researchers to demonstrate that they have cloned the gene of interest rather than another gene.

The first step was to obtain pure SUR protein. As was already known, drugs such as glyburide bind SUR with very high affinity ($K_d < 10$ nM), and SUR has a molecular weight of 140 to 170 kDa. Aguilar-Bryan and coworkers made use of the high-affinity glyburide binding to tag the SUR protein with a radioactive label that would serve as a marker to purify the protein from a cell extract. First, they made a radiolabeled derivative of glyburide, using radioactive iodine (^{125}I):



(d) In preliminary studies, the ^{125}I -labeled glyburide derivative (hereafter, [^{125}I]glyburide) was shown to have the same K_d and binding characteristics as unaltered glyburide. Why was it necessary to demonstrate this? (What alternative possibilities did it rule out?)

Even though [^{125}I]glyburide bound to SUR with high affinity, a significant amount of the labeled drug would probably dissociate from the SUR protein during purification. To prevent this, [^{125}I]glyburide had to be covalently cross-linked to SUR. There are many methods for covalent cross-linking; Aguilar-Bryan and coworkers used UV light. When aromatic molecules are exposed to short-wave UV, they enter an excited state and readily form covalent bonds with nearby molecules. By cross-linking the radiolabeled glyburide to the SUR protein, the researchers could simply track the ^{125}I radioactivity to follow SUR through the purification procedure.

The research team treated hamster HIT cells (which express SUR) with [^{125}I]glyburide and UV light, purified the ^{125}I -labeled 140 kDa protein, and sequenced its 25 residue amino-terminal segment; they found the sequence PLAFCGTENHSAAYRVDQGVLNNGC. The investigators then generated antibodies that bound to two short peptides in this sequence, one binding to PLAFCGTE and the other to HSAAYRVDQGV, and showed that these antibodies bound the purified ^{125}I -labeled 140 kDa protein.

(e) Why was it necessary to include this antibody-binding step?

Next, the researchers designed PCR primers based on the sequences above, and then cloned a gene from a hamster cDNA library that encoded a protein with these sequences (see Chapter 9 on biotechnology methods). The cloned putative SUR cDNA hybridized to an mRNA of the appropriate length that was present in cells known to contain SUR. The putative SUR cDNA did not hybridize to any mRNA fraction of the mRNAs isolated from hepatocytes, which do not express SUR.

(f) Why was it necessary to include this putative SUR cDNA-mRNA hybridization step?

Finally, the cloned gene was inserted into and expressed in COS cells, which do not normally express the SUR gene. The investigators mixed these cells with [^{125}I]glyburide, with or without a large excess of unlabeled glyburide, exposed the cells to UV light, and measured the radioactivity of the 140 kDa protein produced. Their results are shown in the table.

Experiment	Cell type	Added putative SUR cDNA?	Added excess unlabeled glyburide?	^{125}I label in 140 kDa protein
1	HIT	No	No	+++
2	HIT	No	Yes	-
3	COS	No	No	-
4	COS	Yes	No	+++
5	COS	Yes	Yes	-

(g) Why was no ^{125}I -labeled 140 kDa protein found in experiment 2?

(h) How would you use the information in the table to argue that the cDNA encoded SUR?

(i) What other information would you want to collect to be more confident that you had cloned the SUR gene?

Reference

Aguilar-Bryan, L., C.G. Nichols, S.W. Wechsler, J.P. Clement, IV, A.E. Boyd, III, G. González, H. Herrera-Sosa, K. Nguy, J. Bryan, and D.A. Nelson. 1995. Cloning of the β cell high-affinity sulfonylurea receptor: a regulator of insulin secretion. *Science* 268:423-426.

INFORMATION PATHWAYS

PART OUTLINE

- 24** Genes and Chromosomes 885
- 25** DNA Metabolism 914
- 26** RNA Metabolism 960
- 27** Protein Metabolism 1005
- 28** Regulation of Gene Expression 1054

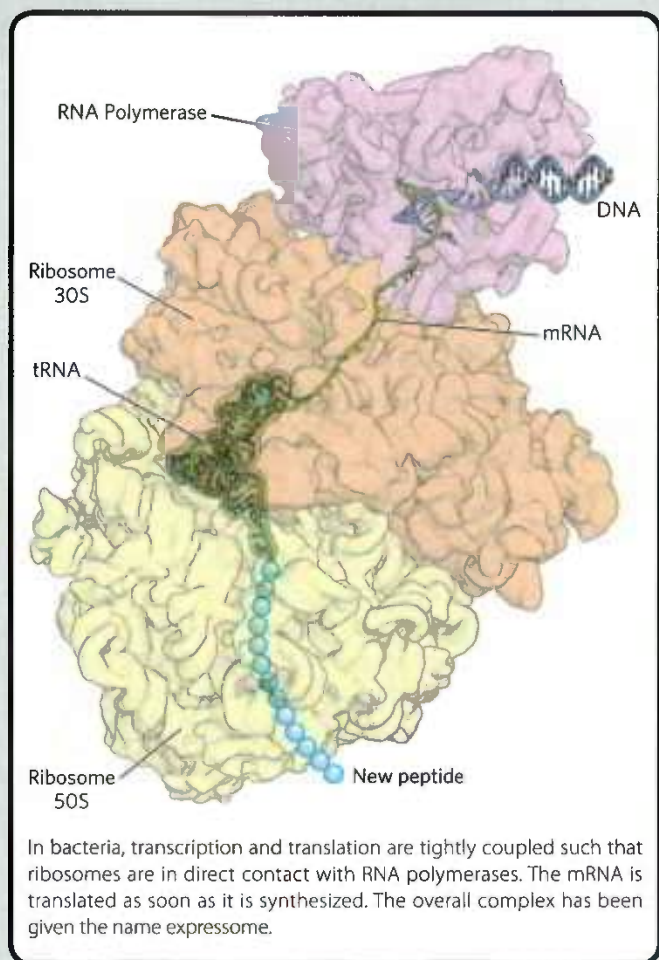
The third and final part of this book explores the biochemical mechanisms underlying the apparently contradictory requirements for both genetic continuity and the evolution of living organisms. What is the molecular nature of genetic material? How is genetic information transmitted from one generation to the next with high fidelity? How do the rare changes in genetic material that are the raw material of evolution arise? How is genetic information ultimately expressed in the amino acid sequences of the astonishing variety of protein molecules in a living cell?

Today's understanding of information pathways has arisen from the convergence of genetics, physics, and chemistry in modern biochemistry. This convergence was epitomized by the discovery of the double-helical structure of DNA, postulated by James Watson and Francis Crick in 1953. Genetic theory contributed the concept of coding by genes. Physics permitted the determination of molecular structure by x-ray diffraction analysis. Chemistry revealed the composition of DNA. The profound impact of the Watson-Crick hypothesis arose from its ability to account for a wide range of observations derived from studies in these diverse disciplines.

This discovery revolutionized our understanding of the structure of DNA and inevitably stimulated questions about its function. The double-helical structure itself clearly suggested how DNA might be copied so that the information

it contains can be transmitted from one generation to the next. Clarification of how the information in DNA is converted into functional proteins came with the discoveries of messenger RNA and transfer RNA and with the deciphering of the genetic code.

These and other major advances gave rise to the central dogma of molecular biology, comprising the three major processes in the cellular utilization of genetic information. The first is replication, the copying of parental DNA to form daughter DNA molecules with identical nucleotide sequences. The second is transcription, the process by which parts of the genetic message encoded in DNA are copied precisely into RNA. The third is translation, whereby the genetic message encoded in messenger RNA is translated on the ribosomes into a polypeptide with a particular sequence of amino acids.



Part III explores these and related processes. In Chapter 24 we examine the structure, topology, and packaging of chromosomes and genes. The processes underlying the central dogma are elaborated in Chapters 25 through 27. Finally, in Chapter 28, we turn to regulation, examining how the expression of genetic information is controlled.

A major theme running through these chapters is the added complexity inherent in the biosynthesis of macromolecules that contain information. Assembling nucleic acids and proteins with particular sequences of nucleotides and amino acids represents nothing less than preserving the faithful expression of the template upon which life itself is based. We might expect the formation of phosphodiester bonds in DNA or peptide bonds in proteins to be a trivial feat for cells, given the arsenal of enzymatic and chemical tools described in Part II. However, the framework of patterns and rules established in our examination of metabolic pathways thus far must be enlarged considerably to take into account molecular information. Bonds must be formed between *particular* subunits in informational biopolymers, avoiding either the occurrence or the persistence of sequence errors. This requirement has an enormous impact on the thermodynamics, chemistry, and enzymology of the biosynthetic processes. Formation of a peptide bond requires an energy input of only about 21 kJ per mole of bonds and can be catalyzed by relatively simple enzymes. But to synthesize a bond between two specific amino acids at a particular place in a polypeptide, the cell invests about 125 kJ/mol while making use of more than 200 enzymes, RNA molecules, and specialized proteins. The chemistry involved in peptide bond formation does not change because of this requirement, but additional processes are layered over the basic reaction to ensure that the peptide bond is formed between particular amino acids. Biological information is expensive.

The dynamic interaction between nucleic acids and proteins is another central theme of Part III. Regulatory and catalytic RNA molecules are gradually taking a more prominent place in our understanding of these pathways (discussed in Chapters 26 through 28). However, most of the processes that make up the pathways of cellular information flow are catalyzed and regulated by proteins. An understanding of these enzymes and other proteins can have practical as well as intellectual rewards, because they form the basis of recombinant DNA technology (introduced in Chapter 9).

Evolution again constitutes an overarching theme. Many of the processes outlined in Part III can be traced back billions of years, and a few can be traced to LUCA, the last universal common ancestor. The ribosome, most of the translational apparatus, and some parts of the transcriptional machinery are shared by every living organism on this planet. Genetic information is a kind of molecular clock that can help define ancestral relationships among species. Shared information pathways connect humans to every other species now living on Earth, and to all species that came before. Exploration of these pathways is allowing scientists to slowly open the curtain on the first act — the events that may have heralded the beginning of life on Earth.

GENES AND CHROMOSOMES

24.1 Chromosomal Elements 885

24.2 DNA Supercoiling 891

24.3 The Structure of Chromosomes 898

DNA molecules are typically the largest macromolecules in any cell, often many orders of magnitude longer than the cells or viral particles that contain them (Fig. 24-1). The extraordinary degree of organization required for the tertiary packaging of DNA into **chromosomes** — the repositories of genetic information, is the focus of this chapter. It also gives rise to the principles around which the chapter is organized:

- P1 Chromosomes include dedicated sequences that ensure their replication, transcription, packaging, and transmission from one generation to the next.** They are more than a long stretch of protein-encoding genes.
- P2 Chromosomes are large.** To constrain them in a small space can require multiple layers and multiple modes of tertiary structure.
- P3 Chromosomes in all cells are maintained in a state of torsional stress.** DNA is underwound relative to the stable B-form structure, facilitating both the packaging of DNA and access to the genetic information contained within it.
- P4 Specialized proteins and RNAs maintain chromosome structure.** Topoisomerases control DNA underwinding. Histones, condensins, cohesins, and other DNA-binding proteins provide scaffolds to organize chromosome structure. Certain long, noncoding RNAs also play important roles in chromosome structure and function.

The chapter begins with an examination of the elements that make up viral and cellular chromosomes, and then considers chromosomal size and organization. We then discuss DNA topology, describing the coiling and supercoiling of DNA molecules. Finally, we consider the protein-DNA interactions that organize chromosomes into compact structures.



FIGURE 24-1 Bacteriophage T2 protein coat surrounded by its single, linear molecule of DNA. The DNA was released by lysing the bacteriophage particle in distilled water and allowing the DNA to spread on the water surface. An undamaged T2 bacteriophage particle consists of a head structure that tapers to a tail by which the bacteriophage attaches itself to the outer surface of a bacterial cell. All the DNA shown in this electron micrograph is normally packaged inside the phage head. [Republished with permission of Elsevier, from A. K. Kleinschmidt et al. (1962), "Preparation and length measurements of the total deoxyribonucleic acid content of T2 bacteriophages," *Biochim. Biophys. Acta* 61:857–864; permission conveyed through Copyright Clearance Center, Inc.]

24.1 Chromosomal Elements

Cellular DNA contains genes and intergenic regions, both of which may serve functions vital to the cell. The more complex genomes, such as those of eukaryotic cells, demand increased levels of chromosomal organization, and this is reflected in the structural features of the chromosomes. We begin by considering the different types of DNA sequences and structural elements within chromosomes.

Genes Are Segments of DNA That Code for Polypeptide Chains and RNAs

Our understanding of genes has evolved tremendously over the past century. Classically, a gene was defined as a portion of a chromosome that determines or affects a single character or **phenotype** (visible property), such as eye color. George Beadle and Edward Tatum proposed a molecular definition of a gene in 1940. After exposing spores of the fungus *Neurospora crassa* to x-rays and other agents now known to damage DNA and cause alterations in DNA sequence (**mutations**), they detected mutant fungal strains that lacked one or another specific enzyme, sometimes resulting in the failure of an entire metabolic pathway. Beadle and Tatum concluded that a gene is a segment of genetic material that determines, or codes for, one enzyme: the **one gene–one enzyme** hypothesis. Later this concept was broadened to **one gene–one polypeptide**, because many genes code for a protein that is not an enzyme or for one polypeptide of a multisubunit protein.

The modern biochemical definition of a gene is even more precise. A **gene** is all the DNA that encodes the primary sequence of some final gene product, which can be either a polypeptide or an RNA with a structural or catalytic function. DNA also contains other segments or sequences that have a purely regulatory function.

Regulatory sequences provide signals that may denote the beginning or the end of genes, or influence the transcription of genes, or function as initiation points for replication or recombination (Chapter 28). Some genes can be expressed in different ways to generate multiple gene products from a single segment of DNA; the special transcriptional and translational mechanisms that allow this are described in Chapters 26 through 28.

We can estimate directly the minimum overall size of genes that encode proteins. As described in detail in Chapter 27, each amino acid of a polypeptide chain is coded for by a sequence of three consecutive nucleotides in a single strand of DNA (Fig. 24-2), with these “codons” arranged in a sequence that corresponds to the sequence of amino acids in the polypeptide that the gene encodes. A polypeptide chain of 350 amino acid residues (an average-size chain) corresponds to 1,050 base pairs (bp) of coding DNA. Many genes in eukaryotes and a few in bacteria and archaea are interrupted by noncoding DNA segments and are therefore considerably longer than this simple calculation would suggest.

How many genes are in a single chromosome? The *Escherichia coli* chromosome is a circular DNA molecule (in the sense of an endless loop rather than a perfect circle) with 4,641,652 bp. These base pairs encode about 4,300 genes for proteins and more than 200 genes for structural or catalytic RNA molecules. Among eukaryotes, the approximately 3.1 billion base pairs of the human genome include approximately 20,000 genes on the 24 different chromosomes.

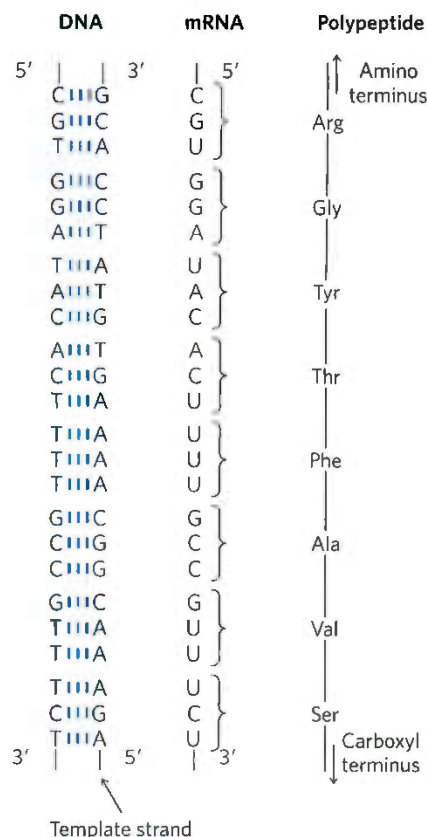


FIGURE 24-2 Colinearity of the coding nucleotide sequences of DNA and mRNA and the amino acid sequence of a polypeptide chain. The triplets of nucleotide units in DNA determine the amino acids in a protein through the intermediary mRNA. One of the DNA strands serves as a template for synthesis of mRNA, which has nucleotide triplets (codons) complementary to those of the DNA. In some bacterial and many eukaryotic genes, coding sequences are interrupted at intervals by regions of noncoding sequences (called introns).

DNA Molecules Are Much Longer than the Cellular or Viral Packages That Contain Them

Chromosomal DNAs are often many orders of magnitude longer than the cells or viruses in which they are located (Fig. 24-1; Table 24-1). This is true of every class of organism or viral parasite.

Viruses Viruses are not free-living organisms; rather, they are infectious parasites that use the resources of a host cell to carry out many of the processes they require to propagate. Many viral particles consist of no more than a genome (usually a single RNA or DNA molecule) surrounded by a protein coat.

Almost all plant viruses and some bacterial and animal viruses have RNA genomes. These genomes tend to be particularly small. For example, the genomes of mammalian retroviruses such as HIV consist of 9,000 nucleotides of single-stranded RNA.

The genomes of DNA viruses vary greatly in size. Many viral DNAs are circular for at least part of their life

TABLE 24-1 The Sizes of DNA and Viral Particles for Some Bacterial Viruses (Bacteriophages)

Virus	Size of viral DNA (bp)	Length of viral DNA (nm)	Long dimension of viral particles (nm)
ϕ X174	5,386	1,939	25
T7	39,936	14,377	78
λ (lambda)	48,502	17,460	190
T4	168,889	60,800	210

Note: Data on size of DNA are for the replicative form (double-stranded). The contour length is calculated assuming that each base pair occupies a length of 3.4 Å (see Fig. 8-13).

cycle. During viral replication within a host cell, specific types of viral DNA called **replicative forms** may appear; for example, many linear DNAs become circular and all single-stranded DNAs become double-stranded. A typical medium-size DNA virus is bacteriophage λ (lambda), which infects *E. coli*. In its replicative form inside cells, λ DNA is a circular double helix. This double-stranded DNA contains 48,502 bp and has a contour length of 17.5 μ m. Bacteriophage ϕ X174 is much smaller; the DNA in the viral particle is a single-stranded circle, and the double-stranded replicative form contains 5,386 bp. **P2** Although viral genomes are small, the contour lengths of their DNAs are typically hundreds of times longer than the long dimensions of the viral particles that contain them (Table 24-1).

Bacteria A single *E. coli* cell contains almost 100 times as much DNA as a bacteriophage λ particle. The chromosome of an *E. coli* cell is a single, double-stranded circular DNA molecule. Its 4,641,652 bp have a contour length of about 1.7 mm, some 850 times the length of the *E. coli* cell (Fig. 24-3). In addition to the very large, circular DNA chromosome in their nucleoid, many bacteria contain one or more small circular DNA molecules that are free in the cytosol. These extrachromosomal elements are called **plasmids** (Fig. 24-4; see also p. 305). Most plasmids are only a few thousand base pairs long, but some contain up to 400,000 bp. They carry genetic information and undergo replication to yield daughter plasmids, which pass into the daughter cells at cell division. Plasmids have been found in yeast and other fungi as well as in bacteria.

In many cases plasmids confer no obvious advantage on their host, and their sole function seems to be self-propagation. However, some plasmids carry genes that are useful to the host bacterium. For example, some plasmid genes make a host bacterium resistant to antibacterial agents. Plasmids carrying the gene for the enzyme β -lactamase confer resistance to β -lactam antibiotics such as penicillin, ampicillin, and amoxicillin (see Fig. 6-34). These and similar plasmids may pass from an antibiotic-resistant cell to an antibiotic-sensitive cell of the same or another bacterial species, making the recipient cell antibiotic-resistant. The extensive use of antibiotics in some human populations and in animal feeds has served as a strong selective force, encouraging

the spread of antibiotic resistance-coding plasmids (as well as transposable elements, described below, that harbor similar genes) in disease-causing bacteria.

Eukaryotes A yeast cell, one of the simplest eukaryotes, has 2.6 times more DNA in its genome than an *E. coli* cell (Table 24-2). Cells of *Drosophila*, the fruit fly used in classical genetic studies, contain more than 35 times as much DNA as *E. coli* cells, and human cells have almost 700 times as much. The cells of many plants and amphibians contain even more. The genetic material of eukaryotic cells is apportioned into chromosomes, the diploid ($2n$) number depending on the species. A human somatic cell, for example, has 46 chromosomes (Fig. 24-5). Each chromosome of a eukaryotic cell, such as that shown in Figure 24-5a, contains a single, very large, duplex DNA molecule. The DNA molecules in the 24 different types of human chromosomes (22 matching pairs of autosomes plus the X and Y sex chromosomes) vary in length over a 25-fold range. Each type of chromosome in eukaryotes carries a characteristic set of genes.

P2 DNA molecules of one human genome (22 chromosomes plus X and Y), placed end to end, would extend for about a meter. Most human cells are diploid, so each cell contains a total of 2 m of DNA. An adult human body contains approximately 10^{14} cells and thus a total DNA length of 2×10^{11} km. Compare this with the circumference of the earth (4×10^4 km) or the distance between the earth and the sun (1.5×10^8 km)—a dramatic illustration of the extraordinary degree of DNA compaction in our cells.

Eukaryotic cells also have organelles, mitochondria and chloroplasts, that contain DNA. Mitochondrial DNA (mtDNA) molecules are much smaller than the nuclear chromosomes. In animal cells, mtDNA contains fewer than 20,000 bp (16,569 bp in human mtDNA) and is a circular duplex. Each mitochondrion typically has 2 to 10 copies of this mtDNA molecule, and the number can rise to hundreds in certain cells of an embryo that is undergoing cell differentiation. Plant cell mtDNA ranges in size from 200,000 to 2,500,000 bp. Chloroplast DNA (cpDNA) also exists as circular duplexes and ranges in size from 120,000 to 160,000 bp. Mitochondrial and chloroplast DNAs have an evolutionary origin in the chromosomes of ancient bacteria that gained

FIGURE 24-3 A bacterial cell and its DNA. The length of the *E. coli* chromosome (1.7 mm), depicted in linear form, relative to the length of a typical *E. coli* cell (2 μm).

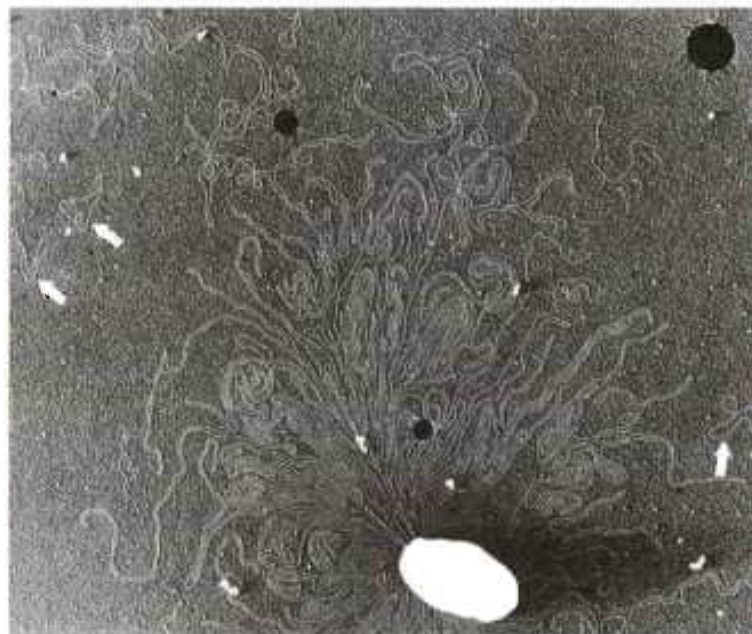
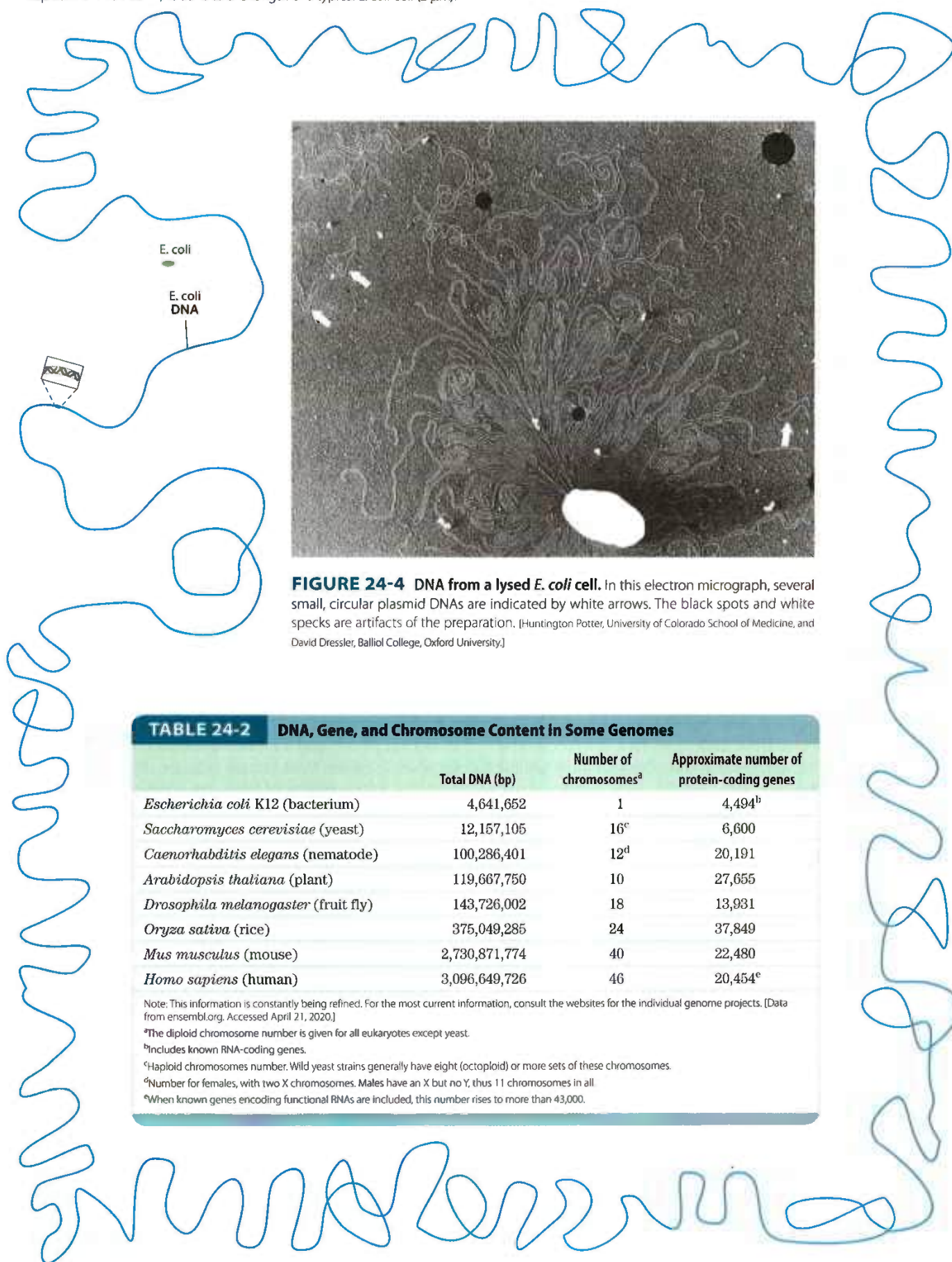


FIGURE 24-4 DNA from a lysed *E. coli* cell. In this electron micrograph, several small, circular plasmid DNAs are indicated by white arrows. The black spots and white specks are artifacts of the preparation. [Huntington Potter, University of Colorado School of Medicine, and David Dressler, Balliol College, Oxford University.]

TABLE 24-2 DNA, Gene, and Chromosome Content in Some Genomes

	Total DNA (bp)	Number of chromosomes ^a	Approximate number of protein-coding genes
<i>Escherichia coli</i> K12 (bacterium)	4,641,652	1	4,494 ^b
<i>Saccharomyces cerevisiae</i> (yeast)	12,157,105	16 ^c	6,600
<i>Caenorhabditis elegans</i> (nematode)	100,286,401	12 ^d	20,191
<i>Arabidopsis thaliana</i> (plant)	119,667,750	10	27,655
<i>Drosophila melanogaster</i> (fruit fly)	143,726,002	18	13,931
<i>Oryza sativa</i> (rice)	375,049,285	24	37,849
<i>Mus musculus</i> (mouse)	2,730,871,774	40	22,480
<i>Homo sapiens</i> (human)	3,096,649,726	46	20,454 ^e

Note: This information is constantly being refined. For the most current information, consult the websites for the individual genome projects. [Data from ensembl.org. Accessed April 21, 2020.]

^aThe diploid chromosome number is given for all eukaryotes except yeast.

^bIncludes known RNA-coding genes.

^cHaploid chromosome number. Wild yeast strains generally have eight (octoploid) or more sets of these chromosomes.

^dNumber for females, with two X chromosomes. Males have an X but no Y, thus 11 chromosomes in all.

^eWhen known genes encoding functional RNAs are included, this number rises to more than 43,000.

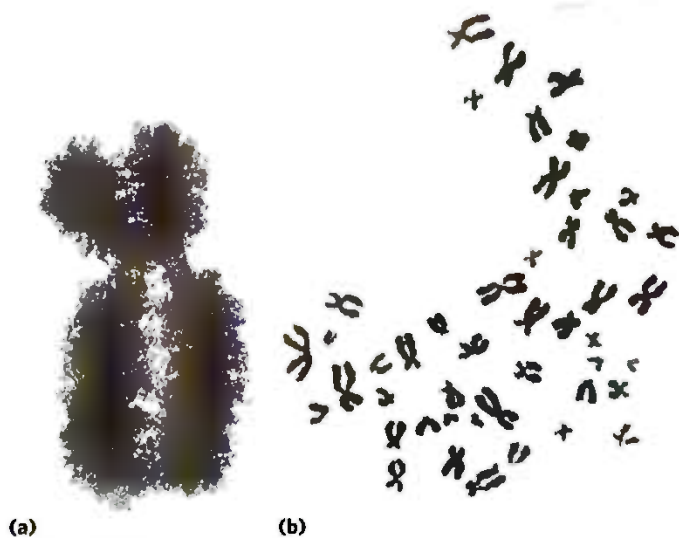


FIGURE 24-5 Eukaryotic chromosomes. (a) A pair of linked and condensed sister chromatids from a Chinese hamster ovary cell. Eukaryotic chromosomes are in this state after replication at metaphase during mitosis. (b) A complete set of chromosomes from a leukocyte from one of the authors. There are 46 chromosomes in every normal human somatic cell. [(a) Don W. Fawcett/Science Source. (b) © Michael M. Cox.]

access to the cytoplasm of host cells and became the precursors of these organelles (see Fig. 1-37). Mitochondrial DNA codes for the mitochondrial tRNAs and rRNAs and for a few mitochondrial proteins. More than 95% of mitochondrial proteins are encoded by nuclear DNA. Mitochondria and chloroplasts divide when the cell divides. Their DNA is replicated before and during division, and the daughter DNA molecules pass into the daughter organelles.

Eukaryotic Genes and Chromosomes Are Very Complex

Many bacterial species have only one chromosome per cell and, in nearly all cases, each chromosome contains only one copy of each gene. A very few genes, such as those for rRNAs, are repeated several times. Genes and regulatory sequences account for almost all the DNA in bacteria. Moreover, almost every gene is precisely colinear with the amino acid sequence (or RNA sequence) it encodes throughout its entire length (Fig. 24-2).

The organization of genes in eukaryotic DNA is structurally and functionally much more complex. Studies of eukaryotic chromosome structure and the sequencing of entire eukaryotic genomes have yielded many surprises.

P1 Many, if not most, eukaryotic genes have a distinctive structural feature: the colinearity of the DNA and amino acid sequence is periodically broken by intervening segments of DNA that do not code for the amino acid sequence of the polypeptide product. Such nontranslated DNA segments in genes are called **introns**, and the coding segments are called **exons**. Few bacterial genes contain introns. In higher eukaryotes, the typical gene has much more intron sequence than sequences devoted to exons. For example, in the gene coding for the single polypeptide chain of ovalbumin, an avian egg protein (Fig. 24-6), the introns are much longer than the exons; altogether, the seven introns make up 85% of the gene's DNA. The gene encoding the hemoglobin β subunit has only two introns, but again they are larger than the exons. Genes for histones seem to have no introns. In many cases the function of introns is not clear. In total, only about 1.5% of human DNA is "coding" or exon DNA, carrying sequence information for protein products. However, when the much larger introns are included as functional elements in the gene and their length is included, as much as 30% of the human genome consists of protein-coding genes.

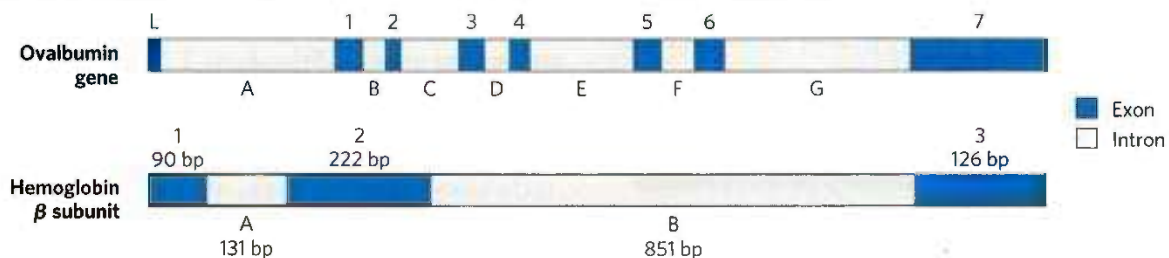


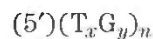
FIGURE 24-6 Introns in two eukaryotic genes. The gene for ovalbumin has seven introns (A to G), splitting the coding sequences into eight exons (L, and 1 to 7). The gene for the β subunit of hemoglobin has two introns and three exons, including one intron that alone contains more than half the base pairs of the gene.

A great deal of work remains to be done to understand the genomic sequences that do not correspond to protein-encoding genes. Much of the DNA that is not within genes is made up of repeated sequences of several kinds. These include transposable elements (transposons), molecular parasites that account for nearly half of the DNA in the human genome (see Fig. 9-25a and Chapters 25 and 26), and genes encoding functional RNA molecules of many types.

Approximately 3% of the human genome consists of **highly repetitive sequences**, also referred to as **simple-sequence DNA** or **simple sequence repeats (SSR)**. These short sequences, generally less than 10 bp long, are sometimes repeated millions of times per cell. The simple-sequence DNA is also called **satellite DNA**, so named because its unusual base composition often causes it to migrate as “satellite” bands (separated from the rest of the DNA) when fragmented cellular DNA samples are centrifuged in a cesium chloride density gradient. Studies suggest that simple-sequence DNA does not encode proteins or RNAs. The functional importance of the highly repetitive DNA has been defined in at least some cases. Much of it is associated with two crucial features of eukaryotic chromosomes: centromeres and telomeres.

P1 The **centromere** (Fig. 24-7) is a sequence of DNA that functions during cell division as an attachment point for proteins that link the chromosome to the mitotic spindle. This attachment is essential for the equal and orderly segregation of chromosome sets to daughter cells. The centromeres of *Saccharomyces cerevisiae* have been isolated and studied. The sequences essential to centromere function are about 130 bp long and are very rich in A=T pairs. The centromeric sequences of higher eukaryotes are much longer and, unlike those of yeast, generally consist of thousands of tandem copies of one or several sequences of 5 to 10 bp, in the same orientation.

P1 **Telomeres** (Greek *telos*, “end”) are sequences at the ends of eukaryotic chromosomes that help stabilize the chromosome. Telomeres end with multiple repeated sequences of the form



where x and y are generally between 1 and 4 (Table 24-3). The number of telomere repeats, n , is in the range of 20 to 100 for most single-celled eukaryotes and is generally

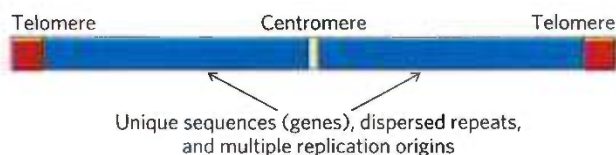


FIGURE 24-7 Important structural elements of a yeast chromosome.

TABLE 24-3 Telomere Sequences

Organism	Telomere repeat sequence
<i>Homo sapiens</i> (human)	(TTAGGG) _n
<i>Tetrahymena thermophila</i> (ciliated protozoan)	(TTGGGG) _n
<i>Saccharomyces cerevisiae</i> (yeast)	(T(G) ₁₋₃ (TG) ₂₋₃) _n
<i>Arabidopsis thaliana</i> (plant)	(TTTAGGG) _n

more than 1,500 in mammals. The ends of a linear DNA molecule cannot be routinely replicated by the cellular replication machinery (which may be one reason why bacterial DNA molecules are circular). Repeated telomeric sequences are added to eukaryotic chromosome ends primarily by the enzyme telomerase (see Fig. 26-35).

Artificial chromosomes (Chapter 9) have been constructed as a means of better understanding the functional significance of many structural features of eukaryotic chromosomes. A reasonably stable artificial linear chromosome requires only three components: a centromere, a telomere at each end, and sequences that allow the initiation of DNA replication. Yeast artificial chromosomes (YACs; see Fig. 9-6) have been developed as a research tool in biotechnology. Similarly, human artificial chromosomes (HACs) are being developed for the treatment of genetic diseases. These may eventually provide a new path to the intracellular replacement of missing or defective gene products, or somatic gene therapy.

SUMMARY 24.1 Chromosomal Elements

- Genes are segments of a chromosome that contain the information for a functional polypeptide or RNA molecule. In addition to genes, chromosomes contain a variety of regulatory sequences involved in replication, transcription, and other processes.
- Genomic DNA and RNA molecules are generally orders of magnitude longer than the viral particles or cells that contain them.
- Many genes in eukaryotic cells (but few in bacteria and archaea) are interrupted by noncoding sequences, or introns. The coding segments separated by introns are called exons. Only about 1.5% of human genomic DNA encodes proteins; even when introns are included, less than one-third of human genomic DNA consists of genes. Much of the remainder consists of repeated sequences of various types. Nucleic acid parasites known as transposons account for about half of the human genome. Eukaryotic chromosomes have two important special-function repetitive DNA sequences: centromeres, which are attachment points for the mitotic spindle, and telomeres, located at the ends of chromosomes.

24.2 DNA Supercoiling

How is cellular or viral DNA compacted into the cells or viral coats that contain it in a way that still permits access to the information in the DNA? The extreme compaction implies a high degree of structural organization. First, the many negative charges of the phosphoryl groups in the DNA backbone must be neutralized. Cations, particularly Mg^{2+} ions, and a class of molecules called polyamines provide multiple positive counterions to permit DNA compaction. Polyamines are derived from the amino acid ornithine (see Box 6-1). The second key to DNA compaction is a DNA structural alteration known as **supercoiling**. All cells maintain their DNA in a state that is **underwound**—having fewer right-handed helical turns per given length of DNA—than B-form DNA. The underwinding places structural strain on the DNA, causing it to twist upon itself. Supercoiling affects and is affected by processes such as replication and transcription. We introduce it here as a prelude to a broader discussion of DNA metabolism.

P3 “Supercoiling” means the coiling of a coil. An old-fashioned telephone cord, for example, was typically a coiled wire. The path taken by the wire between the base of the phone and the receiver often included one or more supercoils (Fig. 24-8). DNA is coiled in the form of a double helix, with both strands of the DNA coiling around an axis (Fig. 24-9) produces DNA supercoiling. As detailed below, DNA supercoiling is generally a manifestation of structural strain. When there is no net bending of the DNA axis upon itself, the DNA is said to be in a **relaxed** state.

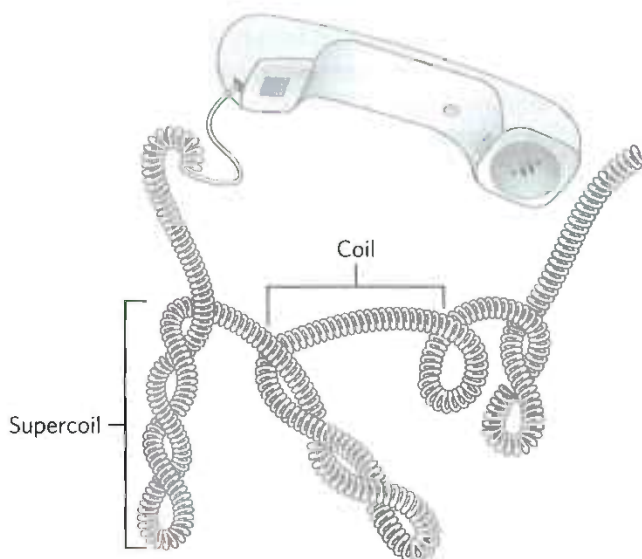


FIGURE 24-8 Supercoils. An old-fashioned phone cord is coiled like a DNA helix, and the coiled cord can itself coil in a supercoil. Examining phone cords helped Jerome Vinograd and his colleagues to the insight that many properties of small circular DNAs can be explained by supercoiling. They first detected DNA supercoiling—in small circular viral DNAs—in 1965.

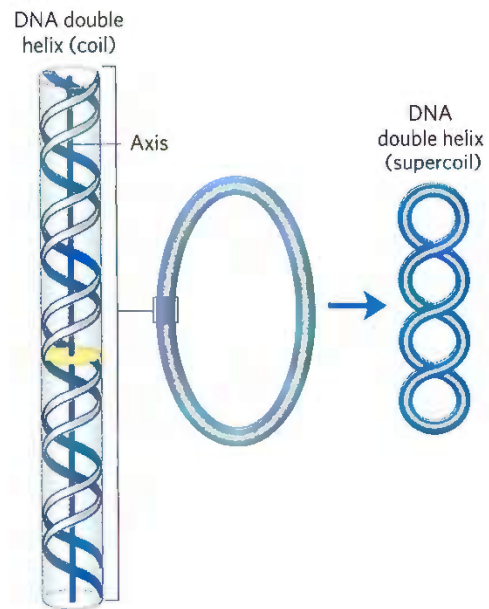


FIGURE 24-9 Supercoiling of DNA. When the axis of the DNA double helix is coiled on itself, it forms a new helix (superhelix). The DNA superhelix is usually called a supercoil.

As we shall see, the supercoiling that occurs in cells reflects underwinding of the DNA, facilitating the separation of strands required for the processes of replication and transcription (Fig. 24-10).

That a DNA molecule would bend on itself and become supercoiled in tightly packaged cellular DNA would seem logical, and perhaps even trivial, were it not for one additional fact: many circular DNA molecules remain highly supercoiled even after they are extracted and purified, freed from protein and other cellular components. **P3** This indicates that supercoiling is an intrinsic property of DNA tertiary structure. It occurs in all cellular DNAs and is highly regulated by each cell.

Several measurable properties of supercoiling have been established, and the study of supercoiling has provided many insights into DNA structure and function. This work has drawn heavily on concepts derived from a branch of mathematics called **topology**, the study of the properties of an object that do not change under continuous deformations. For DNA, continuous deformations include conformational changes due to thermal motion or due to interaction with proteins or other molecules; discontinuous deformations involve DNA strand breakage. For circular DNA molecules, a topological property is one that is unaffected by deformations of the DNA strands as long as no breaks are introduced. Topological properties are changed only by breakage and rejoining of the backbone of one or both DNA strands.

We now examine the fundamental properties and physical basis of supercoiling.

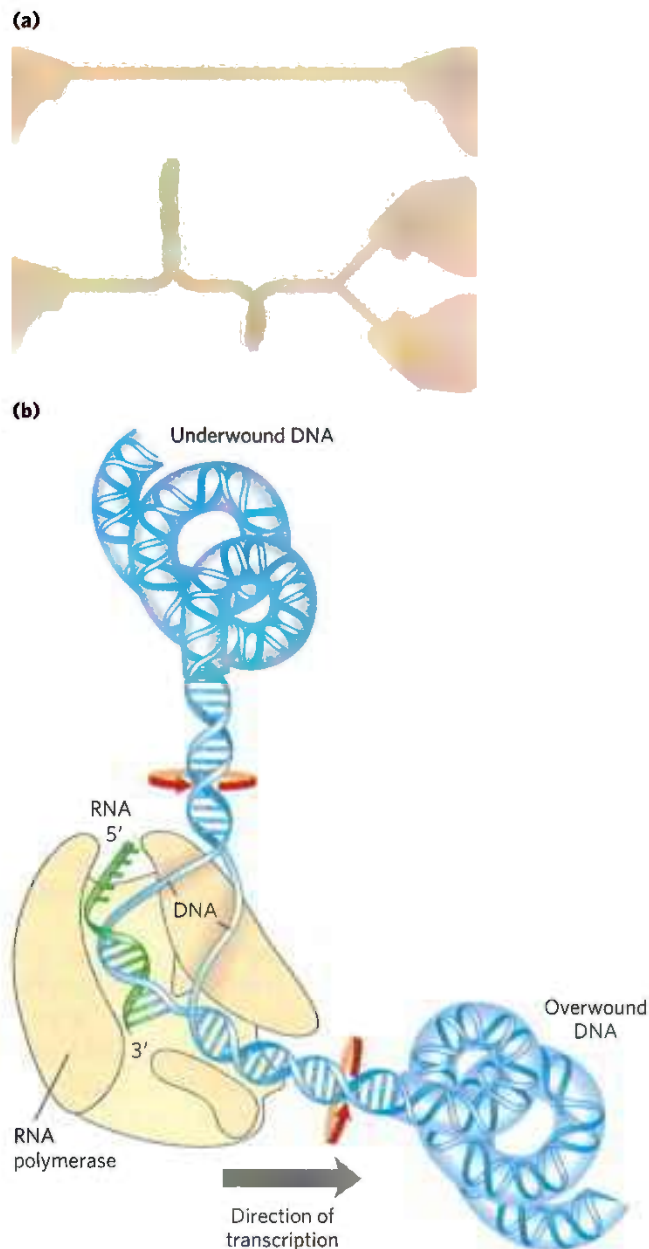


FIGURE 24-10 The effects of replication and transcription on DNA supercoiling. Because DNA is a double-helical structure, strand separation leads to added stress and supercoiling if the DNA is constrained (not free to rotate) ahead of the strand separation. (a) The general effect can be illustrated by twisting two strands of a rubber band about each other to form a double helix. If one end is constrained, separating the two strands at the other end will lead to twisting. (b) In a DNA molecule, the progress of a DNA polymerase or RNA polymerase (as shown here) along the DNA involves separation of the strands. As a result, the DNA becomes overwound ahead of the enzyme (upstream) and underwound behind it (downstream). Red arrows indicate the direction of winding.

Most Cellular DNA Is Underwound

To understand supercoiling, we must first focus on the properties of small circular DNAs such as plasmids and small viral DNAs. When these DNAs have no breaks in either strand, they are referred to as **closed-circular DNAs**. In aqueous solutions, DNA is most stable—that

is, it is in its lowest free-energy form—in the B-form structure. If the DNA of a closed-circular molecule conforms closely to the B-form structure (Watson-Crick structure; see Fig. 8-13), with one turn of the double helix per 10.5 bp, the DNA is relaxed rather than supercoiled (**Fig. 24-11**). Supercoiling results when DNA is subject to some form of structural strain such that it is overwound or underwound. Purified closed-circular DNA is rarely relaxed, regardless of its biological origin. **P3** Furthermore, DNAs derived from a given cellular source have a characteristic degree of supercoiling. DNA structure is therefore strained in a manner that is regulated by the cell to induce the supercoiling.

In almost every instance, the strain is a result of underwinding of the DNA double helix in the closed circle. In other words, the DNA has *fewer* helical turns than would be expected for the B-form structure. The effects of underwinding are summarized in **Figure 24-12**. An 84 bp segment of a circular DNA in the relaxed state would contain eight double-helical turns, one for every 10.5 bp. If one of these turns were removed, there would be $(84 \text{ bp})/7 = 12.0 \text{ bp}$ per turn, rather than the 10.5 found in B-DNA (Fig. 24-12b). This is a deviation from the most stable DNA form, and the molecule is thermodynamically strained as a result. Generally, much of this strain would be accommodated by coiling the axis of the DNA on itself to form a supercoil (Fig. 24-12c; some of the strain in this 84 bp segment would simply become dispersed in the untwisted structure of the larger DNA molecule). In principle, the strain could also be accommodated by separating the two DNA strands over a distance of about 10 bp (Fig. 24-12d). In isolated closed-circular DNA, strain introduced by underwinding is generally accommodated by supercoiling rather than strand separation, because coiling the axis of the DNA usually requires less energy than breaking the hydrogen bonds and disrupting the base stacking that stabilizes paired bases. Note, however, that the underwinding of DNA *in vivo* makes separation of the DNA strands easier, facilitating access to the information they contain.

Every cell actively underwinds its DNA with the aid of enzymatic processes (described below), and the resulting strained state represents a form of stored energy. Cells maintain DNA in an underwound state to facilitate its compaction by coiling. The underwinding of DNA is also important to enzymes of DNA metabolism that must bring about strand separation as part of their function.

The underwound state can be maintained only if the DNA is a closed circle or if it is bound and stabilized by proteins so that the strands are not free to rotate about each other. If there is a break in one strand of an isolated, protein-free circular DNA, free rotation at that point will cause the underwound DNA to revert spontaneously to the relaxed state. In a closed-circular DNA molecule, however, the number of helical turns cannot be changed without at least transiently breaking one of the DNA

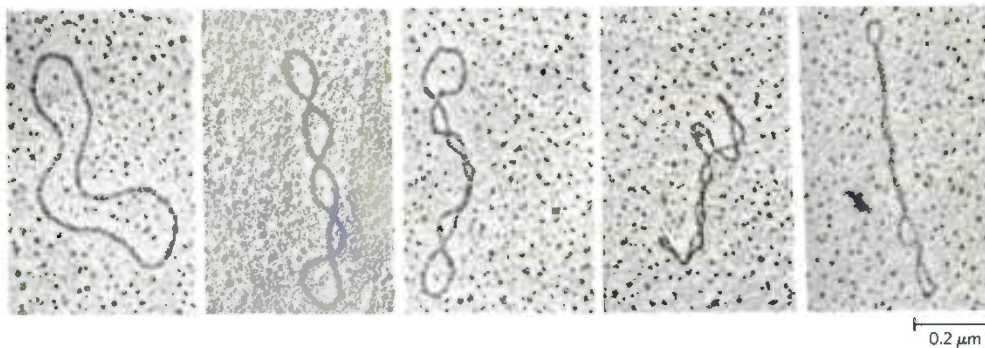


FIGURE 24-11 Relaxed and supercoiled plasmid DNAs. The molecule in the leftmost electron micrograph is relaxed; the degree of supercoiling increases from left to right. [Laurien Polder, from A. Kornberg, *DNA Replication*, p. 29, W. H. Freeman, 1980.]

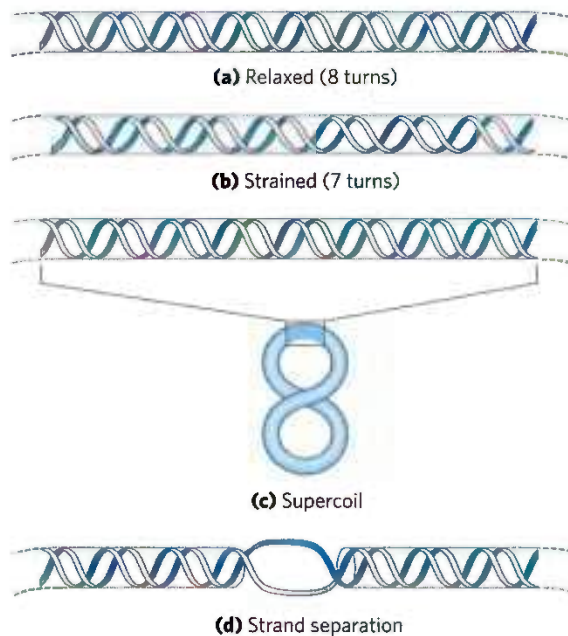


FIGURE 24-12 Effects of DNA underwinding. (a) A segment of DNA in a closed-circular molecule, 84 bp long, in its relaxed form with eight helical turns. (b) Removal of one turn induces structural strain. (c) The strain is generally accommodated by formation of a supercoil. (d) DNA underwinding also makes the separation of strands somewhat easier. In principle, each turn of underwinding should facilitate strand separation over about 10 bp, as shown here. However, the hydrogen-bonded base pairs would generally preclude strand separation over such a short distance, and the effect becomes important only for longer DNAs and higher levels of DNA underwinding.

strands. The number of helical turns in a DNA molecule therefore provides a precise description of supercoiling.

DNA Underwinding Is Defined by Topological Linking Number

The field of topology provides some ideas that are useful to the discussion of DNA supercoiling, particularly the concept of **linking number**. Linking number is a topological property of double-stranded DNA, because it does not vary when the DNA is bent or deformed, as long as both DNA strands remain intact. Linking number (Lk) is illustrated in **Figure 24-13**. As we shall see, all cells have enzymes called topoisomerases that catalyze

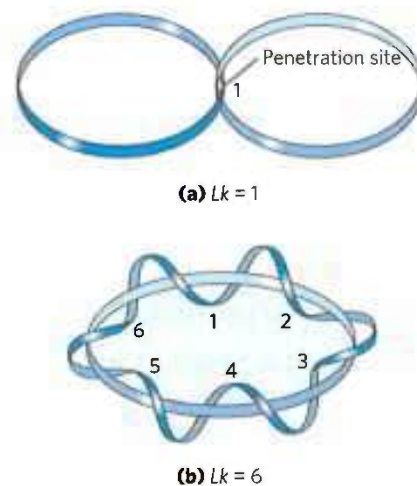


FIGURE 24-13 Linking number, Lk . Here, as usual, each blue ribbon represents one strand of a double-stranded DNA molecule. For the molecule in (a), $Lk = 1$. For the molecule in (b), $Lk = 6$. One of the strands in (b) is kept untwisted for illustrative purposes, to define the border of an imaginary surface (shaded blue). The number of times the twisting strand penetrates this surface provides a rigorous definition of linking number.

changes in the linking number. Because of this critical role, topoisomerases are the targets of many antibiotics and cancer chemotherapy agents.

Let's begin by visualizing the separation of the two strands of a double-stranded circular DNA. If the two strands are linked as shown in **Figure 24-13a**, they are effectively joined by what can be described as a topological bond. Even if all hydrogen bonds and base-stacking interactions were abolished such that the strands were not in physical contact, this topological bond would still link the two strands. Visualize one of the circular strands as the boundary of a surface (such as the soap film framed by the loop of a bubble wand before you blow a bubble). The linking number can be defined as the number of times the second strand pierces this surface. For the molecule in **Figure 24-13a**, $Lk = 1$; for that in **Figure 24-13b**, $Lk = 6$. The linking number for a closed-circular DNA is always an integer. By convention, if the links between two DNA strands are arranged so that the strands are interwound in a right-handed helix,

the linking number is defined as positive (+); for strands interwound in a left-handed helix, the linking number is negative (-). Negative linking numbers are, for all practical purposes, not encountered in DNA.

We can now extend these ideas to a closed-circular DNA with 2,100 bp (Fig. 24-14a). When the molecule is relaxed, the linking number is simply the number of base pairs divided by the number of base pairs per turn, which is close to 10.5; so in this case, $Lk = 200$. The linking number in relaxed DNA is designated Lk_0 . For a circular DNA molecule to have a topological property such as linking number, both strands must be intact, without a break. If there is a break in either strand, the two strands can, in principle, be unraveled and separated completely. In this case, no topological bond exists and Lk is undefined (Fig. 24-14b).

We can now describe DNA underwinding in terms of changes in the linking number. For the molecule shown in Figure 24-14a, $Lk_0 = 200$; if two turns are removed from this molecule, $Lk = 198$. The change can be described by the equation

$$\begin{aligned}\Delta Lk &= Lk - Lk_0 \\ &= 198 - 200 = -2\end{aligned}\quad (24-1)$$

It is often convenient to express the change in linking number in terms of a quantity that is independent of

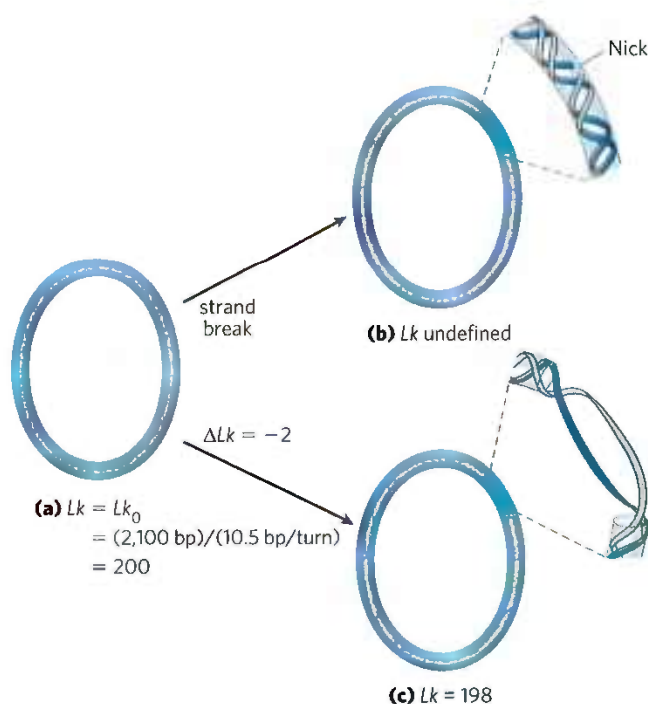


FIGURE 24-14 Linking number applied to closed-circular DNA molecules. A 2,100 bp circular DNA is shown in three forms: (a) relaxed, $Lk = 200$; (b) relaxed with a nick (break) in one strand, Lk undefined; and (c) underwound by two turns, $Lk = 198$. The underwound molecule generally exists as a supercoiled molecule, but underwinding also facilitates the separation of DNA strands.

the length of the DNA molecule. This quantity, called the **specific linking difference** or **superhelical density** (σ), is a measure of the number of turns removed relative to the number present in relaxed DNA:

$$\sigma = \frac{\Delta Lk}{Lk_0} \quad (24-2)$$

In the example in Figure 24-14c, $\sigma = 0.01$, which means that 1% (2 of 200) of the helical turns present in the DNA (in its B form) have been removed. The degree of underwinding in cellular DNAs generally falls in the range of 5% to 7%; that is, $\sigma = -0.05$ to -0.07 . The negative sign indicates that the change in linking number is due to underwinding of the DNA. The supercoiling induced by underwinding is therefore defined as negative supercoiling. Conversely, under some conditions DNA can be overwound, resulting in positive supercoiling. Note that the twisting path taken by the axis of the DNA helix when the DNA is underwound (negative supercoiling) is the mirror image of that taken when the DNA is overwound (positive supercoiling) (Fig. 24-15). Supercoiling is not a random process; the path of the supercoiling is largely prescribed by the torsional strain imparted to the DNA by decreasing or increasing the linking number relative to B-DNA.

P2 P3 An increase in superhelical density brings about an increase in DNA compaction.

Linking number can be changed by ± 1 by breaking one DNA strand, rotating one of the ends 360° about the unbroken strand, and rejoining the broken ends. This change has no effect on the number of base pairs or the number of atoms in the circular DNA molecule. Two forms of a circular DNA that differ only in a topological property such as linking number are referred to as **topoisomers**.

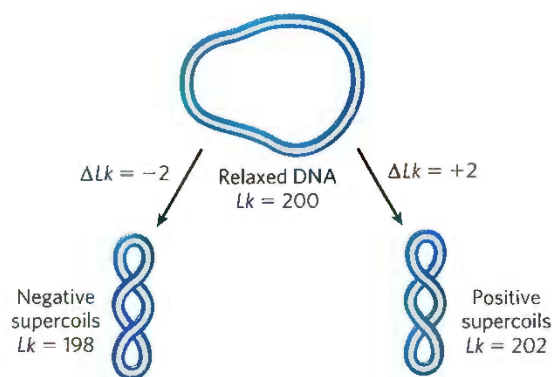


FIGURE 24-15 Negative and positive supercoils. For the relaxed DNA molecule of Figure 24-14a, underwinding or overwinding by two helical turns ($Lk = 198$ or 202) will produce negative or positive supercoiling, respectively. Notice that the DNA axis twists in opposite directions in the two cases.

WORKED EXAMPLE 24-1 Calculation of Superhelical Density

What is the superhelical density (σ) of a closed-circular DNA with a length of 4,200 bp and a linking number (Lk) of 374? What is the superhelical density of the same DNA when $Lk = 412$? Are these molecules negatively or positively supercoiled?

SOLUTION: First, calculate Lk_0 by dividing the length of the closed-circular DNA (in bp) by 10.5 bp/turn: $(4,200 \text{ bp}) / (10.5 \text{ bp/turn}) = 400$. We can now calculate ΔLk from Equation 24-1: $\Delta Lk = Lk - Lk_0 = 374 - 400 = -26$. Substituting the values for ΔLk and Lk_0 into Equation 24-2: $\sigma = \Delta Lk / Lk_0 = -26 / 400 = -0.065$. Because the superhelical density is negative, this DNA molecule is negatively supercoiled.

When the same DNA molecule has an Lk of 412, $\Delta Lk = 412 - 400 = 12$, and $\sigma = 12 / 400 = 0.03$. The superhelical density is positive, and the molecule is positively supercoiled.

In addition to causing supercoiling and making strand separation somewhat easier, the underwinding of DNA facilitates structural changes in the molecule. These are of less physiological importance but they help illustrate the effects of underwinding. Recall that a cruciform (see Fig. 8-19) generally contains a few unpaired bases; DNA underwinding helps to maintain the required strand separation (Fig. 24-16). Underwinding of a right-handed DNA helix also enables the formation of short stretches of left-handed Z-DNA in regions where the base sequence is consistent with the Z form (see Chapter 8).

Topoisomerases Catalyze Changes in the Linking Number of DNA

DNA supercoiling is a precisely regulated process that influences many aspects of DNA metabolism. Every cell has enzymes with the sole function of underwinding and/or relaxing DNA. The enzymes that increase or decrease the extent of DNA underwinding are **topoisomerases**; the property of DNA that they change is the linking number. These enzymes play an especially important role in processes such as replication and DNA packaging. There are two classes of topoisomerases. **Type I topoisomerases** act by transiently breaking one of the two DNA strands, passing the unbroken strand through the break, and rejoining the broken ends; they change Lk in increments of 1. **Type II topoisomerases** break both DNA strands and change Lk in increments of 2.

P2 When a circular DNA is supercoiled, it is twisted upon itself and therefore is more compact than when it is relaxed. The supercoiled molecule will thus migrate faster in a gel matrix (Fig. 24-17). A population of identical plasmid DNAs with the same linking number migrates as a discrete band during agarose gel electrophoresis. Topoisomers with Lk values differing by as little as 1 can be separated by this method, so changes in linking number induced by topoisomerases are readily detected.

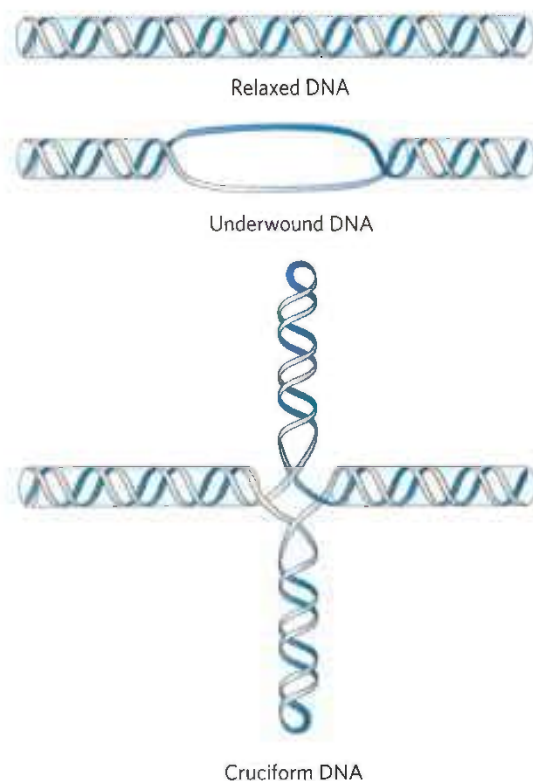


FIGURE 24-16 Promotion of cruciform structures by DNA underwinding. In principle, cruciforms can form at palindromic sequences (see Fig. 8-19), but they seldom occur in relaxed DNA because the linear DNA accommodates more paired bases than does the cruciform structure. Underwinding of the DNA facilitates the partial strand separation needed to promote cruciform formation at appropriate sequences.

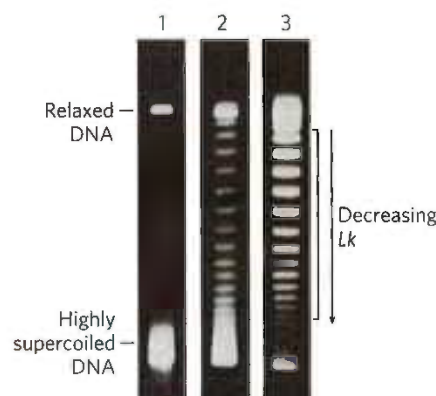


FIGURE 24-17 Visualization of topoisomers. In this experiment, all DNA molecules have the same number of base pairs but exhibit some range in the degree of supercoiling. Because supercoiled DNA molecules are more compact than relaxed molecules, they migrate more rapidly during gel electrophoresis. The gels shown here separate topoisomers (moving from top to bottom) over a limited range of superhelical density. In lane 1, highly supercoiled DNA migrates in a single band, even though different topoisomers are probably present. Lanes 2 and 3 illustrate the effect of treating the supercoiled DNA with a type I topoisomerase; the DNA in lane 3 was treated for a longer time than that in lane 2. As the superhelical density of the DNA is reduced to the point where it corresponds to the range in which the gel can resolve individual topoisomers, distinct bands appear. Individual bands in the region indicated by the bracket next to lane 3 each contain DNA circles with the same linking number; Lk changes by 1 from one band to the next. (Courtesy of Michael Cox Lab.)

E. coli has at least four individual topoisomerases (I through IV). Those of type I (topoisomerases I and III) generally relax DNA by removing negative supercoils (increasing Lk). The way in which bacterial type I topoisomerases change linking number is illustrated in **Figure 24-18**. A bacterial type II enzyme, called either topoisomerase II or DNA gyrase, can introduce negative supercoils (decrease Lk). It uses the energy of ATP to accomplish this. To alter DNA linking number, type II topoisomerases cleave both strands of a DNA molecule and pass another duplex through the break. The overall degree of supercoiling of bacterial DNA is maintained by regulation of the net activity of topoisomerases I and II. Topoisomerases III and IV have more-specialized roles in DNA metabolism.

Eukaryotic cells also have type I and type II topoisomerases. The type I enzymes are topoisomerases I and III. The single type II enzyme has two isoforms in vertebrates, called II α and II β . Most type II enzymes, including a DNA gyrase in archaea, are related and define a family called type IIA. The eukaryotic type II topoisomerases cannot underwind DNA (introduce negative supercoils), but they can relax both positive

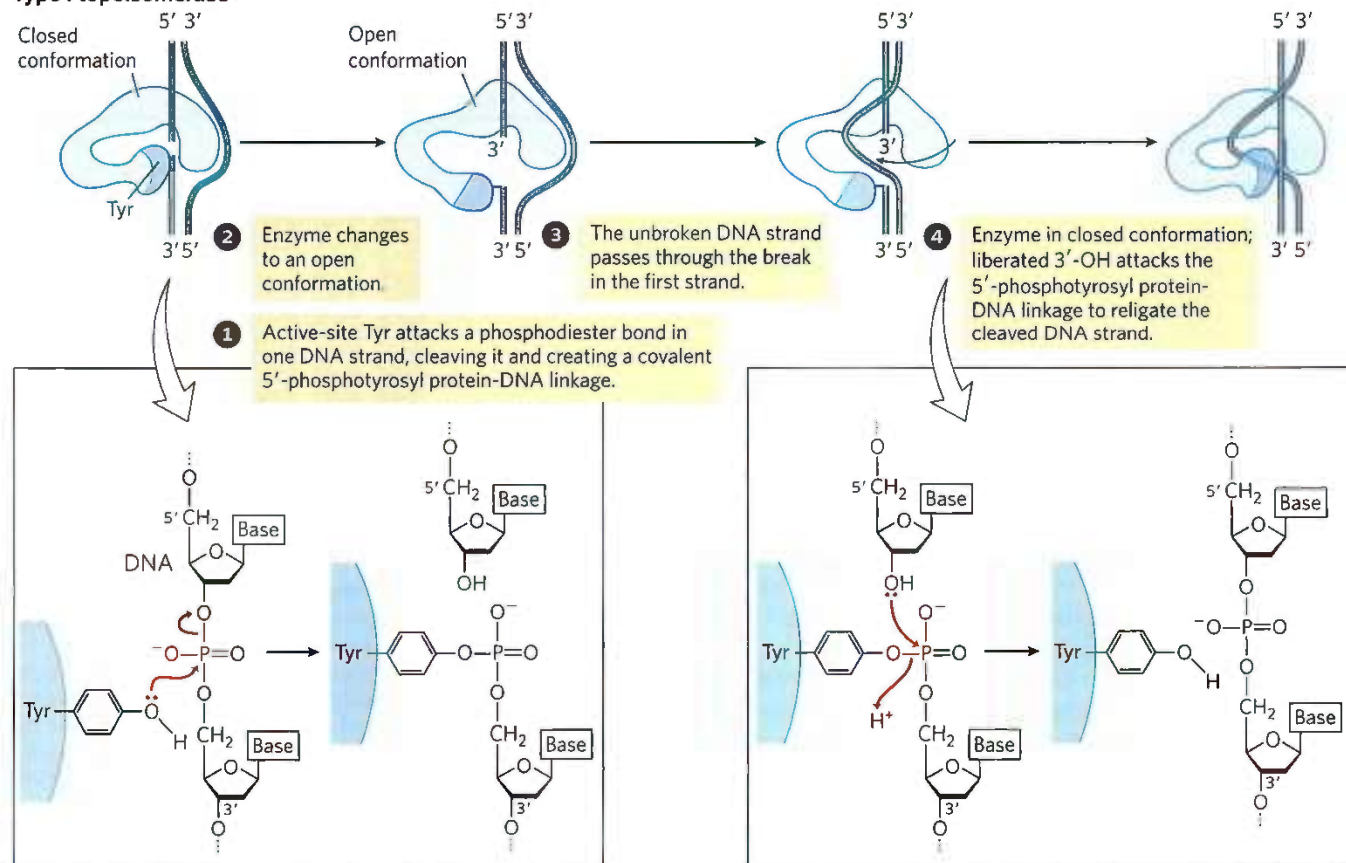
and negative supercoils (**Fig. 24-19a**). The capacity of type II topoisomerases to pass one duplex DNA segment through a double-strand break in another duplex allows these enzymes to untangle **catenanes**, DNA circles that are topologically linked (**Fig. 24-19b**). Some topoisomerases are specialized for decatenation functions. For example, the bacterial type II enzyme called topoisomerase IV is involved in chromosome untangling during cell division (Chapter 25).

Archaea have an unusual enzyme, topoisomerase VI, which alone defines the type IIB family. The full diversity of DNA topoisomerases is illustrated in **Table 24-4**. As we shall show in the next few chapters, topoisomerases play a critical role in every aspect of DNA metabolism, making them important drug targets for the treatment of bacterial infections and cancer.

DNA Compaction Requires a Special Form of Supercoiling

Supercoiled DNA molecules are uniform in several respects. The supercoils are right-handed in a negatively supercoiled DNA molecule (**Fig. 24-15**), and they tend to be extended and narrow rather than compacted, often with multiple

Type I topoisomerase



MECHANISM FIGURE 24-18 The type I topoisomerase reaction. Bacterial topoisomerase I increases Lk by breaking one DNA strand, passing the unbroken strand through the break, then resealing the break.

Nucleophilic attack by the active-site Tyr residue breaks one DNA strand. The ends are ligated by a second nucleophilic attack. At each step, one high-energy bond replaces another. [Information from J. J. Champoux, *Annu. Rev. Biochem.* 70:369, 2001, Fig. 3.]

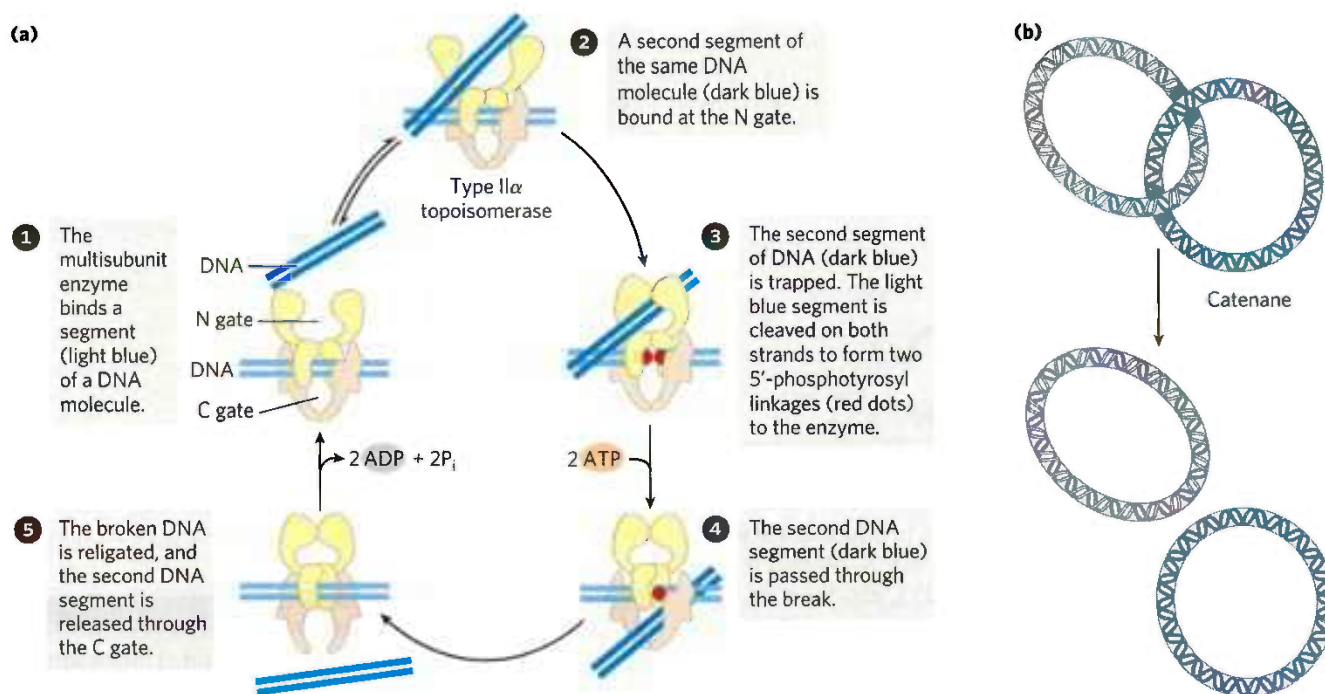


FIGURE 24-19 Alteration of the linking number by a eukaryotic type II α topoisomerase. (a) The general mechanism features passage of one intact duplex DNA segment through a transient double-strand break in another segment. The DNA segment enters and leaves the topoisomerase through gated cavities, called the N gate and the C gate, above and below the bound DNA. Two ATPs are bound and hydrolyzed

during this cycle. The enzyme structure and use of ATP are specific to this reaction. (b) When topologically linked as shown, two DNA circles are referred to as a catenane. By cleaving both strands of one circle and passing a segment of the second circle through the break, a type II topoisomerase can decatenate the circles. [(a) Information from J. J. Champoux, *Annu. Rev. Biochem.* 70:369, 2001, Fig. 11.]

TABLE

Diversity in DNA Topoisomerases

Type	Mechanism	Family (defined by structural class)	Domain(s)	Notes
IA	Strand passage ^a	Topoisomerase I	Bacteria, Eukarya	Relaxes (–)
		Topoisomerase III	Bacteria, Eukarya	Relaxes (–)
		Reverse gyrase	Archaea, Bacteria	Uses ATP to introduce positive supercoils; thermophilic bacteria and archaea only
IB	Swivelase ^b	Topoisomerase IB	Bacteria, Eukarya	A few bacteria; all eukaryotes
IC	Swivelase	Topoisomerase V	Archaea	<i>Methanopyrus</i> only
IIA	Strand passage ^c	Topoisomerase II (DNA gyrase)	Archaea, Bacteria	Introduces negative supercoils (ATPase)
		Topoisomerase II α	Eukarya	Relaxes (+ or –)
		Topoisomerase II β	Eukarya	Relaxes (+ or –)
		Topoisomerase IV	Bacteria	Decatenase ^d
IIB	Strand passage	Topoisomerase VI	Archaea, Bacteria, Eukarya	Among eukaryotes, plants, algae, and protists only

^aSee Figure 24-18.

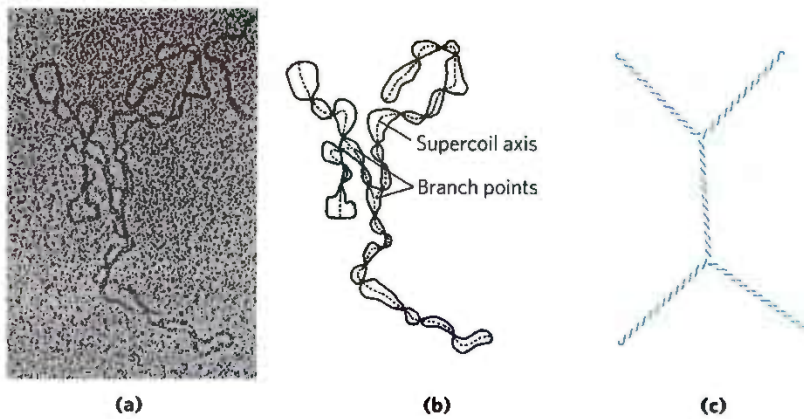
^bA nick is made in one strand, and the other strand is allowed to rotate to relieve topological strain.

^cSee Figure 24-19a.

^dSee Figure 24-19b.

branches (Fig. 24-20). At the superhelical densities normally encountered in cells, the length of the supercoil axis, including branches, is about 40% of the length of the DNA. This type of supercoiling is referred to as **plectonemic** (from the

Greek *plektos*, “twisted,” and *nema*, “thread”). The term can be applied to any structure with strands intertwined in some simple and regular way, and it is a good description of the general structure of supercoiled DNA in solution.

**FIGURE 24-20 Plectonemic supercoiling.**

(a) Electron micrograph of plectonemically supercoiled plasmid DNA and (b) an interpretation of the observed structure. The dotted lines show the axis of the supercoil; notice the branching of the supercoil. (c) An idealized representation of this structure. [(a, b) Republished with permission of Elsevier, from T. C. Boles et al. (1990), "Structure of plectonemically supercoiled DNA," *J. Mol. Biol.* 213:931–951, Fig. 2; permission conveyed through Copyright Clearance Center, Inc.]

Plectonemic supercoiling, the form observed in isolated DNAs in the laboratory, does not produce sufficient compaction to package DNA in the cell. A second form of supercoiling, **solenoidal** (Fig. 24-21), can be adopted by an underwound DNA. Instead of the extended right-handed supercoils characteristic of the plectonemic form, solenoidal supercoiling involves tight left-handed turns, similar to the shape taken up by a garden hose neatly wrapped on a reel. Although their structures are dramatically different, plectonemic and solenoidal supercoiling are two forms of negative supercoiling that can be taken up by the *same* segment of underwound DNA. The two forms are readily interconvertible. Although the plectonemic form is more stable in solution, the solenoidal form can be stabilized by protein binding, as it is in eukaryotic chromosomes; it provides a much greater degree of compaction. Solenoidal supercoiling is a

primary mechanism by which underwinding contributes to DNA compaction in cells.

SUMMARY 24.2 DNA Supercoiling

- Most cellular DNAs are supercoiled. Underwinding decreases the total number of helical turns in DNA relative to the relaxed, B form. To maintain an underwound state, DNA must be either a closed circle or bound to protein.
- Underwinding is quantified by a topological parameter called linking number, Lk . Underwinding is measured in terms of specific linking difference, or superhelical density, σ , which is $(Lk - Lk_0)/Lk_0$. For cellular DNAs, σ is typically -0.05 to -0.07 , which means that approximately 5% to 7% of the helical turns in the DNA have been removed. DNA underwinding facilitates strand separation by enzymes of DNA metabolism.
- DNAs that differ only in linking number are called topoisomers. Topoisomerases, enzymes that underwind and/or relax DNA, catalyze changes in linking number. The two classes of topoisomerases, type I and type II, change Lk in increments of 1 or 2, respectively, per catalytic event.
- Supercoiled DNA in solution, unconstrained by proteins, takes on a plectonemic supercoiling structure. When supercoiled DNA is wrapped around specialized DNA-binding proteins, it forms solenoidal supercoils.

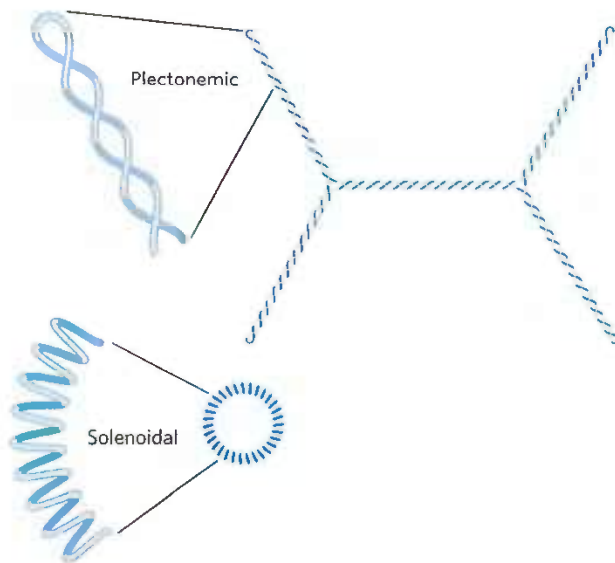


FIGURE 24-21 Plectonemic and solenoidal supercoiling of the same DNA molecule, drawn to scale. Plectonemic supercoiling takes the form of extended right-handed coils. Solenoidal negative supercoiling takes the form of tight left-handed turns about an imaginary tubelike structure. The two forms are readily interconverted, although the solenoidal form is generally not observed unless certain proteins are bound to the DNA. Solenoidal supercoiling provides a much greater degree of compaction.

24.3 The Structure of Chromosomes

The term "chromosome" is used to refer to a nucleic acid molecule that is the repository of genetic information in a virus, a bacterium, an archaeon, a eukaryotic cell, or an organelle. It also refers to the densely colored bodies seen in the nuclei of dye-stained eukaryotic cells undergoing mitosis, as visualized using a light microscope.

Chromatin Consists of DNA, Proteins, and RNA

The eukaryotic cell cycle produces remarkable changes in the structure of chromosomes (Fig. 24-22). In nondividing eukaryotic cells (in the G₀ phase) and those in interphase (G₁, S, and G₂), the chromosomal material,

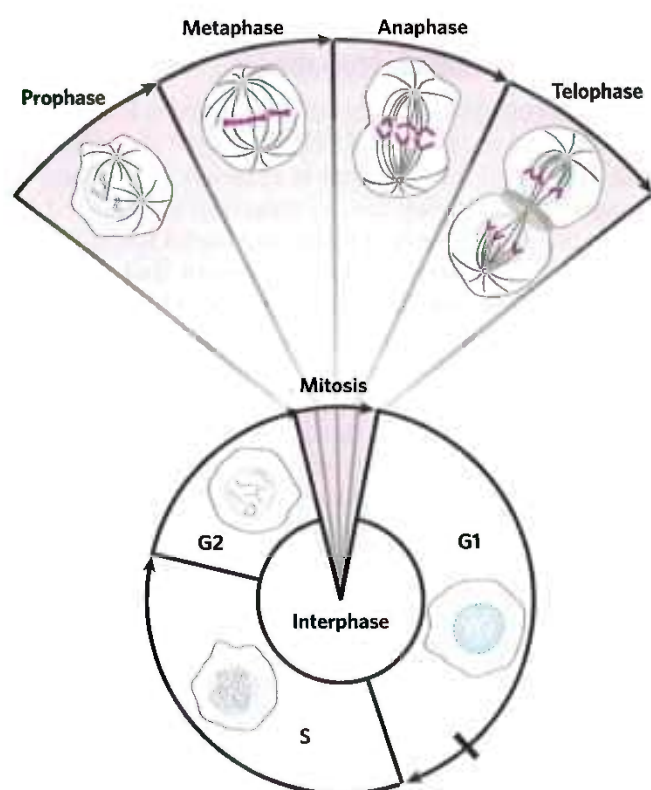


FIGURE 24-22 Changes in chromosome structure during the eukaryotic cell cycle. The relative lengths of the phases shown here are arbitrary. The duration of each phase varies with cell type and with growth conditions (for single-celled organisms) or metabolic state (for multicellular organisms); mitosis is typically the shortest phase. Cellular DNA is uncondensed throughout interphase, as shown in the cartoons of the nucleus. The interphase period can be divided into the G1 (gap) phase; the S (synthesis) phase, when the DNA is replicated; and the G2 phase, throughout which the replicated chromosomes (chromatids) cohere to one another. Mitosis can be divided into four stages. The DNA undergoes condensation in prophase. During metaphase, the condensed chromosomes line up in pairs along the plane halfway between the spindle poles. The two chromosomes of each pair are linked to different spindle poles via microtubules that extend between the spindle and the centromere. The sister chromatids separate at anaphase, each drawn toward the spindle pole to which it is connected. The process is completed in telophase. After cell division, the chromosomes decondense and the cycle begins anew.

chromatin, is amorphous. In the S phase of interphase, the DNA in this amorphous state replicates, each chromosome producing two sister chromosomes (called sister chromatids) that remain associated with each other after replication is complete. The chromosomes become much more condensed during prophase of mitosis, taking the form of a species-specific number of well-defined pairs of sister chromatids (Fig. 24-5).

P4 Chromatin consists of fibers containing protein and DNA in approximately equal proportions (by mass), along with a significant amount of associated RNA. The DNA in the chromatin is very tightly associated with proteins called **histones**, which package and order the DNA into structural units called **nucleosomes** (Fig. 24-23). Also found in chromatin are many nonhistone proteins, some of which help maintain chromosome structure and others that regulate the expression of specific genes (Chapter 28).

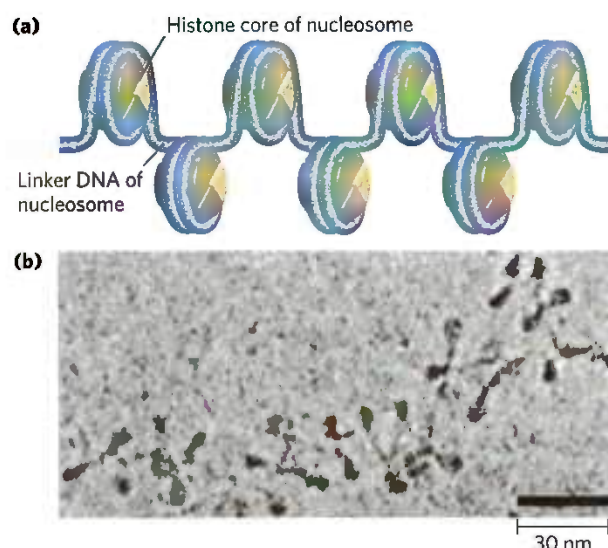


FIGURE 24-23 Nucleosomes. (a) Regularly spaced nucleosomes consist of core histone proteins bound to DNA. (b) In this electron micrograph, the DNA-wrapped histone octamer structures are clearly visible. (b) J. Bednar et al., Nucleosomes, linker DNA, and linker histone form a unique structural motif that directs the higher-order folding and compaction of chromatin, *Proc. Natl. Acad. Sci. USA* vol. 95 no. 24: 14173–14178, November 1998 Cell Biology Fig. 1. © 1998 National Academy of Sciences, U.S.A.]

Beginning with nucleosomes, eukaryotic chromosomal DNA is packaged into a succession of higher-order structures that ultimately yield the compact chromosome seen with the light microscope. We now turn to a description of this structure in eukaryotes and compare it with the packaging of DNA in bacterial cells.

Histones Are Small, Basic Proteins

Found in the chromatin of all eukaryotic cells, histones have molecular weights between 11,000 and 21,000 and are very rich in the basic amino acids arginine and lysine (together these make up about one-fourth of the amino acid residues). All eukaryotic cells have five major classes of histones, differing in molecular weight and amino acid composition (Table 24-5). The H3 histones are nearly identical in amino acid sequence in all eukaryotes, as are the H4 histones, suggesting strict conservation of their functions. For example, only 2 of 102 amino acid residues differ between the H4 histone molecules of peas and cows, and only 8 differ between the H4 histones of humans and yeast. Histones H1, H2A, and H2B show less sequence similarity across eukaryotic species.

Each type of histone is subject to enzymatic modification by methylation, acetylation, ADP-ribosylation, phosphorylation, glycosylation, SUMOylation, or ubiquitination (p. 216 and Fig. 6-38). Such modifications affect the net electric charge, shape, and other properties of histones, as well as the structural and functional properties of the chromatin. The modifications play a role in the regulation of transcription and in chromatin structure at different stages of the cell cycle.

In addition, eukaryotes generally have several variant forms of certain histones, most notably histones

TABLE 24-5 Types and Properties of the Common Histones

Histones	Molecular weight	Number of amino acid residues	Content of basic amino acids (% of total)	
			Lys	Arg
H1 ^a	21,130	223	29.5	11.3
H2A ^a	13,960	129	10.9	19.3
H2B ^a	13,774	125	16.0	16.4
H3	15,273	135	19.6	13.3
H4	11,236	102	10.8	13.7

^aThe sizes of these histones vary somewhat from species to species. The numbers given here are for bovine histones.

H2A and H3, described in more detail below. The variant forms, along with their modifications, have specialized roles in DNA metabolism.

Nucleosomes Are the Fundamental Organizational Units of Chromatin

P4 The eukaryotic chromosome depicted in Figure 24-5 represents the compaction of a DNA molecule about $10^5 \mu\text{m}$ long into a cell nucleus that is typically 5 to $10 \mu\text{m}$ in diameter. This 10,000-fold compaction is achieved by means of several levels of highly organized folding. Subsection of chromosomes to treatments that partially unfold them reveals a structure in which the DNA is bound tightly to beads of protein, often regularly spaced. The beads in this “beads-on-a-string” arrangement are complexes of histones and DNA. The bead plus the connecting DNA that leads to the next bead form the nucleosome, the fundamental unit of organization on which the higher-order packing of chromatin is built (**Fig. 24-24**). The bead of each nucleosome contains eight histone molecules: two copies each of H2A, H2B, H3, and H4.

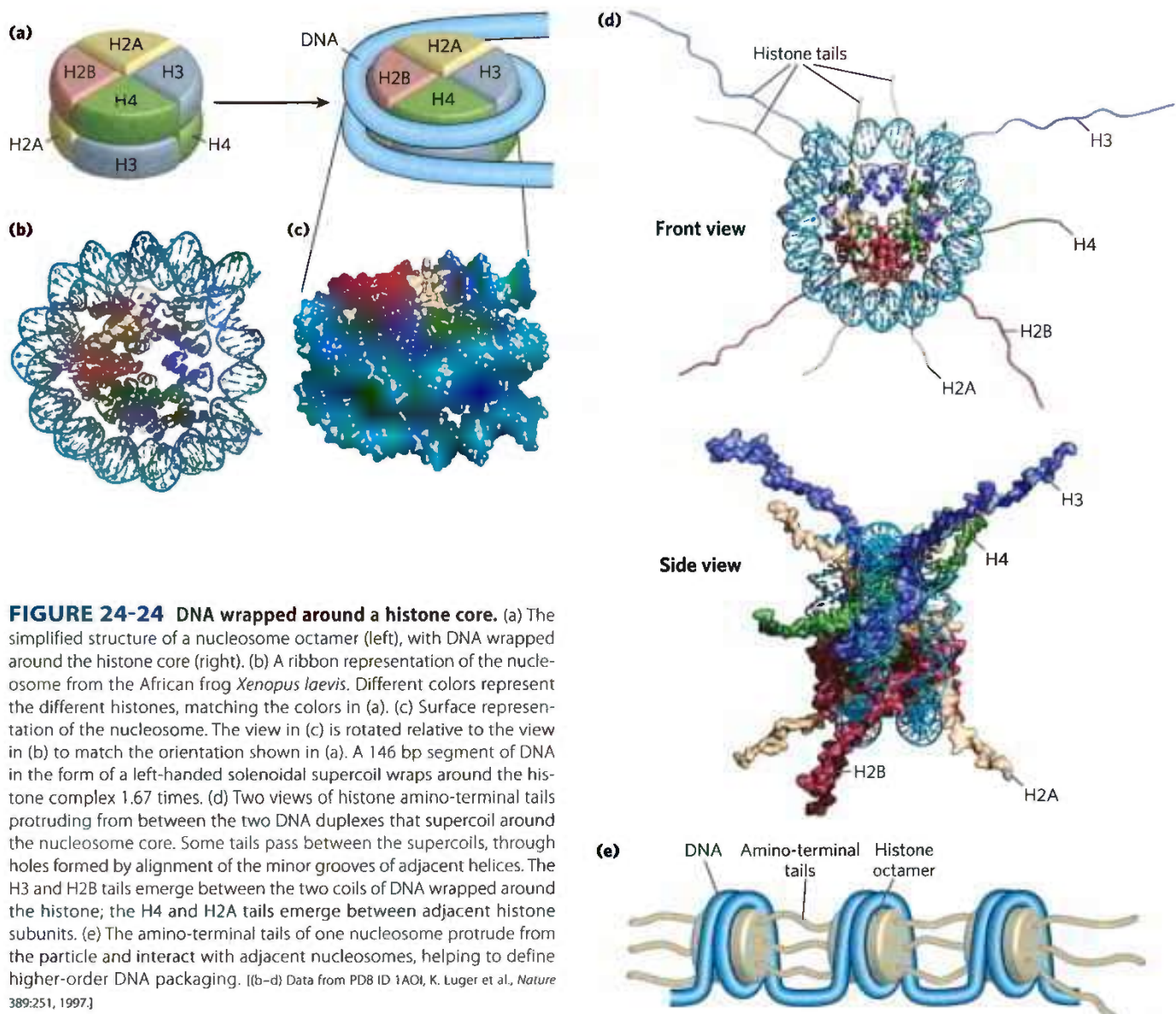


FIGURE 24-24 DNA wrapped around a histone core. (a) The simplified structure of a nucleosome octamer (left), with DNA wrapped around the histone core (right). (b) A ribbon representation of the nucleosome from the African frog *Xenopus laevis*. Different colors represent the different histones, matching the colors in (a). (c) Surface representation of the nucleosome. The view in (c) is rotated relative to the view in (b) to match the orientation shown in (a). A 146 bp segment of DNA in the form of a left-handed solenoidal supercoil wraps around the histone complex 1.67 times. (d) Two views of histone amino-terminal tails protruding from between the two DNA duplexes that supercoil around the histone core. Some tails pass between the supercoils, through holes formed by alignment of the minor grooves of adjacent helices. The H3 and H2B tails emerge between the two coils of DNA wrapped around the histone; the H4 and H2A tails emerge between adjacent histone subunits. (e) The amino-terminal tails of one nucleosome protrude from the particle and interact with adjacent nucleosomes, helping to define higher-order DNA packaging. [(b–d) Data from PDB ID 1AOI, K. Luger et al., *Nature* 389:251, 1997.]

The spacing of the nucleosome beads provides a repeating unit typically of about 200 bp, of which 146 bp are bound tightly around the eight-part histone core and the remainder serve as linker DNA between nucleosome beads. Histone H1 binds to the linker DNA. Brief treatment of chromatin with enzymes that digest DNA causes the linker DNA to degrade preferentially, releasing histone particles containing 146 bp of bound DNA that is protected from digestion.

Researchers have crystallized nucleosome cores obtained in this way, and x-ray diffraction analysis reveals a particle made up of the eight histone molecules with the DNA wrapped around the core in the form of a left-handed solenoidal supercoil (Fig. 24-24; see also Fig. 24-21). Extending out from the nucleosome core are the amino-terminal tails of the histones, which are intrinsically disordered (Fig. 24-24d). Most of the histone modifications occur in these tails. The tails play a key role in forming contacts between nucleosomes in the chromatin (Fig. 24-24e). As the nucleosomes are approximately 10 to 11 nm in diameter, this simple beads-on-a-string structure is sometimes called the 10 nm fiber.

A close inspection of the nucleosome structure reveals why eukaryotic DNA is underwound even though eukaryotic cells lack enzymes that underwind DNA. Recall that the solenoidal wrapping of DNA in nucleosomes is but one form of supercoiling that can be taken up by underwound (negatively supercoiled) DNA. The tight wrapping of DNA around the histone core requires the removal of about one helical turn in the DNA. When the protein core of a nucleosome binds *in vitro* to a relaxed closed-circular DNA, the binding introduces a negative supercoil. Because this binding process does not break the DNA or change the linking number, the formation of a negative solenoidal supercoil must be accompanied by a compensatory positive supercoil in the unbound region of the DNA (Fig. 24-25). As mentioned earlier, eukaryotic topoisomerases can relax positive supercoils. Relaxing the unbound positive supercoil leaves the negative supercoil fixed (through its binding to the nucleosome's histone core) and results in an overall decrease in linking number. Indeed, topoisomerases have proved necessary for assembling chromatin from purified histones and closed-circular DNA *in vitro*.

Another factor that affects the binding of DNA to histones in nucleosome cores is the sequence of the bound DNA. Histone cores do not bind at random positions on the DNA; rather, some locations are more likely to be bound than others. This positioning is not fully understood, but in some cases it seems to depend on a local abundance of A=T base pairs in the DNA helix where it is in contact with the histones (Fig. 24-26). A cluster of two or three A=T base pairs facilitates the compression of the minor groove that is needed for the DNA to wrap tightly around the nucleosome's histone core. Nucleosomes bind particularly well to sequences where AA or AT or TT dinucleotides are staggered at 10 bp intervals,

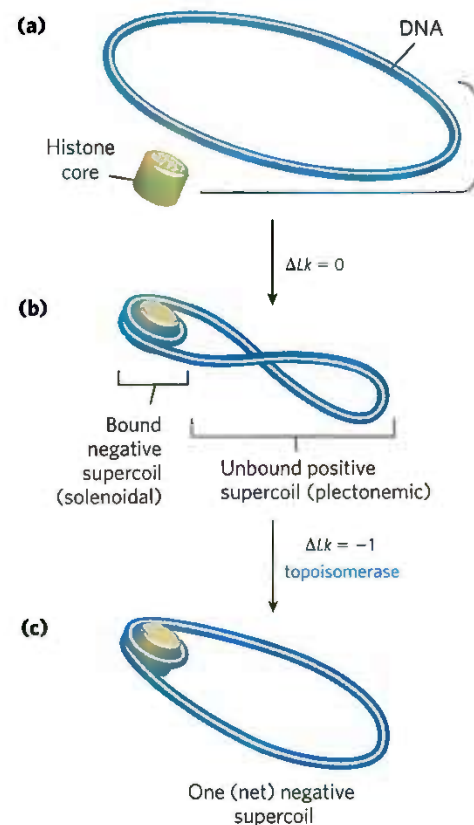


FIGURE 24-25 Chromatin assembly. (a) Relaxed closed-circular DNA. (b) Binding of a histone core to form a nucleosome induces one negative supercoil. In the absence of any strand breaks, a positive supercoil must form elsewhere in the DNA ($\Delta Lk = 0$). (c) Relaxation of this positive supercoil by a topoisomerase leaves one net negative supercoil ($\Delta Lk = -1$).

an arrangement that can account for up to 50% of the positions of bound histones *in vivo*.

Nucleosome cores are deposited on DNA during replication or following other processes that require a transient displacement of nucleosomes. Other, nonhistone proteins are required for the positioning of some nucleosome cores. In several organisms, certain proteins bind to a specific DNA sequence and facilitate the formation of a nucleosome core nearby. Nucleosome cores seem to be deposited stepwise. A tetramer of two H3 and two H4 histones binds first, followed by two H2A–H2B dimers. The incorporation of nucleosomes into chromosomes after chromosomal replication is mediated by a complex of **histone chaperones** conserved in all eukaryotes and described here for the yeast system. These include the proteins CAF1 (chromatin assembly factor 1), RTT106 and RTT109 (regulation of Ty1 transposition), and ASF1 (anti-silencing factor 1). ASF1 binds to newly synthesized H3–H4 dimers and facilitates RTT109-mediated acetylation at Lys⁶⁶ (K56) of histone H3 (H3K56). The H3K56 modification increases the affinity of H3 for CAF1 and RTT, which in turn promote the deposition of H3-containing histone complexes on the DNA after replication. CAF1 binds directly to a key component of the

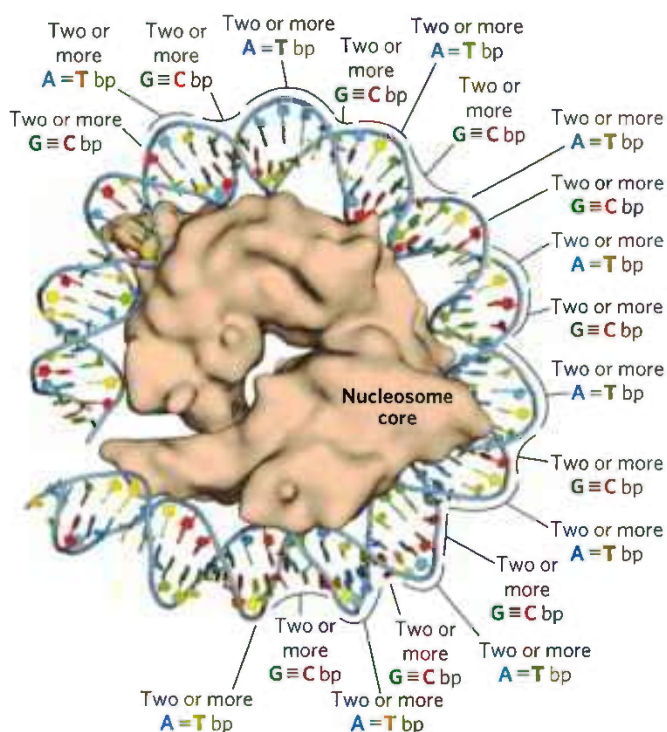


FIGURE 24-26 The effect of DNA sequence on nucleosome binding. Runs of two or more A=T base pairs facilitate the bending of DNA, whereas runs of two or more G=C base pairs have the opposite effect. When spaced at about 10 bp intervals, consecutive A=T base pairs help bend DNA into a circle. When consecutive G=C base pairs are spaced 10 bp apart, offset by 5 bp from runs of A=T base pairs, DNA binding to the nucleosome core is facilitated. [Data from PDB ID 1AOI, K. Luger et al., *Nature* 389:251, 1997.]

replication complex called PCNA (see Chapter 25), so that nucleosome deposition is closely coordinated with replication. Some of the same histone chaperones, or different ones, may help assemble nucleosomes after DNA repair, transcription, or other processes.

Histone exchange factors permit the substitution of histone variants for core histones in contexts other than postreplication. Proper placement of these variant histones is important. Studies show that mice lacking one of the variant histones die as early embryos (**Box 24-1**). Precise positioning of nucleosome cores also plays a role in the expression of some eukaryotic genes (Chapter 28).

Nucleosomes Are Packed into Highly Condensed Chromosome Structures

Wrapping of DNA around a nucleosome core compacts the DNA length about sevenfold. The overall compaction in a chromosome, however, is greater than 10,000-fold—ample evidence for even higher orders of structural organization. The condensation does not follow a rigid organization, but it is also not random and occurs so as to avoid the formation of knots.

P4 The higher levels of folding are not yet fully understood, but certain regions of DNA seem to associate with a chromosomal scaffold (**Fig. 24-27**) that contains many proteins. Given the need to fold and compact the chromosome without creating knots, topoisomerase II is one of the most abundant proteins in the chromosome, further emphasizing the relationship between DNA underwinding and chromatin structure. **P3** Topoisomerase II is so important to the maintenance of chromatin structure that inhibitors of this enzyme can kill rapidly dividing cells. Several drugs used in cancer chemotherapy are topoisomerase II inhibitors that allow the enzyme to promote strand breakage but not the resealing of the breaks (**Box 24-2**, p. 906).

There are additional layers of organization in the eukaryotic nucleus. Just before cell division during mitosis, chromosomes can be seen as highly condensed and organized structures (**Figure 24-28a**). During interphase, chromosomes appear dispersed (**Fig. 24-28b**,

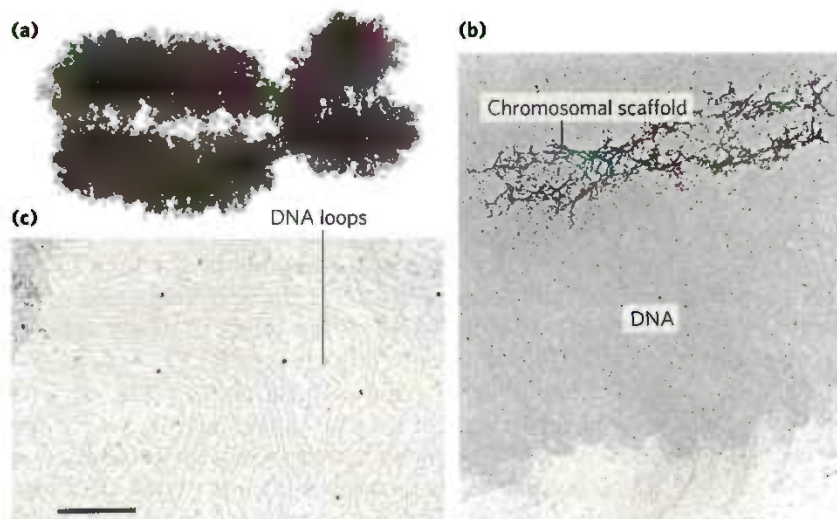


FIGURE 24-27 Loops of DNA attached to a chromosomal scaffold. (a) A swollen mitotic chromosome, produced in a buffer of low ionic strength, as seen in the electron microscope. Notice the appearance of chromatin loops at the margins. (b) Extraction of the histones leaves a proteinaceous chromosomal scaffold surrounded by naked DNA. (c) The DNA appears to be organized in loops attached at their base to the scaffold in the upper left corner; scale bar = 1 μ m. The three images are at different magnifications. [(a, b) Don W. Fawcett/ Science Source. (c) U. K. Laemmli et al., "Metaphase chromosome structure: The role of nonhistone proteins," *Cold Spring Harb. Symp. Quant. Biol.* 42:351, 1978. © Cold Spring Harbor Laboratory Press.]

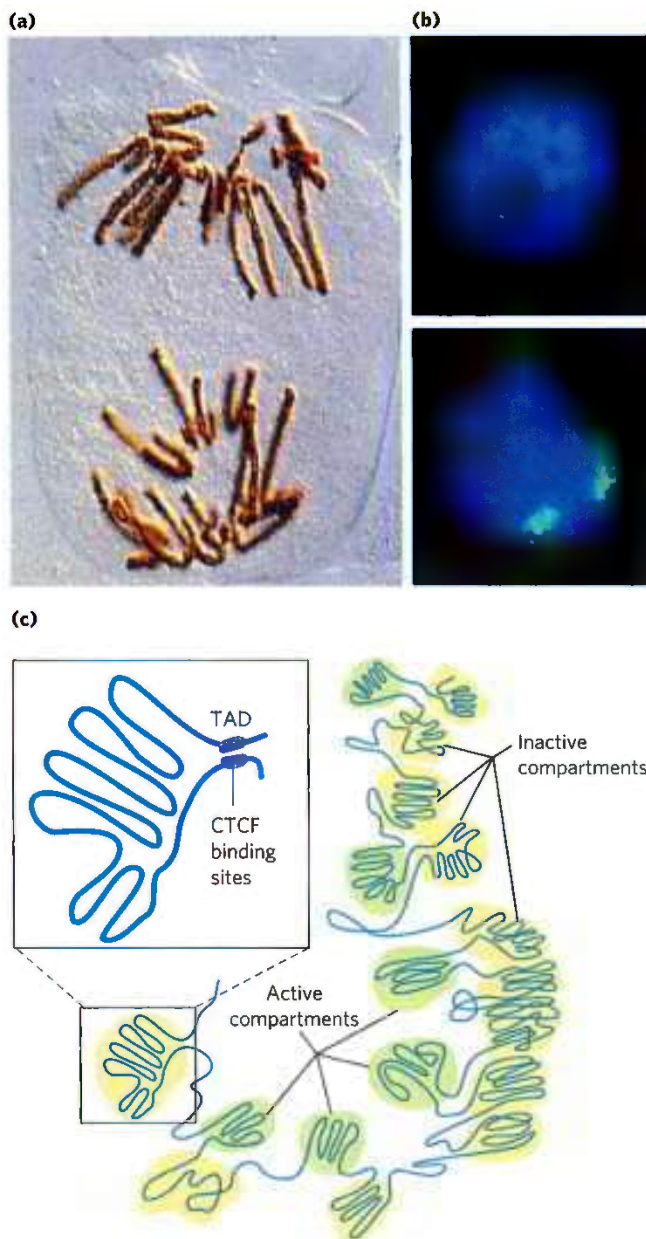


FIGURE 24-28 Chromosomal organization in the eukaryotic nucleus. (a) Condensed chromosomes at the mitotic anaphase in cells of the bluebell (*Endymion* sp.). (b) Interphase nuclei of human breast epithelial cells. The nucleus on the bottom has been treated so that its two copies of chromosome 11 fluoresce green. (c) Chromosomes are organized into active compartments, in which actively transcribed genes are clustered, and inactive compartments, made of heterochromatin within which genes are silenced. In both cases, the compartments feature large DNA loops called topologically associating domains (TADs), many constrained at their bases by DNA-binding proteins such as CTCF. Certain lncRNAs (Fig. 24-29) also play a role in defining loops within chromatin. Constraining a loop at its base not only provides a boundary for the loop, but also allows supercoiling within the loop to be controlled. [(a) Pr. G. Giménez-Martín/Science Source. (b) Karen Meaburn and Tom Misteli/National Cancer Institute.]

top), but they do not meander randomly in nuclear space (Fig. 24-28b, bottom). Each chromosome appears to be organized with two sets of compartments, one set that is transcriptionally active and the other that is transcriptionally inactive. The level of chromatin condensation is reduced

in the transcriptionally active compartments. The highly condensed DNA in transcriptionally inactive regions or in regions lacking genes is also called **heterochromatin**. Within each compartment, large segments of DNA are organized in loops called **topologically associating domains**, or **TADs**. The TADs, which typically average about 800,000 bp, are often bordered by DNA sites recognized by the CCCTC-binding factor (CTCF). The binding by CTCF brings together sites that are otherwise quite distant in the linear DNA sequence, tethering the base of the loop (Fig. 24-28c).

P4 Another important component defining the structure of chromosomes is RNA, particularly a class of RNAs called **long noncoding RNAs (lncRNAs)**. RNA has the potential to take up a variety of structures (see Chapter 8), and can interact with DNA, proteins, or other RNA molecules. The lncRNAs, as the name implies, are functional RNAs, generally over 200 nucleotides long, that do not necessarily encode proteins. Many lncRNAs are now known and more are being discovered rapidly. Many of them provide a scaffold for proteins that both bind to the RNA and affect chromosome structure and function. Some proteins that bind to lncRNAs also have binding sites on DNA, and the RNAs provide a link that tethers distant parts of the chromosome together (Fig. 24-29). Other proteins that bind to lncRNAs help to position nucleosomes, modify histones,

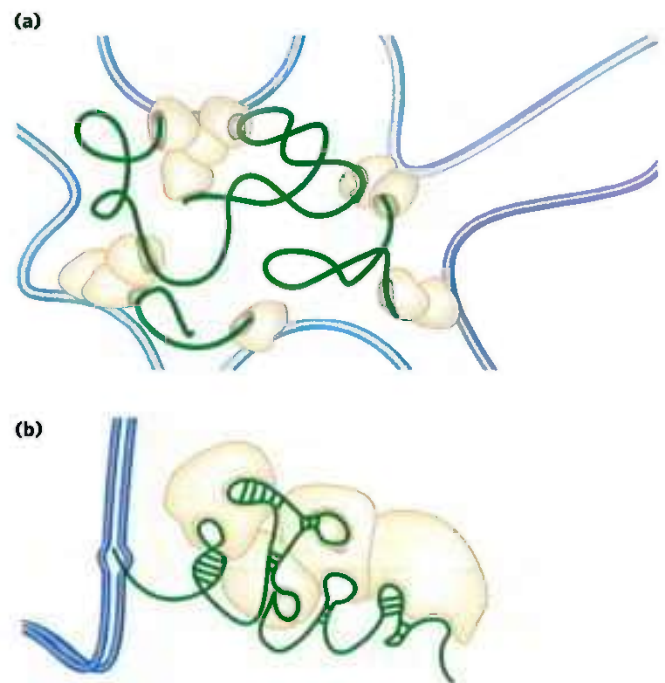


FIGURE 24-29 Effects of lncRNAs on chromosome architecture and gene expression. (a) Various lncRNAs can interact with DNA-binding proteins and in some cases with DNA to tether otherwise distant segments of DNA. (b) A transcribed lncRNA can interact with multiple proteins that have gene regulatory roles, suppressing or activating transcription of nearby genes. [Information from J. M. Engreitz et al., *Nat. Rev. Mol. Cell Biol.* 17:756, 2016, Fig. 5.]

Epigenetics, Nucleosome Structure, and Histone Variants

Information that is passed from one generation to the next—to daughter cells at cell division or from parent to offspring—but is not encoded in DNA sequences is referred to as **epigenetic** information. Much of it is in the form of covalent modification of histones and/or the placement of histone variants in chromosomes. Understanding that placement in the context of a chromosome encompassing millions of base pairs is the focus of some powerful technologies.

Chromatin regions where active gene expression (transcription) is occurring tend to be partially decondensed and are called **euchromatin**. In these regions, histones H3 and H2A are often replaced by the histone variants H3.3 and H2AZ, respectively (Fig. 1). The complexes that deposit nucleosome cores containing histone variants on the DNA are similar to those that deposit nucleosome cores with the more common histones. Nucleosome cores containing histone H3.3 are

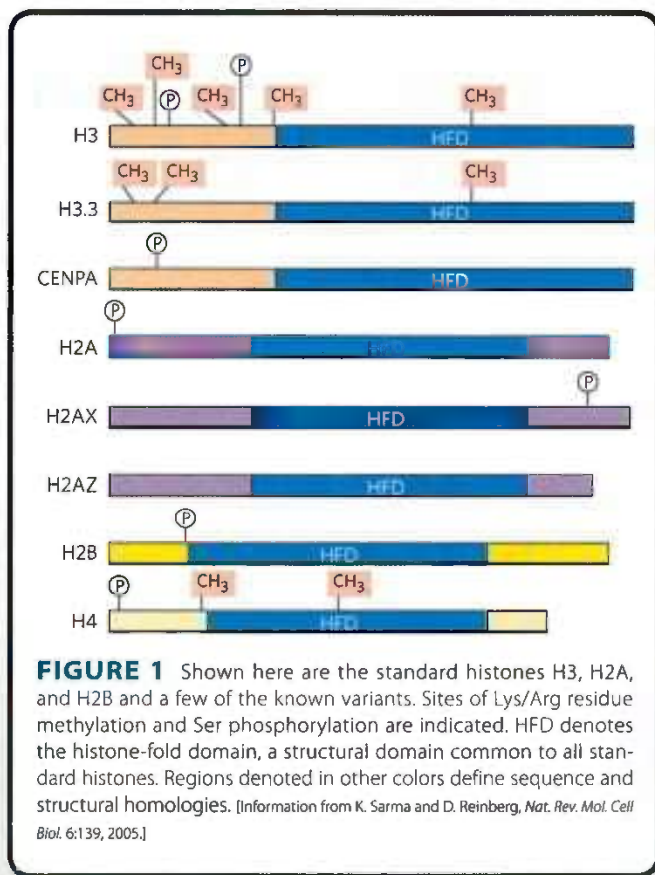
deposited by a complex in which CAF1 (chromatin assembly factor 1) is replaced by the protein HIRA (a name derived from a class of proteins called HIR, for *histone repressor*). Both CAF1 and HIRA can be considered histone chaperones, helping to ensure the proper assembly and placement of nucleosomes. Histone H3.3 differs in sequence from H3 by only four amino acid residues, but these residues all play key roles in histone deposition.

Like histone H3.3, H2AZ is associated with a distinct nucleosome deposition complex, and it is generally associated with chromatin regions involved in active transcription. Incorporation of H2AZ stabilizes the nucleosome octamer, but it impedes some cooperative interactions between nucleosomes that are needed to compact the chromosome. This leads to a more open chromosome structure that enables the expression of genes in the region where H2AZ is located. The gene encoding H2AZ is essential in mammals. In fruit flies, loss of H2AZ prevents development beyond the larval stage.

Another H2A variant is H2AX, which is associated with DNA repair and genetic recombination. In mice, the absence of H2AX results in genome instability and male infertility. Modest amounts of H2AX seem to be scattered throughout the genome. When a double-strand break occurs, nearby molecules of H2AX become phosphorylated at Ser¹³⁹ in the carboxyl-terminal region. If this phosphorylation is blocked experimentally, formation of the protein complexes necessary for DNA repair is inhibited.

The H3 histone variant known as CENPA (*centromere protein A*) is associated with the repeated DNA sequences in centromeres. The chromatin in the centromere region contains the histone chaperones CAF1 and HIRA, and both proteins could be involved in the deposition of nucleosome cores containing CENPA. Elimination of the gene for CENPA is lethal in mice.

The function and positioning of the histone variants can be studied by an application of technologies used in genomics. One useful technology is **chromatin immunoprecipitation**, or **chromatin IP (ChIP)**. Nucleosomes containing a particular histone variant are precipitated by an antibody that binds specifically to this variant. These nucleosome cores can be studied in isolation from their DNA, but more commonly the DNA associated with them is included in the study to determine where the nucleosome cores of interest bind. The DNA can be sequenced, yielding a map of genomic sequences to which those particular nucleosome cores bind. This technique is called a **ChIP-Seq** experiment (Fig. 2).



methyrate DNA at various locations to alter gene transcription, and generally affect chromosome structure in many different ways. Some well-studied examples include particular lncRNAs that play a major role in X chromosome inactivation in mammals (Box 24-3).

The entire structure of each chromosome is constrained within a subnuclear domain called a **chromosome territory** (Fig. 24-30). There is little or no intermingling of chromosomal DNA in different territories. The exact location of chromosome territories

The histone variants, along with the many covalent modifications that histones undergo, help define and isolate the functions of chromatin. They mark the chromatin, facilitating or suppressing specific functions such as chromosome segregation,

transcription, and DNA repair. The histone modifications do not disappear at cell division or during meiosis, and thus they become part of the information transmitted from one generation to the next in all eukaryotic organisms.

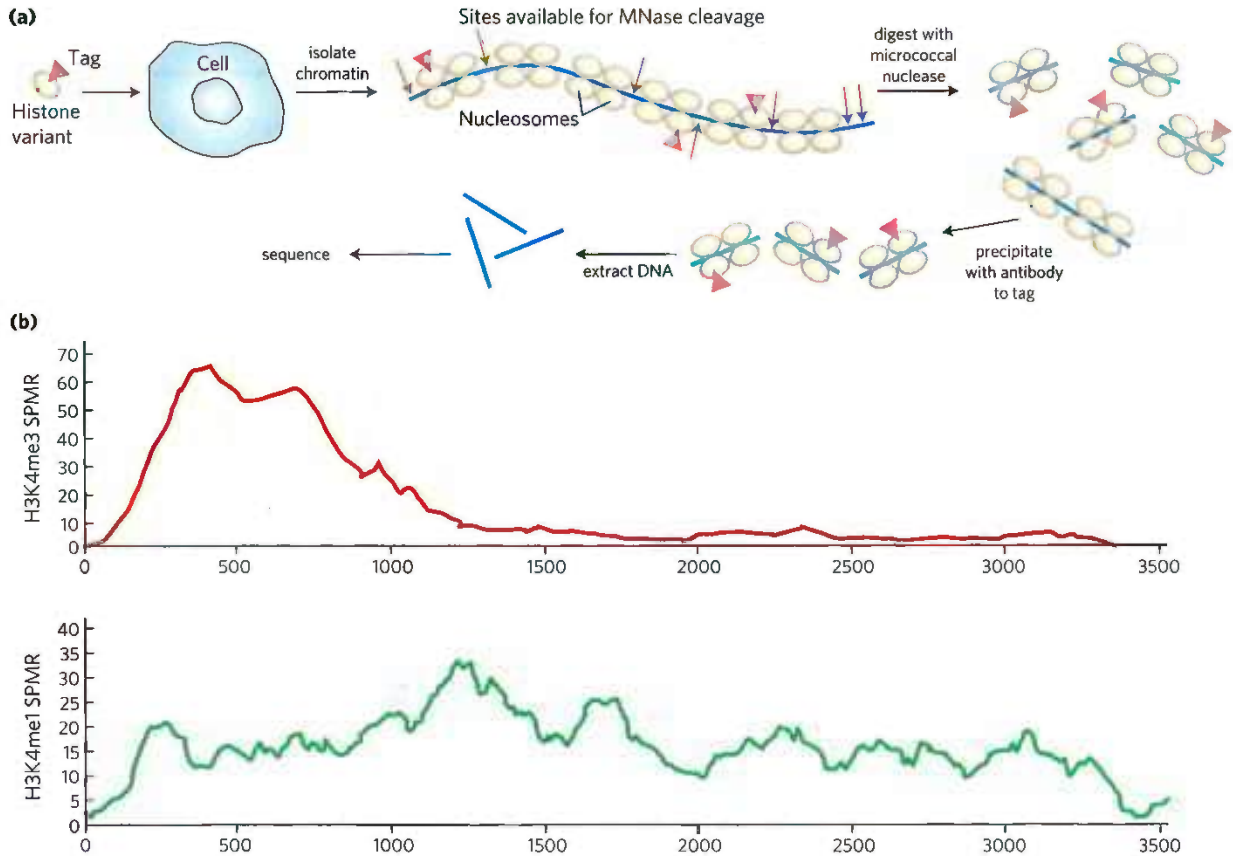


FIGURE 2 A ChIP-Seq experiment is designed to reveal the genomic DNA sequences to which a particular histone variant binds. (a) A histone variant with an epitope tag (a protein or chemical structure recognized by an antibody; see Chapters 5 and 9) is introduced into a particular cell type, where it is incorporated into nucleosomes. (In some cases, an epitope tag is unnecessary because antibodies are available that bind directly to the histone modification of interest.) Chromatin is isolated from the cells and digested briefly with micrococcal nuclease (MNase). The DNA bound in nucleosomes is protected from digestion, but the linker DNA is cleaved, releasing segments of DNA bound to one or two nucleosomes. An antibody that binds to the epitope tag is

added, and the nucleosomes containing the epitope-tagged histone variant are selectively precipitated. The DNA in these nucleosomes is extracted from the precipitate and sequenced in depth to reveal the locations where the nucleosomes are bound. (b) In this example, the binding of different modified versions of histone H3 is characterized along the gene *YLR249W* from the yeast *S. cerevisiae*. Numbers along the bottom correspond to numbered nucleotide positions in this chromosomal segment. The top and bottom panels show the distributions of H3K4me3 (H3 with Lys⁴ methylated three times) and H3K4me1, respectively. SPMR is sequence tags per million reads. [Information from L. M. Soares et al., *Mol. Cell* 68:773, 2017, Fig. 1D.]

varies from cell to cell in an organism, but some spatial patterns are evident. Some chromosomes have a higher density of genes than others (for example, human chromosomes 1, 16, 17, 19, and 22), and these tend to have territories in the center of the nucleus. Chromosomes

with more heterochromatin tend to be located on the nuclear periphery. Spaces between chromosomes are often sites where transcriptional machinery and transcriptionally active genes on adjacent chromosomes are concentrated.

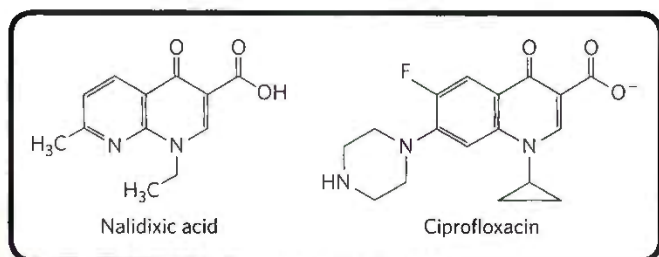


Curing Disease by Inhibiting Topoisomerases

P3 The topological state of cellular DNA is intimately connected with its function. Without topoisomerases, cells cannot replicate or package their DNA, or express their genes — and they die. Inhibitors of topoisomerases have therefore become important pharmaceutical agents, targeted at infectious agents and malignant cells.

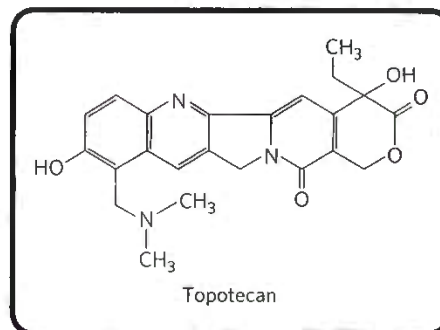
Two classes of bacterial topoisomerase inhibitors have been developed as antibiotics. The coumarins, including novobiocin and coumermycin A1, are natural products derived from *Streptomyces* species. They inhibit the ATP binding of the bacterial type II topoisomerases, DNA gyrase and topoisomerase IV. These antibiotics are not often used to treat infections in humans, but research continues to identify clinically effective variants.

The quinolone antibiotics, also inhibitors of bacterial DNA gyrase and topoisomerase IV, first appeared in 1962 with the introduction of nalidixic acid. This compound had limited effectiveness and is no longer used clinically in the United States, but the continued development of this class of drugs eventually led to the introduction of the fluoroquinolones, exemplified by ciprofloxacin (Cipro). The quinolones act by blocking the last step of the topoisomerase reaction, the resealing of the DNA strand breaks. Ciprofloxacin is a broad-spectrum antibiotic. It is one of the few antibiotics reliably effective in treating anthrax infections and is considered a valuable agent in protection against possible bioterrorism. Quinolones are selective for the bacterial topoisomerases, inhibiting the eukaryotic enzymes only at concentrations several orders of magnitude greater than the therapeutic doses.

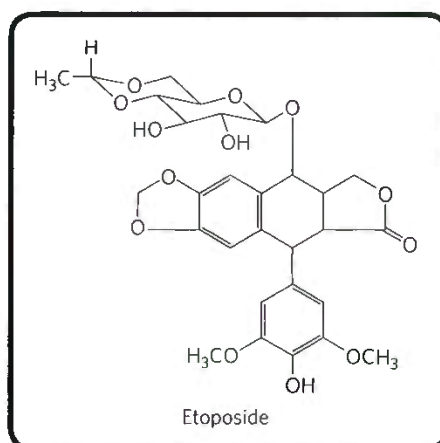


Some of the most important chemotherapeutic agents used in cancer treatment are inhibitors of human topoisomerases. Topoisomerases are generally present at elevated levels in tumor cells, and agents targeted to these enzymes are much more toxic to the tumors than to most other tissue types. Inhibitors of both type I and type II topoisomerases have been developed as anticancer drugs.

Camptothecin, isolated from a Chinese ornamental tree and first tested clinically in the 1970s, is an inhibitor of



eukaryotic type I topoisomerases. Clinical trials indicated limited effectiveness, despite its early promise in preclinical work on mice. However, two effective derivatives, irinotecan (Campto) and topotecan (Hycamtin) — used to treat colorectal cancer and ovarian cancer, respectively — were developed in the 1990s. All of these drugs act by trapping the topoisomerase-DNA complex in which the DNA is cleaved, inhibiting religation.



The human type II topoisomerases are targeted by a variety of antitumor drugs, including doxorubicin (Adriamycin), etoposide (Etopophos), and ellipticine. Doxorubicin, effective against several kinds of human tumors, is an anthracycline in clinical use. Most of these drugs stabilize the covalent topoisomerase-DNA (cleaved) complex.

All of these anticancer agents generally increase the levels of DNA damage in the targeted, rapidly growing tumor cells. However, noncancerous tissues can also be affected, leading to a more general toxicity and unpleasant side effects that must be managed during therapy. As cancer therapies become more effective and survival statistics for cancer patients improve, the independent appearance of new tumors is becoming a greater problem. In the continuing search for new cancer therapies, topoisomerases are likely to remain prominent targets for research.

X Chromosome Inactivation by an lncRNA: Preventing Too Much of a Good (or Bad) Thing

In higher eukaryotes, the Y chromosome contains only a few genes. The X chromosome, in contrast, often contains more than a thousand genes, many essential for cell function and organismal development. Males have just one X chromosome, whereas females have two. Females thus could end up with a double dose of gene products, leading to a variety of potentially toxic outcomes. Mechanisms of gene dosage compensation vary among different groups of eukaryotes. In most mammals, dosage compensation is accomplished by X chromosome inactivation, randomly affecting just one of the two X chromosomes in a given cell. The effects of this can be seen in calico cats, all of which are female. Fur pigmentation genes are found on the X chromosome. When a cat inherits chromosomes with different pigmentation alleles on its two X chromosomes, its coloration patterns reflect the random inactivation of one or the other X chromosome in different sets of cells.



A calico cat. [krlbokhin/Getty Images]

An lncRNA called Xist (X-inactive specific transcript) is uniquely expressed from its gene on the inactive X chromosome. Xist is approximately 17,000 nucleotides long, encodes no proteins, and is expressed only in cells with two X chromosomes (not in males). Xist is an essential component of the X inactivation process. This lncRNA interacts with many proteins and transcription factors (Fig. 1). As it migrates in the region immediately proximal to the gene from which it is transcribed, Xist gradually encompasses more and more DNA as that X chromosome undergoes a major condensation, giving rise to the inactive form called a **Barr body**.

The active X chromosome also transcribes the gene that produces Xist, but it does so in the opposite direction to produce a longer lncRNA called Tsix (Xist spelled backward). Tsix is synthesized using a different RNA polymerase binding site (promoter; Chapter 26) and is 40,000 nucleotides long. A large part of Tsix is perfectly complementary to Xist, and it antagonizes the function of any Xist that may appear near the active X chromosome.

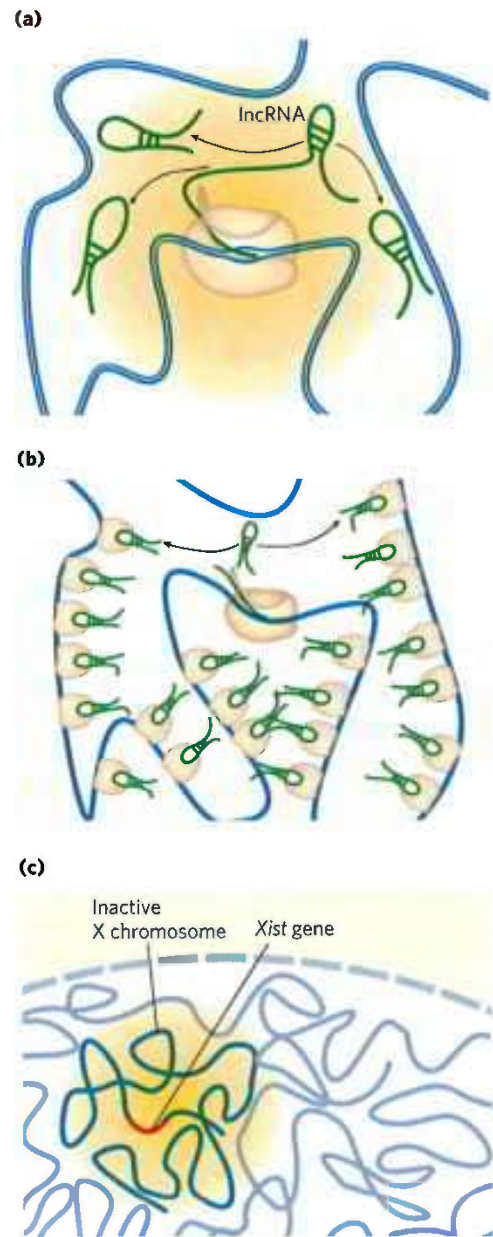


FIGURE 1 (a) As Xist RNA is transcribed, it migrates to nearby regions within an X chromosome, (b) binding to proteins including SAFA (scaffold attachment factor A). Xist binding spreads through the chromosome, leading to condensation and other architectural changes to form a Barr body. Other than the Xist gene itself, gene expression is suppressed throughout the chromosome. (c) The effects are quite local; Xist does not spread within the nucleus beyond the chromosome territory occupied by the inactivated X chromosome.

[Information from J. M. Engreitz et al., *Nat. Rev. Mol. Cell Biol.* 17:756, 2016, Fig. 5.]

Random X inactivation occurs in the cells of a female embryo long before birth. The processes that initiate the inactivation of one, but not both, of the X chromosomes in a particular cell are still being elucidated.

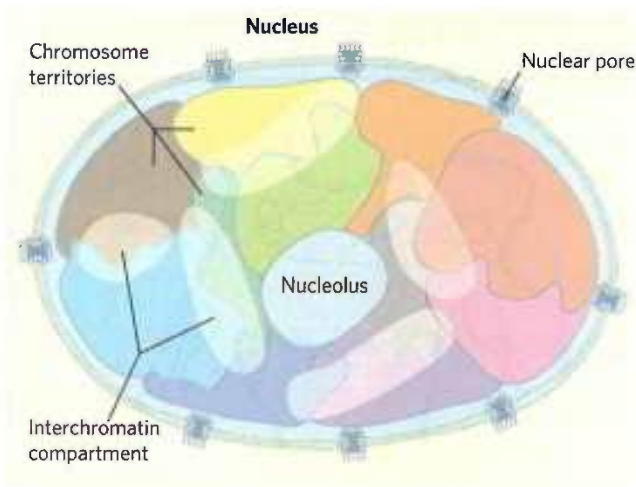


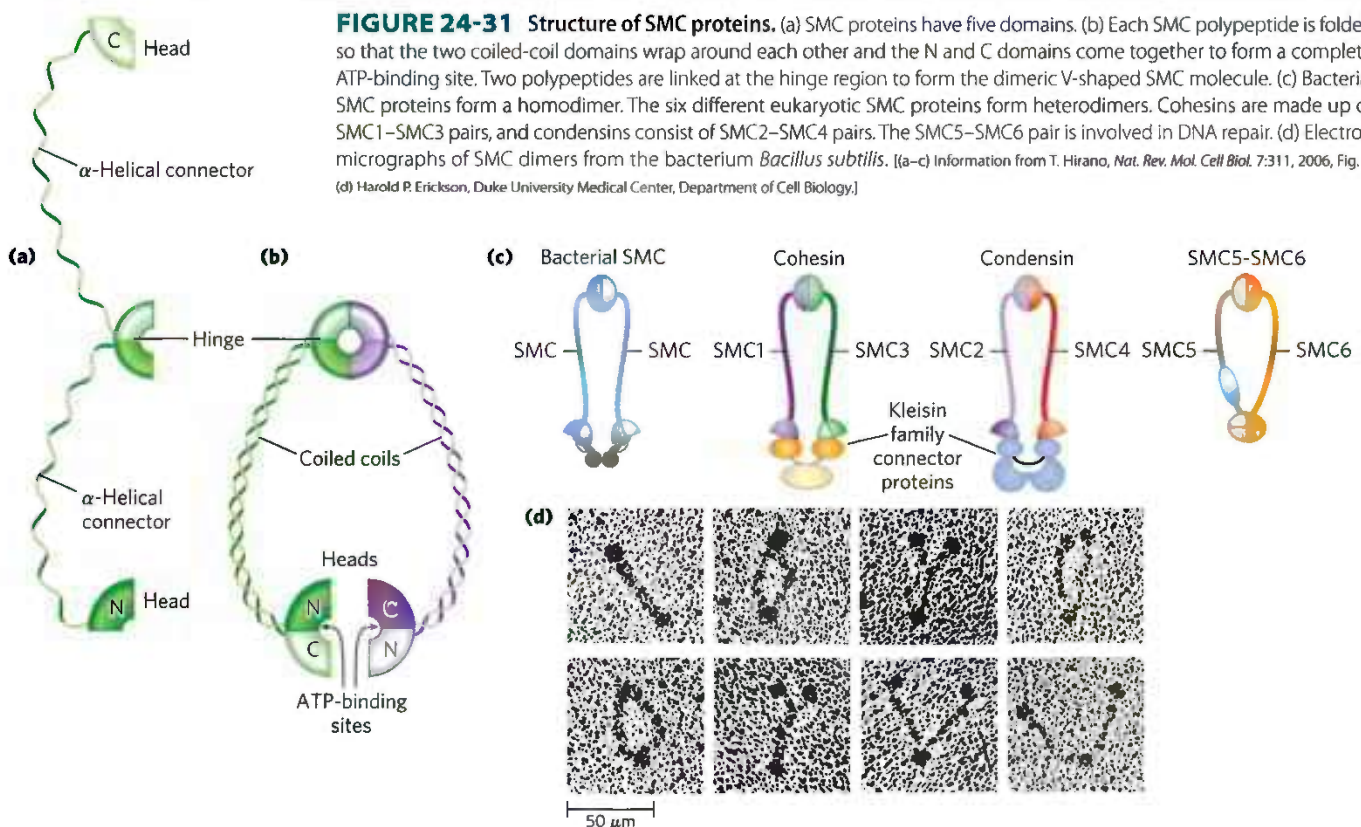
FIGURE 24-30 Chromosome territories. Cartoon showing chromosome territories in a eukaryotic nucleus. The interchromatin compartments are enriched in transcriptional machinery and have abundant actively transcribed genes. The nucleolus is a suborganelle within the nucleus where ribosomes are synthesized and assembled (Chapter 27).

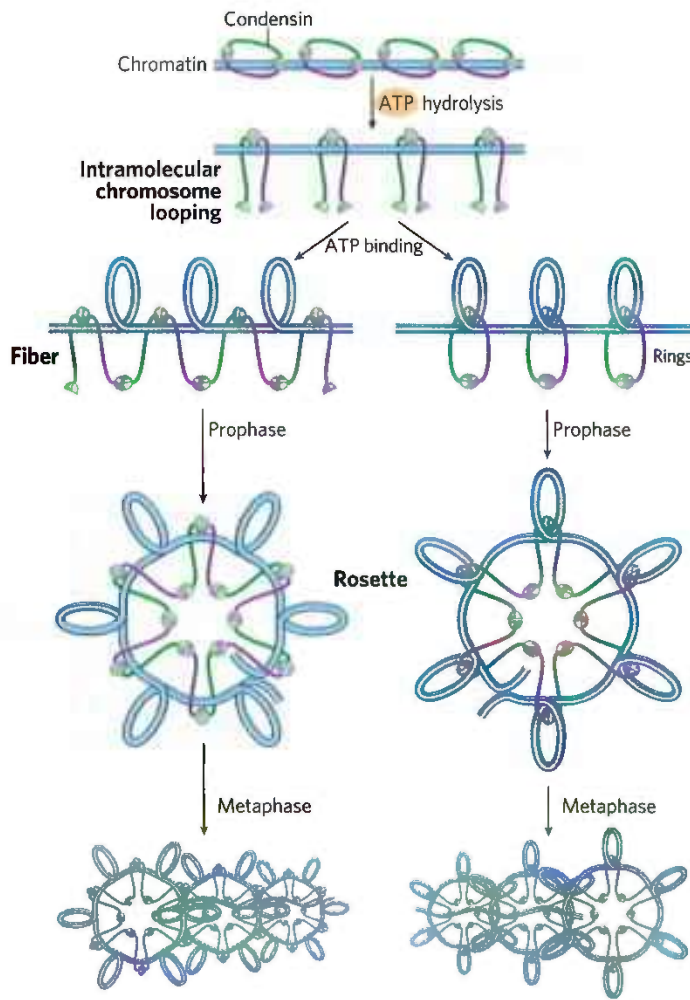
Condensed Chromosome Structures Are Maintained by SMC Proteins

P4 SMC proteins (structural maintenance of chromosomes), the third major class of chromatin protein in addition to the histones and topoisomerases, are responsible for maintaining the structure and integrity of

chromosomes following replication. The primary structure of SMC proteins consists of five distinct domains (Fig. 24-31a). The amino- and carboxyl-terminal globular domains, N and C, each of which contains part of an ATP-hydrolytic site, are connected by two regions of α -helical coiled-coil motifs (see Fig. 4-10) joined by a hinge domain. The proteins are generally dimeric, forming a V-shaped complex that is thought to be tied together through the protein's hinge domains. One N domain and one C domain come together like tweezers to form a complete ATP-hydrolytic site at each free end of the V (Fig. 24-31b).

Proteins in the SMC family are found in all types of organisms, from bacteria to humans. Eukaryotes have two major types, cohesins and condensins, both of which are bound by regulatory and accessory proteins (Fig. 24-31c, d). **Cohesins** play a substantial role in linking together sister chromatids immediately after replication and keeping them together as the chromosomes condense to metaphase. This linkage is essential if chromosomes are to segregate properly at cell division. Cohesins, along with a third protein, kleisin, are thought to form a ring around the replicated chromosomes that ties them together until separation is required. The ring may expand and contract in response to ATP hydrolysis. **Condensins** are essential to the condensation of chromosomes as cells enter mitosis. In the laboratory, condensins bind to DNA in a manner that creates positive supercoils; that is, condensin binding causes the DNA to become overwound, in contrast to the underwinding





induced by the binding of nucleosomes. A model for the role of condensins in chromatin compaction is presented in Figure 24-32. In brief, as DNA is compacted to

FIGURE 24-32 Two current models of the possible role of condensins in chromatin condensation. Initially, the DNA is bound at the hinge region of the SMC protein, in the interior of what can become an intramolecular SMC ring. ATP binding leads to head-to-head association, forming supercoiled loops in the bound DNA. Subsequent rearrangement of the head-to-head interactions to form rosettes condenses the DNA. Condensins may organize the looping of the chromosome segments in several ways. [Information from T. Hirano, *Nat. Rev. Mol. Cell Biol.* 7:311, 2006, Fig. 6.]

form tighter and tighter loops, the condensins stabilize the loops by binding at the base of each one. Cohesins and condensins are essential in orchestrating the many changes in chromosome structure during the eukaryotic cell cycle (Fig. 24-33).

Bacterial DNA Is Also Highly Organized

We now turn briefly to the structure of bacterial chromosomes. **P2** Bacterial DNA is compacted in a structure called the **nucleoid**, which can occupy a significant fraction of the cell volume (Fig. 24-34). The DNA seems to be attached at one or more points to the inner surface of the plasma membrane. Much less is known about the structure of the nucleoid than of eukaryotic chromatin, but a complex organization is slowly being revealed. **P3** In *E. coli*, a scaffoldlike structure seems to organize the *circular* chromosome into a series of about 500 looped domains, each encompassing, on average, 10,000 bp (Fig. 24-35), as described above for chromatin. The domains are topologically constrained; for example, if the DNA is cleaved in one domain, only the DNA within that domain will be relaxed. The domains do not have fixed end points. Instead, the boundaries are most likely in constant motion along the DNA, coordinated with DNA replication.

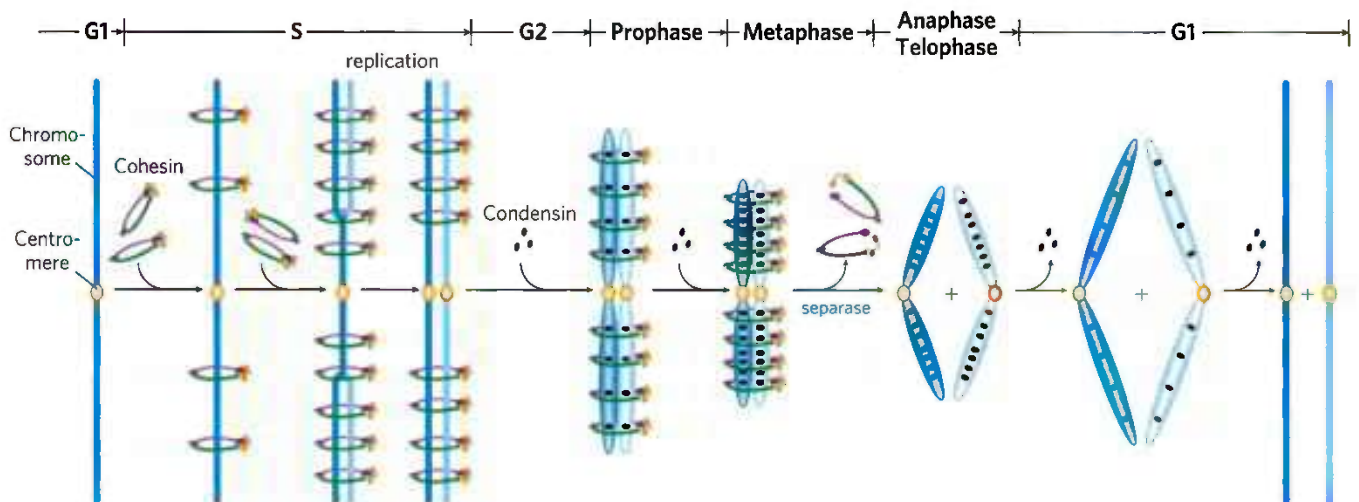


FIGURE 24-33 The roles of cohesins and condensins in the eukaryotic cell cycle. Cohesins are loaded onto the chromosomes during G1 (see Fig. 24-22), tying the sister chromatids together during replication. At the onset of mitosis, condensins bind and maintain the chromatids

in a condensed state. During anaphase, the enzyme separase removes the cohesin links. Once the chromatids separate, condensins begin to unload and the daughter chromosomes return to the uncondensed state. [Information from D. P. Bazett-Jones et al., *Mol. Cell* 9:1183, 2002, Fig. 5.]



FIGURE 24-34 *E. coli* nucleoids. The DNA of these cells is stained with a dye that fluoresces blue when exposed to UV light. The blue areas define the nucleoids. Notice that some cells have replicated their DNA but have not yet undergone cell division and hence have multiple nucleoids. (Lars Renner)

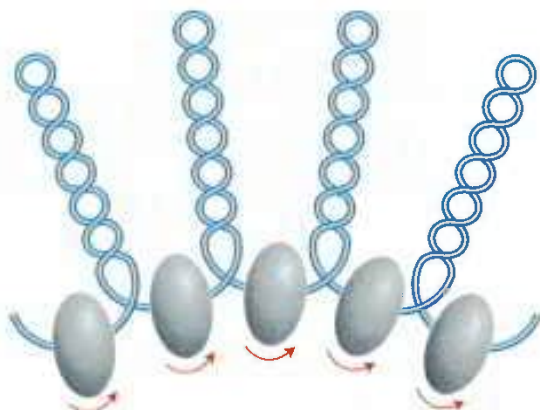


FIGURE 24-35 Looped domains of the *E. coli* chromosome. Each domain is about 10,000 bp long. The domains are not static but move along the DNA as replication proceeds. Barriers at the boundaries of the domains, of unknown composition, prevent the relaxation of DNA beyond the boundaries of the domain where a strand break occurs. The putative boundary complexes are shown as gray-shaded ovoids. The arrows denote movement of DNA through the boundary complexes.

Bacterial DNA does not seem to have any structure comparable to the local organization provided by nucleosomes in eukaryotes. **P4** Histone-like proteins are abundant in *E. coli* — the best-characterized example is a two-subunit protein called HU (M_r 19,000) — but these proteins bind and dissociate within minutes, and no regular, stable DNA-histone structure has been found. The dynamic structural changes in the bacterial chromosome may reflect a requirement for more ready access to its genetic information. The bacterial cell division cycle can be as short as 15 min, whereas a typical eukaryotic cell may not divide for hours or even months. In addition, a much greater fraction of bacterial DNA is used to encode RNA and/or protein products. Higher rates of cellular metabolism in bacteria mean that a much higher proportion of the DNA is being transcribed or replicated at a given time than in most eukaryotic cells.

SUMMARY 24.3 The Structure of Chromosomes

- A eukaryotic chromosome is made of DNA, protein, and RNA, forming a structure called chromatin.
- Histones are small, basic DNA-binding proteins. Complexes of histones form nucleosomes, the fundamental structural unit of chromatin.
- The nucleosome consists of histones and a 200 bp segment of DNA. A core protein particle containing eight histone molecules (two copies each of histones H2A, H2B, H3, and H4) is encircled by a segment of DNA (about 146 bp) in the form of a left-handed solenoidal supercoil.
- Higher-order folding of chromosomes involves attachment to a chromosomal scaffold. Transcriptionally active and inactive regions of chromosomes are separated into compartments, each featuring large loops of DNA, each loop constrained at its base by proteins and lncRNAs. Individual chromosomes are constrained within nuclear subdomains called territories. Histone H1, topoisomerase II, and SMC proteins play organizational roles in chromosomes.
- The SMC proteins, principally cohesins and condensins, have important roles in keeping the chromosomes organized during each stage of the cell cycle.
- The bacterial chromosome is extensively compacted into the nucleoid, but the chromosome seems to be much more dynamic and irregular in structure than eukaryotic chromatin, reflecting the shorter cell cycle and very active metabolism of a bacterial cell.

KEY TERMS

Terms in bold are defined in the glossary.

chromosome 885	topoisomerases 895
phenotype 886	catenane 896
mutation 886	plectonemic 897
gene 886	solenoidal 898
regulatory sequence 886	chromatin 899
plasmid 887	histones 899
intron 889	nucleosome 899
exon 889	histone exchange factors 902
simple-sequence DNA 890	heterochromatin 903
satellite DNA 890	topologically associating domains (TADs) 903
centromere 890	long noncoding RNA (lncRNA) 903
telomere 890	epigenetic 904
supercoil 891	euchromatin 904
relaxed DNA 891	chromosome territory 904
topology 891	Barr body 907
linking number 893	SMC proteins 908
specific linking difference 894	cohesins 908
superhelical density (σ) 894	condensins 908
topoisomers 894	nucleoid 909

PROBLEMS

1. Packaging of DNA in a Virus Bacteriophage T2 has a DNA of molecular weight 120×10^6 contained in a head about 210 nm long. Calculate the length of the DNA and compare it with the length of the T2 head. Assume the molecular weight of a nucleotide base pair is 650 and that the DNA is in the B form and relaxed.

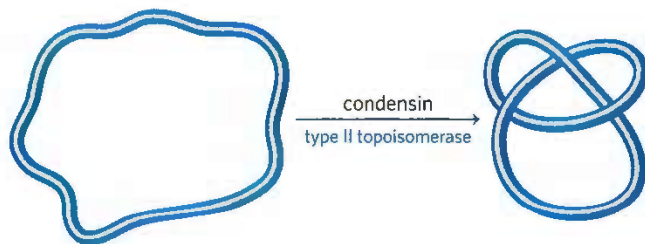
2. The DNA of Phage M13 The base composition of the DNA within a bacteriophage M13 viral particle is A, 23%; T, 36%; G, 21%; C, 20%. What does this tell you about this DNA molecule?

3. The *Mycoplasma* Genome The complete genome of the simplest bacterium known, *Mycoplasma genitalium*, is a circular DNA molecule with 580,070 bp. Calculate the molecular weight and contour length (when relaxed) of this molecule. What is Lk_0 for the *Mycoplasma* chromosome? If $\sigma = -0.06$, what is Lk ?

4. Size of Eukaryotic Genes An enzyme isolated from rat liver has 192 amino acid residues and is encoded by a gene with 1,440 bp. Explain the relationship between the number of amino acid residues in the enzyme and the number of nucleotide pairs in its gene.

5. Linking Number A closed-circular DNA molecule in its relaxed form has an Lk of 500. Approximately how many base pairs are in this DNA? How is Lk altered (increases, decreases, doesn't change, becomes undefined) when (a) a protein complex binds to form a nucleosome, (b) one DNA strand is broken, (c) DNA gyrase and ATP are added to the DNA solution, or (d) the double helix is denatured by heat?

6. DNA Topology In the presence of a eukaryotic condensin and a bacterial type II topoisomerase, the Lk of a relaxed closed-circular DNA molecule does not change. However, the DNA becomes highly knotted.

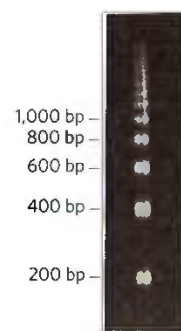


Formation of the knots requires breakage of the DNA, passage of a segment of DNA through the break, and religation by the topoisomerase. Given that every reaction of the topoisomerase would be expected to result in a change in linking number, how can Lk remain the same?

7. Superhelical Density Bacteriophage λ infects *E. coli* by integrating its DNA into the bacterial chromosome. The success of this recombination depends on the topology of the *E. coli* DNA. When the superhelical density (σ) of the *E. coli* DNA is greater than -0.045 , the probability of integration is less than 20%; when σ is less than -0.06 , the probability is $>70\%$. Plasmid DNA isolated from an *E. coli* culture is found to have a length of 13,800 bp and an Lk of 1,222. Calculate σ for this DNA and predict the likelihood that bacteriophage λ will be able to infect this culture.

8. Altering Linking Number (a) What is the Lk of a 5,000 bp circular duplex DNA molecule with a nick in one strand? (b) What is the Lk of the molecule in (a) when the nick is sealed (relaxed)? (c) How would the Lk of the molecule in (b) be affected by the action of a single molecule of *E. coli* topoisomerase I? (d) What is the Lk of the molecule in (b) after eight enzymatic turnovers by a single molecule of DNA gyrase in the presence of ATP? (e) What is the Lk of the molecule in (d) after four enzymatic turnovers by a single molecule of bacterial type I topoisomerase? (f) What is the Lk of the molecule in (d) after binding of one nucleosome core?

9. Chromatin The agarose gel shown, in which the thick bands represent DNA, helped researchers define nucleosome structure. They generated this result by briefly treating chromatin with an enzyme that degrades DNA, then removing all protein and subjecting the purified DNA to electrophoresis. Numbers at the side of the gel denote the position to which a linear DNA of the indicated size would migrate. What does this gel demonstrate about chromatin structure? Why are the DNA bands thick and spread out rather than sharply defined?



[Courtesy Dr. Roger D. Kornberg, Stanford University School of Medicine.]

10. DNA Structure Explain how the underwinding of a B-DNA helix might facilitate or stabilize the formation of Z-DNA (see Fig. 8-17).

11. Maintaining DNA Structure (a) Describe two structural features required for a DNA molecule to maintain a negatively supercoiled state. (b) List three structural changes that become more favorable when a DNA molecule is negatively supercoiled. (c) What *E. coli* enzyme, with the aid of ATP, can generate negative superhelicity in DNA? (d) Describe the physical mechanism by which this enzyme acts.

12. Yeast Artificial Chromosomes (YACs) Researchers use YACs to clone large pieces of DNA in yeast cells. What three types of DNA sequence do researchers require to ensure proper replication and propagation of a YAC in a yeast cell?

13. Nucleoid Structure in Bacteria In bacteria, DNA topology affects the transcription of a subset of genes, with expression increasing or (more often) decreasing when the DNA is relaxed. Following cleavage of a bacterial chromosome at a specific site by a restriction enzyme (one that cuts at a long, and thus rare, sequence), only nearby genes (within 10,000 bp) exhibit either an increase or a decrease in expression. The transcription of genes elsewhere in the chromosome is unaffected. Explain. (Hint: See Fig. 24-35.)

14. DNA Topology When DNA is subjected to electrophoresis in an agarose gel, shorter molecules migrate faster than

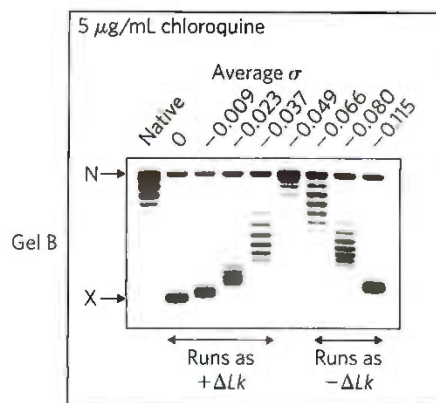
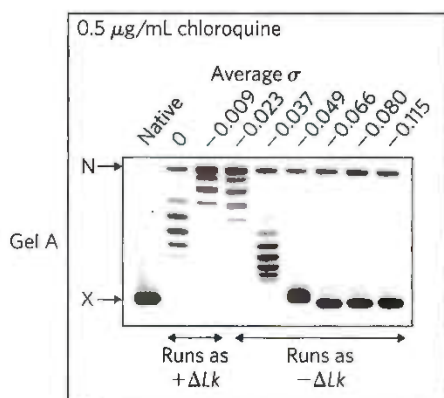
longer ones. Closed-circular DNAs of the same size but with different linking numbers also can be separated on an agarose gel: topoisomers that are more supercoiled, and thus more condensed, migrate faster through the gel. In the gel shown, purified plasmid DNA has migrated from top to bottom. There are two bands, with the faster band much more prominent.



[Michael Cox Laboratory.]

- (a) What are the DNA species in the two bands? (b) If topoisomerase I were added to a solution of this DNA, what would happen to the upper and lower bands after electrophoresis? (c) If DNA ligase were added to the DNA, would the appearance of the bands change? Explain your answer. (d) If DNA gyrase plus ATP were added to the DNA after the addition of DNA ligase, how would the band pattern change?

15. DNA Topoisomers When DNA is subjected to electrophoresis in an agarose gel, shorter molecules migrate faster than longer ones. Closed-circular DNAs of the same size but different linking number also can be separated on an agarose gel: topoisomers that are more supercoiled, and thus more condensed, migrate faster through the gel—from top to bottom in the two gels shown. An investigator added a dye, chloroquine, to these gels. Chloroquine intercalates between base pairs and stabilizes a more underwound DNA structure. When the dye binds to a relaxed closed-circular DNA, the DNA is underwound where the dye binds, and unbound regions take on positive supercoils to compensate. In the experiment shown here, an investigator used topoisomerases to make preparations of the same DNA circle with different superhelical densities (σ). Completely relaxed DNA migrated to the position labeled N (for nicked), and highly supercoiled DNA (above the limit where individual topoisomers can be distinguished) migrated to the position labeled X.



[R. P. Bowater (2005), "Supercoiled DNA: structure," in *Encyclopedia of Life Sciences*, John Wiley & Sons/Wiley InterScience, www.els.net.]

- (a) In gel A, why does the $\sigma = 0$ lane (i.e., DNA prepared so that $\sigma = 0$, on average) have multiple bands? (b) In gel B, is the DNA from the $\sigma = 0$ preparation positively or negatively supercoiled in the presence of the intercalating dye? (c) In both gels, the $\sigma = -0.115$ lane has two bands, one a highly supercoiled DNA and one relaxed. Propose a reason for the presence of relaxed DNA in these lanes (and others). (d) The native DNA (leftmost lane in each gel) is the same DNA circle isolated from bacterial cells and untreated. What is the approximate superhelical density of this native DNA?

16. Nucleosomes The human genome comprises just over 3.1 billion base pairs. Assuming it contains nucleosomes that are spaced as described in this chapter, how many molecules of histone H2A are present in one somatic human cell? (Ignore reductions in H2A due to its replacement in some regions by H2A variants.) How would the number change after DNA replication but before cell division?

17. Bacterial DNA Topoisomerase IV The gene encoding topoisomerase IV in *E. coli* is essential, even though another type II topoisomerase (topoisomerase II or gyrase) is present. Suggest a reason for the requirement for topoisomerase IV.

18. Chromosome Topology Eukaryotic chromosomes are linear DNA molecules, yet the DNA of a chromosome retains a high level of underwinding (supercoiling) throughout its length. How does the organization of chromosomal DNA into loops called TADs contribute to the maintenance of supercoiling?

DATA ANALYSIS PROBLEM

19. Defining the Functional Elements of Yeast Chromosomes Figure 24-7 shows the major structural elements of a chromosome of budding yeast (*S. cerevisiae*). Heiter, Mann, Snyder, and Davis (1985) determined the properties of some of these elements. They based their study on the finding that in yeast cells, plasmids (which have genes and an origin of replication) act differently from chromosomes (which have these elements plus centromeres and telomeres) during mitosis. The plasmids are not manipulated by the mitotic apparatus and segregate randomly between daughter cells. Without a selectable

marker to force the host cells to retain them (see Fig. 9-4), these plasmids are rapidly lost. In contrast, chromosomes, even without a selectable marker, are manipulated by the mitotic apparatus and are lost at a very low frequency (about 10^{-6} per cell division).

Heiter and colleagues set out to determine the important components of yeast chromosomes by constructing plasmids with various parts of chromosomes and observing whether these "synthetic chromosomes" segregated properly during mitosis. To measure the frequencies of different types of failed chromosome segregation, the researchers needed a rapid assay to determine the number of copies of synthetic chromosomes present in different cells. The assay took advantage of the fact that wild-type yeast colonies are white whereas certain adenine-requiring (*ade⁻*) mutants yield red colonies on nutrient media; *ade2⁻* cells lack functional AIR carboxylase (the enzyme of step 6a in Fig. 22-35) and accumulate AIR (5-aminoimidazole ribonucleotide) in their cytoplasm, and the excess AIR is converted to a conspicuous red pigment. The other part of the assay involved the gene *SUP11*, which encodes an ochre suppressor (a type of nonsense suppressor in which a termination codon specifies an amino acid) that suppresses the phenotype of some *ade2⁻* mutants.

Heiter and coworkers started with a diploid strain of yeast homozygous for *ade2⁻*; these cells are red. When the mutant cells contain one copy of *SUP11*, the metabolic defect is partly suppressed and the cells are pink. When the cells contain two or more copies of *SUP11*, the defect is completely suppressed and the cells are white.

The researchers inserted one copy of *SUP11* into synthetic chromosomes containing various elements thought to be important in chromosome function, and then observed how well these chromosomes were passed from one generation to the next. These pink cells were plated on nonselective media, and the behavior of the synthetic chromosomes was observed. Heiter and coworkers looked for colonies in which the synthetic chromosomes segregated improperly at the first division after plating, giving rise to a colony that was half one genotype and half the other. Because yeast cells are nonmotile, this would be a sectorized colony, with one half one color and the other half another color.

(a) One way for the mitotic process to fail is *nondisjunction*: the chromosome replicates but the sister chromatids fail to separate, so both copies of the chromosome end up in the same daughter cell. Explain how nondisjunction of the synthetic chromosome would give rise to a colony that is half red and half white.

(b) Another way for the mitotic process to fail is *chromosome loss*: the chromosome does not enter the daughter nucleus or is not replicated. Explain how loss of the synthetic chromosome would give rise to a colony that is half red and half pink.

By counting the frequency of the different colony types, Heiter and colleagues could estimate the frequency of these aberrant mitotic events with different types of synthetic

chromosome. First, they explored the requirement for centromeric sequences by constructing synthetic chromosomes with DNA fragments of different sizes containing a known centromere. Their results are shown here.

Synthetic chromosome	Size of centromere-containing fragment (kbp)	Chromosome loss (%)	Nondisjunction (%)
1	None	—	>50
2	0.63	1.6	1.1
3	1.6	1.9	0.4
4	3.0	1.7	0.35
5	6.0	1.6	0.35

(c) Based on these data, what can you conclude about the size of the centromere required for normal mitotic segregation? Explain your reasoning.

(d) All the synthetic chromosomes created in these experiments were circular and lacked telomeres. Explain how they could be replicated more-or-less properly.

Heiter and colleagues next constructed a series of linear synthetic chromosomes that included the functional centromeric sequence and telomeres, and they measured the total mitotic error frequency (% loss + % nondisjunction) as a function of size.

Synthetic chromosome	Size (kbp)	Total error frequency (%)
6	15	11.0
7	55	1.5
8	95	0.44
9	137	0.14

(e) Based on these data, what can you conclude about the chromosome size required for normal mitotic segregation? Explain your reasoning.

(f) Normal yeast chromosomes are linear, range from 250 kbp to 2,000 kbp in length, and, as noted above, have a mitotic error frequency of about 10^{-5} per cell division. Extrapolating the results from (e), do the centromeric and telomeric sequences used in these experiments explain the mitotic stability of normal yeast chromosomes, or must other elements be involved? Explain your reasoning. (Hint: A plot of log of error frequency vs. length will be helpful.)

Reference

Heiter, P., C. Mann, M. Snyder, and R.W. Davis. 1985. Mitotic stability of yeast chromosomes: A colony color assay that measures nondisjunction and chromosome loss. *Cell* 40:381-392.

DNA METABOLISM

25.1 DNA Replication 915

25.2 DNA Repair 930

25.3 DNA Recombination 940

As the repository of genetic information, DNA occupies a unique and central place among biological macromolecules. The nucleotide sequences of DNA encode the primary structures of all cellular RNAs and proteins and, through enzymes, indirectly affect the synthesis of all other cellular constituents. This passage of information from DNA to RNA and protein guides the size, shape, and functioning of every living thing.

DNA is a marvelous device for the stable storage of genetic information. The phrase “stable storage,” however, conveys a static and misleading picture. It fails to capture the complexity of processes by which genetic information is preserved in an uncorrupted state and then transmitted from one generation of cells to the next. DNA metabolism comprises both the process that gives rise to faithful copies of DNA molecules (replication) and the processes that affect the inherent structure of the information (repair and recombination). Together, these activities are both the focus of this chapter and the underpinning for several guiding principles:

P1 Along with catalysis, biological information is one of the two key prerequisites for life. The faithful maintenance and transmission of genetic information from one generation to another ensures continuity within each species.

P2 Information is expensive. The chemistry of joining one nucleotide to the next in DNA replication is elegant and simple, almost deceptively so. However, the enzymatic and thermodynamic commitment to linking one nucleotide to another in DNA far exceeds what would normally be required to successfully form a phosphodiester bond. It is not enough to synthesize a phosphodiester bond; that bond must accurately link two *particular* nucleotides.

P3 The fidelity of genome maintenance and transmission is not perfect. DNA damage

happens, often by spontaneous processes. DNA replication and repair deal with the vast majority of DNA lesions, providing a high degree of genetic fidelity and stability. The few DNA damage events that slip through uncorrected provide fuel for evolution.

P4 Although considered separately, the processes of replication, repair, and recombination of DNA are not distinct. These processes are highly integrated in cells, and they are required for proper genome maintenance.

We give the *enzymes* of DNA metabolism special emphasis in this chapter. These enzymes are not only intrinsically important biologically; they are also important in medicine and biochemical technologies. Many of the seminal discoveries in DNA metabolism have been made with *Escherichia coli*, so its well-understood enzymes are generally used to illustrate the ground rules.

KEY CONVENTION Before taking a closer look at replication, we make a short digression into the use of abbreviations in naming genes and proteins — you will encounter many of these in this and later chapters. Bacterial genes generally are named using three italicized, lowercase letters that often reflect a gene's apparent function. For example, the *dna*, *uvr*, and *rec* genes affect DNA replication, resistance to the damaging effects of UV radiation, and recombination, respectively. Where several genes affect the same process, the letters *A*, *B*, *C*, and so forth, are added — as in *dnaA*, *dnaB*, *dnaQ*, for example — usually reflecting their order of discovery rather than the order of their gene products in a reaction sequence. Similar conventions exist for naming eukaryotic genes, although the exact form of the abbreviations may vary with the species, and no single convention applies to all eukaryotic systems. For example, in the budding yeast *Saccharomyces cerevisiae*, gene names are generally three uppercase letters followed by a number, all italicized (e.g., the gene *COX1* encodes a subunit of cytochrome oxidase). Gene names that predate current conventions may differ in format.

The use of abbreviations in naming proteins is less straightforward. During genetic investigations, the protein product of each gene is usually isolated and characterized. Many bacterial genes were identified and named before the roles of their protein products were understood in detail. Sometimes the gene product was found to be a previously isolated protein, and some renaming

occurred. Often, however, the product turned out to be an as yet unknown protein, with an activity not easily described by a simple enzyme name.

Bacterial proteins often retain the name of their genes. When referring to the protein product of an *E. coli* gene, we use roman type and capitalize the first letter: for example, the *dnaA* and *recA* gene products are the DnaA and RecA proteins, respectively. Conventions for eukaryotic proteins are again complex. For yeast, some proteins have long common names (such as cytochrome oxidase). Others have the same name as the gene, in which case the protein name usually has one uppercase and two lowercase letters, followed by a number and the letter “p,” all in roman type (such as Rad51p). The “p” is to emphasize that this is a protein and to prevent confusion with naming conventions for other organisms. ■

25.1 DNA Replication

P1 Long before the structure of DNA was known, scientists wondered at the ability of organisms to create faithful copies of themselves and, later, at the ability of cells to produce many identical copies of large, complex macromolecules. Speculation about these problems centered around the concept of a **template**, a structure that would allow molecules to be lined up in a specific order and joined to create a macromolecule with a unique sequence and function. The 1940s brought the revelation that DNA was the genetic molecule, but not until James Watson and Francis Crick deduced its structure did the way in which DNA could act as a template for the replication and transmission of genetic information become clear: *one strand is the complement of the other*. The strict base-pairing rules mean that each strand provides the template for a new strand with a predictable and complementary sequence (see Figs. 8-14, 8-15). The fundamental properties of the DNA replication process and the mechanisms used by the enzymes that catalyze it have proved to be essentially identical in all species.

DNA Replication Follows a Set of Fundamental Rules

Early research on bacterial DNA replication and its enzymes helped to establish several basic properties that have proven applicable to DNA synthesis in every organism.

DNA Replication Is Semiconservative **P1** Each DNA strand serves as a template for the synthesis of a new strand, producing two new DNA molecules, each with one new strand and one old strand. This is **semiconservative replication**. The semiconservative nature of replication was established by Matthew Meselson and Frank Stahl in 1957.

Replication Begins at an Origin and Usually Proceeds Bidirectionally Following the confirmation of a semiconservative mechanism of replication, a host of questions

arose. Are the parent DNA strands completely unwound before each is replicated? Does replication begin at random places or at a unique point? After initiation at any point in the DNA, does replication proceed in one direction or both?

Photographic images of tritium (^3H)-labeled bacterial DNA made by John Cairns revealed that the intact chromosome of *E. coli* is a single huge circle, 1.7 mm long. Radioactive DNA isolated from cells during replication showed an extra loop (Fig. 25-1). Cairns concluded that the loop resulted from the formation of two radioactive daughter strands, each complementary to a parent

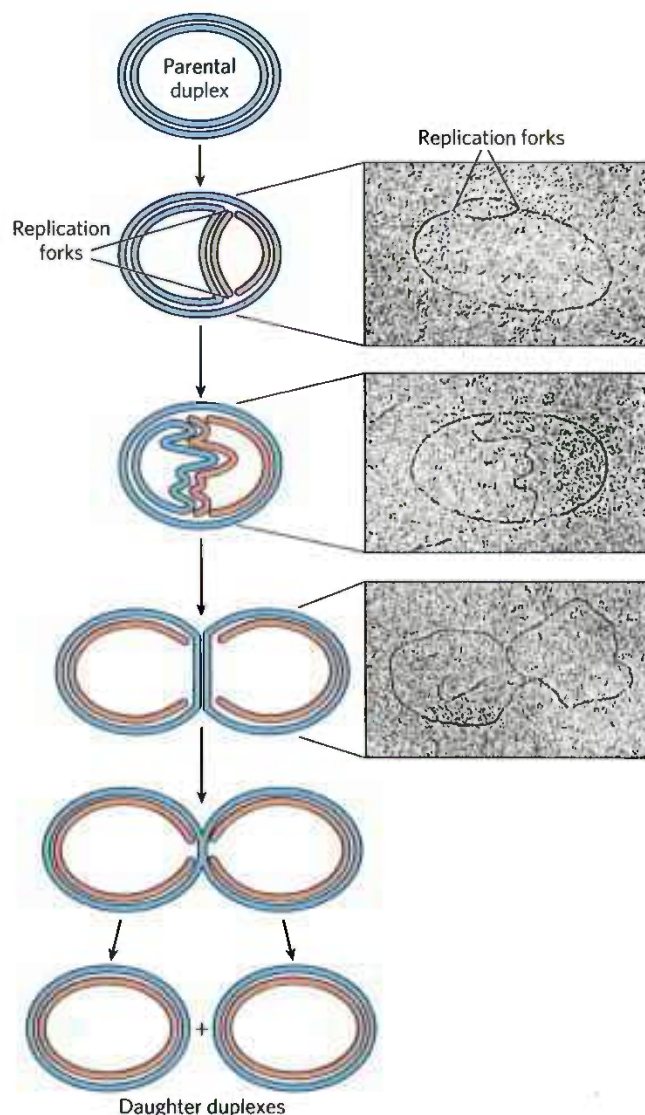


FIGURE 25-1 Visualization of DNA replication. Stages in the replication of circular DNA molecules have been visualized by electron microscopy. Replication of a circular chromosome produces a structure resembling the Greek letter theta, θ , as both strands are replicated simultaneously (new strands shown in light red). The electron micrographs show images of plasmid DNA being replicated from a single replication origin. [Electron micrographs: Cairns, J. (1963), *The Chromosome of Escherichia coli*, Cold Spring Harbor Symp Quant Biol, 28, 43–46. © Cold Spring Harbor Laboratory Press.]

strand. One or both ends of the loop are dynamic points, termed **replication forks**, where parent DNA is being unwound and the separated strands quickly replicated. Cairns's results demonstrated that both DNA strands are replicated simultaneously, and variations on his experiment showed that replication of bacterial chromosomes is bidirectional: both ends of the loop have active replication forks.

Determination of whether the replication loops originate at a unique point in the DNA required landmarks along the DNA molecule. These landmarks were provided by a technique called **denaturation mapping**, developed by Ross Inman and colleagues. Using the 48,502 bp chromosome of bacteriophage λ , Inman showed that DNA could be selectively denatured at sequences unusually rich in A=T base pairs, generating a reproducible pattern of single-strand bubbles (see Fig. 8-28). Using the denatured regions as points of reference, investigators have subsequently been able to measure the position and progress of the replication forks. Inman and colleagues found that the replication loops always initiated at a unique point, which was termed an **origin**. They also confirmed the earlier observation that replication is usually bidirectional. For circular DNA molecules, the two replication forks meet at a point on the side of the circle opposite to the origin. Specific origins of replication have since been identified and characterized in bacteria and eukaryotes.

DNA Synthesis Proceeds in a 5'→3' Direction and Is Semidiscontinuous A new strand of DNA is always synthesized in the 5'→3' direction, with the free 3'-OH as the point at which the DNA is elongated. (Recall from Chapter 8 that the 5' end lacks a nucleotide attached to the 5' position, and the 3' end lacks a nucleotide attached to the 3' position.) Because the two DNA strands are antiparallel, the strand serving as the template is read from its 3' end toward its 5' end.

If synthesis always proceeds in the 5'→3' direction, how can both strands be synthesized simultaneously? If both strands were synthesized *continuously* while the replication fork moved, one strand would have to undergo 3'→5' synthesis. This problem was resolved by Reiji Okazaki and colleagues in the 1960s. Okazaki found that one of the new DNA strands is synthesized in short pieces, now called **Okazaki fragments**. Thus, one strand is synthesized continuously and the other discontinuously (Fig. 25-2). The continuous strand, or **leading strand**, is the one for which 5'→3' synthesis proceeds in the *same* direction that the replication fork moves. The discontinuous strand, or **lagging strand**, is the one in which 5'→3' synthesis proceeds in the direction *opposite* to the direction of fork movement. Okazaki fragments are typically 150 to 200 nucleotides long in eukaryotes, and 1,000 to 2,000 nucleotides long in bacteria. As we shall see, leading and lagging strand syntheses are tightly coordinated.

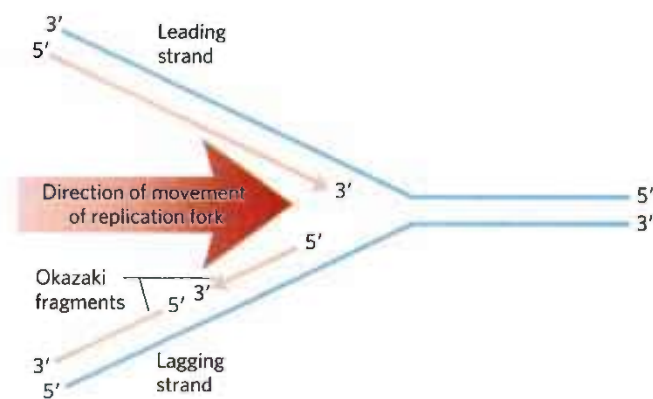


FIGURE 25-2 Defining DNA strands at the replication fork.

A new DNA strand (light red) is always synthesized in the 5'→3' direction. The template is read in the opposite direction, 3'→5'. The leading strand is continuously synthesized in the direction taken by the replication fork. The other strand, the lagging strand, is synthesized discontinuously in short pieces (Okazaki fragments) in a direction opposite to that in which the replication fork moves. The Okazaki fragments are spliced together by DNA ligase. In bacteria, Okazaki fragments are ~1,000 to 2,000 nucleotides long. In eukaryotic cells, they are 150 to 200 nucleotides long.

DNA Is Degraded by Nucleases

To explain the enzymology of DNA replication, we first introduce the enzymes that degrade DNA rather than synthesize it. These enzymes are known as **nucleases**, or **DNases** if they are specific for DNA rather than RNA. Every cell contains several different nucleases, belonging to two broad classes: exonucleases and endonucleases. **Exonucleases** degrade nucleic acids from one end of the molecule. Many operate in only the 5'→3' direction or the 3'→5' direction, removing nucleotides only from the 5' end or the 3' end, respectively, of one strand of a double-stranded nucleic acid or of a single-stranded DNA. **Endonucleases** can begin to degrade at specific internal sites in a nucleic acid strand or molecule, reducing it to smaller and smaller fragments. A few exonucleases and endonucleases degrade only single-stranded DNA. A few important classes of endonucleases cleave only at specific nucleotide sequences (such as the restriction endonucleases that are so important in biotechnology; see Chapter 9, Fig. 9-2). You will encounter many types of nucleases in this and subsequent chapters.

DNA Is Synthesized by DNA Polymerases

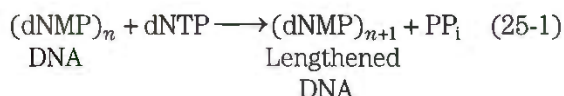


Arthur Kornberg, 1918–2007
[World History Archive/Alamy]

The search for an enzyme that could synthesize DNA began in 1955. Work by Arthur Kornberg and colleagues led to the purification and characterization of a **DNA polymerase** from *E. coli* cells, a single-polypeptide enzyme now called **DNA polymerase I** (M_r 103,000; encoded by the *polA* gene). Much later, investigators found that *E. coli* contains at least

four additional distinct DNA polymerases, described below.

Detailed studies of DNA polymerase I revealed features of the DNA synthetic process that are now known to be common to all DNA polymerases. The fundamental reaction is a phosphoryl group transfer. The nucleophile is the 3'-hydroxyl group of the nucleotide at the 3' end of the growing strand. Nucleophilic attack occurs at the α phosphorus of the incoming deoxynucleoside 5'-triphosphate (Fig. 25-3). Inorganic pyrophosphate is released in the reaction. The general reaction is



where dNMP and dNTP are a deoxynucleoside 5'-monophosphate and 5'-triphosphate, respectively. Catalysis by virtually all DNA polymerases prominently involves two Mg^{2+} ions at the active site. One of these helps to deprotonate the 3'-hydroxyl group, rendering it a more effective nucleophile. The other binds to the incoming dNTP and facilitates departure of the pyrophosphate.

P2 The reaction seems to proceed with only a minimal change in free energy, given that one phosphodiester bond is formed at the expense of a somewhat less stable phosphate anhydride. However, noncovalent base-stacking

and base-pairing interactions provide additional stabilization to the lengthened DNA product relative to the free nucleotide. Also, the formation of products is facilitated in the cell by the 19 kJ/mol generated in the subsequent hydrolysis of the pyrophosphate product by the enzyme pyrophosphatase (p. 485).

Early work on DNA polymerase I led to the definition of two central requirements for DNA polymerization (Fig. 25-4). First, all DNA polymerases require a template. The polymerization reaction is guided by a template DNA strand according to the base-pairing rules predicted by Watson and Crick: where a guanine is present in the template, a cytosine deoxynucleotide is added to the new strand, and so on. **P1** This was a particularly important discovery, not only because it provided a chemical basis for accurate semiconservative DNA replication, but also because it represented the first example of the use of a template to guide a biosynthetic reaction.

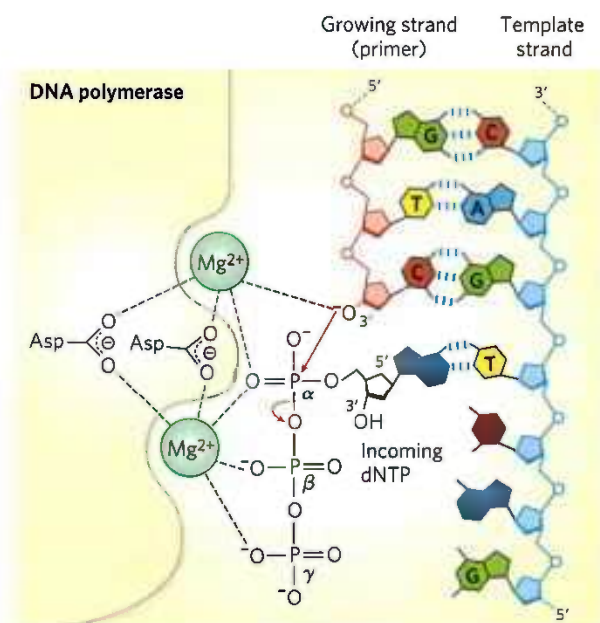
Second, the polymerases require a **primer**. A primer is a strand segment (complementary to the template) with a free 3'-hydroxyl group to which a nucleotide can be added; the free 3' end of the primer is called the **primer terminus** (Fig. 25-4a). In other words, part of the new strand must already be in place: all DNA polymerases can add nucleotides only to a preexisting strand. Many primers are oligonucleotides of RNA rather than DNA, and specialized enzymes synthesize primers when and where they are required.

A DNA polymerase active site has two parts (Fig. 25-4a). The incoming nucleotide is initially positioned in the **insertion site**. Once the phosphodiester bond is formed, the polymerase slides forward on the DNA and the new base pair is positioned in the **postinsertion site**. These sites are located in a pocket that resembles the palm of a hand (Fig. 25-4b, c).

After adding a nucleotide to a growing DNA strand, a DNA polymerase either dissociates or moves along the template and adds another nucleotide. Dissociation and reassociation of the polymerase can limit the overall polymerization rate—the process is generally faster when a polymerase adds more nucleotides without dissociating from the template. The average number of nucleotides added before a polymerase dissociates defines its **processivity**. DNA polymerases vary greatly in processivity; some add just a few nucleotides before dissociating, whereas others add many thousands.

Replication Is Very Accurate

Replication proceeds with an extraordinary degree of fidelity. In *E. coli*, a mistake is made only once for every 10^9 to 10^{10} nucleotides added. For the *E. coli* chromosome of $\sim 4.6 \times 10^6$ bp, this means that an error occurs only once per 1,000 to 10,000 replications. During polymerization, discrimination between correct and incorrect nucleotides relies not just on the hydrogen bonds that specify the correct pairing between complementary bases but also on the common geometry of the standard $\text{A}=\text{T}$ and $\text{G}=\text{C}$ base pairs (Fig. 25-5). The active site of DNA polymerase I



MECHANISM FIGURE 25-3 The DNA polymerase reaction. The catalytic mechanism for addition of a new nucleotide by DNA polymerase involves two Mg^{2+} ions, coordinated to the phosphate groups of the incoming nucleotide triphosphate, the 3'-hydroxyl group that will act as a nucleophile, and three Asp residues, two of which are highly conserved in all DNA polymerases. The Mg^{2+} ion depicted at the top facilitates attack of the 3'-hydroxyl group of the primer on the α phosphate of the nucleotide triphosphate; the other Mg^{2+} ion facilitates displacement of the pyrophosphate. Both ions stabilize the structure of the pentacoordinate transition state. RNA polymerases use a similar mechanism.

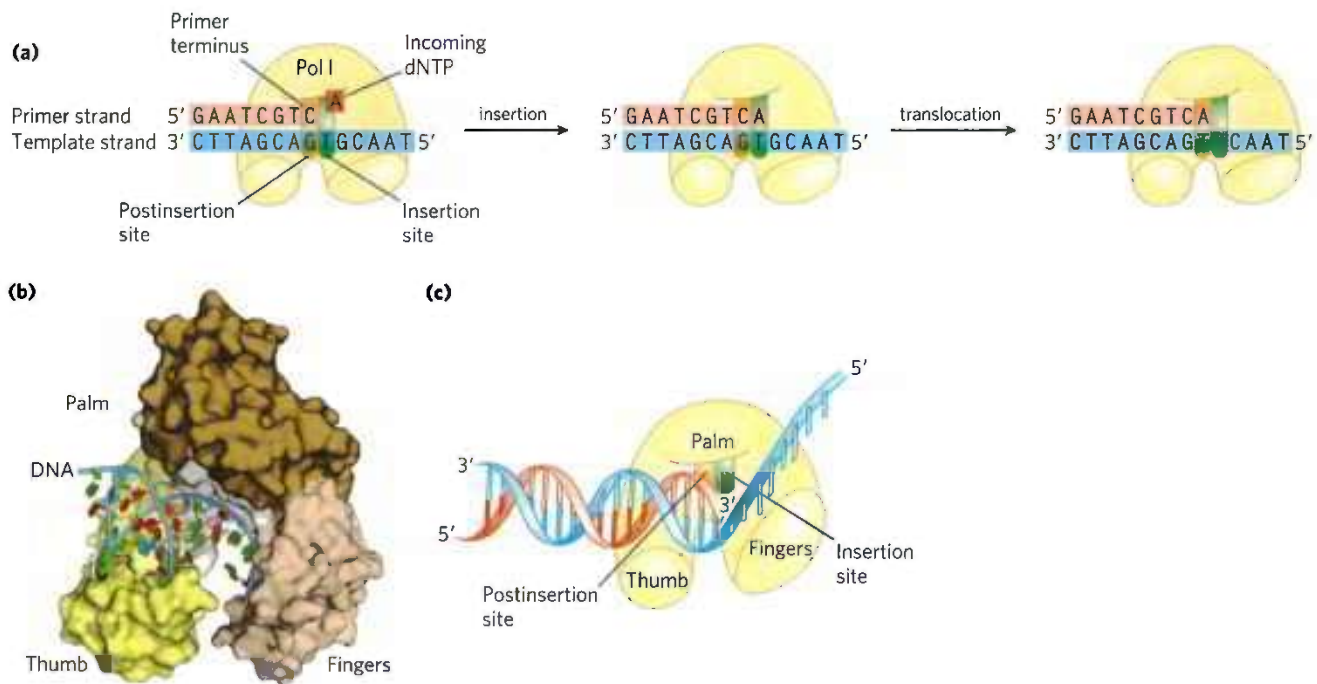
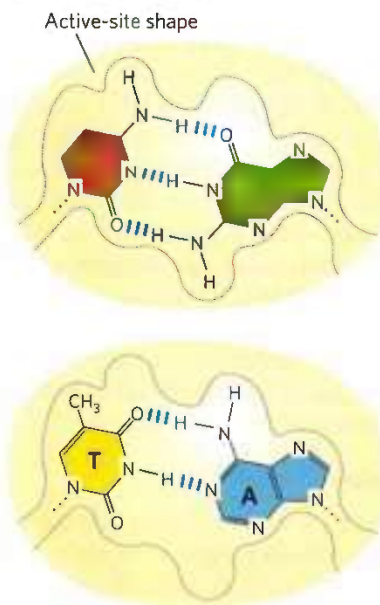


FIGURE 25-4 Elongation of a DNA chain. (a) DNA polymerase I activity requires a single unpaired strand to act as template and a primer strand to provide the free hydroxyl group at the 3' end to which the new nucleotide unit is added. Each incoming complementary nucleotide is bound selectively, in part by base-pairing to the appropriate nucleotide in the template strand. The reaction product has a new free 3' hydroxyl, allowing the addition of another nucleotide. The newly formed base pair

translocates to make the active site available to the next pair to be formed. (b) The core of most DNA polymerases is shaped like a human hand that wraps around the active site. The structure shown here is DNA polymerase I of *Thermus aquaticus*, bound to DNA. (c) A cartoon interpretation shows the insertion site, where the nucleotide addition occurs, and the postinsertion site, to which the newly formed base pair translocates. [(b) Data from PDB ID 4KTQ, Y. Li et al., *EMBO J.* 17:7514, 1998.]

(a) Correct base pairs



(b) Incorrect base pairs

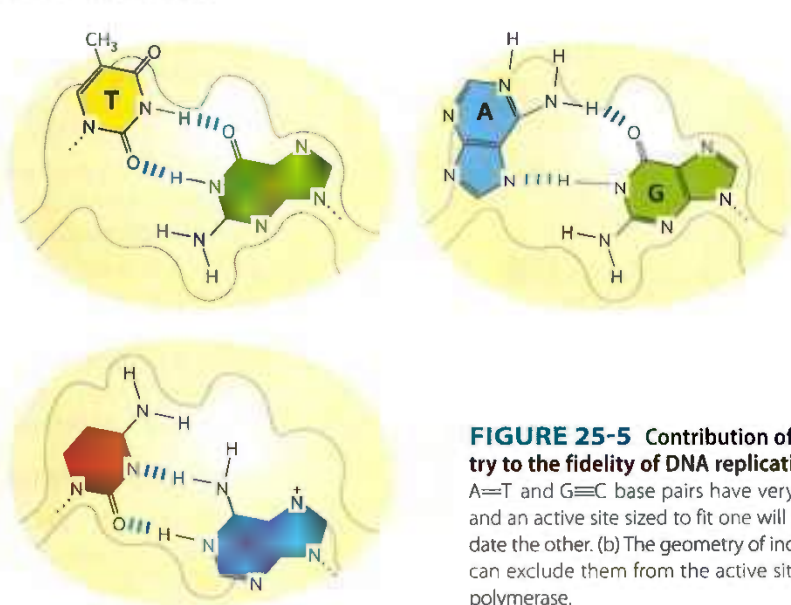


FIGURE 25-5 Contribution of base-pair geometry to the fidelity of DNA replication. (a) The standard A=T and G≡C base pairs have very similar geometries, and an active site sized to fit one will generally accommodate the other. (b) The geometry of incorrectly paired bases can exclude them from the active site, as occurs on DNA polymerase.

accommodates only base pairs with this geometry. An incorrect nucleotide may be able to hydrogen-bond with a base in the template, but it generally will not fit into the active site. Incorrect bases can be rejected before the phosphodiester bond is formed.

The accuracy of the polymerization reaction itself, however, is insufficient to account for the high degree of fidelity in replication. Careful measurements in vitro have shown that DNA polymerases insert one incorrect nucleotide for every 10^4 to 10^5 correct ones. These

mistakes sometimes occur because a base is briefly in an unusual tautomeric form (see Fig. 8-9), allowing it to hydrogen-bond with an incorrect partner. In vivo, the error rate is reduced by additional enzymatic mechanisms.

One mechanism intrinsic to many DNA polymerases is a separate 3'→5' exonuclease activity that double-checks each nucleotide after it is added. This nuclease activity permits the enzyme to remove a newly added nucleotide and is highly specific for mismatched base pairs (Fig. 25-6). If the polymerase has added the wrong nucleotide, translocation of the enzyme to the position where the next nucleotide is to be added is inhibited. This kinetic pause provides the opportunity for a correction. The

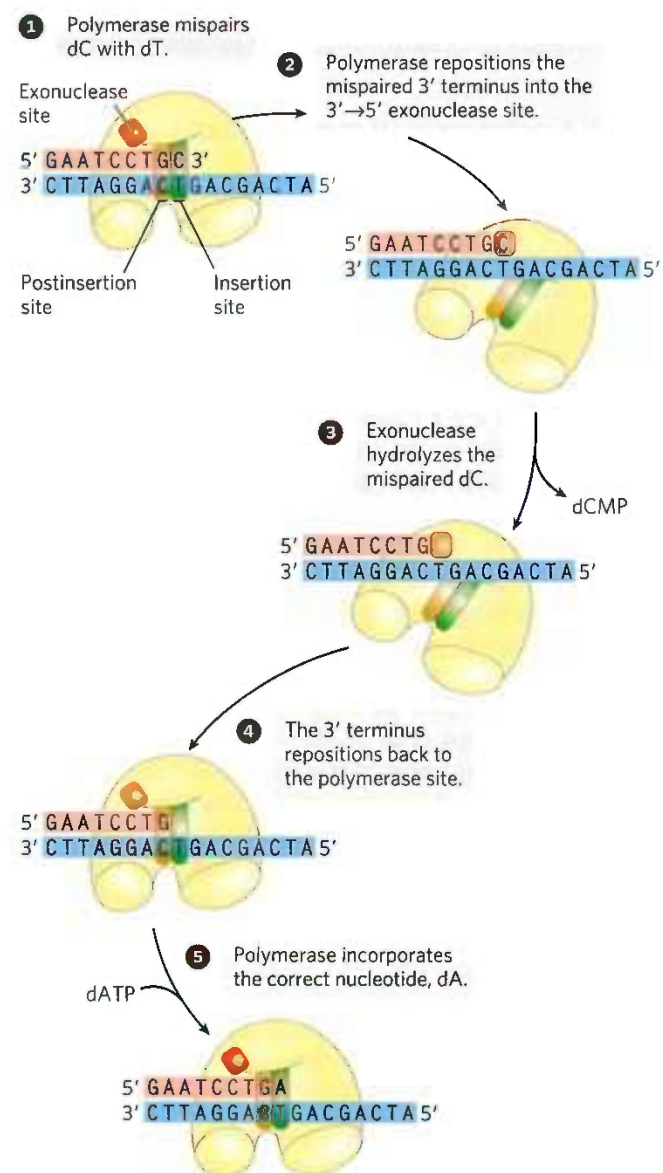


FIGURE 25-6 An example of error correction by the 3'→5' exonuclease activity of DNA polymerase I. Structural analysis has located the exonuclease activity behind the polymerase activity as the enzyme is oriented in its movement along the DNA. A mismatched base (here, a C-T mismatch) impedes translocation of DNA polymerase I (Pol I) to the next site.

3'→5' exonuclease activity cleaves the most recently added phosphodiester bond and removes the mispaired nucleotide; the polymerase then adds another nucleotide to begin active synthesis again. This activity, known as **proofreading**, is not simply the reverse of the polymerization reaction (Eqn 25-1). **P2** Instead, replacement of the incorrect nucleotide requires the expenditure of three high-energy bonds. The polymerizing and proofreading activities of a DNA polymerase can be measured separately. Proofreading improves the inherent accuracy of the polymerization reaction 10^2 - to 10^3 -fold. In the monomeric DNA polymerase I, the polymerizing and proofreading activities have separate active sites within the same polypeptide.

P3 When base selection and proofreading are combined, DNA polymerase leaves behind one net error for every 10^6 to 10^8 bases added. Yet the measured accuracy of replication in *E. coli* is higher still. The additional accuracy is provided by a separate enzyme system that repairs the mismatched base pairs remaining after replication. We describe this mismatch repair, along with other DNA repair processes, in Section 25.2.

E. coli Has at Least Five DNA Polymerases

More than 90% of the DNA polymerase activity observed in *E. coli* extracts can be accounted for by DNA polymerase I. Soon after the isolation of this enzyme in 1955, however, evidence began to accumulate that it is not suited for replication of the large *E. coli* chromosome. First, the rate at which it adds nucleotides (600 nucleotides/min) is too slow (by a factor of 100 or more) to account for the rates at which the replication fork moves in the bacterial cell. Second, DNA polymerase I has a relatively low processivity. Third, genetic studies have demonstrated that many genes, and therefore many proteins, are involved in replication: DNA polymerase I clearly does not act alone. Fourth, and most important, in 1969 John Cairns isolated a bacterial strain with an altered gene for DNA polymerase I that produced an inactive enzyme. Although this strain was abnormally sensitive to agents that damaged DNA, it was nevertheless viable.

A search for other DNA polymerases led to the discovery of *E. coli* **DNA polymerase II** and **DNA polymerase III** in the early 1970s. DNA polymerase II is an enzyme involved in one type of DNA repair (Section 25.3). DNA polymerase III is the principal replication enzyme in *E. coli*. DNA polymerases IV and V, identified in 1999, are involved in an unusual form of DNA repair (Section 25.2). The properties of these five DNA polymerases are compared in Table 25-1.

DNA polymerase I, then, is not the primary enzyme of replication; instead, it performs a host of cleanup functions during replication, recombination, and repair. The polymerase's special functions are enhanced by its 5'→3' exonuclease activity. This activity, distinct from the 3'→5' proofreading exonuclease (Fig. 25-6), is

TABLE 25-1 Comparison of the Five DNA Polymerases of *E. coli*

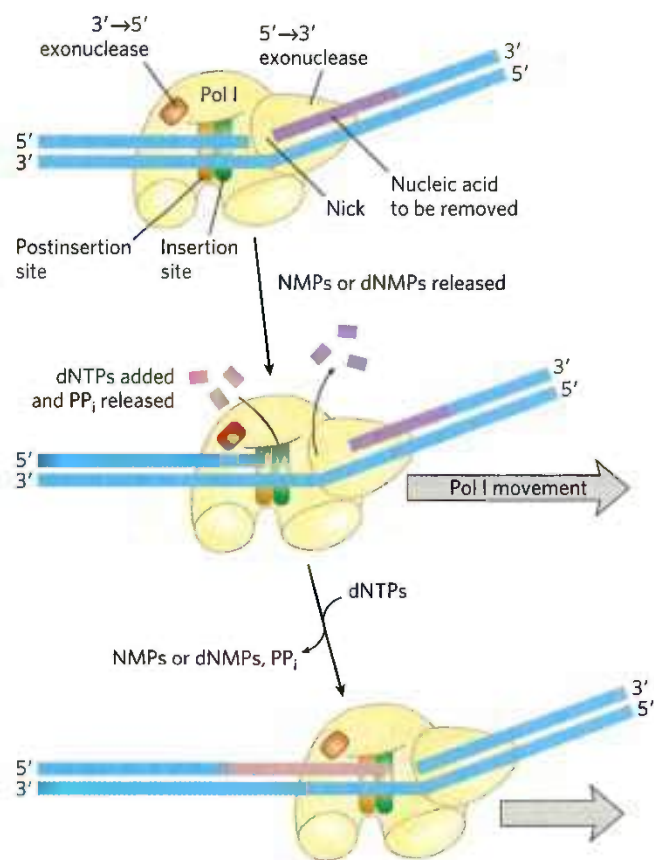
	DNA polymerase				
	I	II ^a	III	IV ^a	V ^a
Structural gene ^b	<i>polA</i>	<i>polB</i>	<i>polC (dnaE)</i>	<i>dinB</i>	<i>umuC</i>
Subunits (number of different types)	1	7	9	1	3
M_r	103,000	88,000 ^c	1,065,400	39,100	110,000
3'→5' exonuclease (proofreading)	Yes	Yes	Yes	No	No
5'→3' exonuclease	Yes	No	No	No	No
Polymerization rate (nucleotides/s)	10–20	40	250–1,000	2–3	1
Processivity (nucleotides added before polymerase dissociates)	3–200	1,500	≥500,000	1	6–8

^aTranslesion (mutagenic) DNA polymerases. For DNA polymerase IV, processivity is increased substantially by association with a β clamp. These polymerases are slowed when a DNA lesion is present in the DNA template strand.

^bFor enzymes with more than one subunit, the gene listed here encodes the subunit with polymerization activity. Note that *dnaE* is an earlier designation for the gene now referred to as *polC*.

^cPolymerization subunit only. DNA polymerase II shares several subunits with DNA polymerase III, including the β , δ , δ' , χ , and ψ subunits (see Table 25-2).

located in a structural domain that can be separated from the rest of the enzyme by mild protease treatment. When the 5'→3' exonuclease domain is removed, the remaining fragment (M_r 68,000), the **large fragment** or **Klenow fragment**, retains the polymerization and proofreading activities. The 5'→3' exonuclease activity of intact DNA polymerase I can replace a segment of DNA (or RNA) paired to the template strand, in a process known as nick translation (Fig. 25-7). Most other DNA polymerases lack a 5'→3' exonuclease activity.



DNA polymerase III is much more complex than DNA polymerase I, with nine different kinds of subunits (Table 25-2). Its polymerization and proofreading activities reside in its α and ϵ subunits, respectively. The θ subunit associates with α and ϵ to form a core polymerase, which can polymerize DNA but with limited processivity. Up to three core polymerases can be linked by a clamp-loading complex consisting of five subunits of three different types, $\tau_3\delta\delta'$. The core polymerases are linked through the τ (tau) subunits. Two additional subunits, χ (chi) and ψ (psi), are bound to the clamp-loading complex. The entire assembly of 16 protein subunits (eight different types) is called DNA polymerase III* (Fig. 25-8a).

DNA polymerase III* can polymerize DNA, but with a much lower processivity than one would expect for the organized replication of an entire chromosome. The necessary increase in processivity is provided by the addition of the β subunits. The β subunits associate in pairs to form donut-shaped structures that encircle the DNA and act like clamps (Fig. 25-8b). Each dimer associates with a core subassembly of polymerase III* (one dimeric clamp per active core subassembly) and slides along the DNA as replication proceeds. The β sliding clamp prevents the dissociation of DNA polymerase III from DNA, dramatically increasing processivity—to greater than 500,000

FIGURE 25-7 Nick translation. The bacterial DNA polymerase I has three domains, catalyzing its DNA polymerase, 5'→3' exonuclease, and 3'→5' exonuclease activities. The 5'→3' exonuclease domain is in front of the enzyme as it moves along the DNA and is not shown in Figure 25-4. By degrading the DNA strand ahead of the enzyme and synthesizing a new strand behind, DNA polymerase I can promote nick translation, in which a break or nick in the DNA is effectively moved along with the enzyme. This process has a role in DNA repair and in the removal of RNA primers during replication (both described later in this chapter). The strand of nucleic acid to be removed (either DNA or RNA) is shown in purple, the replacement strand in red. DNA synthesis begins at a nick (a broken phosphodiester bond, leaving a free 3' hydroxyl and a free 5' phosphate). A nick remains where DNA polymerase I eventually dissociates, and the nick is later sealed by another enzyme.

TABLE 25-2 Subunits of DNA Polymerase III of *E. coli*

Subunit	Number of subunits per holoenzyme	M_r of subunit	Gene	Function of subunit	
α	3	129,900	<i>polC</i> (<i>dnaE</i>)	Polymerization activity	Core polymerase
ϵ	3	27,500	<i>dnaQ</i> (<i>mutD</i>)	3' $\rightarrow$ 5' proofreading exonuclease	
θ	3	8,600	<i>holE</i>	Stabilization of ϵ subunit	
τ	3	71,100	<i>dnaX</i>	Stable template binding; core enzyme dimerization	Clamp-loading (γ) complex that loads β subunits on lagging strand at each Okazaki fragment ^a
δ	1	38,700	<i>holA</i>	Clamp opener	
δ'	1	36,900	<i>holB</i>	Clamp loader	
χ	1	16,600	<i>holC</i>	Interaction with SSB	
ψ	1	15,200	<i>holD</i>	Interaction with τ and χ	
β	6	40,600	<i>dnaN</i>	DNA clamp required for optimal processivity	

^aThe γ subunit is encoded by a portion of the gene for the τ subunit (*dnaX*), such that the amino-terminal 66% of the τ subunit has the same amino acid sequence as the γ subunit. The γ subunit is generated by a translational frameshifting mechanism (p. 1013) that leads to premature translational termination. The γ subunit shares the clamp-loading functions of τ but lacks the protein segments that interact with the core polymerase or with DnaB helicase. Clamp-loading complexes incorporating γ subunits may operate independently of the DNA polymerase III holoenzyme, promoting the unloading of β clamps discarded on the lagging strand as the replication fork progresses. They may also promote loading of β clamps for some DNA repair processes that require DNA synthesis away from the replication fork.

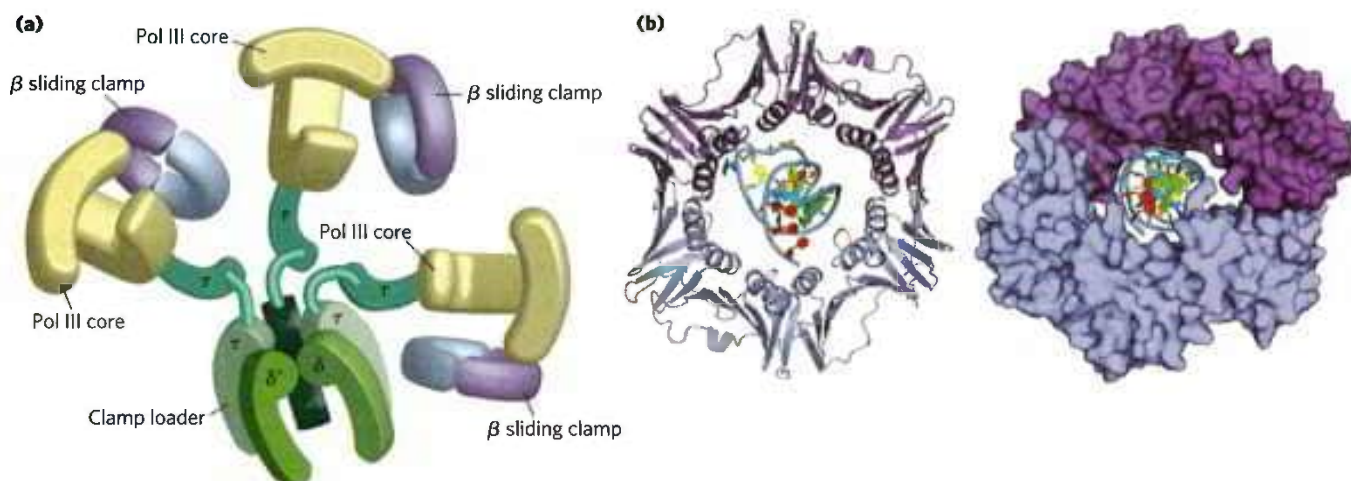


FIGURE 25-8 DNA polymerase III. (a) Architecture of bacterial DNA polymerase III (Pol III). Three core domains, composed of subunits α , ϵ , and θ , are linked by a five-subunit clamp-loading complex, with the composition $\tau_3\delta\delta'$. The core subunits and clamp-loading complex constitute DNA polymerase III*. The other two subunits of DNA polymerase III*, χ and ψ (not shown), also bind to the clamp-loading complex. Three β clamps interact with the three core subassemblies, each clamp a dimer of the β subunit. The complex interacts with the DnaB helicase (described later in the text)

through the τ subunits. (b) Two β subunits of *E. coli* polymerase III form a circular clamp that surrounds the DNA. The clamp slides along the DNA molecule, increasing the processivity of the polymerase III holoenzyme to more than 500,000 nucleotides by preventing its dissociation from the DNA. The two β subunits are shown in two shades of purple as ribbon structures (left) and surface contour images (right), surrounding the DNA. [(a) Information from N. Yao and M. O'Donnell, *Mol. Biosyst.* 4:1075, 2008. (b) Data from PDB ID 2POL, X.-P. Kong et al., *Cell* 69:425, 1992.]

(Table 25-1). The addition of the β subunits converts DNA polymerase III* to DNA polymerase III holoenzyme.

DNA Replication Requires Many Enzymes and Protein Factors

Replication in *E. coli* requires not just a single DNA polymerase but 20 or more different enzymes and proteins, each performing a specific task. The entire complex has

been termed the **DNA replicase system** or **replisome**. The enzymatic complexity of replication reflects the constraints imposed by the structure of DNA and by the requirements for accuracy. The main classes of replication enzymes are considered here in terms of the problems they overcome.

The DNA must be separated into two strands that each act as a template. **P2** This is generally accomplished by **helicases**, enzymes that move along the DNA

and separate the strands, using chemical energy from ATP. Strand separation creates topological stress in the helical DNA structure (see Fig. 24-11), which is relieved by the action of **topoisomerases**. The separated strands are stabilized by **DNA-binding proteins**. As noted earlier, before DNA polymerases can begin synthesizing DNA, primers must be present on the template—generally, short segments of RNA synthesized by enzymes known as **primases**. Ultimately, the RNA primers are removed and replaced by DNA; in *E. coli*, this is one of the many functions of DNA polymerase I. A specialized nuclease that degrades RNA in RNA-DNA hybrids, called RNase H1, also removes some RNA primers. After an RNA primer is removed and the gap is filled in with DNA, a nick remains in the DNA backbone in the form of a broken phosphodiester bond. These nicks are sealed by **DNA ligases**. All these processes require coordination and regulation best characterized in the *E. coli* system.

Replication of the *E. coli* Chromosome Proceeds in Stages

The synthesis of a DNA molecule can be divided into three stages: initiation, elongation, and termination, distinguished both by the reactions taking place and by the enzymes required. As you will find here and in the next two chapters, synthesis of the major information-containing biological polymers—DNAs, RNAs, and proteins—can be understood in terms of these same three stages, with the stages of each pathway having unique characteristics. The events described below reflect information derived primarily from *in vitro* experiments using purified *E. coli* proteins, although the principles are highly conserved in all replication systems.

Initiation The *E. coli* replication origin, *oriC*, consists of 245 bp and contains DNA sequence elements that are highly conserved among bacterial replication origins. The general arrangement of the conserved sequences is illustrated in **Figure 25-9**. Two types of sequences are of special interest: five repeats of a 9 bp sequence (R sites) that serve as binding sites for the key initiator protein, DnaA, and a region rich in A=T base pairs called the **DNA unwinding element (DUE)**. There are three additional DnaA-binding sites (I sites), and binding sites for the proteins IHF (integration host factor) and FIS (factor for

inversion stimulation). These two proteins were discovered as necessary components of certain recombination reactions described later in this chapter, and their names reflect those roles. Another DNA-binding protein, HU (a histonelike bacterial protein originally dubbed factor U), also participates but does not have a specific binding site.

At least 10 different enzymes or proteins (summarized in **Table 25-3**) participate in the initiation phase of replication. They open the DNA helix at the origin and establish a prepriming complex for subsequent reactions. The crucial component in the initiation process is the DnaA protein, a member of the **AAA+ ATPase** protein family (ATPases associated with diverse cellular activities). Many AAA+ ATPases, including DnaA, form oligomers and hydrolyze ATP relatively slowly. This ATP hydrolysis acts as a switch that mediates interconversion of the protein between two states. In the case of DnaA, the ATP-bound form is active and the ADP-bound form is inactive.

Eight DnaA protein molecules, all in the ATP-bound state, assemble to form a helical complex encompassing the R and I sites in *oriC* (**Fig. 25-10**). DnaA has a higher affinity for R sites than I sites, and it binds R sites equally well in its ATP- or ADP-bound form. The I sites, which bind only the ATP-bound DnaA, allow discrimination between the active and inactive forms of DnaA. The tight right-handed wrapping of the DNA around this complex introduces a positive supercoil (see Chapter 24). The associated strain in the nearby DNA, combined with the binding of additional DnaA protein to the DUE region, leads to strand separation in the A=T-rich DUE. The complex formed at the replication origin also includes several DNA-binding proteins—HU, IHF, and FIS—that facilitate DNA bending.

The DnaC protein, another AAA+ ATPase, then loads the DnaB protein onto the separated DNA strands in the denatured region. A hexamer of DnaC, each subunit bound to ATP, forms a tight complex with the ring-shaped, hexameric DnaB helicase. This DnaC-DnaB interaction opens the DnaB ring, the process being aided by a further interaction between DnaB and DnaA. Two of the ring-shaped DnaB hexamers are loaded in the DUE, one onto each DNA strand. The ATP bound to DnaC is hydrolyzed, releasing the DnaC and leaving the DnaB bound to the DNA.

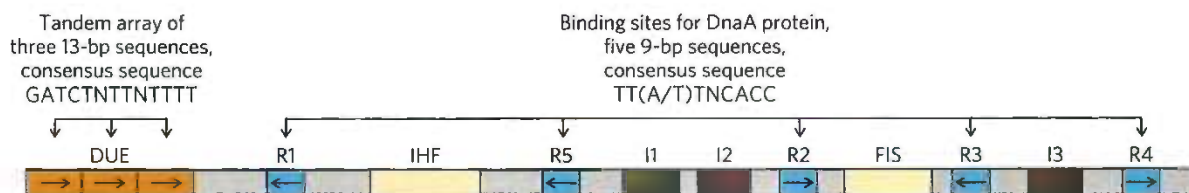


FIGURE 25-9 Arrangement of sequences in the *E. coli* replication origin, *oriC*. Conserved sequences for key repeated elements are shown. N represents any of the four nucleotides. The horizontal arrows indicate the orientations of the nucleotide sequences (left-to-right arrow denotes

a sequence in the top strand; right-to-left, in the bottom strand). FIS and IHF are binding sites for proteins described in the text. R sites are bound by DnaA. I sites are additional DnaA-binding sites (with different sequences, labeled I1, I2, and I3), for DnaA only when the protein is complexed with ATP.

TABLE 25-3 Proteins Required to Initiate Replication at the *E. coli* Origin

Protein	M_r	Number of subunits	Function
DnaA protein	52,000	1	Recognizes <i>oriC</i> sequence; opens duplex at specific sites in origin
DnaB protein (helicase)	300,000	6 ^a	Unwinds DNA
DnaC protein	174,000	6 ^a	Required for DnaB binding at origin
HU	19,000	2	Histonelike protein; DNA-binding protein; stimulates initiation
FIS	22,500	2 ^a	DNA-binding protein; stimulates initiation
IHF	22,000	2	DNA-binding protein; stimulates initiation
Primase (DnaG protein)	60,000	1	Synthesizes RNA primers
Single-stranded DNA-binding protein (SSB)	75,600	4 ^a	Binds single-stranded DNA
DNA gyrase (DNA topoisomerase II)	400,000	4	Relieves torsional strain generated by DNA unwinding
Dam methylase	32,000	1	Methylates (5')GATC sequences at <i>oriC</i>

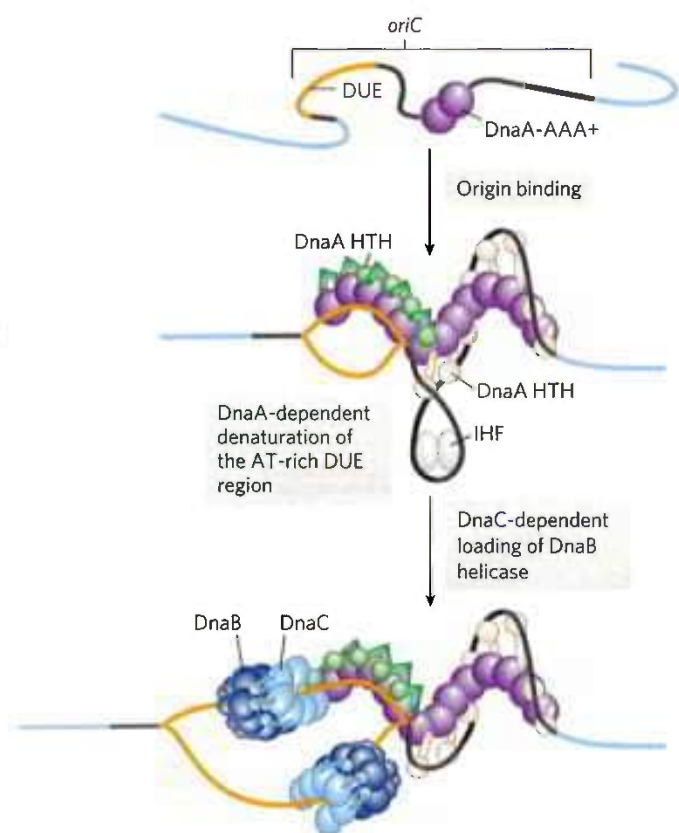
^aSubunits in these cases are identical.

FIGURE 25-10 Model for initiation of replication at the *E. coli* origin, *oriC*. DnaA protein molecules bind initially at the five specific R sites. Upon ATP binding, additional DnaA molecules bind at the I sites in the origin, forming a right-handed helical complex and drawing in more DnaA molecules that continue the helix into the DUE region (see Fig. 25-9). The DNA is wrapped around this complex. DnaA molecules bound at the R and I sites bind the DNA with an HTH domain (referring to a DNA binding motif called helix-turn-helix). The A=T-rich DUE region is denatured as a result of the strain imparted by the DnaA binding. Within the DUE, single-stranded DNA is bound by the ATPase domain of DnaA rather than by the HTH. Formation of the helical DnaA complex is facilitated by the proteins HU, IHF, and FIS. The detailed structural roles of these proteins are not known, but IHF may stabilize a transient DNA loop, as shown here. Hexamers of the DnaB protein bind to each strand, with the aid of DnaC protein. The DnaB helicase activity further unwinds the DNA in preparation for priming and DNA synthesis. [Information from J. P. Erzberger et al., *Nat. Struct. Mol. Biol.* 13:676, 2006]

Loading of the DnaB helicase is the key step in replication initiation. As a replicative helicase, DnaB migrates along the single-stranded DNA in the 5'→3' direction, unwinding the DNA as it travels. The DnaB helicases loaded onto the two DNA strands thus travel in opposite directions, creating two potential replication forks. All other proteins at the replication fork are linked directly or indirectly to DnaB. The DNA polymerase III holoenzyme is linked through its τ subunits; additional DnaB interactions are described below. As replication begins and the DNA strands are separated at the fork, many molecules of single-stranded DNA-binding protein (SSB) bind to and stabilize the separated strands, and DNA gyrase (DNA topoisomerase II) relieves the topological stress induced ahead of the fork by the unwinding reaction.

Initiation is the only phase of DNA replication that is known to be regulated, and it is regulated such that replication occurs only once in each cell cycle. The mechanism of regulation is not yet entirely understood, but genetic and biochemical studies have provided insights into several separate regulatory mechanisms.

Once DNA polymerase III has been loaded onto the DNA, along with the β subunits (signaling completion of the initiation phase), the protein Hda binds to the β subunits and interacts with DnaA to stimulate hydrolysis of its bound ATP. Hda is yet another AAA+ ATPase closely related to DnaA (its name is derived from *homologous* to DnaA). This ATP hydrolysis leads to disassembly of the DnaA complex at the origin. Slow release of ADP by DnaA and rebinding of ATP cycles the protein between its inactive (with bound ADP) and active (with bound ATP) forms on a time scale of 20 to 40 minutes.

The timing of replication initiation is affected by DNA methylation and interactions with the bacterial plasma membrane. The *oriC* DNA is methylated by the Dam methylase (Table 25-3), which methylates the N^6 position of adenine within the palindromic sequence (5')GATC. (Dam is not a biochemical expletive; it stands

for DNA adenine methylation.) The *oriC* region of *E. coli* is highly enriched in GATC sequences—it has 11 in its 245 bp; the average frequency of GATC in the *E. coli* chromosome as a whole is 1 in 256 bp.

Immediately after replication, the DNA is hemimethylated: the parent strands have methylated *oriC* sequences but the newly synthesized strands do not. The hemimethylated *oriC* sequences are now sequestered by interaction with the plasma membrane (the mechanism is unknown) and by binding of the protein SeqA. After a time, *oriC* is released from the plasma membrane, SeqA dissociates, and the DNA must be fully methylated by Dam methylase before it can again bind DnaA and initiate a new round of replication.

Elongation The elongation phase of replication includes two distinct but related operations: leading strand synthesis and lagging strand synthesis. Several enzymes at the replication fork are important to the synthesis of both strands. Parent DNA is first unwound by DNA helicases, and the resulting topological stress is relieved by topoisomerases. Each separated strand is then stabilized by SSB. From this point, synthesis of leading and lagging strands is sharply different.

Leading strand synthesis, the more straightforward of the two, begins with the synthesis by primase (DnaG protein) of a short (10 to 60 nucleotide) RNA primer at the replication origin. DnaG interacts with DnaB helicase

to carry out this reaction, and the primer is synthesized in the direction opposite to that in which the DnaB helicase is moving. In effect, the DnaB helicase moves along the strand that becomes the lagging strand in DNA synthesis; however, the first primer laid down in the first DnaG-DnaB interaction serves to prime leading strand DNA synthesis in the opposite direction. Deoxyribonucleotides are added to this primer by a DNA polymerase III complex linked to the DnaB helicase tethered to the opposite DNA strand. Leading strand synthesis then proceeds continuously, keeping pace with the unwinding of DNA at the replication fork.

Lagging strand synthesis, as we have noted, is accomplished in short Okazaki fragments (Fig. 25-11a). First, an RNA primer is synthesized by primase, and, as in leading strand synthesis, DNA polymerase III binds to the RNA primer and adds deoxyribonucleotides (Fig. 25-11b). On this level, the synthesis of each Okazaki fragment seems straightforward, but the details are quite complex. The complexity lies in the *coordination* of leading and lagging strand synthesis. Both strands are produced by a *single* asymmetric DNA polymerase III dimer; this is accomplished by looping the DNA of the lagging strand as shown in Figure 25-12, bringing together the two points of polymerization.

The synthesis of Okazaki fragments on the lagging strand entails some elegant enzymatic choreography.

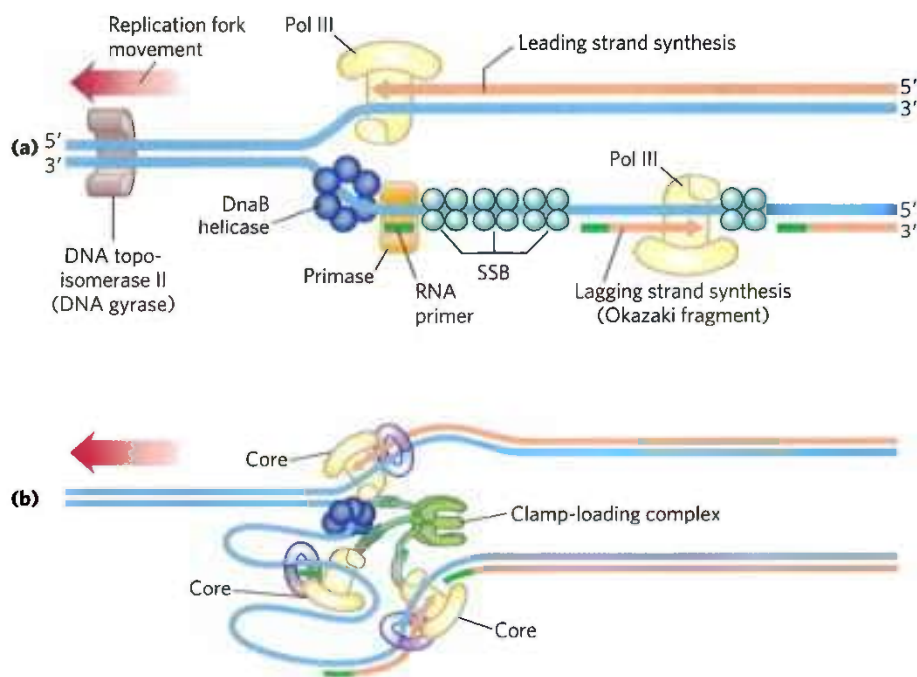
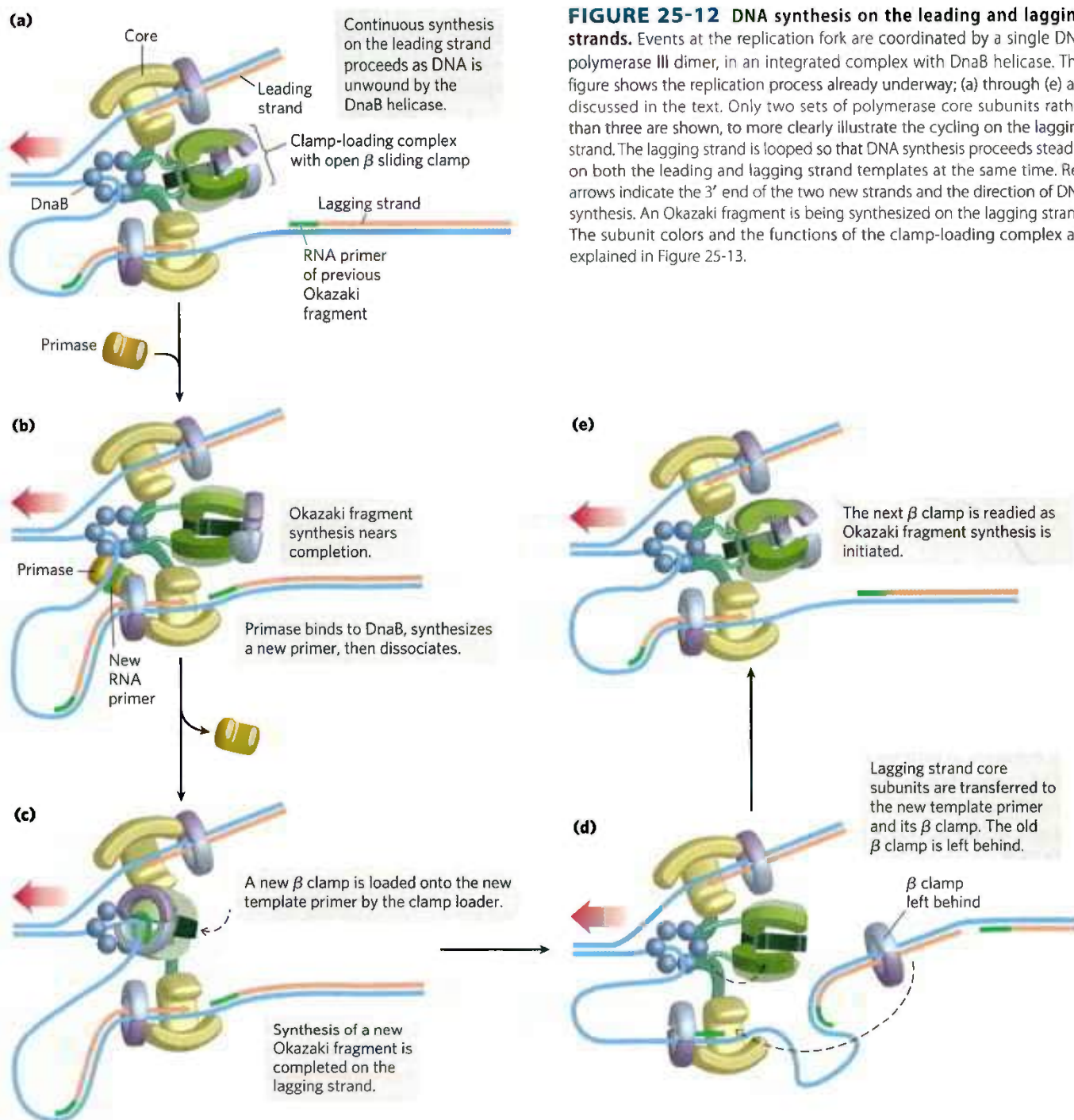


FIGURE 25-11 Synthesis of Okazaki fragments. (a) At intervals, primase synthesizes an RNA primer for a new Okazaki fragment. If we consider the two template strands as lying side by side, lagging strand synthesis formally proceeds in the opposite direction from fork movement. Each primer is extended by DNA polymerase III. DNA synthesis continues until the fragment extends as far as the primer of the previously added Okazaki fragment.

A new primer is synthesized near the replication fork to begin the process again. (b) In the replisome complex, DNA synthesis on the leading and lagging strands is tightly coordinated. Each DNA polymerase III holoenzyme has three sets of core subunits (yellow), linked together with a single clamp-loading complex, so one or two Okazaki fragments can be synthesized simultaneously, along with the leading strand.

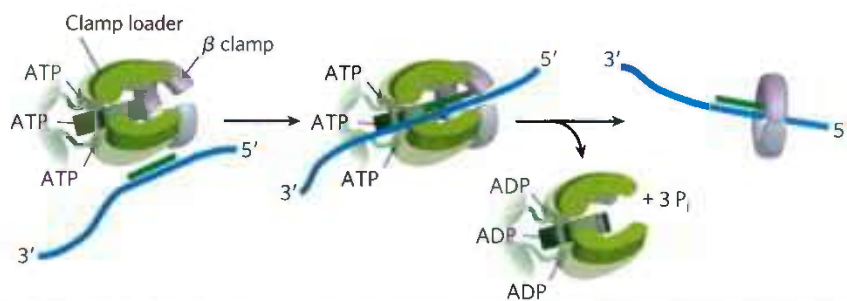


DNA polymerase III uses one set of its core subunits (the core polymerase) to synthesize the leading strand continuously, while the other two sets of core subunits cycle from one Okazaki fragment to the next on the looped lagging strand. In vitro, a DNA polymerase III holoenzyme with only two sets of core subunits can synthesize both leading and lagging strands. However, a third set of core subunits increases the efficiency of lagging strand synthesis as well as the processivity of the overall replisome.

DnaB helicase, bound in front of DNA polymerase III, unwinds the DNA at the replication fork (Fig. 25-12a) as it travels along the lagging strand template in the 5'→3' direction. DnaG primase occasionally associates with DnaB helicase and synthesizes a short RNA primer (Fig. 25-12b). A new β sliding clamp is then positioned at the primer by the clamp-loading complex of DNA polymerase III (Fig. 25-12c). When synthesis of an Okazaki fragment has been completed, replication halts, and the core subunits of DNA polymerase III dissociate from their

TABLE 25-4 Proteins of the *E. coli* Replisome

Protein	M_r	Number of subunits	Function
SSB	75,600	4	Binding to single-stranded DNA
Helicase (DnaB protein)	300,000	6	DNA unwinding
Primase (DnaG protein)	60,000	1	RNA primer synthesis
DNA polymerase III	1,065,400	17	New strand elongation
DNA polymerase I	103,000	1	Filling of gaps; excision of primers
DNA ligase	74,000	1	Ligation
DNA gyrase (DNA topoisomerase II)	400,000	4	Supercoiling

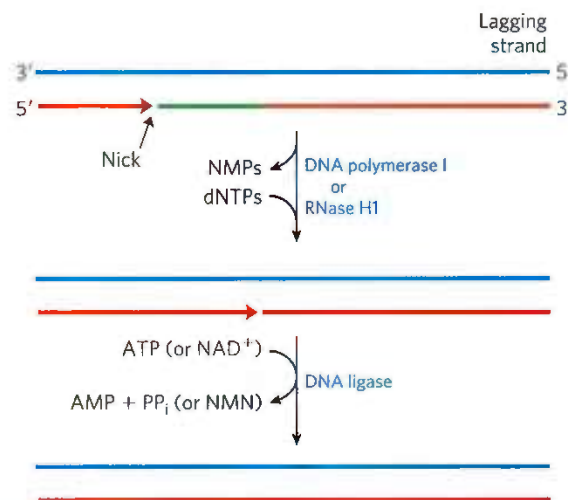
**FIGURE 25-13** The DNA polymerase III clamp loader. The five subunits of the clamp-loading (γ) complex are the δ and δ' subunits and the amino-terminal domain of each of the three τ subunits (see Fig. 25-8). The complex binds to three molecules of ATP and to a dimeric β clamp. This binding forces the β clamp open at one of its two subunit interfaces. Hydrolysis of the bound ATP allows the β clamp to close again around the DNA.

β sliding clamp (and from the completed Okazaki fragment) and associate with the new clamp (Fig. 25-12d, e). This initiates synthesis of a new Okazaki fragment. Two sets of core subunits may be engaged in the synthesis of two different Okazaki fragments at the same time. The proteins acting at the replication fork are summarized in **Table 25-4**.

The clamp-loading complex of DNA polymerase III, consisting of parts of the three τ subunits along with the δ and δ' subunits, is also an AAA+ ATPase. This complex binds to ATP and to the new β sliding clamp. The binding imparts strain on the dimeric clamp, opening up the ring at one subunit interface (**Fig. 25-13**). The newly primed lagging strand is slipped into the ring through the resulting break. The clamp loader then hydrolyzes ATP, releasing the β sliding clamp and allowing it to close around the DNA.

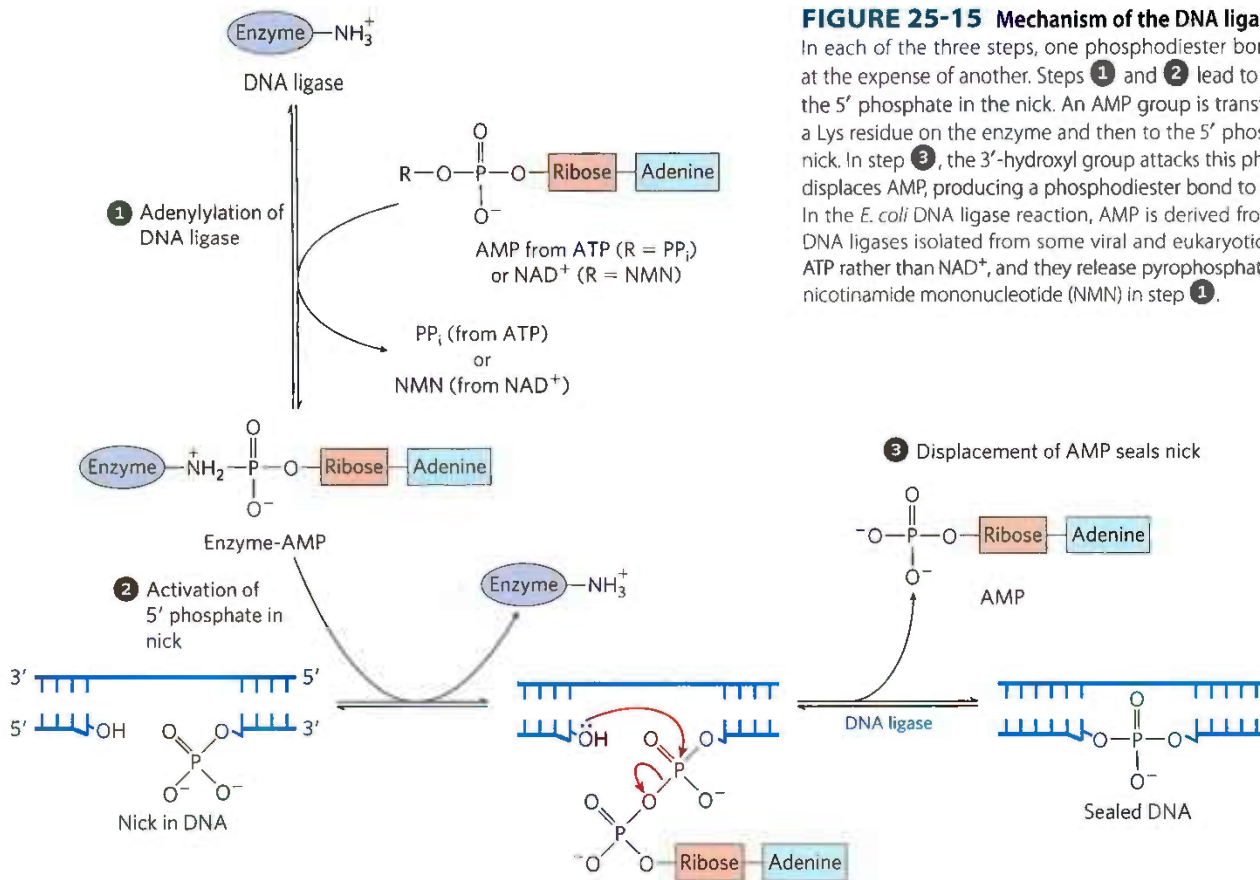
The replisome promotes rapid DNA synthesis, adding ~1,000 to 2,000 nucleotides to each strand (leading and lagging). Once an Okazaki fragment has been completed, its RNA primer is removed by DNA polymerase I or RNase H1 and replaced with DNA by the polymerase; the remaining nick is sealed by DNA ligase (**Fig. 25-14**).

DNA ligase catalyzes the formation of a phosphodiester bond between a 3' hydroxyl at the end of one DNA strand and a 5' phosphate at the end of another strand. The phosphate must be activated by adenylation. DNA ligases isolated from viruses and eukaryotes use ATP for this purpose. DNA ligases from bacteria are unusual in that many use NAD^+ —a cofactor that usually functions

**FIGURE 25-14** Final steps in the synthesis of lagging strand segments. RNA primers in the lagging strand are removed by the 5'→3' exonuclease activity of DNA polymerase I or RNase H1, and then replaced with DNA by DNA polymerase I. The remaining nick is sealed by DNA ligase. The role of ATP or NAD^+ is shown in Figure 25-15.

in hydride transfer reactions (see Fig. 13-24)—as the source of the AMP activating group (**Fig. 25-15**). DNA ligase is another enzyme of DNA metabolism that has become an important reagent in recombinant DNA experiments (see Fig. 9-1).

Termination Eventually, the two replication forks of the circular *E. coli* chromosome meet at a terminus region

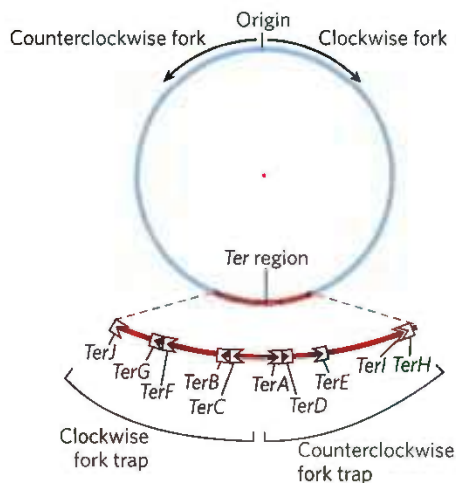
**FIGURE 25-15 Mechanism of the DNA ligase reaction.**

In each of the three steps, one phosphodiester bond is formed at the expense of another. Steps 1 and 2 lead to activation of the 5' phosphate in the nick. An AMP group is transferred first to a Lys residue on the enzyme and then to the 5' phosphate in the nick. In step 3, the 3'-hydroxyl group attacks this phosphate and displaces AMP, producing a phosphodiester bond to seal the nick. In the *E. coli* DNA ligase reaction, AMP is derived from NAD^+ . The DNA ligases isolated from some viral and eukaryotic sources use ATP rather than NAD^+ , and they release pyrophosphate rather than nicotinamide mononucleotide (NMN) in step 1.

containing multiple copies of a 20 bp sequence called Ter (Fig. 25-16). The Ter sequences are arranged on the chromosome to create a trap that a replication fork can enter but cannot leave. The Ter sequences function as binding sites for the protein Tus (terminus utilization substance). The Tus-Ter complex can arrest a replication fork from

only one direction. Only one Tus-Ter complex functions per replication cycle—the complex first encountered by either replication fork. Given that opposing replication forks generally halt when they collide, Ter sequences would not seem to be essential, but they may prevent overreplication by one fork in the event that the other is delayed or halted by an encounter with DNA damage or some other obstacle.

So, when either replication fork encounters a functional Tus-Ter complex, it halts; the other fork halts when it meets the first (arrested) fork. The final few hundred base pairs of DNA between these large protein complexes are then replicated (by an as yet unknown mechanism), completing two topologically interlinked (catenated) circular chromosomes (Fig. 25-17). DNA circles linked in this way are known as **catenanes**. Separation of the catenated circles in *E. coli* requires topoisomerase IV (a type II topoisomerase). The separated chromosomes then segregate into daughter cells at cell division. The terminal phase of replication of other circular chromosomes, including many of the DNA viruses that infect eukaryotic cells, is similar.

**FIGURE 25-16 Termination of chromosome replication in *E. coli*.** The Ter sequences (*TerA* through *TerJ*) are positioned on the chromosome in two clusters with opposite orientations. The overall Ter region encompasses about 9% of the circular chromosome.

Replication in Eukaryotic Cells Is Similar but More Complex

The DNA molecules in eukaryotic cells are considerably larger than those in bacteria and are organized into

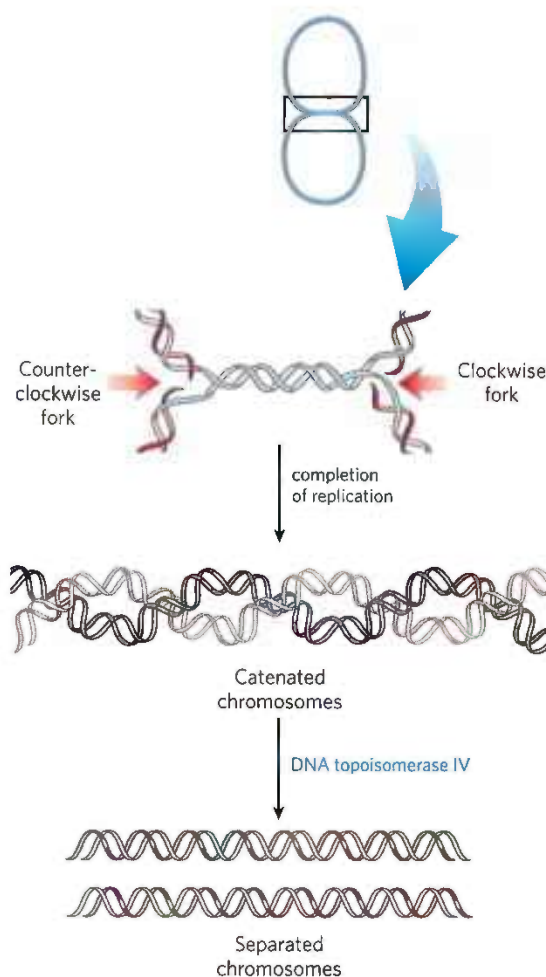


FIGURE 25-17 Role of topoisomerases in replication termination.

Replication of the DNA separating opposing replication forks leaves the completed chromosomes joined as catenanes, or topologically interlinked circles. The circles are not covalently linked, but because they are interwound and each is covalently closed, they cannot be separated — except by the action of topoisomerases. In *E. coli*, a type II topoisomerase known as DNA topoisomerase IV plays the primary role in separating catenated chromosomes, transiently breaking both DNA strands of one chromosome and allowing the other chromosome to pass through the break.

complex nucleoprotein structures (chromatin; p. 898). The essential features of DNA replication are the same in eukaryotes and bacteria, and many of the protein complexes are functionally and structurally conserved. However, eukaryotic replication is regulated and coordinated with the cell cycle, and must function within the complexities of chromatin structure.

Origins of replication have a well-characterized structure in some lower eukaryotes, but they are much less defined in higher eukaryotes. In both cases, replication begins in short nucleosome-free regions. Yeast (*S. cerevisiae*) has about 400 defined replication origins called autonomously replicating sequences (ARs), or **replicators**. Yeast replicators span ~150 bp and contain several essential, conserved sequences. There are about 30,000 to 50,000 replication origins in human chromosomes.

The replication origins of vertebrates in general may be defined by some aspect of DNA secondary structure, as yet unknown.

Regulation ensures that all cellular DNA is replicated once per cell cycle. Much of this regulation involves proteins called cyclins and the cyclin-dependent kinases (CDKs) with which they form complexes (see Section 12.8). The cyclins are rapidly destroyed by ubiquitin-dependent proteolysis at the end of the M phase (mitosis), and the absence of cyclins allows the establishment of **prereplicative complexes (pre-RCs)** on replication initiation sites. In rapidly growing cells, the pre-RC forms at the end of M phase. In slow-growing cells, it does not form until the end of G1. Formation of the pre-RC renders the cell competent for replication, an event sometimes called **licensing**.

As in bacteria, the key event in the initiation of replication in all eukaryotes is the loading of the replicative helicase, a heterohexameric complex of **minichromosome maintenance (MCM) proteins** (MCM2 to MCM7). The ring-shaped MCM2–7 helicase functions in some ways like the bacterial DnaB helicase, although it translocates 3'→5' along the leading strand template. It is loaded onto the DNA in steps (**Fig. 25-18**). The origin is recognized and bound first by another six-protein complex, called **ORC (origin recognition complex)**, followed by the protein CDC6 (cell division cycle), which recruits CDT1 (CDC10-dependent transcript 1). Together, they facilitate the loading of two inactive MCM2–7 complexes (the pre-RC). The ORC-CDC6 complex and CDT1 dissociate, leaving behind the pre-RC. ORC has five AAA+ ATPase domains among its subunits and is functionally analogous to the bacterial DnaA. The yeast CDC6 is yet another AAA+ ATPase that forms a complex with the ORC subunits. Following pre-RC formation, another set of proteins, CDC45 and the GINS, bind to and activate the MCM2–7 helicase, triggering DNA denaturation. (GINS refers to the first letters of the numbers 5-1-2-3 in Japanese, *go-ichi-ni-san*, providing a somewhat cryptic callout to the four protein subunits of the complex: SLD5, PSF1, PSF2, and PSF3.) The replication proteins then bind to form a replisome, and bidirectional replication begins.

Commitment to replication requires the synthesis and activity of S-phase cyclin-CDK complexes (such as the cyclin E-CDK2 complex; see Fig. 12-36) and CDC7-DBF4. Both types of complexes help to activate replication by binding to and phosphorylating several subunits of the pre-RC. Other cyclins and CDKs function to inhibit the formation of more pre-RC complexes once replication has been initiated. For example, CDK2 binds to cyclin A as cyclin E levels decline during S phase, inhibiting CDK2 and preventing the licensing of additional pre-RC complexes.

The rate of movement of the replication fork in eukaryotes (~50 nucleotides/s) is only one-twentieth that observed in *E. coli*. At this rate, replication of an average

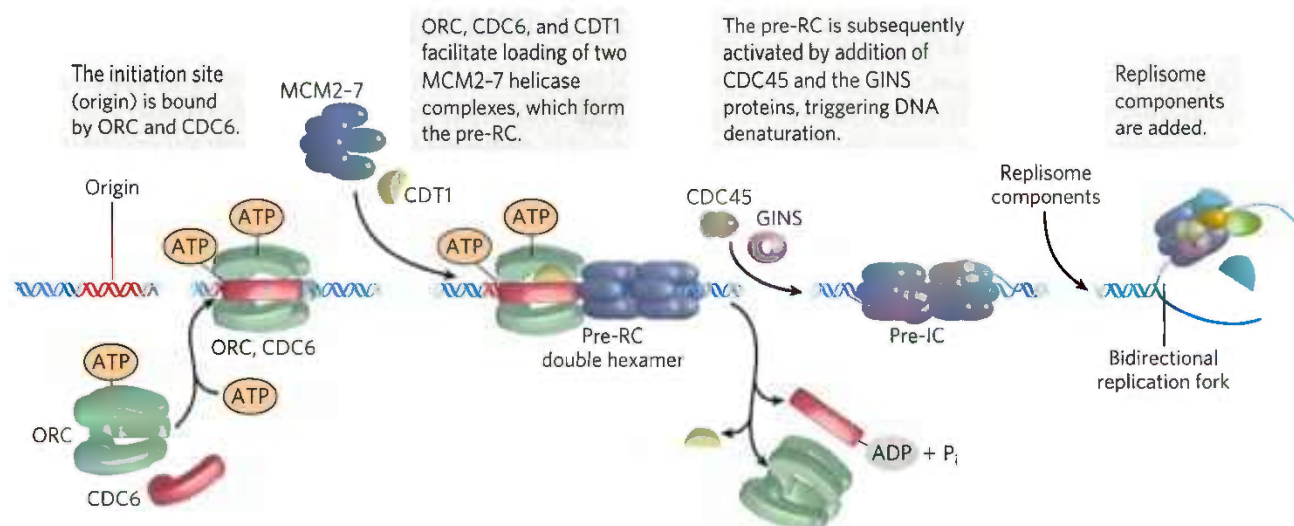


FIGURE 25-18 Assembly of a prereplicative complex at a eukaryotic replication origin. The initiation site (origin) is bound by ORC, CDC6, and CDT1. These proteins, many of them AAA+ ATPases, promote loading of two MCM2-7 helicase complexes, in a reaction analogous to the loading of the bacterial DnaB

helicase by DnaC protein. The two loaded but inactive MCM2-7 complexes comprise the prereplicative complex, or pre-RC. The pre-RC is subsequently activated by addition of CDC45 and the GINS proteins, followed by addition of the replisome components. [Information from M.W. Parker et al., *Crit. Rev. Biochem. Mol. Biol.* 52:107, 2017.]

human chromosome proceeding from a single origin would take more than 500 hours, making the requirement for many origins evident.

Like bacteria, eukaryotes have several types of DNA polymerases. Some have been linked to particular functions, such as the replication of mitochondrial DNA. The replication of nuclear chromosomes primarily involves three multisubunit DNA polymerases. The highly processive **DNA polymerase ϵ** synthesizes the leading strand, and **DNA polymerase δ** synthesizes the lagging strand. Both enzymes have 3'→5' proofreading exonuclease activities. **DNA polymerase α** , a DNA polymerase/primase, synthesizes RNA primers and also extends them by about 10 nucleotides of DNA to initiate synthesis of each Okazaki fragment on the lagging strand. One subunit of DNA polymerase α has a primase activity, and the largest subunit (M_r ~180,000) contains the polymerization activity. However, this polymerase has no proofreading 3'→5' exonuclease activity, making it unsuitable for high-fidelity DNA replication.

DNA polymerases ϵ and δ are associated with and stimulated by proliferating cell nuclear antigen (PCNA; M_r 29,000), a protein found in large amounts in the nuclei of proliferating cells. The three-dimensional structure of PCNA is remarkably similar to that of the β subunit of *E. coli* DNA polymerase III (Fig. 25-8b), although primary sequence homology is not evident. PCNA has a function analogous to that of the β subunit, forming a circular clamp that enhances the processivity of the two polymerases.

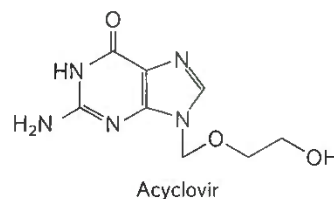
Two other protein complexes also function in eukaryotic DNA replication. RPA (replication protein A) is a single-stranded DNA-binding protein, equivalent in function to the *E. coli* SSB protein. RFC (replication factor C)

is a clamp loader for PCNA and facilitates the assembly of active replication complexes. The subunits of the RFC complex have significant sequence similarity to the subunits of the bacterial clamp-loading (γ) complex.

Termination of replication on linear eukaryotic chromosomes occurs when replication forks operating from nearby origins converge. As in bacteria, there are successive steps of final replication, replisome dissociation, and decatenation of the DNA products. All of the steps are mediated by additional protein complexes, with some parts of the process still undefined.

Viral DNA Polymerases Provide Targets for Antiviral Therapy

Many DNA viruses encode their own DNA polymerases, and some of these have become targets for pharmaceuticals. For example, the DNA polymerase of the herpes simplex virus is inhibited by acyclovir, a compound developed by Gertrude Elion and George Hitchings (p. 836). Acyclovir consists of guanine attached to an incomplete ribose ring.



It is phosphorylated by a virally encoded thymidine kinase; acyclovir binds to this viral enzyme with an affinity 200-fold greater than its binding to the cellular thymidine kinase. This ensures that phosphorylation occurs mainly

in virus-infected cells. Cellular kinases convert the resulting acyclo-GMP to acyclo-GTP, which is both an inhibitor and a substrate of DNA polymerases; acyclo-GTP competitively inhibits the herpes DNA polymerase more strongly than cellular DNA polymerases. Because it lacks a 3' hydroxyl, acyclo-GTP also acts as a chain terminator when incorporated into DNA. Thus viral replication is inhibited at several steps. ■

SUMMARY 25.1 DNA Replication

- Replication of DNA follows a set of universal rules. Replication is semiconservative, each strand acting as template for a new daughter strand. It is carried out in three identifiable phases: initiation, elongation, and termination. The process starts at a single origin in bacteria and usually proceeds bidirectionally. DNA is synthesized in the 5'→3' direction by DNA polymerases. At the replication fork, the leading strand is synthesized continuously in the same direction as replication fork movement; the lagging strand is synthesized discontinuously as Okazaki fragments, which are subsequently ligated.
- Nucleases are enzymes that degrade DNA. Endonucleases cleave within a DNA polymer; exonucleases degrade DNA from the end of one strand (either 5'→3' or 3'→5').
- DNA polymerases are complex enzymes that synthesize DNA, and often possess additional activities, including exonuclease functions.
- DNA is replicated with very high fidelity. Accuracy is maintained by (1) base selection by the polymerase, (2) a 3'→5' proofreading exonuclease activity that is part of many DNA polymerases, and (3) specific repair systems for mismatches left behind after replication.
- Most cells have several DNA polymerases. In *E. coli*, DNA polymerase III is the primary replication enzyme. DNA polymerase I is responsible for special functions during replication, recombination, and repair.
- Replication requires an array of enzymes and protein factors in addition to DNA polymerases. Many of these proteins belong to the AAA+ ATPase family.
- Replication initiation occurs when replicative helicases are loaded onto replication origins in stepwise fashion. Elongation is achieved by an active replisome—a supramolecular complex of nucleic acids and many proteins, including polymerases. Termination occurs when replisomes proceeding in opposite directions converge. It requires decatenation of the replication products when replication is complete.
- The major replicative DNA polymerases in eukaryotes are DNA polymerases ϵ and δ . DNA polymerase α synthesizes primers.
- Viral DNA replication is a drug target.

25.2 DNA Repair

Most cells have only one or two sets of genomic DNA.

P1 Damaged proteins and RNA molecules can be quickly replaced by using information encoded in the DNA, but DNA molecules themselves are irreplaceable. Maintaining the integrity of the information in DNA is a cellular imperative, supported by an elaborate set of DNA repair systems. DNA can become damaged by a variety of processes, some spontaneous, others catalyzed by environmental agents (Chapter 8). Replication itself can very occasionally damage the information content in DNA when polymerase errors create mismatched base pairs (such as G paired with T).

The chemistry of DNA damage is diverse and complex. The cellular response to this damage includes enzymatic systems that catalyze some of the most interesting chemical transformations in DNA metabolism. We first examine the effects of alterations in DNA sequence and then consider specific repair systems.

Mutations Are Linked to Cancer



The best way to illustrate the importance of DNA repair is to consider the effects of *unrepaired* DNA damage (a lesion). **P3** The most serious outcome is a change in the base sequence of the DNA, which, if replicated and transmitted to future generations of cells, becomes permanent. A permanent change in the nucleotide sequence of DNA is called a **mutation**. Mutations can involve the replacement of one base pair with another (substitution mutation) or the addition or deletion of one or more base pairs (insertion or deletion mutations). If the mutation affects nonessential DNA or if it has a negligible effect on the function of a gene, it is known as a **silent mutation**. **P1** **P3** Rarely, a mutation confers some biological advantage. Most nonsilent mutations, however, are neutral or deleterious.

In mammals there is a strong correlation between the accumulation of mutations and cancer. A simple test developed by Bruce Ames in the 1970s measures the potential of a given chemical compound to promote certain easily detected mutations in a specialized bacterial strain (**Fig. 25-19**). Few of the chemicals that we encounter in daily life score as mutagens in this test. However, of the compounds known to be carcinogenic from extensive animal trials, more than 90% are also found to be mutagenic in the Ames test. Because of this strong correlation between mutagenesis and carcinogenesis, the Ames test for bacterial mutagens is still widely used as a rapid and inexpensive screen for potential human carcinogens.

The genomic DNA in a typical mammalian cell accumulates many thousands of lesions during a 24-hour period. However, as a result of DNA repair, fewer than 1 in 1,000 become a mutation. DNA is a relatively stable molecule, but in the absence of repair systems, the cumulative effect of many infrequent but damaging reactions would make life impossible. ■

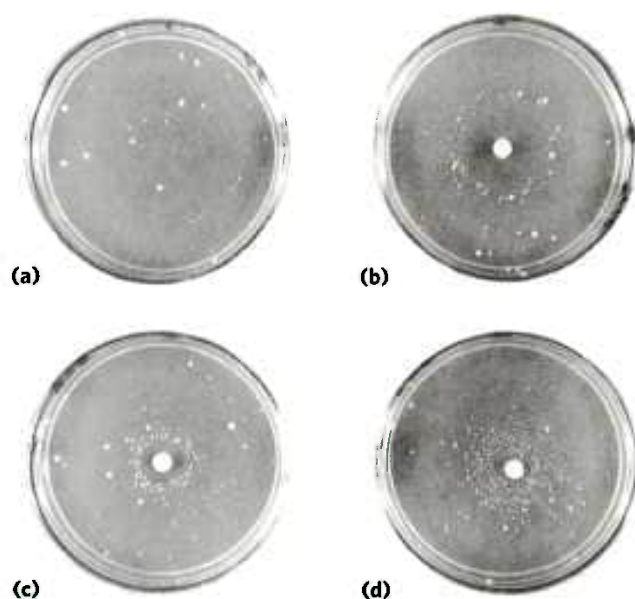


FIGURE 25-19 Ames test for carcinogens, based on their mutagenicity. A strain of *Salmonella typhimurium* having a mutation that inactivates an enzyme of the histidine biosynthetic pathway is plated on a histidine-free medium. Few cells grow. (a) The few small colonies of *S. typhimurium* that do grow on a histidine-free medium carry spontaneous mutations that permit the histidine biosynthetic pathway to operate. Three identical nutrient plates (b), (c), and (d) have been inoculated with an equal number of cells. Each plate then receives a disk of filter paper containing progressively lower concentrations of a mutagen. The mutagen greatly increases the rate of back-mutation and hence the number of colonies. The clear areas around the filter paper indicate where the concentration of mutagen is so high that it is lethal to the cells. As the mutagen diffuses away from the filter paper, it is diluted to sublethal concentrations that promote back-mutation. Mutagens can be compared on the basis of their effect on mutation rate. Because many compounds undergo a variety of chemical transformations after entering cells, compounds are sometimes tested for mutagenicity after first incubating them with a liver extract. Some substances have been found to be mutagenic only after this treatment. [Bruce N. Ames, University of California, Berkeley, Department of Biochemistry and Molecular Biology.]

All Cells Have Multiple DNA Repair Systems

The number and diversity of repair systems reflect both the importance of DNA repair to cell survival and the diverse sources of DNA damage (Table 25-5). Some common types of lesions, such as pyrimidine dimers (see Fig. 8-30), can be repaired by several distinct systems. Nearly 200 genes in the human genome encode proteins dedicated to DNA repair. In many cases, the loss of function of one of these proteins results in genomic instability and an increased occurrence of oncogenesis (Box 25-1).

P2 Many DNA repair processes also seem to be extraordinarily inefficient energetically—an exception to the pattern observed in the vast majority of metabolic pathways, where every ATP is generally accounted for and used optimally. When the integrity of the genetic information is at stake, the amount of chemical energy invested in a repair process seems almost irrelevant.

Accurate DNA repair is possible largely because the DNA molecule consists of two complementary

TABLE 25-5 Types of DNA Repair Systems in *E. coli*

Enzymes/proteins	Type of damage
Mismatch repair	
Dam methylase	} Mismatches
MutH, MutL, MutS proteins	
DNA helicase II	
SSB	
DNA polymerase III	
Exonuclease I	
Exonuclease VII	
RecJ nuclease	
Exonuclease X	
DNA ligase	
Base-excision repair	
DNA glycosylases	} Abnormal bases (uracil, hypoxanthine, xanthine); alkylated bases; in some other organisms, pyrimidine dimers
AP endonucleases	
DNA polymerase I	
DNA ligase	
Nucleotide-excision repair	
ABC excinuclease	} DNA lesions that cause large structural change (e.g., pyrimidine dimers)
DNA polymerase I	
DNA ligase	
Direct repair	
DNA photolyases	Pyrimidine dimers
<i>O</i> ⁶ -Methylguanine-DNA methyltransferase	<i>O</i> ⁶ -Methylguanine
AlkB protein	1-Methylguanine, 3-methylcytosine

strands. Damaged DNA in one strand can be removed and replaced, without introducing mutations, by using the undamaged complementary strand as a template. We consider here the principal types of repair systems, beginning with those that repair the rare nucleotide mismatches that are left behind by replication.

Mismatch Repair Correction of the rare mismatches left after replication in *E. coli* improves the overall fidelity of replication by an additional factor of 10^2 to 10^3 . The mismatches are nearly always corrected to reflect the information in the old (template) strand, which the repair system can distinguish from the newly synthesized strand by the presence of methyl group tags on the template DNA. The mismatch repair system of *E. coli* includes at least 10 protein components (Table 25-5) that function either in strand discrimination or in the repair process itself. The functions of many of these were first worked out by Paul Modrich and colleagues in the 1980s.

The strand discrimination mechanism has not been determined for most bacteria or eukaryotes, but it is well understood for *E. coli* and some closely related bacterial species. In these bacteria, strand discrimination is based on the action of Dam methylase, which, as you will recall, methylates DNA at the *N*⁶ position of all adenines within (5')GATC sequences. Immediately after passage of the replication fork, there is a short period (a few seconds or minutes) during which the template strand is methylated



DNA Repair and Cancer

Human cancers develop when genes that regulate normal cell division (oncogenes and tumor suppressor genes; see Chapter 12) fail to function, are activated at the wrong time, or are altered. As a consequence, cells may grow out of control and form a tumor. The genes controlling cell division can be damaged by spontaneous mutation or overridden by the invasion of a tumor virus (Chapter 26). Not surprisingly, alterations in DNA repair genes that result in a higher rate of mutation can greatly increase an individual's susceptibility to cancer. Defects in the genes encoding the proteins involved in nucleotide-excision repair, mismatch repair, recombinational repair, and error-prone translesion DNA synthesis have all been linked to human cancers. Clearly, DNA repair can be a matter of life and death.

Nucleotide-excision repair requires a larger number of proteins in humans than in bacteria, although the overall pathways are very similar. Genetic defects that inactivate nucleotide-excision repair have been associated with several genetic diseases, the best-studied of which is xeroderma pigmentosum (XP). Because nucleotide-excision repair is the sole repair pathway for pyrimidine dimers in humans, people with XP are extremely sensitive to light and readily develop sunlight-induced skin cancers. Most people with XP also have neurological abnormalities, presumably because of their inability to repair certain lesions caused by the high rate of oxidative metabolism in neurons. Defects in the genes encoding any of at least seven different protein components of the nucleotide-excision repair system can result in XP, giving rise to seven different genetic groups, denoted XPA to XPG. Note that XPC and XPE are parts of complexes that recognize damaged DNA, whereas XPA, XPB, XPD, XPF, and XPG are all components of a much larger multisubunit complex that represents the human excinuclease depicted in Figure 25-24. These proteins are involved in making the DNA incisions and removing the 29mer segment of DNA.

Most microorganisms have redundant pathways for the repair of cyclobutane pyrimidine dimers — making use of DNA photolyases and sometimes base-excision repair as alternatives to nucleotide-excision repair — but humans and other placental mammals do not. This lack of a backup for nucleotide-excision repair for removing pyrimidine dimers has led to speculation that early mammals were small, furry, nocturnal animals with little need to repair UV damage. However, mammals do have a pathway for the translesion bypass of cyclobutane pyrimidine dimers, which involves

DNA polymerase η . This enzyme preferentially inserts two A residues opposite a T-T pyrimidine dimer, minimizing mutations. People with a genetic condition in which DNA polymerase η function is missing exhibit an XP-like illness known as XP-variant, or XP-V. Clinical manifestations of XP-V are similar to those of the classic XP diseases, although mutation levels are higher in XP-V when cells are exposed to UV light. Apparently, the nucleotide-excision repair system works in concert with DNA polymerase η in normal human cells, repairing and/or bypassing pyrimidine dimers as needed to keep cell growth and DNA replication going. Exposure to UV light introduces a heavy load of pyrimidine dimers, and some must be bypassed by translesion synthesis to keep replication on track. When one system is missing, it is partly compensated for by the other. A loss of DNA polymerase η activity leads to stalled replication forks and bypass of UV lesions by different, more mutagenic, translesion synthesis (TLS) polymerases. As when other DNA repair systems are absent, the resulting increase in mutations often leads to cancer.

One of the most common inherited cancer-susceptibility syndromes is hereditary nonpolyposis colon cancer (HNPCC). This syndrome has been traced to defects in mismatch repair. Human and other eukaryotic cells have several proteins analogous to the bacterial MutL and MutS proteins (see Fig. 25-21). Defects in at least five different mismatch repair genes can give rise to HNPCC. The most prevalent are defects in the *hMLH1* (human MutL homolog 1) and *hMSH2* (human MutS homolog 2) genes. In individuals with HNPCC, cancer generally develops at an early age, with colon cancers being most common.

Most human breast and ovarian cancer occurs in women with no known predisposition. However, about 10% of cases are associated with inherited defects in two genes, *BRCA1* and *BRCA2*. Human *BRCA1* and *BRCA2* are large proteins (1,834 and 3,418 amino acid residues, respectively) that interact with a wide range of other proteins involved in transcription, chromosome maintenance, DNA repair, and control of the cell cycle. *BRCA2* has been implicated in the recombinational DNA repair of double-strand breaks. One of the key roles of *BRCA2* is to load the human RecA homolog, called Rad51, onto DNA at the sites of double-strand breaks. *BRCA1* has as yet imperfectly defined roles in the repair of double-strand breaks, transcription, and some other processes of DNA metabolism. Women with defects in either the *BRCA1* or *BRCA2* gene have a high (~70%) chance of developing breast cancer.

but the newly synthesized strand is not (Fig. 25-20). The transient unmethylated state of GATC sequences in the newly synthesized strand permits the new strand to be distinguished from the template strand. Replication mismatches in the vicinity of a hemimethylated GATC sequence are then repaired according to the information

in the methylated parent (template) strand. If both strands are methylated at a GATC sequence, few mismatches are repaired; if neither strand is methylated, repair occurs but does not favor either strand. The methyl-directed mismatch repair system of *E. coli* efficiently repairs mismatches up to 1,000 bp from a hemimethylated GATC sequence.

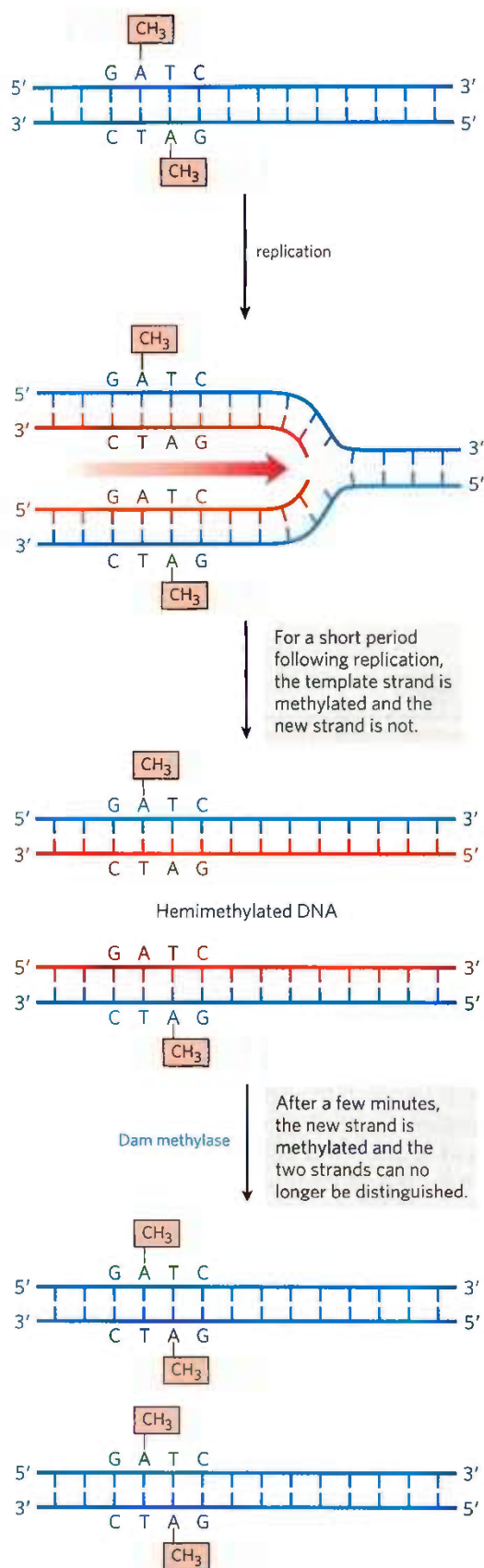


FIGURE 25-20 Methylation and mismatch repair. Methylation of DNA strands can serve to distinguish parent (template) strands from newly synthesized strands in *E. coli* DNA, a function that is critical to mismatch repair. The methylation occurs at the N^6 of adenines in (5')GATC sequences. This sequence is a palindrome, present in opposite orientations on the two strands.

How is the mismatch correction process directed by relatively distant GATC sequences? **Figure 25-21** illustrates one mechanism. MutS scans the DNA and forms a clamplike complex upon encountering a lesion. The complex binds to all mismatched base pairs (except C–C). MutL protein forms a complex with MutS protein, and the MutSL complex slides along the DNA to find a hemimethylated GATC sequence. MutH binds to MutL, and the MutSLH complex moves in either direction at random along the DNA. MutH has a site-specific endonuclease activity that is inactive until the complex encounters a hemimethylated GATC sequence. At this site, MutH catalyzes cleavage of the unmethylated strand on the 5' side of the G in GATC, which marks the strand for repair. Further steps in the pathway depend on where the mismatch is located relative to this cleavage site (**Fig. 25-22**).

When the mismatch is on the 5' side of the cleavage site (**Fig. 25-22**, right side), the unmethylated strand is unwound and degraded in the 3'→5' direction from the

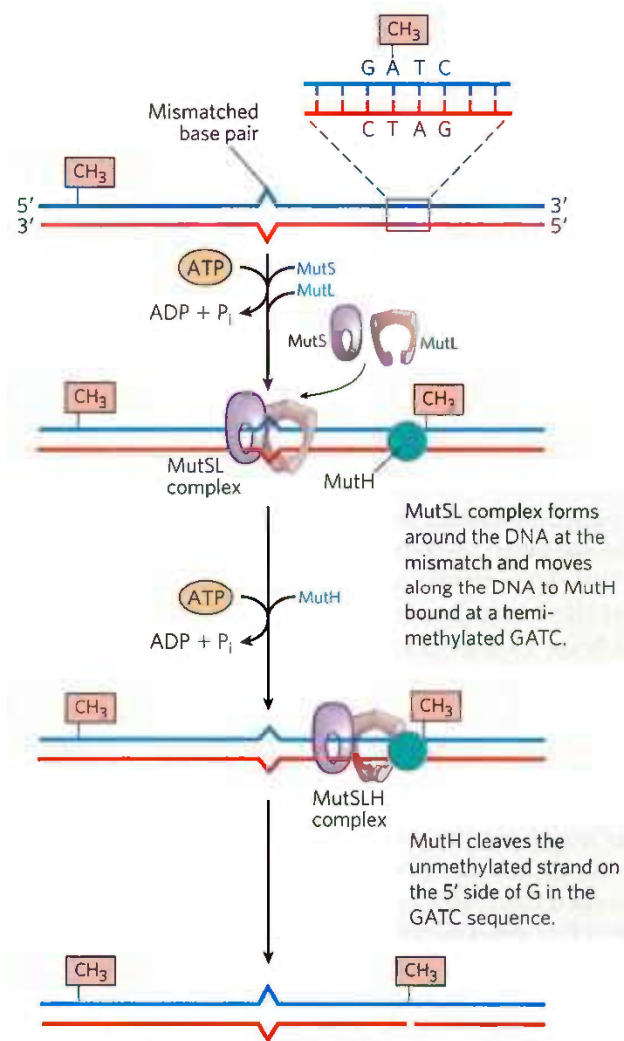


FIGURE 25-21 A model for the early steps of methyl-directed mismatch repair. Recognition of the sequence (5')GATC and of the mismatch are specialized functions of the MutH and MutS proteins, respectively.

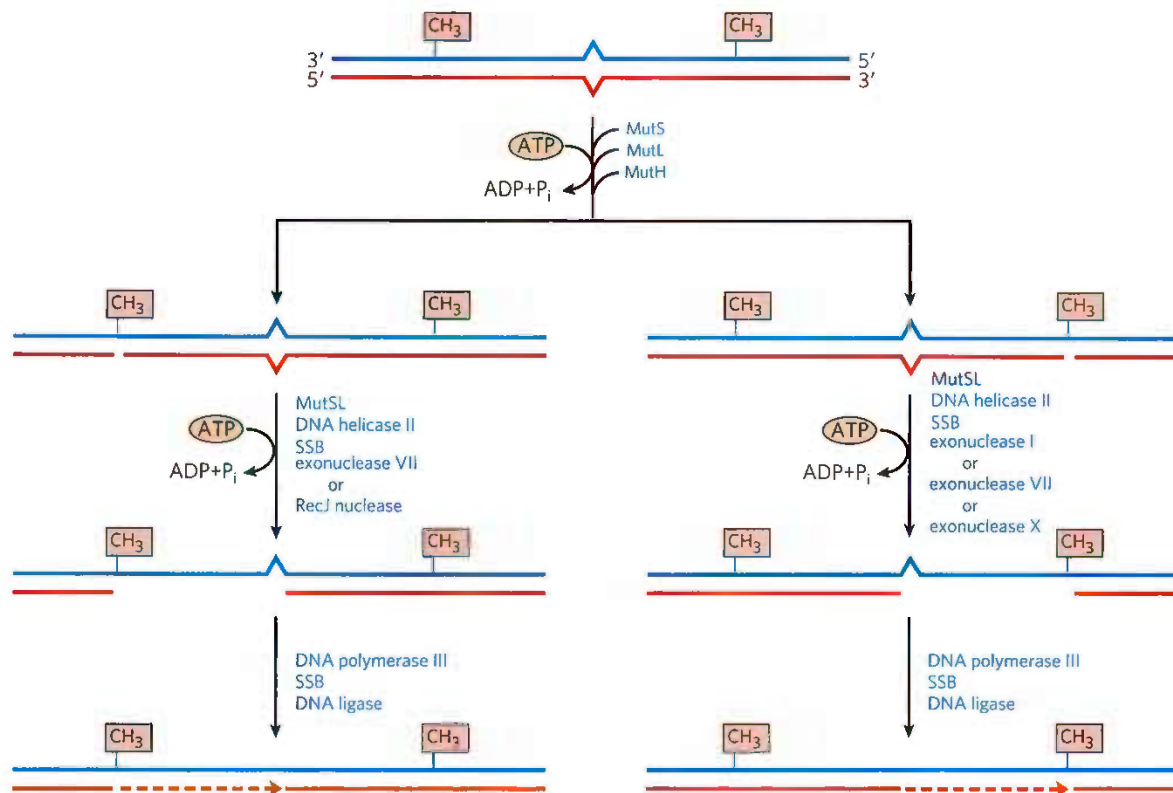


FIGURE 25-22 Completion of methyl-directed mismatch repair. The combined action of DNA helicase II, SSB, and one of four different exonucleases removes a segment of the new strand between the MutH cleavage site and a point just beyond the mismatch. The particular

exonuclease depends on the location of the cleavage site relative to the mismatch, as shown by the alternative pathways here. The resulting gap is filled in (dashed line) by DNA polymerase III, and the nick is sealed by DNA ligase (not shown).

cleavage site through the mismatch, and this segment is replaced with new DNA. This process requires the combined action of DNA helicase II (also called UvrD helicase), SSB, exonuclease I or exonuclease X (both of which degrade strands of DNA in the 3'→5' direction) or exonuclease VII (which degrades single-stranded DNA in either direction), DNA polymerase III, and DNA ligase. The pathway for repair of mismatches on the 3' side of the cleavage site is similar (Fig. 25-22, left), except that the exonuclease is either exonuclease VII or RecJ nuclease (which degrades single-stranded DNA in the 5'→3' direction).

P2 Mismatch repair is particularly costly for *E. coli* in terms of energy expended. The mismatch may occur 1,000 or more base pairs from the GATC sequence. The degradation and replacement of a strand segment of this length require an enormous investment in activated deoxynucleotide precursors to repair a *single* mismatched base. This again underscores the importance to the cell of genomic integrity.

Eukaryotic cells also have mismatch repair systems, with several proteins structurally and functionally analogous to the bacterial MutS and MutL (but not MutH) proteins. Alterations in human genes encoding proteins of this type produce some of the most common inherited cancer-susceptibility syndromes (see Box 25-1), further

demonstrating the value to the organism of DNA repair systems. The main MutS homologs in most eukaryotes, from yeast to humans, are MSH2 (MutS homolog), MSH3, and MSH6. Heterodimers of MSH2 and MSH6 generally bind to single base-pair mismatches, and they bind less well to slightly longer mispaired loops. In many organisms, the longer mismatches (2 to 6 bp) may be bound instead by a heterodimer of MSH2 and MSH3, or are bound by both types of heterodimers in tandem. Homologs of MutL, predominantly a heterodimer of MLH1 (MutL homolog) and PMS1 (post-meiotic segregation), bind to and stabilize the MSH complexes. Many details of the subsequent events in eukaryotic mismatch repair remain to be worked out. In particular, we do not know how newly synthesized DNA strands are identified, although research reveals that this process does not involve GATC sequences.

Base-Excision Repair Every cell has a class of enzymes called **DNA glycosylases** that recognize particularly common DNA lesions (such as the products of cytosine and adenine deamination; see Fig. 8-29a) and remove the affected base by cleaving the *N*-glycosyl bond. The repair pathway is called **base-excision repair**, as the first step involves only the removal of the base rather than an entire nucleotide. The cleavage creates an apurinic or

aprimidinic site in the DNA, commonly referred to as an **AP site** or **abasic site**. Each DNA glycosylase is generally specific for one type of lesion.

Uracil DNA glycosylases, for example, found in most cells, specifically remove from DNA the uracil that results from spontaneous deamination of cytosine. Mutant cells that lack this enzyme have a high rate of G≡C to A≡T mutations. This glycosylase does not remove uracil residues from RNA or thymine residues from DNA. The capacity to distinguish thymine from uracil, the product of cytosine deamination—necessary for the selective repair of the latter—may be one reason why DNA evolved to contain thymine instead of uracil (p. 280).

Most bacteria have just one type of uracil DNA glycosylase, whereas humans have at least four types, with different specificities—an indicator of the importance of removing uracil from DNA. The most abundant human uracil glycosylase, UNG, is associated with the replisome, where it eliminates the occasional U residue inserted in place of a T during replication. The deamination of C residues is 100-fold faster in single-stranded DNA than in double-stranded DNA, and humans have an enzyme, hSMUG1, that removes any U residues occurring in single-stranded DNA during replication or transcription. Two other human DNA glycosylases, TDG and MBD4, remove either U or T residues paired with G, which are generated by deamination of cytosine or 5-methylcytosine, respectively.

Other DNA glycosylases recognize and remove a variety of damaged bases, including formamidopyrimidine and 8-hydroxyguanine (both arising from purine oxidation), hypoxanthine (from adenine deamination), and alkylated bases such as 3-methyladenine and 7-methylguanine. Glycosylases that recognize other lesions, including pyrimidine dimers, have also been identified in some classes of organisms. Remember that AP sites also arise from slow, spontaneous hydrolysis of the *N*-glycosyl bonds in DNA (see Fig. 8-29b).

Once an AP site has been formed by a DNA glycosylase, another type of enzyme must repair it. The repair is *not* made by simply inserting a new base and re-forming the *N*-glycosyl bond. Instead, the deoxyribose 5'-phosphate left behind is removed and replaced with a new nucleotide. This process begins with one of the **AP endonucleases**, enzymes that cut the DNA strand containing the AP site. The position of the incision relative to the AP site (5' or 3' to the site) depends on the type of AP endonuclease. A segment of DNA including the AP site is then removed, DNA polymerase I replaces the DNA, and DNA ligase seals the remaining nick (Fig. 25-23). In eukaryotes, nucleotide replacement is carried out by specialized polymerases, as described below.

Nucleotide-Excision Repair DNA lesions that cause large distortions in the helical structure of DNA generally are repaired by the nucleotide-excision system, a repair pathway critical to the survival of all free-living organisms. In nucleotide-excision repair (Fig. 25-24),

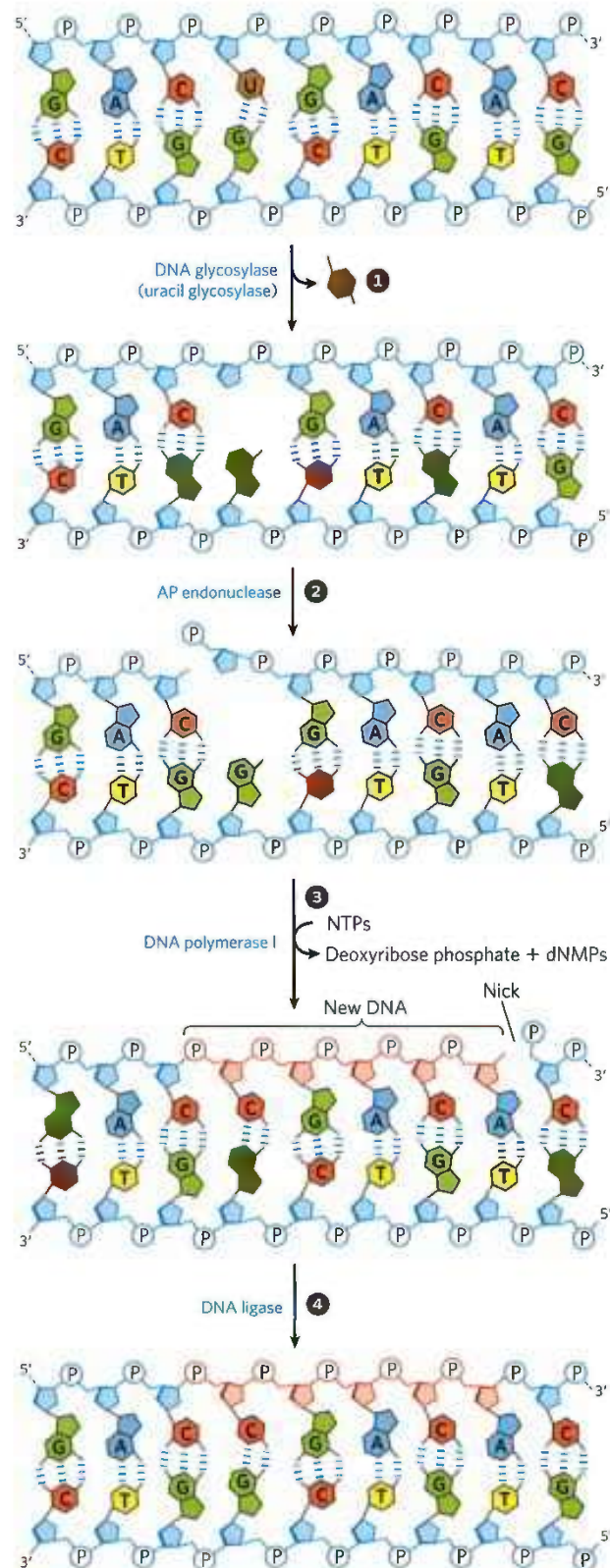


FIGURE 25-23 DNA repair by the base-excision repair pathway.

1 A DNA glycosylase recognizes a damaged base (in this case, a uracil) and cleaves between the base and deoxyribose in the backbone. 2 An AP endonuclease cleaves the phosphodiester backbone near the AP site. 3 DNA polymerase I initiates repair synthesis from the free 3' hydroxyl at the nick, removing (with its 5'→3' exonuclease activity) and replacing a portion of the damaged strand. 4 The nick remaining after DNA polymerase I has dissociated is sealed by DNA ligase.

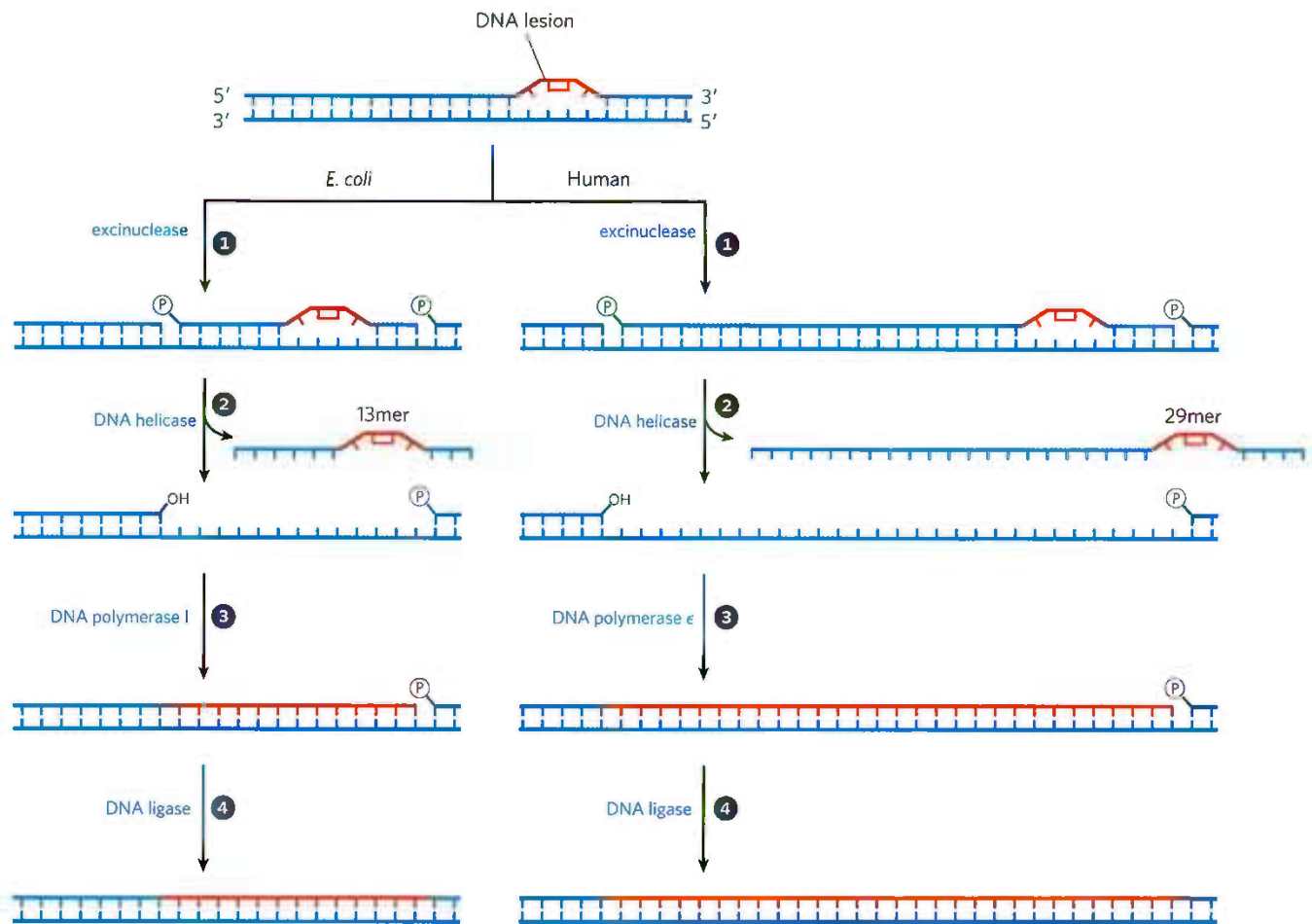


FIGURE 25-24 Nucleotide-excision repair in *E. coli* and humans. The general pathway of nucleotide-excision repair is similar in all organisms. ① An excinuclease binds to DNA at the site of a bulky lesion and cleaves the damaged DNA strand on either side of the lesion. ② The

DNA segment — of 13 nucleotides (13mer) or 29 nucleotides (29mer) — is removed with the aid of a helicase. ③ The gap is filled in by DNA polymerase, and ④ the remaining nick is sealed with DNA ligase. (Information from a figure provided by Aziz Sancar.)

a multisubunit enzyme (excinuclease) hydrolyzes two phosphodiester bonds, one on either side of the distortion caused by the lesion. In *E. coli* and other bacteria, the enzyme system hydrolyzes the fifth phosphodiester bond on the 3' side and the eighth phosphodiester bond on the 5' side to generate a fragment of 12 to 13 nucleotides (depending on whether the lesion involves one or two bases). In humans and other eukaryotes, the enzyme system hydrolyzes the sixth phosphodiester bond on the 3' side and the twenty-second phosphodiester bond on the 5' side, producing a fragment of 27 to 29 nucleotides. Following the dual incision, the excised oligonucleotides are released from the duplex and the resulting gap is filled — by DNA polymerase I in *E. coli* and DNA polymerase ϵ in humans. DNA ligase seals the nick.

In *E. coli*, the key enzymatic complex is the ABC excinuclease, which has three protein components, UvrA (M_r 104,000), UvrB (M_r 78,000), and UvrC (M_r 68,000). The term “excinuclease” is used to describe the unique capacity of this enzyme complex to catalyze two specific

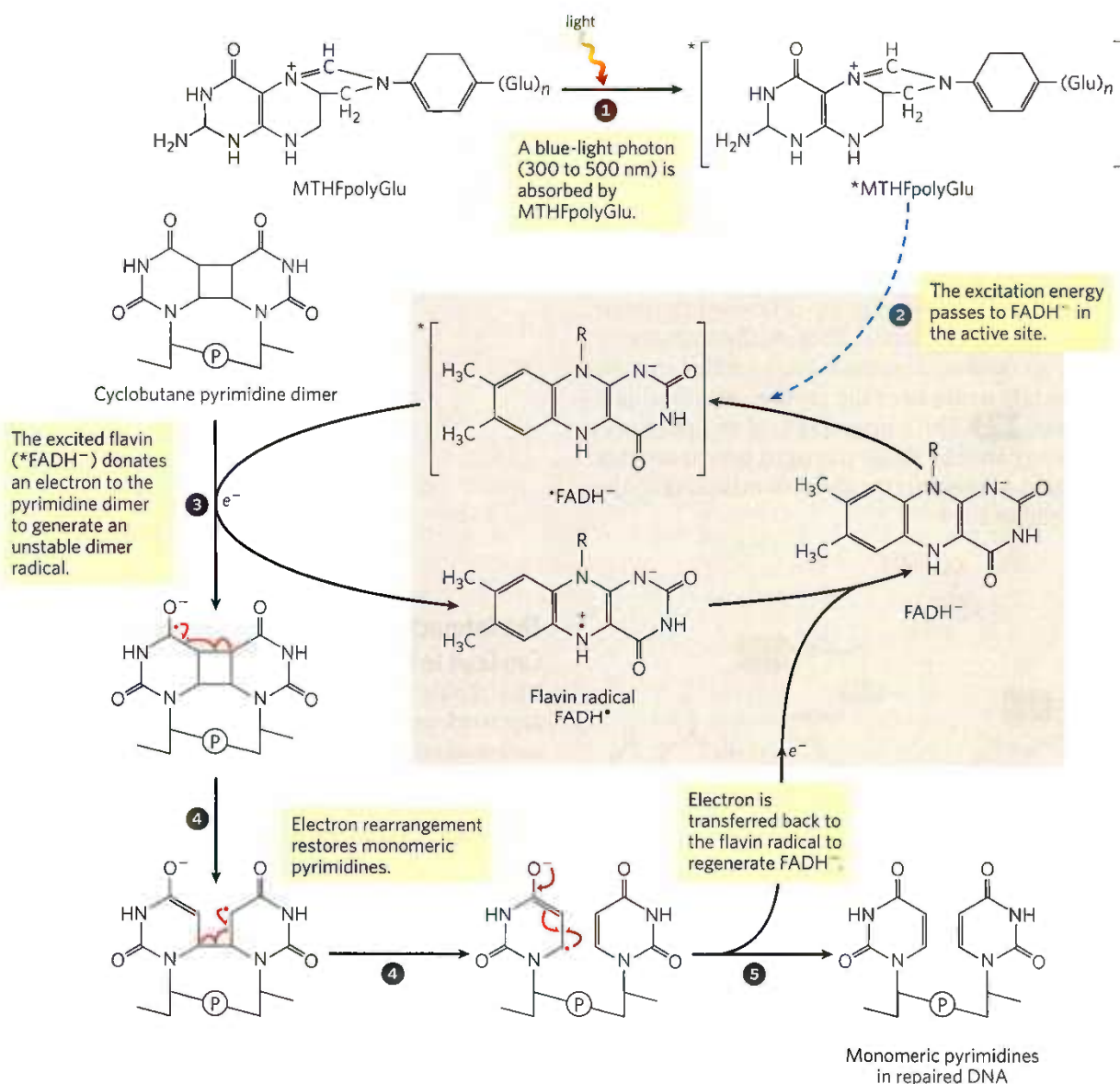
endonucleolytic cleavages, distinguishing this activity from that of standard endonucleases. A dimeric UvrA protein (an ATPase) scans the DNA and binds to the site of a lesion. A UvrB protein can bind to UvrA either before or after an encounter with the lesion. At the lesion, the UvrA dimer dissociates, leaving a tight UvrB-DNA complex. UvrC protein then binds to UvrB, and UvrB makes an incision at the fifth phosphodiester bond on the 3' side of the lesion. This is followed by a UvrC-mediated incision at the eighth phosphodiester bond on the 5' side. The resulting fragment, consisting of 12 to 13 nucleotides, is removed by UvrD helicase. The short gap thus created is filled in by DNA polymerase I and DNA ligase. This pathway (Fig. 25-24, left) is a primary repair route for many types of lesions, including cyclobutane pyrimidine dimers, 6-4 photoproducts (see Fig. 8-30), and several other types of base adducts, including benzo [a]pyrene-guanine, which is formed in DNA by exposure to cigarette smoke. The nucleolytic activity of the ABC excinuclease is novel in the sense that two cuts are made in the DNA.

The mechanism of eukaryotic excinucleases is quite similar to that of the bacterial enzyme, although at least 16 polypeptides with no similarity to the *E. coli* excinuclease subunits are required for the dual incision. Some of the nucleotide-excision repair and base-excision repair in eukaryotes is closely tied to transcription (see Chapter 26). Genetic deficiencies in nucleotide-excision repair in humans give rise to a variety of serious diseases (see Box 25-1).

Direct Repair Several types of damage are repaired without removing a base or nucleotide. The best-characterized example is direct photoreactivation of cyclobutane pyrimidine dimers, a reaction promoted by **DNA photolyases**. Pyrimidine dimers result from a UV-induced reaction.

Through a mechanism worked out by Aziz Sancar and colleagues, photolyases use energy derived from absorbed light to reverse the damage (**Fig. 25-25**). Photolyases generally contain two cofactors that serve as light-absorbing agents, or chromophores: in all organisms, one is FADH_2 ; in *E. coli* and yeast, the other is a folate. The reaction mechanism entails the generation of free radicals. DNA photolyases are not found in humans and other placental mammals.

Additional examples are seen in the repair of nucleotides with alkylation damage. The modified nucleotide O^6 -methylguanine forms in the presence of alkylating agents and is a common and highly mutagenic lesion. It tends to pair with thymine rather than cytosine during replication, and therefore causes $\text{G} \equiv \text{C}$ to $\text{A} = \text{T}$ mutations (**Fig. 25-26**).



MECHANISM FIGURE 25-25 Repair of pyrimidine dimers with photolyase. Energy derived from absorbed light is used to reverse the photoreaction that caused the lesion. The two chromophores in *E. coli* photolyase (M_r 54,000), N^5,N^{10} -methenyltetrahydrofolylpolyglutamate (MTHFpolyGlu)

and FADH_2 , perform complementary functions. MTHFpolyGlu functions as a photoantenna to absorb blue-light photons. The excitation energy passes to FADH_2 , and the excited flavin ($^*\text{FADH}_2^-$) donates an electron to the pyrimidine dimer, leading to the rearrangement as shown.

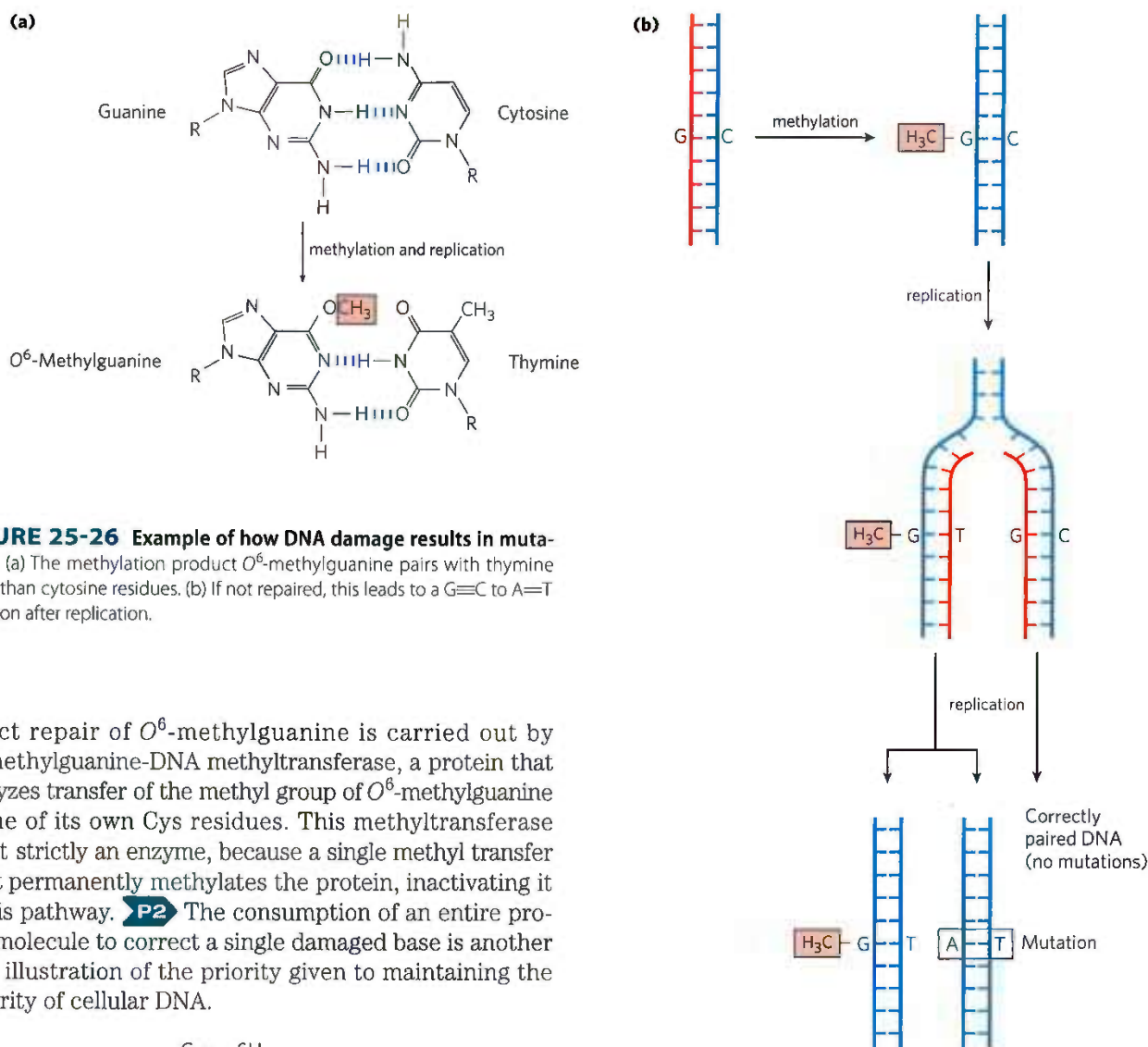
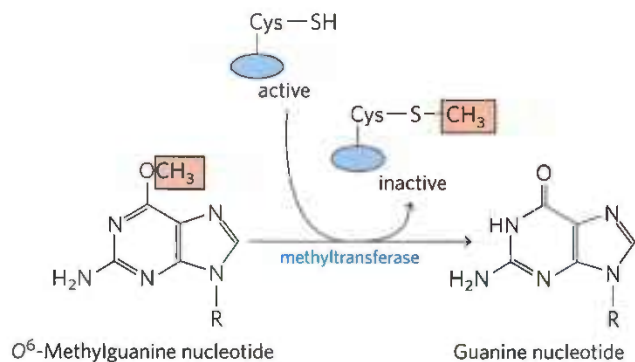


FIGURE 25-26 Example of how DNA damage results in mutations. (a) The methylation product O⁶-methylguanine pairs with thymine rather than cytosine residues. (b) If not repaired, this leads to a G≡C to A≡T mutation after replication.

Direct repair of O⁶-methylguanine is carried out by O⁶-methylguanine-DNA methyltransferase, a protein that catalyzes transfer of the methyl group of O⁶-methylguanine to one of its own Cys residues. This methyltransferase is not strictly an enzyme, because a single methyl transfer event permanently methylates the protein, inactivating it in this pathway. **P2** The consumption of an entire protein molecule to correct a single damaged base is another vivid illustration of the priority given to maintaining the integrity of cellular DNA.



A very different but equally direct mechanism is used to repair 1-methyladenine and 3-methylcytosine. The amino groups of A and C residues are sometimes methylated when the DNA is single-stranded, and the methylation directly affects proper base pairing. In *E. coli*, oxidative demethylation of these alkylated nucleotides is mediated by the AlkB protein, a member of the α -ketoglutarate-Fe²⁺-dependent dioxygenase superfamily (Fig. 25-27). (See Box 4-2 for a description of proline hydroxylation, catalyzed by another member of this enzyme family.)

The Interaction of Replication Forks with DNA Damage Can Lead to Error-Prone Translesion DNA Synthesis

The repair pathways considered to this point generally work only for lesions in double-stranded DNA, the undamaged strand providing the correct genetic information to restore the damaged strand to its original state. However, in certain types of lesions, such as double-strand breaks, double-strand cross-links, or lesions in a single-stranded DNA, the complementary strand is itself damaged or is absent. **P3** Double-strand breaks and lesions in single-stranded DNA most often arise when a replication fork encounters an unrepaired DNA lesion (Fig. 25-28). Such lesions and DNA cross-links can also result from ionizing radiation and oxidative reactions.

At a stalled bacterial replication fork, there are two avenues for repair. In the absence of a second strand, the information required for accurate repair must come from a separate, homologous chromosome. The repair system thus involves homologous genetic recombination. This recombinational DNA repair is considered in

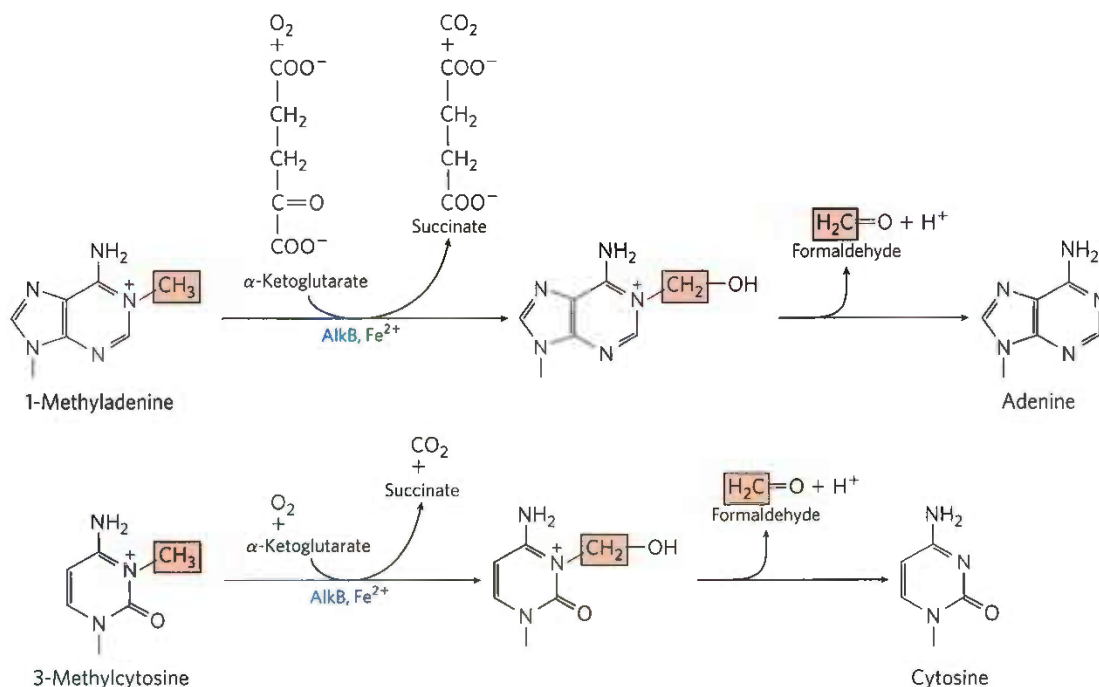


FIGURE 25-27 Direct repair of alkylated bases by AlkB. The AlkB protein is an α -ketoglutarate- Fe^{2+} -dependent hydroxylase (see Box 4-2). It catalyzes the oxidative demethylation of 1-methyladenine and 3-methylcytosine residues.

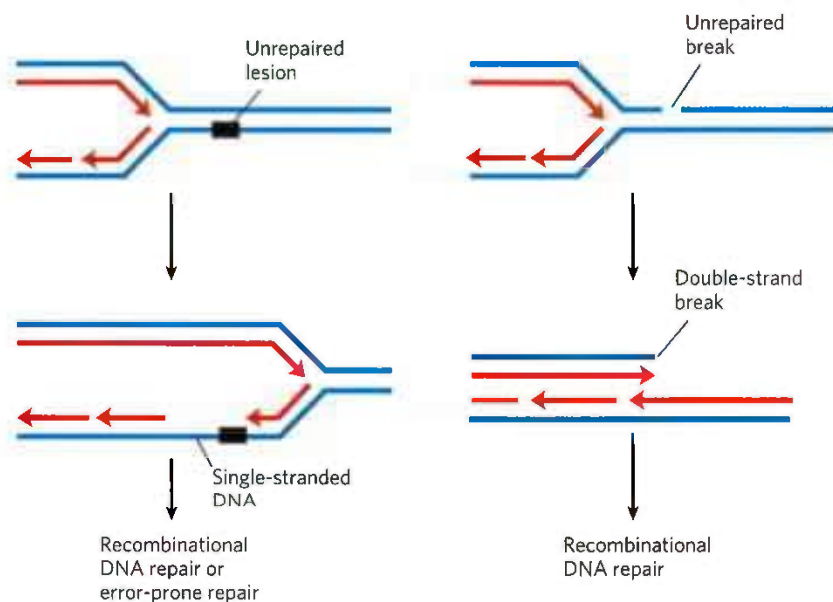


FIGURE 25-28 DNA damage and its effect on DNA replication. If the replication fork encounters an unrepaired lesion or strand break, the DNA polymerase sometimes disengages and re-initiates downstream. The lesion remains in an unreplicated, single-stranded gap that is left behind the replication fork (left). In other cases, a replication fork may encounter a lesion that is actively undergoing repair such that a transient break is present in one of the template strands. When the replication fork encounters it, the single-strand break becomes a double-strand break (right). In each case, the damage to

one strand cannot be repaired by mechanisms described earlier in this chapter, because the complementary strand required to direct accurate repair is damaged or absent. There are at least two possible avenues for repair: recombinational DNA repair or, when lesions are unusually numerous, error-prone repair. The latter mechanism involves translesion DNA polymerases such as DNA polymerase V, encoded by the *umuC* and *umuD* genes and activated by the RecA protein that can replicate, albeit inaccurately, over many types of lesions. The repair mechanism is "error-prone" because mutations often result.

detail in Section 25.3. Under some conditions, a second repair pathway, **error-prone translesion DNA synthesis** (often abbreviated TLS), becomes available. When this pathway is active, DNA repair is significantly

less accurate, and a high mutation rate can result. In bacteria, error-prone translesion DNA synthesis is part of a cellular stress response to extensive DNA damage known, appropriately enough, as the **SOS response**.

Some of the 40 or more SOS proteins, such as the UvrA and UvrB proteins involved in the error-free nucleotide-excision repair already described, are normally present in the cell but are induced to higher levels as part of the SOS response. Additional SOS proteins participate in the pathway for error-prone repair; these include the UmuC and UmuD proteins (*unmutable*; lack of the *umu* gene eliminates error-prone repair). The UmuD protein is cleaved in an SOS-regulated process to a shorter form called UmuD', which forms a complex with UmuC and a protein called RecA (described in Section 25.3) to create a specialized DNA polymerase, DNA polymerase V (UmuD'₂UmuCRecA), which can replicate past many of the DNA lesions that would normally block replication. Proper base pairing is often impossible at the site of such a lesion, so this translesion replication is error-prone.

Given the emphasis on the importance of genomic integrity throughout this chapter, the existence of a system that increases the rate of mutation may seem incongruous. However, we can think of this system as a desperation strategy. The *umuC* and *umuD* genes are fully induced only late in the SOS response, and they are not activated for translesion synthesis initiated by UmuD cleavage unless the levels of DNA damage are particularly high and all replication forks are blocked. The mutations resulting from DNA polymerase V-mediated replication kill some cells and create deleterious mutations in others, but this is the biological price a species pays to overcome an otherwise insurmountable barrier to replication, as it permits at least a few mutant daughter cells to survive.

P3 The resultant mutations contribute to evolution.

Yet another DNA polymerase, DNA polymerase IV, is also induced during the SOS response. Replication by DNA polymerase IV, a product of the *dinB* gene, is also highly error-prone. The bacterial DNA polymerases IV and V (Table 25-1) are part of a family of TLS polymerases found in all organisms. These enzymes lack a proofreading exonuclease and have a more open active site than other DNA polymerases, one that accommodates damaged template nucleotides. With these enzymes, the fidelity of base selection during replication may be reduced by a factor of 10^2 , lowering overall replication fidelity to one error in ~1,000 nucleotides.

P4 Mammals have many low-fidelity DNA polymerases of the TLS polymerase family. However, the presence of these enzymes does not necessarily translate into an unacceptable mutational burden, because most of the enzymes also have specialized functions in DNA repair. DNA polymerase η (eta), for example, found in all eukaryotes, promotes translesion synthesis primarily across cyclobutane T-T dimers. Few mutations result, because the enzyme preferentially inserts two A residues across from the linked T residues. Several other low-fidelity polymerases, including DNA polymerases β , ι (iota), and λ , have specialized roles in eukaryotic base-excision repair. Each of these enzymes has a 5'-deoxyribose phosphate lyase activity in addition to its polymerase activity. After base removal by a glycosylase and backbone cleavage by

an AP endonuclease, these polymerases remove the abasic site (a 5'-deoxyribose phosphate) and fill in the very short gap. The frequency of mutation due to DNA polymerase η activity is minimized by the short lengths (often one nucleotide) of DNA synthesized.

What emerges from research into cellular DNA repair systems is a picture of a DNA metabolism that maintains genomic integrity with multiple and often redundant systems. Most of the major DNA repair systems occur in all organisms. **P4** These repair systems are often integrated with the DNA replication systems and are complemented by recombination systems, which we turn to next.

SUMMARY 25.2 DNA Repair

- Mutations are genomic changes that alter the information in DNA. When mutations occur in genes encoding enzymes involved in DNA repair, the loss of function can lead to cancer.
- Cells have many systems for DNA repair. Major repair systems present in all organisms include mismatch repair, base excision repair, nucleotide-excision repair, and direct repair.
- In bacteria, TLS DNA polymerases respond to very heavy DNA damage with error-prone translesion DNA synthesis. In eukaryotes, similar polymerases have specialized roles in DNA repair that minimize the introduction of mutations.

25.3 DNA Recombination

The rearrangement of genetic information within and among DNA molecules encompasses a variety of processes, collectively placed under the heading of genetic recombination. The practical applications of DNA rearrangements in altering the genomes of increasing numbers of organisms are now being explored (Chapter 9).



Barbara McClintock,
1902–1992
[AP Photo.]

Genetic recombination events fall into at least three general classes. **Homologous genetic recombination** (also called general recombination) involves genetic exchanges between any two DNA molecules (or segments of the same molecule) that share an extended region of nearly identical sequence. The actual sequence of bases is irrelevant, as long as it is similar in the two DNAs. In **site-specific recombination**, the exchanges occur only at a *particular* DNA sequence. **DNA transposition** is distinct from both other classes in that it usually involves a short segment of DNA with the remarkable capacity to move from one location in a chromosome to another. These “jumping genes” were first observed in maize in the 1940s by Barbara McClintock. There are also unusual genetic rearrangements for which no mechanism or purpose has yet been proposed. Here we focus on the three general classes.

Homologous genetic recombination is largely a pathway to repair double-strand breaks in DNA. An alternative process for double-strand break repair that does not entail recombination, called **nonhomologous end joining (NHEJ)**, is also described here. Genetic recombination systems have functions as varied as their mechanisms. They include roles in specialized DNA repair systems, specialized activities in DNA replication, regulation of expression of certain genes, facilitation of proper chromosome segregation during eukaryotic cell division, maintenance of genetic diversity, and implementation of programmed genetic rearrangements during embryonic development. In most cases, genetic recombination is closely integrated with other processes in DNA metabolism, and this becomes a theme of our discussion.

Bacterial Homologous Recombination Is a DNA Repair Function

P4 In bacteria, homologous genetic recombination is primarily a DNA repair process, and in this context

(as noted in Section 25.2) it is referred to as **recombinational DNA repair**. It is usually directed at the reconstruction of replication forks that have stalled or collapsed at the site of DNA damage. Homologous genetic recombination can also occur during conjugation (mating), when chromosomal DNA is transferred from one bacterial cell (donor) to another (recipient). Recombination during conjugation, although rare in wild bacterial populations, contributes to genetic diversity.

P4 When a replication fork encounters DNA damage, many pathways may resolve the conflict. A common feature of the DNA repair pathways illustrated in Figures 25-21 to 25-24 is that they introduce a transient break into one of the DNA strands. If a replication fork encounters a damaged site under repair near a break in one of the template strands, one arm of the replication fork becomes disconnected by a double-strand break and the fork collapses (**Fig. 25-29**). The end of that break is processed by degrading the 5'-ending strand. The resulting 3' single-stranded extension is bound by a recombinase

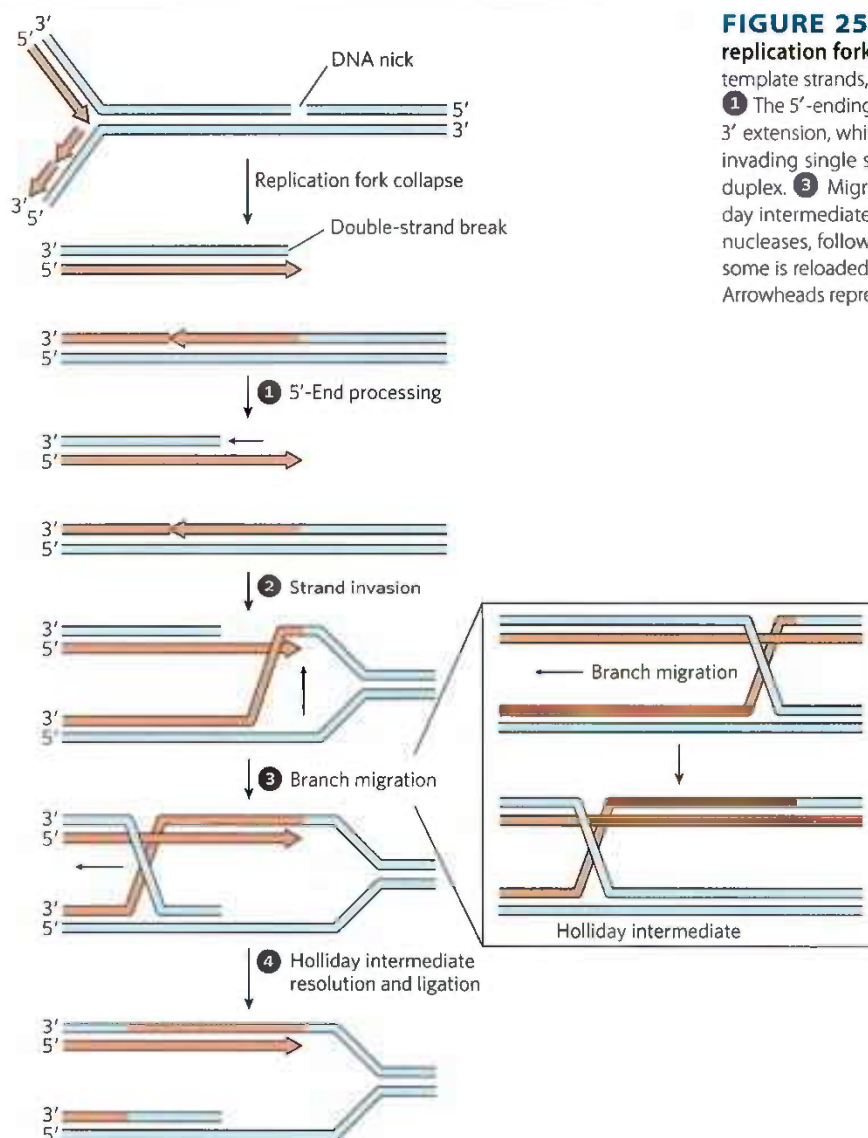


FIGURE 25-29 Recombinational DNA repair at a collapsed replication fork. When a replication fork encounters a break in one of the template strands, one arm of the fork is lost and the replication fork collapses. **1** The 5'-ending strand at the break is degraded to create a single-stranded 3' extension, which is then used in **2** a strand invasion process, pairing the invading single strand with its complementary strand within the adjacent duplex. **3** Migration of the branch (shown in the box) can create a Holliday intermediate. **4** Cleavage of the Holliday intermediate by specialized nucleases, followed by ligation, restores a viable replication fork. The replisome is reloaded onto this structure (not shown), and replication continues. Arrowheads represent 3' ends.

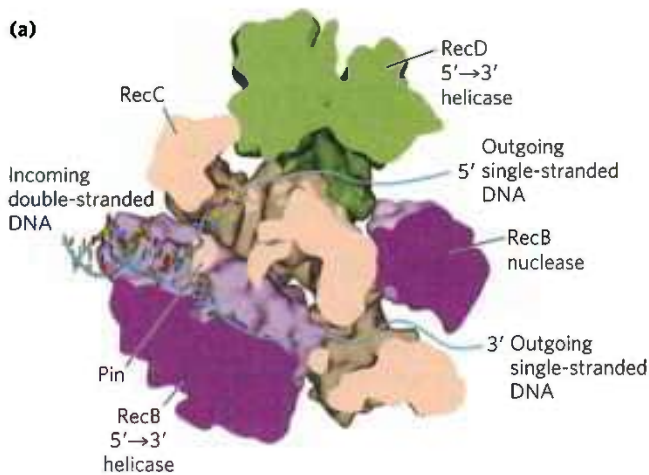
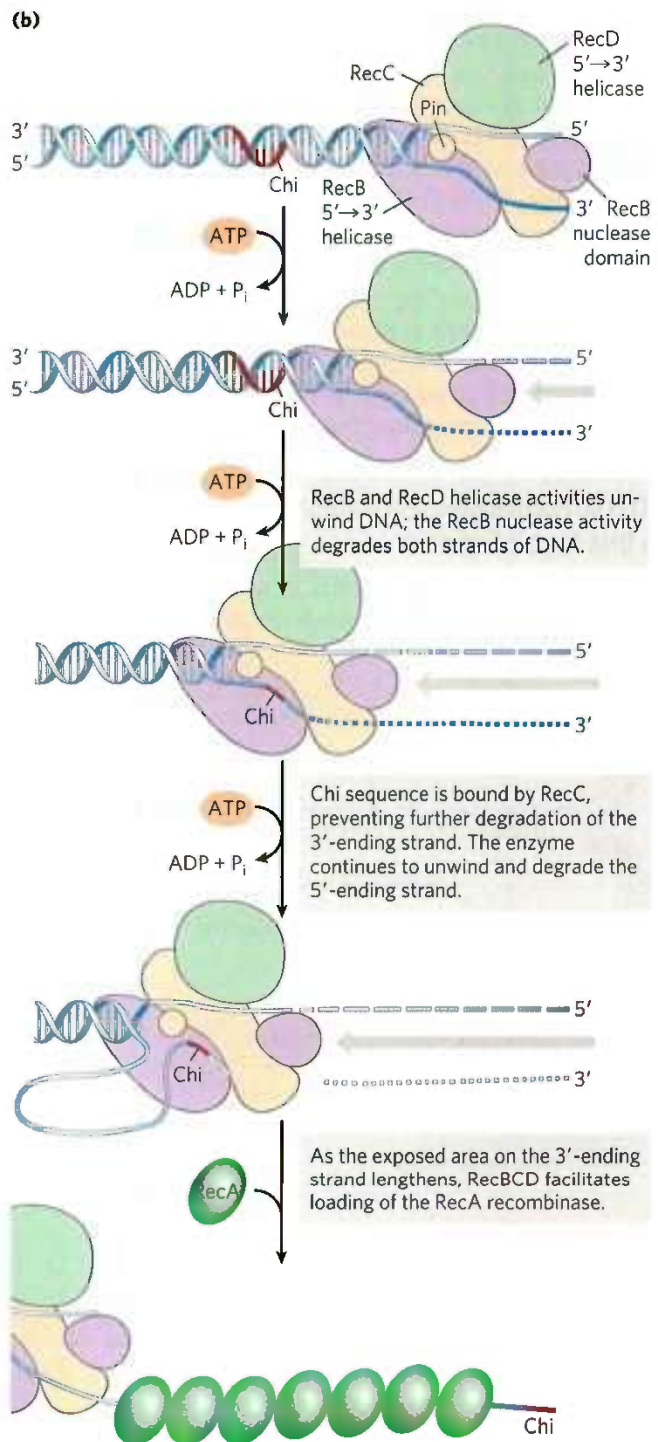


FIGURE 25-30 The RecBCD helicase/nuclease. (a) A cutaway view of the RecBCD enzyme structure as it is bound to DNA. The subunits are shown in different colors; the DNA is entering from the left, and the unwound DNA strands (not part of the solved structure) are shown exiting to the right. A bulbous protein structure called a pin, part of the RecC subunit, facilitates the separation of strands. (b) Activities of the RecBCD enzyme at a DNA end. [(a) Data from PDB ID 1W36, M. R. Singleton et al., *Nature* 432:187, 2004.]

that uses it to promote strand invasion: the 3' end invades the intact duplex DNA connected to the other arm of the fork and pairs with its complementary sequence. This creates a branched DNA structure (a point where three DNA segments come together). The DNA branch can be moved in a process called **branch migration** to create an X-like crossover structure known as a **Holliday intermediate**, named after researcher Robin Holliday, who first postulated its existence. The Holliday intermediate is cleaved, or "resolved," by a special class of nucleases. The overall process reconstructs the replication fork.

In *E. coli*, the DNA end-processing is promoted by the RecBCD nuclease/helicase. The RecBCD enzyme binds to linear DNA at a free (broken) end and moves inward along the double helix, unwinding and degrading the DNA in a reaction coupled to ATP hydrolysis (Fig. 25-30). The RecB and RecD subunits are helicase motors, with RecB moving 3' → 5' along one strand, and RecD moving 5' → 3' along the other strand. The activity of the enzyme is altered when it interacts with a sequence referred to as **chi**, (5')GCTGGTGG, which binds tightly to a site on the RecC subunit. From that point, degradation of the strand with a 3' terminus is greatly reduced, but degradation of the 5'-terminal strand is increased. This process creates a single-stranded DNA with a 3' end, which is used during subsequent steps in recombination. The 1,009 chi sequences scattered throughout the *E. coli* genome enhance the frequency of recombination about 5- to 10-fold within 1,000 bp of each chi site. The enhancement declines as the distance from chi increases. Sequences that enhance recombination frequency have also been identified in several other organisms.



The bacterial recombinase is the RecA protein. RecA is unusual among the proteins of DNA metabolism in that its active form is an ordered, helical filament of up to several thousand subunits that assemble cooperatively on DNA (Fig. 25-31). This filament usually forms on single-stranded DNA, such as that produced by the RecBCD enzyme. Its formation is not as straightforward as shown in Figure 25-31, because the single-stranded DNA-binding protein (SSB) is normally present and

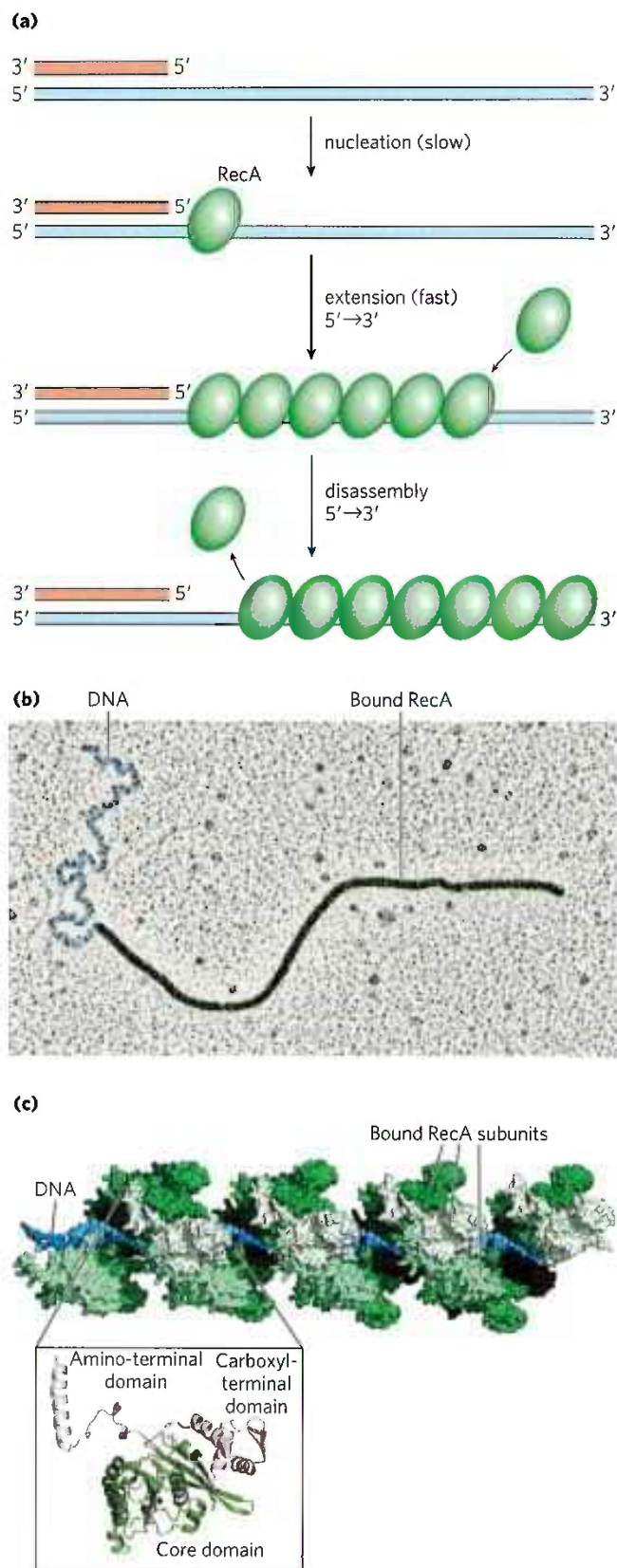


FIGURE 25-31 RecA protein filaments. RecA and other recombinases in this class function as filaments of nucleoprotein. (a) Filament formation proceeds in discrete nucleation and extension steps. Nucleation is the addition of the first few RecA subunits. Extension occurs by adding RecA subunits so that the filament grows in the 5'→3' direction. When disassembly occurs, subunits are subtracted from the trailing end. (b) Colored electron micrograph of a RecA filament bound to DNA. (c) Segment of a RecA filament with four helical turns (24 RecA subunits). Notice the bound double-stranded DNA in the center. The core domain of RecA is structurally related to the motor domains of helicases. [(b) By permission of the Estate of Ross Inman. Special thanks to Kim Voss. (c) Data from PDB ID 3CMX, Z. Chen et al., *Nature* 453:489, 2008.]

with SSB. The filaments assemble and disassemble predominantly in a 5'→3' direction. Many other bacterial proteins regulate the formation and disassembly of RecA filaments, including an alternative set of RecA loading proteins called RecF, RecO, and RecR. RecA protein promotes the central steps of homologous recombination, including the DNA strand invasion step of Figure 25-29, as well as other strand exchange reactions occurring *in vitro*. Once a Holliday intermediate has been created via branch migration, it can be cleaved by specialized nucleases such as the bacterial RuvC protein (Fig. 25-32), and nicks are sealed by DNA ligase. A viable replication fork structure is thus reconstructed, as outlined in Figure 25-29.

After the recombination steps are completed, the replication fork reassembles in a process called **origin-independent restart of replication**. Different combinations of four proteins (PriA, PriB, PriC, and DnaT) act with DnaC in several pathways to load DnaB helicase onto the reconstructed replication fork. The DnaG primase then synthesizes an RNA primer, and DNA polymerase III reassembles on DnaB to restart DNA synthesis. Complexes that include some combination of the PriA, PriB, PriC, and DnaT, along with DnaB, DnaC, and DnaG proteins, are called **replication restart primosomes**. In this way, the process of recombination is tightly intertwined with replication. One process of DNA metabolism supports the other.

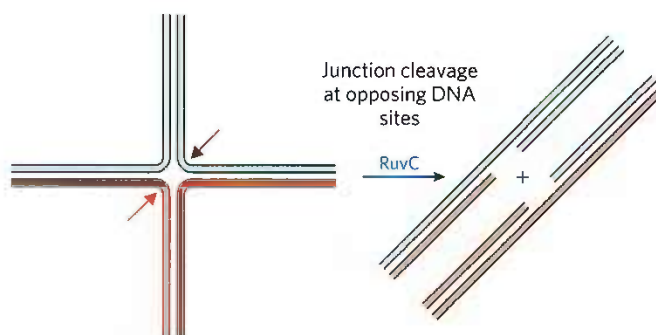
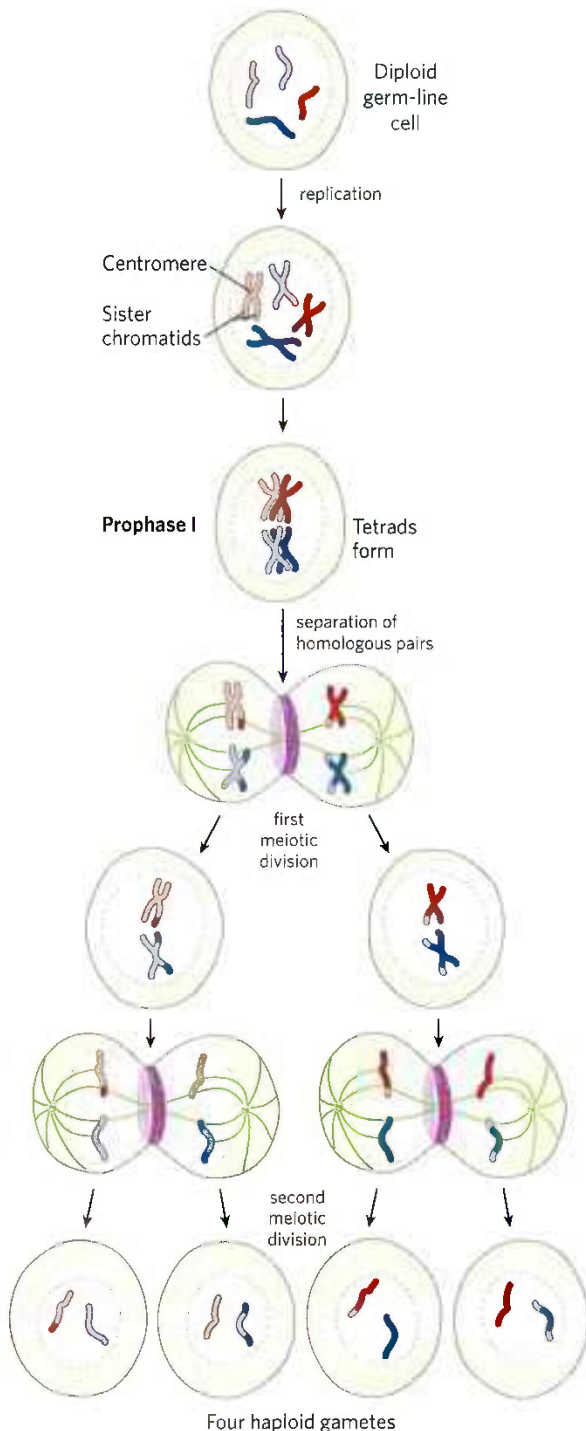


FIGURE 25-32 Resolution of a Holliday intermediate by the RuvC protein. RuvC is a specialized nuclease that binds to the RuvAB complex and cleaves the Holliday intermediate on opposing sides of the crossover junction (red arrows), so that two contiguous DNA arms remain in each product.

specifically impedes the binding of the first few subunits to DNA (filament nucleation). The RecBCD enzyme acts directly as a RecA loader, facilitating the nucleation of a RecA filament on single-stranded DNA that is coated

Eukaryotic Homologous Recombination Is Required for Proper Chromosome Segregation during Meiosis

In eukaryotes, homologous genetic recombination has roles in replication and cell division, including the repair of stalled replication forks. Recombination occurs with the highest frequency during **meiosis**, the process by which diploid germ-line cells with two sets of chromosomes divide to produce haploid gametes (sperm cells or ova) in animals (haploid spores in plants) — each gamete having only one member of each chromosome pair (Fig. 25-33).



Meiosis begins with replication of the DNA in the germ-line cell so that each DNA molecule is present in four copies. Each set of four homologous chromosomes (tetrad) exists as two pairs of sister chromatids, and the sister chromatids remain associated at their centromeres. The cell then goes through two rounds of cell division without an intervening round of DNA replication. In the first cell division, the two pairs of sister chromatids are segregated into daughter cells. In the second cell division, the two chromosomes in each sister chromatid pair are segregated into new daughter cells. In each division, the chromosomes to be segregated are drawn into the daughter cells by spindle fibers attached to opposite poles of the dividing cell. The two successive divisions reduce the DNA content to the haploid level in each gamete.

Proper chromosome segregation into daughter cells requires that physical links exist between the homologous chromosomes to be segregated. As the spindle fibers attach to the centromeres of chromosomes and start to pull, the links between homologous chromosomes create tension. This tension, sensed by cellular mechanisms not yet understood, signals that this pair of chromosomes or sister chromatids is properly aligned for segregation. Once the tension is sensed, the links are gradually dissolved and segregation proceeds. If improper spindle fiber attachment occurs (e.g., if the centromeres of a chromosome pair are attached to the same cellular pole), a cellular kinase senses the lack of tension and activates a system that removes the spindle attachments, allowing the cell to try again.

During the second meiotic division, the centromeric attachments between sister chromatids, augmented by cohesins deposited during replication (see Fig. 24-33), provide the physical links that are needed to guide segregation. However, during the first meiotic cell division, the two pairs of sister chromatids to be segregated are not related by a recent replication event and are not linked by cohesins or any other physical association. Instead, the homologous pairs of sister chromatids are aligned and new links are created by recombination, a process involving the breakage and rejoining of DNA (Fig. 25-34).

FIGURE 25-33 Meiosis in animal germ-line cells. The chromosomes of a hypothetical diploid germ-line cell (four chromosomes; two homologous pairs) replicate and are held together at their centromeres. Each replicated double-stranded DNA molecule is called a chromatid (sister chromatid). In prophase I, just before the first meiotic division, the two homologous sets of chromatids align to form tetrads, held together by covalent links at homologous junctions (chiasmata). Crossovers occur within the chiasmata (see Fig. 25-34). These transient associations between homologs ensure that the two tethered chromosomes segregate properly in the next step, when attached spindle fibers pull them toward opposite poles of the dividing cell in the first meiotic division. The products of this division are two daughter cells, each with two pairs of different sister chromatids. The pairs now line up across the equator of the cell in preparation for separation of the chromatids (now called chromosomes). The second meiotic division produces four haploid daughter cells that can serve as gametes. Each has two chromosomes, half the number of the diploid germ-line cell. The chromosomes have re-sorted and recombined.

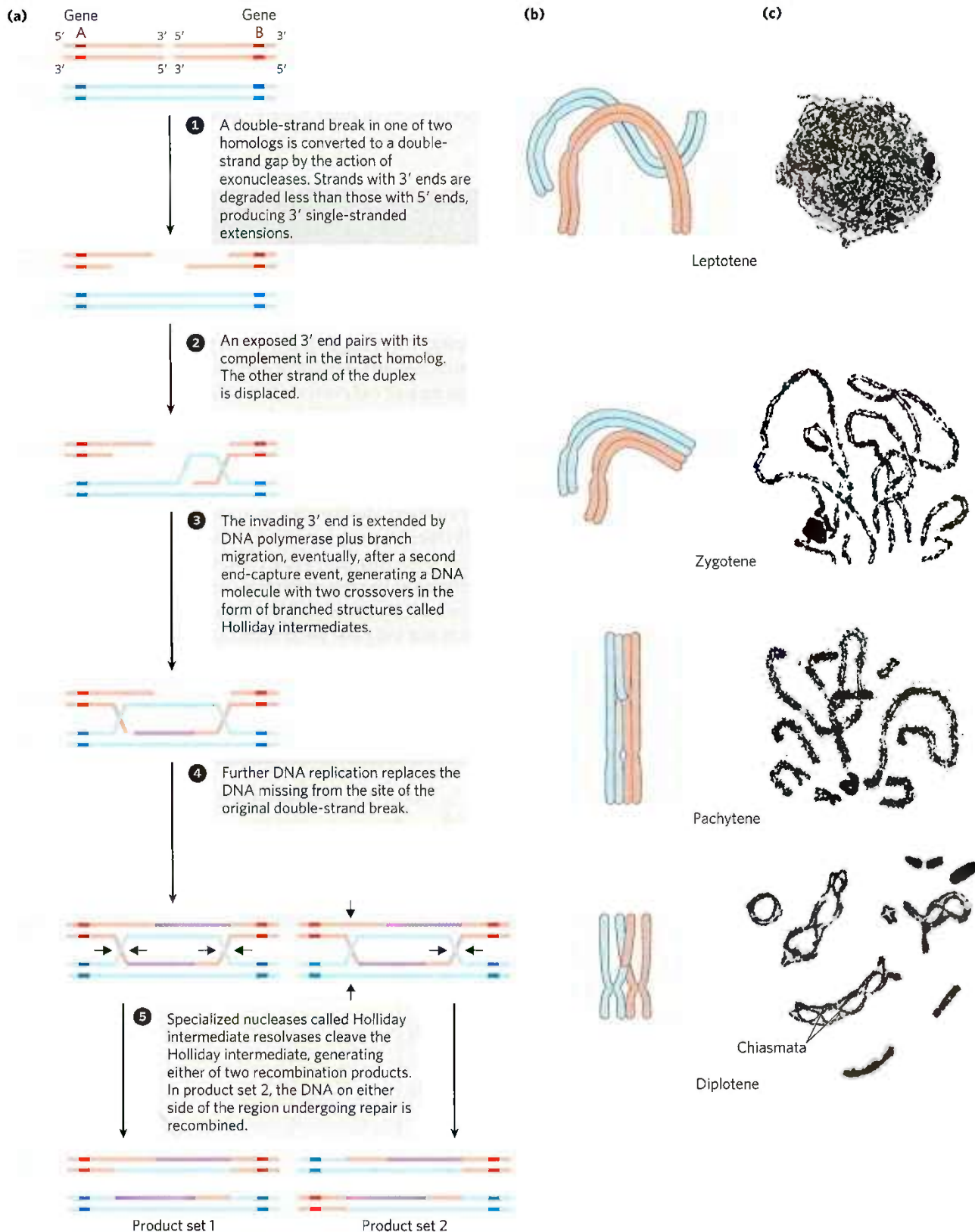


FIGURE 25-34 Recombination during prophase I in meiosis. (a) A model of double-strand break repair for homologous genetic recombination. The two homologous chromosomes (one shown in red, the other blue) involved in this recombination event have identical or very nearly identical sequences. Each of the two genes shown has different alleles on the two chromosomes. The steps are described in the text. (b) Crossing over occurs during prophase I of meiosis I. The several stages of prophase I are aligned with the recombination processes shown in (a). Double-strand breaks are introduced and processed in

the leptotene stage. The strand invasion and completion of crossover occur later. As homologous sequences in the two pairs of sister chromatids are aligned in the zygotene stage, synaptonemal complexes form and strand invasion occurs. The homologous chromosomes are tightly aligned by the pachytene stage. (c) Homologous chromosomes of a grasshopper, viewed at successive stages of meiotic prophase I. The chiasmata become visible in the diplotene stage. [(c) B. John, *Meiosis*, Figs 2.1a, 2.2a, 2.2b, 2.3a, Cambridge University Press, 1990. Reprinted with the permission of Cambridge University Press.]

This exchange, also referred to as crossing over, can be observed with the light microscope. Crossing over links the two pairs of sister chromatids together at points called chiasmata (singular, chiasma). Also during crossing over, genetic material is exchanged between the pairs

of sister chromatids. These exchanges increase genetic diversity in the resulting gametes. The importance of meiotic recombination to proper chromosome segregation is well illustrated by the physiological and societal consequences of their failure (**Box 25-2**).

BOX 25-2



MEDICINE

Why Proper Segregation of Chromosomes Matters

When chromosomal alignment and recombination are not correct and complete in meiosis I, segregation of chromosomes can go awry. One result may be aneuploidy, a condition in which a cell has the wrong number of chromosomes. The haploid products of meiosis (gametes or spores) may have no copies or two copies of a chromosome. When a gamete having two copies of a chromosome joins with a gamete having one copy of a chromosome during fertilization, cells in the resulting embryo have three copies of that chromosome (they are trisomic).

In *S. cerevisiae*, aneuploidy resulting from errors in meiosis occurs at a rate of about 1 in 10,000 meiotic events. In fruit flies, the rate is about 1 in a few thousand. Rates of aneuploidy in mammals are considerably higher. In mice, the rate is 1 in 100, and it is even higher in other mammals. The rate of aneuploidy in fertilized human eggs has been estimated as 10% to 30%; this is almost certainly an underestimate. Most of these aneuploid cells are monosomies (they have a single copy of a chromosome) or trisomies. Most trisomies are lethal, and many result in miscarriage long before the pregnancy is detected. Almost all monosomies are fatal in the early stages of fetal development. Aneuploidy is the leading cause of pregnancy loss. The few trisomic fetuses that survive to birth generally have three copies of chromosome 13, 18, or 21 (trisomy 21 is Down syndrome). Abnormal complements of the sex chromosomes are also found in the human population. The societal consequences of aneuploidy in humans are considerable. Aneuploidy is the leading genetic cause of developmental and mental disabilities. At the heart of these high rates is a feature of meiosis in female mammals that has special significance for the human species.

In a human male, germ-line cells begin to undergo meiosis at puberty, and each meiotic event requires a relatively short time. In contrast, meiosis in the germ-line cells of human females is a highly protracted process. The production of an egg begins before a female is born, with the onset of meiosis in the fetus, at 12 to 13 weeks of gestation. Meiosis is initiated in all the developing fetal germ-line cells over a period of a few weeks. The cells proceed through much of meiosis I. Chromosomes line up and generate crossovers, continuing just beyond the pachytene stage (see Fig. 25-34) — and then the process stops. The chromosomes enter an arrested phase called the dictyate stage, with the crossovers in place, a kind of suspended animation where they remain as the female matures — so, typically

remaining in this stage for anywhere from about 13 to 50 years. At sexual maturity, individual germ-line cells continue through the two meiotic cell divisions to produce egg cells.

Between the onset of the dictyate stage and the final completion of meiosis, something may happen that disrupts or damages the crossovers linking homologous chromosomes in the germ-line cells. As a woman ages, the rate of trisomy in the egg cells she produces increases, dramatically so as she approaches menopause (Fig. 1). There are many hypotheses on why this occurs, and several different factors may play a role. However, most of the hypotheses are centered on recombination crossovers in meiosis I and their stability over the protracted dictyate stage.

It is not yet clear what medical steps could be taken to reduce the incidence of aneuploidy in women of child-bearing age. What is revealed is the inherent importance of recombination and generation of crossovers in human meiosis.

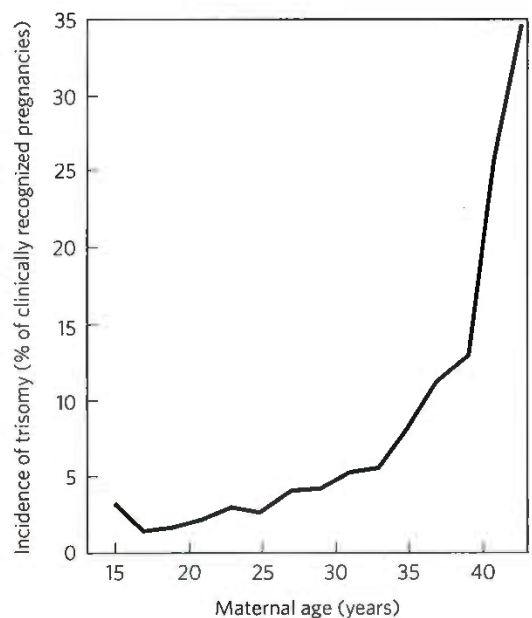


FIGURE 1 The increasing incidence of human trisomy with increasing age of the mother. [Data from T. Hassold and P. Hunt, *Nat. Rev. Genet.* 2:280, 2001, Fig. 6.]

A likely pathway for homologous recombination during meiosis is outlined in Figure 25-34a. The model has four key features. First, homologous chromosomes align. Second, a double-strand break is created in a DNA molecule, and the exposed ends are processed by an exonuclease, leaving a single-stranded extension with a free 3'-hydroxyl group at the broken end (step ①). Third, the exposed 3' ends invade the intact duplex DNA of the homolog, and this is followed by branch migration and/or replication to create a pair of Holliday intermediates (steps ② to ④). Fourth, cleavage of the two crossovers creates either of two pairs of complete recombinant products (step ⑤). Notice the similarity of these steps to the bacterial recombinational repair processes outlined in Figure 25-29. The DNA strand invasion in eukaryotes is catalyzed by RecA-like recombinases called Rad51 and Dmc1. Loading of Rad51 onto DNA is promoted by Rad51 loading protein BRCA2 (analogous to the bacterial RecF, RecO, and RecR proteins).

In this **double-strand break repair model** for recombination, the 3' ends are used to initiate the genetic exchange. Once paired with the complementary strand on the intact homolog, a region of hybrid DNA is created that contains complementary strands from two different parent DNAs (the product of step ② in Fig. 25-34a). Each of the 3' ends can then act as a primer for DNA replication. Meiotic homologous recombination can vary in many details from one species to another, but most of the steps outlined above are generally present in some form. There are two ways to resolve the Holliday intermediate with a RuvC-like nuclease so that the two products carry genes in the same linear order as in the substrates—the original, unrecombined chromosomes (step ⑤). If cleaved one way,

the DNA flanking the region containing the hybrid DNA is not recombined; if cleaved the other way, the flanking DNA is recombined. Both outcomes are observed *in vivo*.

The homologous recombination illustrated in Figure 25-34 is an elaborate process that is essential to accurate chromosome segregation. Its molecular consequences for the generation of genetic diversity are subtle. To understand how this process contributes to diversity, we should keep in mind that the two homologous chromosomes that undergo recombination are not necessarily *identical*. The linear array of genes may be the same, but the base sequences in some of the genes may differ slightly (in different alleles). In a human, for example, one chromosome may contain the allele for hemoglobin A (normal hemoglobin) while the other contains the allele for hemoglobin S (the sickle cell mutation). The difference may consist of no more than one base pair among millions.

Crossing over is not an entirely random process, and “hot spots” have been identified on many eukaryotic chromosomes. However, the assumption that crossing over can occur with equal probability at almost any point along the length of two homologous chromosomes remains a reasonable approximation in many cases, and it is this assumption that permits the mapping of genes on a particular chromosome. The frequency of homologous recombination in any region separating two points on a chromosome is roughly proportional to the distance between the points, and this allows determination of the relative positions of different genes and the distances between those genes. The independent assortment of unlinked genes on different chromosomes (Fig. 25-35)

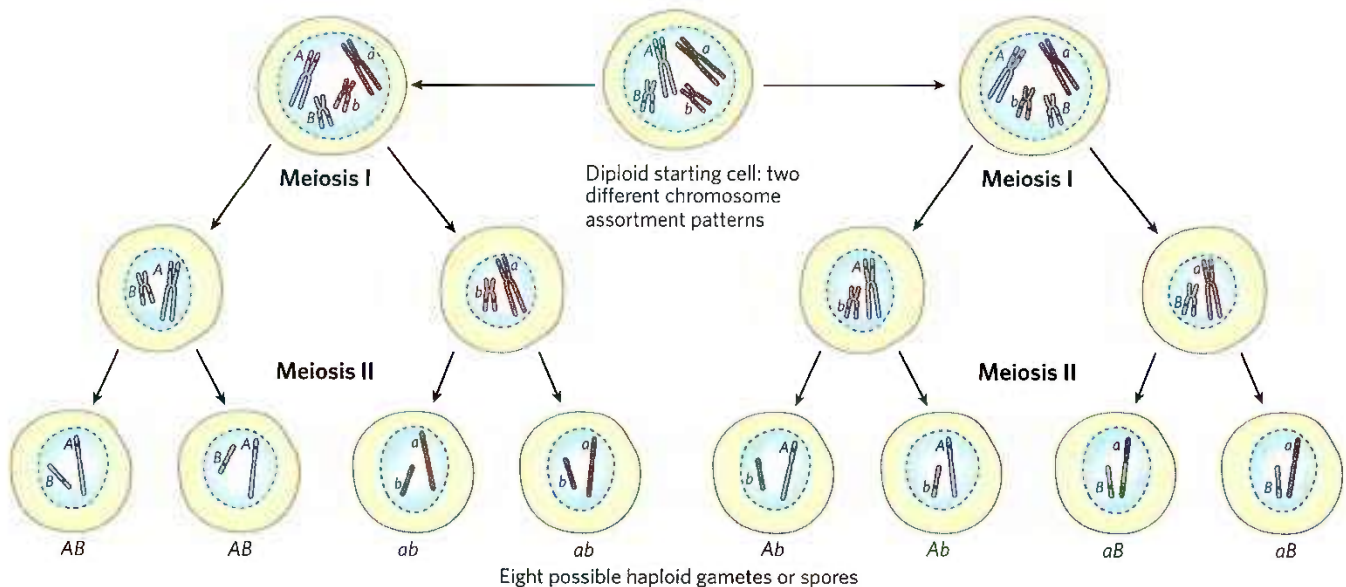


FIGURE 25-35 The contribution of independent assortment to genetic diversity. In this example, the two chromosomes have already been replicated to create two pairs of sister chromatids. Blue and red distinguish the sister chromatids of each pair. One gene on each chromosome is highlighted,

with different alleles (A or a, B or b) in the homologs. Independent assortment can lead to gametes with any combination of the alleles present on the two different chromosomes. Crossing over (not shown here; see Fig. 25-34) would also contribute to genetic diversity in a typical meiotic sequence.

makes another major contribution to the genetic diversity of gametes. These genetic realities guide many of the modern applications of genomics, such as defining haplotypes (see Fig. 9-26) or searching for disease genes in the human genome (see Fig. 9-30).

As in bacteria, this recombination process is used to repair double-strand breaks that arise anywhere in the genome. In eukaryotes, these systems operate in the context of chromatin, rendering additional complexities to their regulation and damage detection mechanisms (Box 25-3). **P3** Homologous recombination thus serves at least three identifiable functions in eukaryotes: (1) it contributes to the repair of several types of DNA damage; (2) it provides, in eukaryotic cells, a transient physical link between chromatids that promotes the orderly segregation of chromosomes at the first meiotic cell division; and (3) it enhances genetic diversity in a population.

Some Double-Strand Breaks Are Repaired by Nonhomologous End Joining

Double-strand breaks sometimes occur when recombinational DNA repair is not feasible, such as during phases of the cell cycle when no replication is occurring and no sister chromatids are present. At these times, another path is needed to avoid the cell death that would result from a broken chromosome. That alternative is provided by **nonhomologous end joining (NHEJ)**. The broken chromosome ends are simply processed and ligated back together.

Nonhomologous end joining is an important pathway for double-strand break repair in all eukaryotes and has also been detected in some bacteria. The importance of NHEJ increases with genomic complexity, and the process accounts for most double-strand break

BOX 25-3



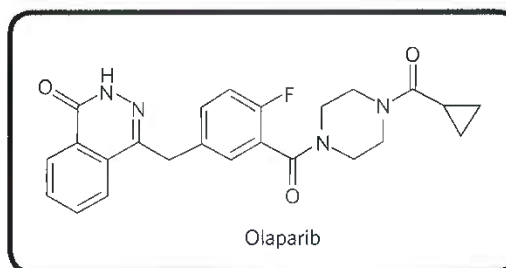
MEDICINE

How a DNA Strand Break Gets Attention

Each human chromosome contains many millions of DNA base pairs, all bound up in an elaborate chromatin structure (Chapter 24). If a strand break occurs somewhere in the DNA, how do the many proteins needed for its repair actually find it? The answer lies, at least in part, in a protein called poly-ADP ribose polymerase 1, or PARP1. PARP1 is a first responder, scanning the DNA for DNA damage and in particular for single-strand breaks. When it finds such sites, it binds and synthesizes an elaborate branched poly-ADP ribose polymer from an NAD precursor (Fig. 1). The polymers are attached to the PARP1 enzyme and also linked to some nearby proteins through Glu, Asp, or Lys residues. The resulting structure is a kind of signal, marking the chromosomal location of damage. A large number of DNA repair proteins bind to and are thus recruited to the poly-ADP ribose polymers, effecting DNA repair. If PARP1 activity is absent, repair is compromised and the number of single-strand breaks in all chromosomes increases. When the chromosome is replicated, the single-strand breaks become double-strand breaks (see Fig. 25-29).

As we saw in Box 25-1, many malignant tumors have a defect in a DNA repair pathway. For example, breast or ovarian cancer is often associated with defects in double-strand break repair (e.g., in the genes encoding BRCA1 or BRCA2 or other proteins in the pathway). In these cells, the further loss of PARP1 activity is especially toxic, as single-strand breaks build up and chromosomes become

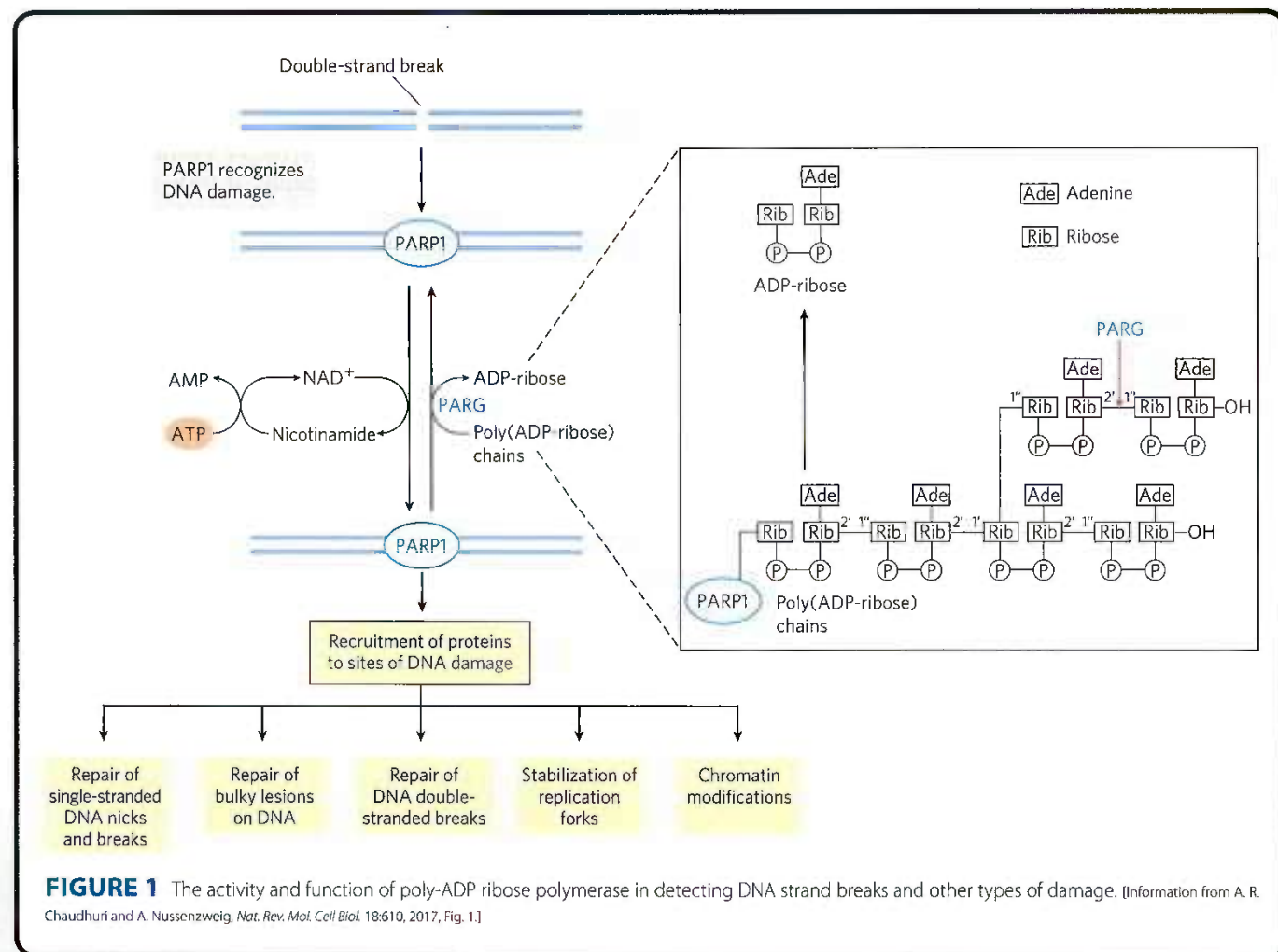
broken during replication. This has led to the development of PARP1 inhibitors as a treatment for tumors in which double-strand break repair is defective. The first such pharmaceutical agent, olaparib, was approved for use in the United States in 2014. Many more PARP1 inhibitors have since been approved or are undergoing clinical trials. The effects have often been dramatic. For women with breast or ovarian tumors displaying deficiencies in BRCA1 or BRCA2 that have responded to more traditional therapies, subsequent maintenance treatment with PARP1 inhibitors has led to a fourfold increase in progression-free survival. PARP1 inhibitors are also showing promise for use with other breast and ovarian tumors, as well as other types of tumors, most of which have DNA repair deficiencies of some kind. As research continues, the use of PARP inhibitors is becoming an important part of the standard of care for a growing list of cancers.



repair outside meiosis in mammals. In yeast, most double-strand breaks are repaired by recombination, and only a few by NHEJ. NHEJ is a mutagenic process, and a smaller genome, such as that of yeast, has relatively little tolerance for the loss of information. The small genomic alterations may be tolerable in mammalian somatic cells, because they are balanced by the undamaged information on the homolog in each diploid cell, and in these non-germ-line cells the mutations are not inherited. In vertebrates, a loss of the genes encoding NHEJ function can produce a predisposition to cancer.

Unlike homologous recombinational repair, NHEJ does not conserve the original DNA sequence. The pathway in eukaryotes is illustrated in **Figure 25-36**. The reaction is initiated at the broken ends of a double-strand break by the binding of a heterodimer consisting of the proteins Ku70 and Ku80 ("KU" being the initials of the

individual with scleroderma whose serum autoantibodies were used to identify this protein complex; the numbers refer to the approximate molecular weights of the subunits). The Ku proteins are conserved in almost all eukaryotes and act as a kind of molecular scaffold to assemble the other protein components. Ku70-Ku80 interacts with another protein complex containing a protein kinase called DNA-PKcs and a nuclease known as Artemis. Once the complex is assembled, the two broken DNA ends are synapsed (held together). DNA-PKcs autophosphorylates in several locations and also phosphorylates Artemis. Artemis, when phosphorylated, acquires an endonuclease function that can remove 5' or 3' single-stranded extensions or hairpins that might be present at the ends. The DNA ends are then separated with the aid of a helicase, and strands from the two different ends are annealed at locations



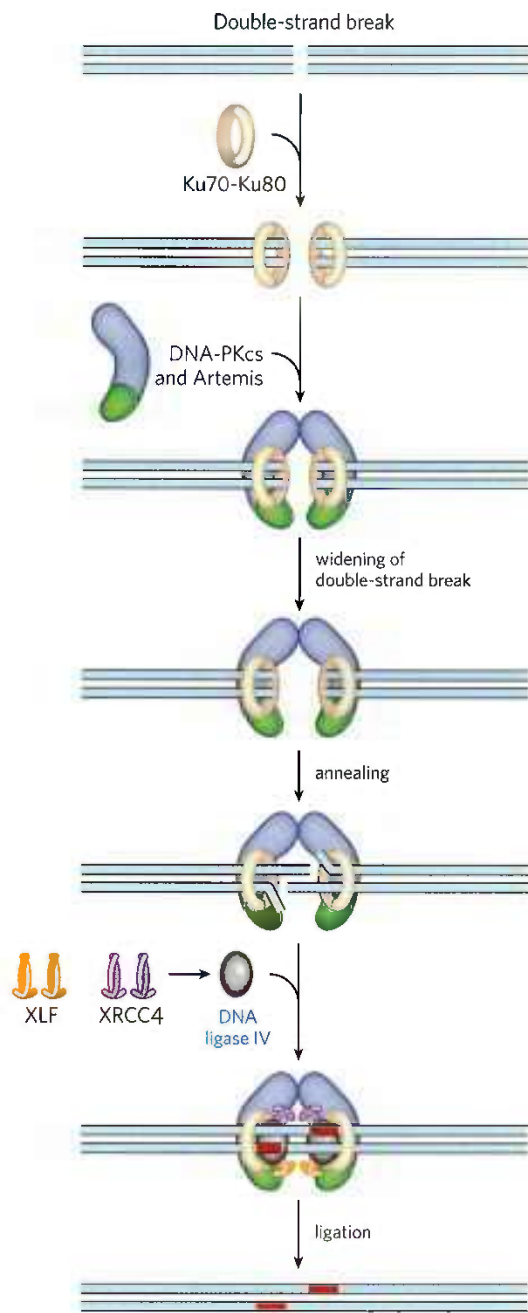


FIGURE 25-36 Nonhomologous end joining. The Ku70-Ku80 complex is the first to bind the DNA ends, followed by a complex including DNA-PKcs and the nuclease Artemis. These proteins then recruit a complex consisting of XRCC4, XLF, and DNA ligase IV. Either of two DNA polymerases, Pol μ or Pol λ (not shown), subsequently extends the annealed DNA strands, as needed, before ligation. [Information from J. M. Sekiguchi and D. O. Ferguson, *Cell* 124:260, 2006, Fig. 1.]

where short regions of complementarity are encountered. Artemis cleaves any unpaired DNA segments that are created. Small DNA gaps are filled by a DNA polymerase, Pol μ or Pol λ . Finally, the nicks are sealed by a protein complex consisting of XRCC4 (*x-ray cross complementation group*), XLF (*XRCC4-like factor*), and DNA ligase IV.

DNA ends are not joined randomly by NHEJ. Instead, when a double-strand break occurs, the ends are generally constrained by the structure of chromatin and thus remain close together. Very rare events linking end sequences that are normally far apart in the chromosome, or are on different chromosomes, may be responsible for occasional dramatic and usually deleterious genomic rearrangements.

Site-Specific Recombination Results in Precise DNA Rearrangements

Homologous genetic recombination can involve any two homologous sequences. The second general type of recombination, site-specific recombination, is a very different type of process: recombination is limited to specific sequences. Recombination reactions of this type occur in virtually every cell, filling specialized roles that vary greatly from one species to another. Examples include regulation of the expression of certain genes and promotion of programmed DNA rearrangements in embryonic development or in the replication cycles of some viral and plasmid DNAs. Each site-specific recombination system consists of an enzyme called a recombinase and a short (20 to 200 bp), unique DNA sequence where the recombinase acts (the recombination site). One or more auxiliary proteins may regulate the timing or outcome of the reaction.

There are two general classes of site-specific recombination systems, which rely on either Tyr or Ser residues in the active site. In vitro studies of many site-specific recombination systems in the tyrosine class have elucidated some general principles, including the fundamental reaction pathway (Fig. 25-37a). Several of these enzymes have been crystallized, revealing structural details of the reaction. A separate recombinase recognizes and binds to each of two recombination sites on two different DNA molecules or within the same DNA. One DNA strand in each site is cleaved at a specific point within the site, and the recombinase becomes covalently linked to the DNA at the cleavage site through a phosphotyrosine bond (step 1). The transient protein-DNA linkage preserves the phosphodiester bond that is lost in cleaving the DNA, so high-energy cofactors such as ATP are unnecessary in subsequent steps. The cleaved DNA strands are rejoined to new partners to form a Holliday intermediate, with new phosphodiester bonds created at the expense of the protein-DNA linkage (step 2). An isomerization then occurs (step 3), and the process is repeated at a second point within each of the two recombination sites (steps 4 and 5). In systems that employ an active-site Ser residue, both strands of each recombination site are cut concurrently and rejoined to new partners without the Holliday intermediate. In both types of systems, the exchange is always reciprocal and precise, regenerating the recombination sites when the reaction is complete. We can view a recombinase as a site-specific endonuclease and ligase in one package.

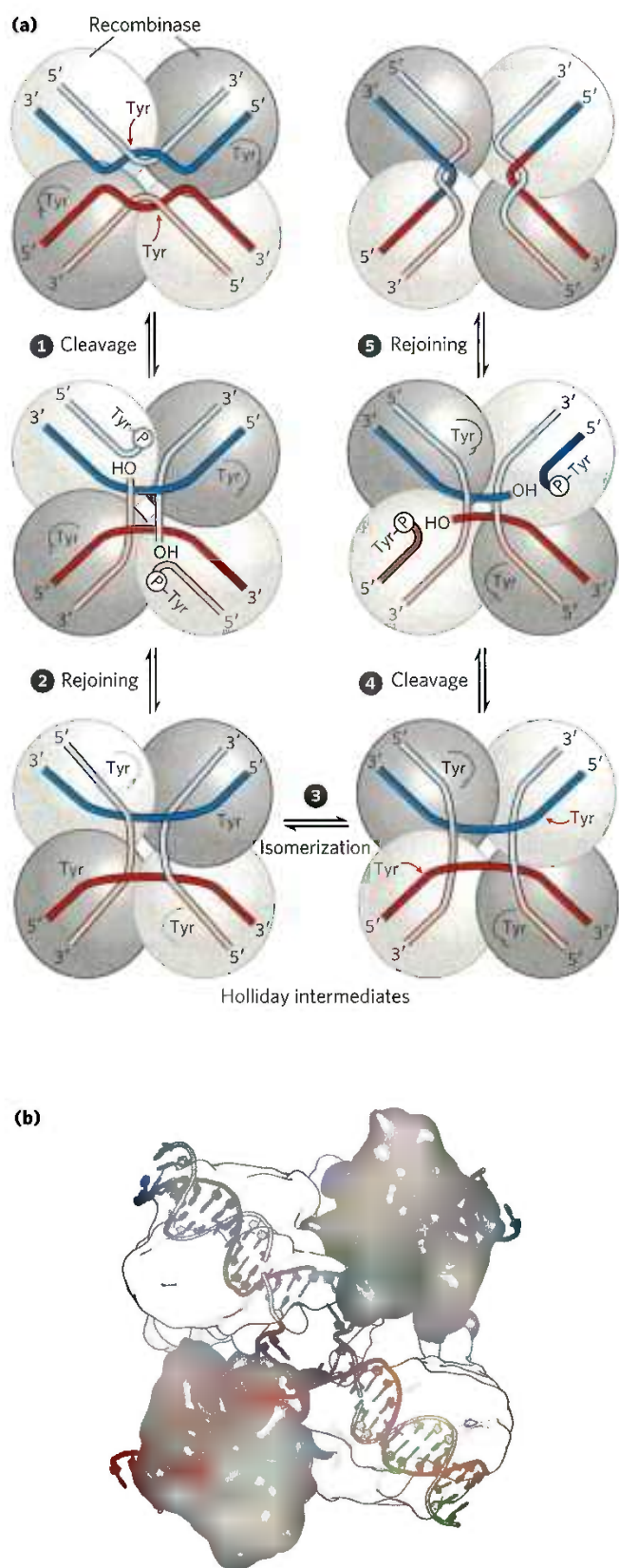


FIGURE 25-37 A site-specific recombination reaction. (a) The reaction shown here is for a common class of site-specific recombinases called integrase-class recombinases (named after bacteriophage λ integrase, the first recombinase characterized). These enzymes use Tyr residues as nucleophiles at the active site. The reaction is carried out within a tetramer of identical subunits. Recombinase subunits bind to a specific sequence, the recombination site. Two dimeric complexes, each bound to a single site in the DNA, come together to form the tetrameric complex shown here. **1** One strand in each DNA is cleaved at particular points in the sequence. The nucleophile is the —OH group of an active-site Tyr residue, and the product of rejoining **2** is a covalent phosphotyrosine link between protein and DNA. After isomerization **3**, the cleaved strands join to new partners, producing a Holliday intermediate. Steps **4** and **5** complete the reaction by a process similar to the first two steps. The original sequence of the recombination site is regenerated after recombining the DNA flanking the site. These steps occur within a complex of multiple recombinase subunits that sometimes includes other proteins not shown here. (b) Surface contour model of a four-subunit integrase-class recombinase called the FLP recombinase, bound to a Holliday intermediate (shown with light blue and dark blue helix strands). The protein has been rendered transparent so that the bound DNA is visible. Another group of recombinases, called the resolvase/invertase family, use a Ser residue as nucleophile at the active site. [Data from PDB ID 1P4E, P. A. Rice and Y. Chen, *J. Biol. Chem.* 278:24800, 2003.]

The outcome depends on the location and orientation of the recombination sites (**Fig. 25-38**). If the two sites are on the same DNA molecule, the reaction either inverts or deletes the intervening DNA, determined by whether the recombination sites have the opposite or the same orientation, respectively. If the sites are on different DNAs, the recombination is intermolecular; if one or both DNAs are circular, the result is an insertion. Some recombinase systems are highly specific for one of these reaction types and act only on sites with particular orientations.

Complete chromosomal replication can require site-specific recombination. Recombinational DNA repair of a circular bacterial chromosome, while essential, sometimes generates deleterious byproducts. The resolution of a Holliday intermediate at a replication fork by a nuclease such as RuvC, followed by completion of replication, can give rise to one of two products: the usual two monomeric chromosomes or a contiguous dimeric chromosome (**Fig. 25-39**). In the latter case, the covalently linked chromosomes cannot be segregated to daughter cells at cell division, and the dividing cells become “stuck.” A specialized site-specific recombination system in *E. coli*, the XerCD system, converts the dimeric chromosomes to monomeric chromosomes so that cell division can proceed. The reaction is a site-specific deletion (**Fig. 25-38b**). **P4** This is another example of the close coordination between DNA recombination processes and other aspects of DNA metabolism.

Transposable Genetic Elements Move from One Location to Another

We now consider the third general type of recombination system: recombination that allows the movement of transposable elements, or **transposons**. These segments of DNA, found in virtually all cells, move, or “jump,” from

The sequences of the recombination sites recognized by site-specific recombinases are partially asymmetric (nonpalindromic), and the two recombining sites align in the same orientation during the recombinase reaction.

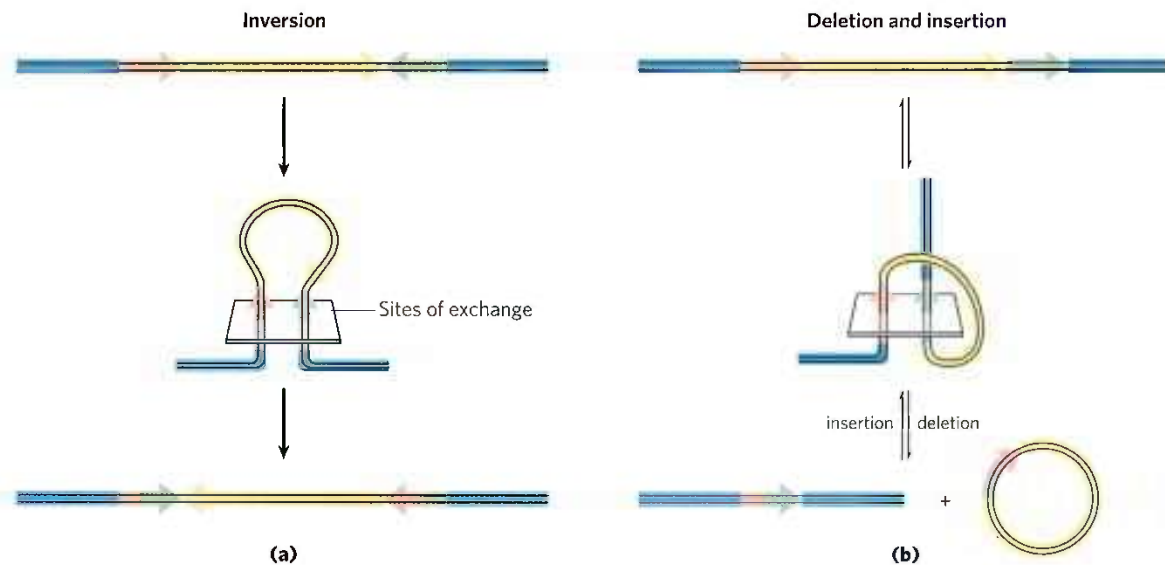


FIGURE 25-38 Effects of site-specific recombination. The outcome of site-specific recombination depends on the location and orientation of the recombination sites (red and green) in a double-stranded DNA molecule. Orientation here (shown by arrowheads) refers to the order of nucleotides in

the recombination site, not the 5'→3' direction. (a) Recombination sites with opposite orientation in the same DNA molecule. The result is an inversion. (b) Recombination sites with the same orientation, either on one DNA molecule, producing a deletion, or on two DNA molecules, producing an insertion.

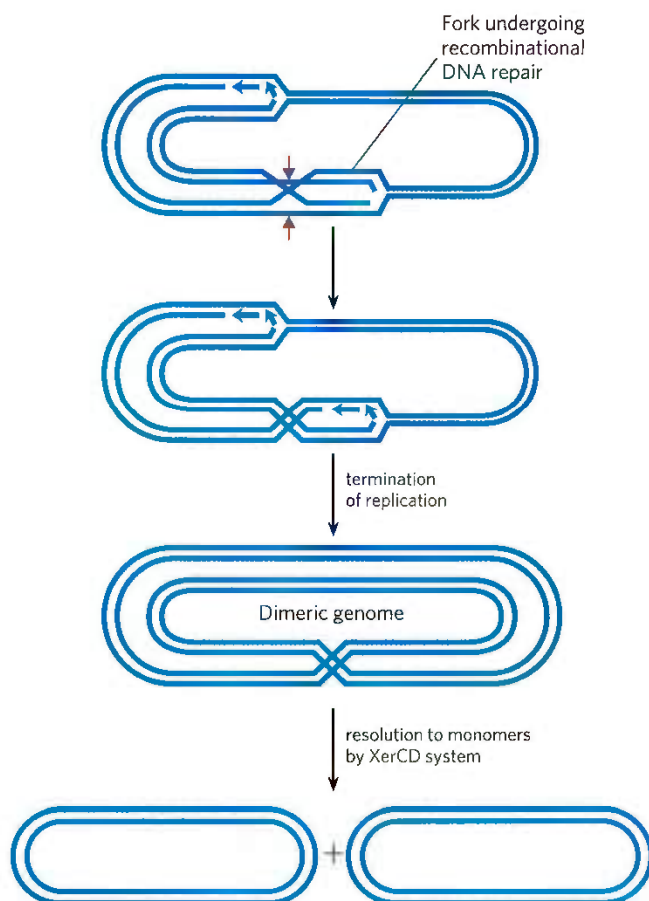


FIGURE 25-39 DNA deletion to undo a deleterious effect of recombinational DNA repair. The resolution of a Holliday intermediate during recombinational DNA repair (if cut at the points indicated by the red arrows) can generate a contiguous dimeric chromosome. A specialized site-specific recombinase in *E. coli*, XerCD, converts the dimer to monomers, allowing chromosome segregation and cell division to proceed.

one place on a chromosome (the donor site) to another on the same or a different chromosome (the target site). DNA sequence homology is not usually required for this movement, called **transposition**; the new location is determined more or less randomly. Insertion of a transposon in an essential gene could kill the cell, so transposition is tightly regulated and usually very infrequent. Transposons are perhaps the simplest of molecular parasites, adapted to replicate passively within the chromosomes of host cells. In some cases they carry genes that are useful to the host cell, and thus exist in a kind of symbiosis with the host.

Bacteria have two classes of transposons. **Insertion sequences** (simple transposons) contain only the sequences required for transposition and the genes for the proteins (transposases) that promote the process. **Complex transposons** contain one or more genes in addition to those needed for transposition. These extra genes might, for example, confer resistance to antibiotics and thus enhance the survival chances of the host cell. The spread of antibiotic-resistance elements among disease-causing bacterial populations that is rendering some antibiotics ineffectual (p. 887) is mediated to a large degree by transposition. ■

Bacterial transposons vary in structure, but most have short repeated sequences at each end that serve as binding sites for the transposase. When transposition occurs, a short sequence at the target site (5 to 10 bp) is duplicated to form an additional short repeated sequence that flanks each end of the inserted transposon (Fig. 25-40). These duplicated segments result from the cutting mechanism used to insert a transposon into the DNA at a new location.

There are two general pathways for transposition in bacteria. In direct (or simple) transposition (Fig. 25-41,

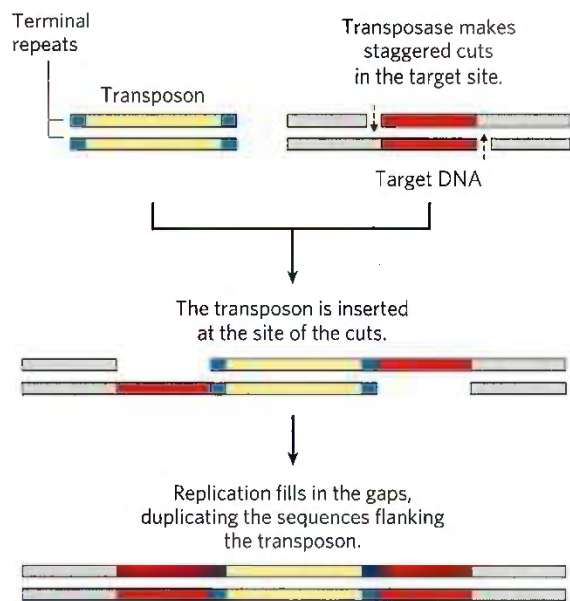


FIGURE 25-40 Duplication of the DNA sequence at a target site when a transposon is inserted. The sequences duplicated following transposon insertion are shown in red. These sequences are generally only a few base pairs long, so their size relative to that of a typical transposon is greatly exaggerated in this drawing.

left), cuts on each side of the transposon excise it, and the transposon moves to a new location. This leaves a double-strand break in the donor DNA that must be repaired. At the target site, a staggered cut is made (as in Fig. 25-40), the transposon is inserted into the break, and DNA replication fills in the gaps to duplicate the target-site sequence. In replicative transposition (Fig. 25-41, right), the entire transposon is replicated, leaving a copy behind at the donor location. A **cointegrate** is an intermediate in this process, consisting of the donor region covalently linked to DNA at the target site. Two complete copies of the transposon are present in the cointegrate, both having the same relative orientation in the DNA. In some well-characterized transposons, the cointegrate intermediate is converted to products by site-specific recombination, in which specialized recombinases promote the required deletion reaction.

Eukaryotes also have transposons, structurally similar to bacterial transposons, and some use similar transposition mechanisms. In other cases, however, the mechanism of transposition seems to involve an RNA intermediate. Evolution of these transposons is intertwined with the evolution of certain classes of RNA viruses. Both are described in the next chapter. As illustrated in Figure 9-25, nearly half of the human genome is made up of various types of transposable elements.

Immunoglobulin Genes Assemble by Recombination

Some DNA rearrangements are a programmed part of development in eukaryotic organisms. An important example is the generation of complete immunoglobulin genes from separate gene segments in vertebrate genomes.

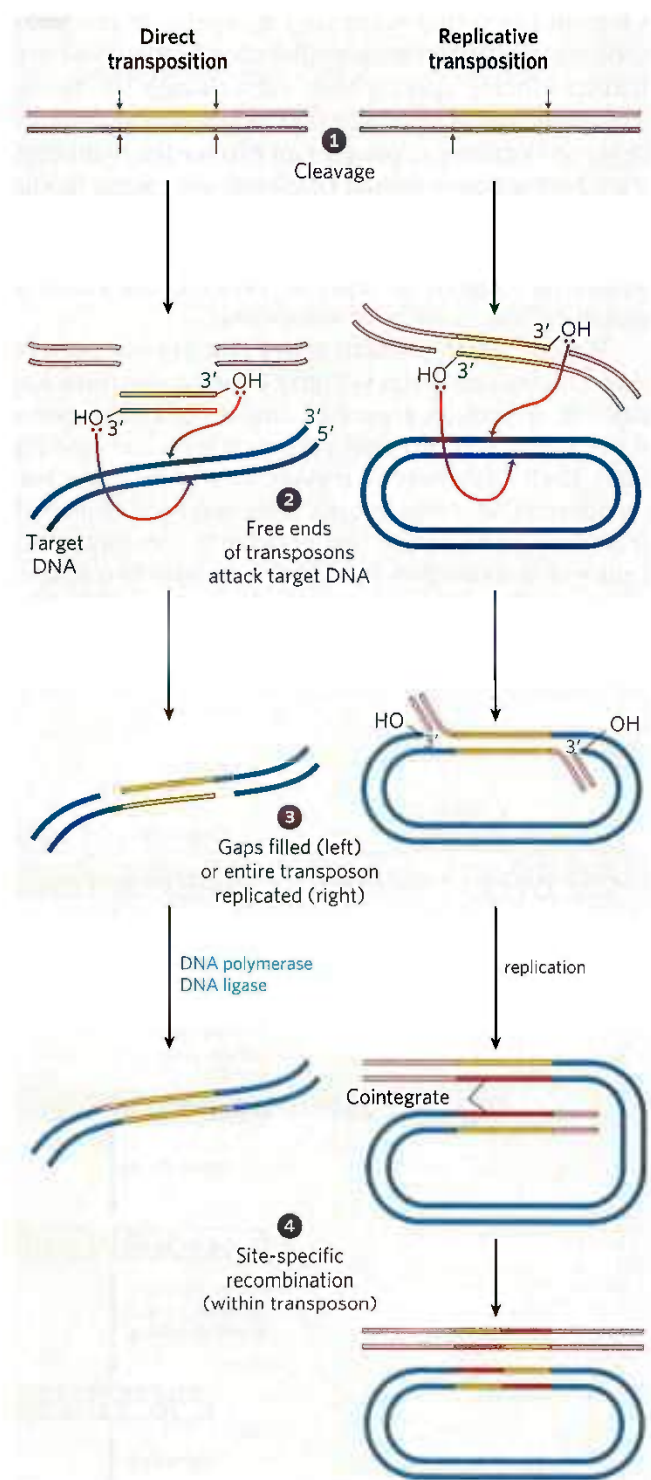


FIGURE 25-41 Two general pathways for transposition: direct (simple) and replicative. ❶ The DNA is first cleaved on each side of the transposon, at the sites indicated by arrows. ❷ The liberated 3'-hydroxyl groups at the ends of the transposon act as nucleophiles in a direct attack on phosphodiester bonds in the target DNA. The target phosphodiester bonds are staggered (not directly across from each other) in the two DNA strands. ❸ The transposon is now linked to the target DNA. In direct transposition (left), replication fills in gaps at each end to complete the process. In replicative transposition (right), the entire transposon is replicated to create a cointegrate intermediate. ❹ The cointegrate is often resolved later, with the aid of a separate site-specific recombination system. The cleaved host DNA left behind after direct transposition is either repaired by DNA end joining or degraded (not shown); the latter outcome can be lethal to the organism.

A human (like other mammals) is capable of producing *millions* of different immunoglobulins (antibodies) with distinct binding specificities, even though the human genome contains only ~20,000 genes. Recombination allows an organism to produce an extraordinary diversity of antibodies from a limited DNA-coding capacity. Studies of the recombination mechanism reveal a close relationship to DNA transposition and suggest that this system for generating antibody diversity may have evolved from an ancient cellular invasion by transposons.

We can use the human genes that encode proteins of the immunoglobulin G (IgG) class to illustrate how antibody diversity is generated. Immunoglobulins consist of two heavy and two light polypeptide chains (see Fig. 5-20). Each chain has two regions: a variable region, with a sequence that differs greatly from one immunoglobulin to another, and a region that is virtually constant within a class of immunoglobulins. There are also two distinct families of light chains, kappa and lambda, which differ somewhat in the sequences of their constant regions. For all three types of polypeptide chains (heavy chain, and

kappa and lambda light chains), diversity in the variable regions is generated by a similar mechanism. The genes for these polypeptides are divided into segments, and the genome contains clusters with multiple versions of each segment. The joining of one version of each gene segment creates a complete gene.

Figure 25-42 depicts the organization of the DNA encoding the kappa light chains of human IgG and shows how a mature kappa light chain is generated. In undifferentiated cells, the coding information for this polypeptide chain is separated into three segments. The V (variable) segment encodes the first 95 amino acid residues of the variable region, the J (joining) segment encodes the remaining 12 residues of the variable region, and the C segment encodes the constant region. The genome contains 40 different V segments, 5 different J segments, and 1 C segment.

As a stem cell in the bone marrow differentiates to form a mature B lymphocyte, one V segment and one J segment are brought together by a specialized recombination system (Fig. 25-42). During this programmed

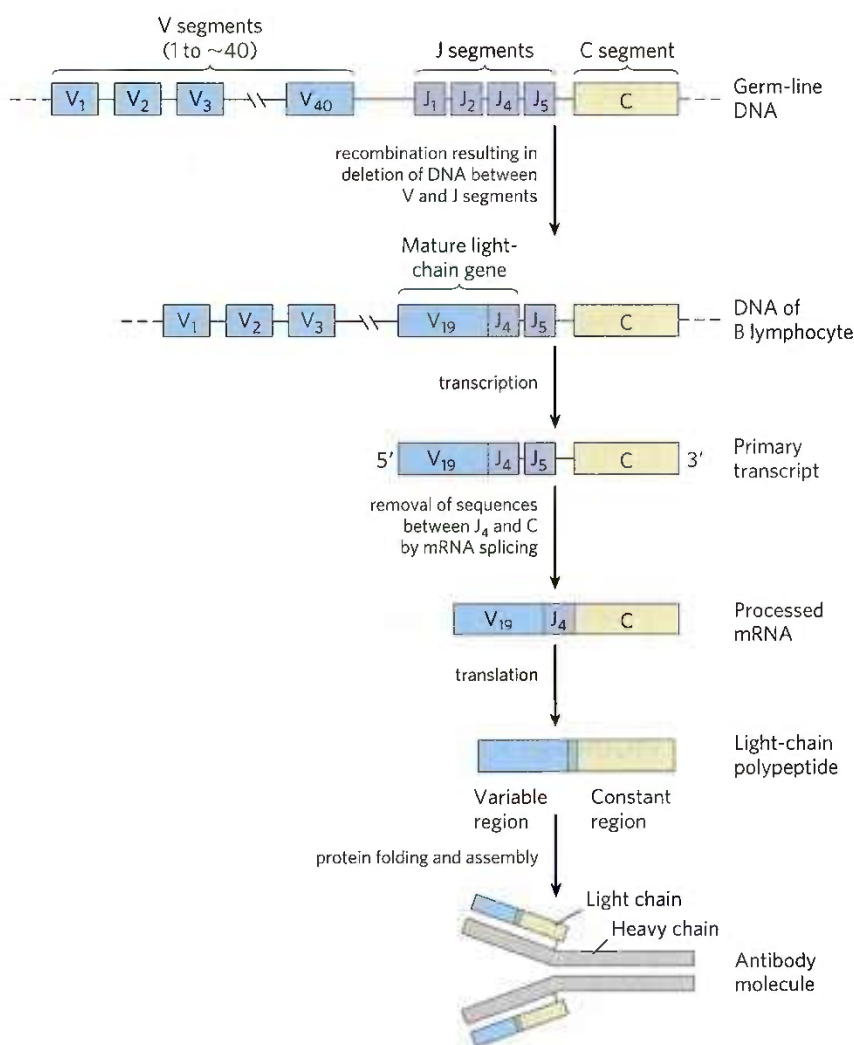


FIGURE 25-42 Recombination of the V and J gene segments of the human IgG kappa light chain. At the top is shown the arrangement of IgG-coding sequences in a stem cell of the bone marrow. Recombination deletes the DNA between a particular V segment and a J segment. Transcription and RNA splicing, as described in Chapter 26, produces the light-chain polypeptide. The light chain can combine with any of 5,000 possible heavy chains to produce an antibody molecule.

DNA deletion, the intervening DNA is discarded. There are about $40 \times 5 = 200$ possible V–J combinations. The recombination process is not as precise as the site-specific recombination described earlier, so additional variation occurs in the sequence at the V–J junction. This increases the overall variation by a factor of at least 2.5, so the cells can generate about $2.5 \times 200 = 500$ different V–J combinations. The final joining of the V–J combination to the C region is accomplished by an RNA-splicing reaction after transcription, a process described in Chapter 26.

The recombination mechanism for joining the V and J segments is illustrated in **Figure 25-43**. Just beyond each V segment and just before each J segment lie recombination signal sequences (RSSs). These are bound by proteins called RAG1 and RAG2 (products of the *recombination activating gene*). The RAG proteins catalyze the formation of a double-strand break between the signal sequences and the V (or J) segments to be joined. The V and J segments are then joined with the aid of a second complex of proteins.

The genes for the heavy chains and the lambda light chains form by similar processes. Heavy chains have more gene segments than light chains, with more than

5,000 possible combinations. Because any heavy chain can combine with any light chain to generate an immunoglobulin, each human has at least $500 \times 5,000 = 2.5 \times 10^6$ possible IgGs. And additional diversity is generated by high mutation rates (of unknown mechanism) in the V sequences during B-lymphocyte differentiation. Each mature B lymphocyte produces only one type of antibody, but the range of antibodies produced by the B lymphocytes of an individual organism is clearly enormous.

Did the immune system evolve in part from ancient transposons? The mechanism for generation of the double-strand breaks by RAG1 and RAG2 mirrors several reaction steps in transposition (Fig. 25-43). In addition, the deleted DNA, with its terminal RSSs, has a sequence structure found in most transposons. In the test tube, RAG1 and RAG2 can associate with this deleted DNA and insert it, transposonlike, into other DNA molecules (probably a rare reaction in B lymphocytes). Although we cannot know for certain, the properties of the immunoglobulin gene rearrangement system suggest an intriguing origin in which the distinction between host and parasite has become blurred by evolution.

SUMMARY 25.3 DNA Recombination

- DNA sequences are rearranged in recombination reactions, usually in processes tightly coordinated with DNA replication or repair.
- Homologous genetic recombination can take place between any two DNA molecules that share sequence homology. In bacteria, recombination serves mainly as a DNA repair process, focused on reactivating stalled or collapsed replication forks or on the general repair of double-strand breaks.
- In eukaryotes, recombination is essential to ensure accurate chromosome segregation during the first meiotic cell division. It also helps to create genetic diversity in the resulting gametes.
- Nonhomologous end joining provides an alternative mechanism for the repair of double-strand breaks, especially in eukaryotic cells.
- Site-specific recombination occurs only at specific target sequences, and this process can also involve a Holliday intermediate. Recombinases cleave the DNA at specific points and ligate the strands to new partners. This type of recombination is found in virtually all cells, and its many functions include DNA integration and regulation of gene expression.
- In almost all cells, transposons use recombination to move within or between chromosomes.
- In vertebrates, a programmed recombination reaction related to transposition joins immunoglobulin gene segments to form immunoglobulin genes during B-lymphocyte differentiation.

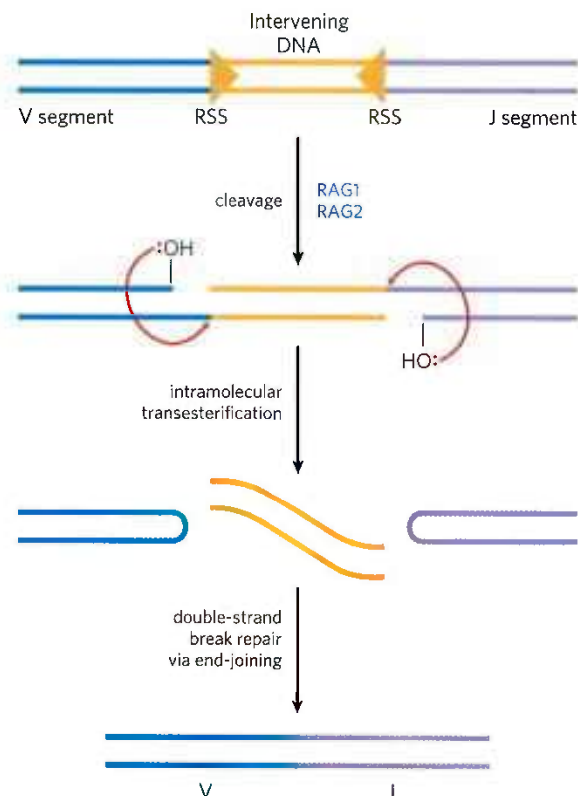


FIGURE 25-43 Mechanism of immunoglobulin gene rearrangement. The RAG1 and RAG2 proteins bind to the recombination signal sequences (RSSs) and cleave one DNA strand between the RSS and the V (or J) segments to be joined. The liberated 3' hydroxyl then acts as a nucleophile, attacking a phosphodiester bond in the other strand to create a double-strand break. The resulting hairpin bends on the V and J segments are cleaved, and the ends are covalently linked by a complex of proteins specialized for end-joining repair of double-strand breaks.

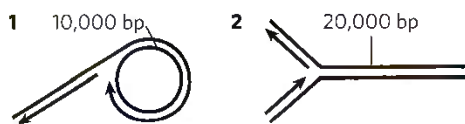
KEY TERMS

Terms in bold are defined in the glossary.

template 915	DNA polymerase ϵ 929
semiconservative	DNA polymerase δ 929
replication 915	DNA polymerase α 929
replication fork 916	mutation 930
origin 916	DNA glycosylases 934
Okazaki fragment 916	base-excision repair 934
leading strand 916	AP site 935
lagging strand 916	abasic site 935
nucleases 916	AP endonucleases 935
exonucleases 916	DNA photolyases 937
endonucleases 916	error-prone translesion DNA synthesis 939
DNA polymerases 916	SOS response 939
DNA polymerase I 916	homologous genetic recombination 940
primer 917	site-specific recombination 940
primer terminus 917	DNA transposition 940
processivity 917	recombinational DNA repair 941
proofreading 919	branch migration 942
DNA polymerase III 919	Holliday intermediate 942
replisome 921	replication restart
helicases 921	primosome 943
topoisomerases 922	meiosis 944
primases 922	double-strand break repair model 947
DNA ligases 922	nonhomologous end joining (NHEJ) 948
DNA unwinding element (DUE) 922	transposon 951
AAA+ ATPases 922	transposition 952
catenane 927	insertion sequence 952
prereplicative complex (pre-RC) 928	
licensing 928	
minichromosome maintenance (MCM) protein 928	
ORC (origin recognition complex) 928	

PROBLEMS

1. DNA Replication An investigator adds DNA polymerase III holoenzyme, DNA primase (DnaG), single-stranded DNA-binding protein (SSB), an ATP-dependent DNA ligase, and the replicative helicase (DnaB) to each of the DNA substrates shown, along with ATP and all four dNTPs. The length of DNA in the circle in structure **1** is 10,000 bp. The linear (unbranched) part of structure **2** is 20,000 bp long.



(a) If precursor (dNTP) concentrations are not limiting, Okazaki fragments are 2,000 nucleotides long, and replication proceeds at 1,000 nucleotides/s for 30 seconds, which DNA substrate will generate a longer replication product?

(b) For structure **1**, draw the product of this 30 second replication. (c) Draw the expected product if DnaG were left out of the reaction. (d) Draw the expected product if DNA ligase were instead left out of the reaction.

2. Heavy Isotope Analysis of DNA Replication A researcher switches a culture of *E. coli* growing in a medium containing $^{15}\text{NH}_4\text{Cl}$ to a medium containing $^{14}\text{NH}_4\text{Cl}$ for three generations (an eightfold increase in population). What is the molar ratio of hybrid DNA (^{15}N - ^{14}N) to light DNA (^{14}N - ^{14}N) at this point?

3. Replication of the *E. coli* Chromosome The *E. coli* chromosome contains 4,641,652 bp.

(a) How many turns of the double helix must be unwound during replication of the *E. coli* chromosome?

(b) Using the data in this chapter, how long would it take to replicate the *E. coli* chromosome at 37 °C if two replication forks proceeded from the origin? Assume replication occurs at a rate of 1,000 bp/s. Under some conditions, *E. coli* cells can divide every 20 min. How might this be possible?

(c) In the replication of the *E. coli* chromosome, about how many Okazaki fragments would be formed? What factors are required to link together newly synthesized Okazaki fragments in the lagging strand?

4. Base Composition of DNAs Made from Single-Stranded Templates Predict the base composition of the total DNA synthesized by DNA polymerase on templates provided by an equimolar mixture of the two complementary strands of bacteriophage ϕX174 DNA (a circular DNA molecule). The base composition of one strand is A, 24.7%; G, 24.1%; C, 18.5%; and T, 32.7%. What assumption is necessary to answer this problem?

5. DNA Replication Kornberg and his colleagues incubated soluble extracts of *E. coli* with a mixture of dATP, dTTP, dGTP, and dCTP, all labeled with ^{32}P in the α -phosphate group. After a time, they treated the incubation mixture with trichloroacetic acid, which precipitates the DNA but not the nucleotide precursors. They then collected the precipitate and determined the extent of precursor incorporation into DNA from the amount of radioactivity present in the precipitate.

(a) If any one of the four nucleotide precursors were omitted from the incubation mixture, would radioactivity be found in the precipitate? Explain.

(b) Would ^{32}P be incorporated into the DNA if only dTTP were labeled? Explain.

(c) Would radioactivity be found in the precipitate if ^{32}P labeled the β phosphate or γ phosphate rather than the α phosphate of the deoxyribonucleotides? Explain.

6. The Chemistry of DNA Replication All DNA polymerases synthesize new DNA strands in the 5'→3' direction. In some respects, replication of the antiparallel strands of duplex DNA would be simpler if there were also a second type of polymerase, one that synthesized DNA in the 3'→5' direction. The two types of polymerase could, in principle, coordinate DNA synthesis without the complicated mechanisms required for lagging strand replication. However, no such 3'→5'-synthesizing enzyme has been found. Suggest two possible mechanisms for 3'→5' DNA synthesis. Pyrophosphate

should be one product of both proposed reactions. Could one or both mechanisms be supported in a cell? Why or why not? (Hint: You may suggest the use of DNA precursors not actually present in extant cells.)

7. Activities of DNA Polymerases You are characterizing a new DNA polymerase. When you incubate the enzyme with ^{32}P -labeled DNA and no dNTPs, you observe the release of [^{32}P]dNMPs. The addition of unlabeled dNTPs prevents this release. Explain the reactions that most likely underlie these observations. What would you expect to observe if you added pyrophosphate instead of dNTPs?

8. Leading and Lagging Strands Prepare a table that lists the names and compares the functions of the precursors, enzymes, and other proteins needed to make the leading strand versus the lagging strand during DNA replication in *E. coli*.

9. Function of DNA Ligase Some *E. coli* mutants contain defective DNA ligase. When researchers expose these mutants to ^3H -labeled thymine and then sediment the DNA produced on an alkaline sucrose density gradient, two radioactive bands appear. One corresponds to a high molecular weight fraction, the other to a low molecular weight fraction. Explain.

10. Fidelity of Replication of DNA What factors promote the fidelity of replication during synthesis of the leading strand of DNA? Would you expect the lagging strand to be made with the same fidelity? Give reasons for your answers.

11. Importance of DNA Topoisomerases in DNA Replication DNA unwinding, such as that occurring in replication, affects the superhelical density of DNA. In the absence of topoisomerases, the DNA would become overwound ahead of a replication fork as the DNA is unwound behind it. A bacterial replication fork will stall when the superhelical density (σ) of the DNA ahead of the fork reaches +0.14 (see Chapter 24).

An investigator initiates bidirectional replication at the origin of a 6,000 bp plasmid *in vitro*, in the absence of topoisomerases. The plasmid initially has a σ of -0.06. How many base pairs will be unwound and replicated by each replication fork before the forks stall? Assume that both forks travel at the same rate and that each includes all components necessary for elongation except topoisomerase.

12. The Ames Test In a nutrient medium that lacks histidine, a thin layer of agar containing $\sim 10^9$ *Salmonella typhimurium* histidine auxotrophs (mutant cells that require histidine to survive) produces ~ 13 colonies over a two-day incubation period at 37 °C (see Fig. 25-19). How do these colonies arise in the absence of histidine? When investigators repeat the experiment in the presence of 0.4 μg of 2-aminoanthracene, the number of colonies produced over two days exceeds 10,000. What does this indicate about 2-aminoanthracene? What can you surmise about its carcinogenicity?

13. DNA Repair Mechanisms Vertebrate and plant cells often methylate cytosine in DNA to form 5-methylcytosine (see Fig. 8-5a). In these same cells, a specialized repair system recognizes G-T mismatches and repairs them to G $\equiv$ C base pairs. How might this repair system be advantageous to the cell? (Explain in terms of the presence of 5-methylcytosine in the DNA.)

14. The Energetic Cost of Mismatch Repair In an *E. coli* cell, DNA polymerase III makes a rare error and inserts a G opposite an A residue at a position 650 bp away from the nearest GATC sequence. The mismatch repair system accurately repairs the mismatch. How many phosphodiester bonds derived from deoxynucleotides (dNTPs) does this repair expend? This process also uses ATP molecules. Which enzyme(s) consume the ATP?



15. DNA Repair in People with Xeroderma Pigmentosum The condition known as xeroderma pigmentosum (XP) arises from mutations in at least seven different human genes (see Box 25-1). The deficiencies are generally in genes encoding enzymes involved in some part of the pathway for human nucleotide-excision repair. The various types of XP are denoted A through G (XPA, XPB, etc.), with a few additional variants lumped under the label XPV.

Investigators irradiate cultures of fibroblasts from healthy individuals and from patients with XPG with ultraviolet light. After isolating and denaturing the DNA, they characterize the resulting single-stranded DNA by analytical ultracentrifugation.

(a) Samples from the normal fibroblasts show a significant reduction in the average molecular weight of the single-stranded DNA after irradiation, but samples from the XPG fibroblasts show no such reduction. Why might this be?

(b) If you assume that a nucleotide-excision repair system is operative in fibroblasts, which step might be defective in the cells from the patients with XPG? Explain.

16. DNA Repair and Cancer Many pharmaceuticals used for tumor chemotherapy are DNA damaging agents. What is the rationale behind actively damaging DNA to address tumors? Why do such treatments often have a greater effect on a tumor than on healthy tissue?

17. Direct Repair Cells normally repair the lesion $\text{O}^6\text{-meG}$ by directly transferring the methyl group to the protein $\text{O}^6\text{-methylguanine-DNA methyltransferase}$. For the nucleotide sequence $\text{AAC}(\text{O}^6\text{-meG})\text{TGCAC}$, with a damaged (methylated) G residue, what would be the sequence of both strands of double-stranded DNA resulting from replication in each of the situations listed?

(a) Replication occurs before repair.

(b) Replication occurs after repair.

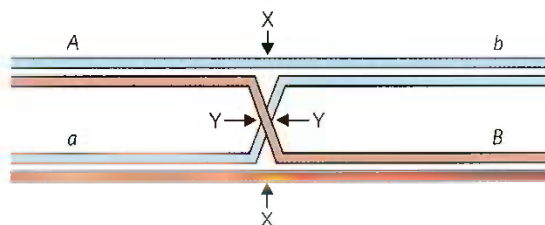
(c) Two rounds of replication occur, followed by repair.

18. Strand Invasion in Recombination A key step in many homologous recombination reactions is strand invasion (see step 2 in Fig. 25-29). In almost every case, strand invasion proceeds with a single strand that has a free 3' end rather than a 5' end. What DNA metabolic advantage is inherent with the use of a free 3' end for strand invasion?

19. Holliday Intermediates How does the formation of Holliday intermediates in homologous genetic recombination differ from their formation in site-specific recombination?

20. Cleavage of Holliday Intermediates A Holliday intermediate forms between two homologous chromosomes, at a point between genes A and B, as shown. The chromosomes have different alleles of the two genes (A and a, B and b). Where would the Holliday intermediate have to be cleaved (points X

and/or Y) to generate a chromosome that would carry (a) an *Ab* genotype or (b) an *ab* genotype?



21. A Connection between Replication and Site-Specific Recombination Most wild strains of *S. cerevisiae* have multiple copies of the circular DNA plasmid 2μ (named for its contour length of about $2\mu\text{m}$), which has ~6,300 bp. For its replication, the plasmid uses the host replication system, under the same strict control as the host cell chromosomes, replicating only once per cell cycle. Replication of the plasmid is bidirectional, with both replication forks initiating at a single, well-defined origin. However, one replication cycle of a 2μ plasmid can result in more than two copies of the plasmid, allowing amplification of the plasmid copy number (number of plasmid copies per cell) whenever plasmid segregation at cell division leaves one daughter cell with fewer than the normal complement of plasmid copies. Amplification requires a site-specific recombination system encoded by the plasmid, which serves to invert one part of the plasmid relative to the other. Explain how a site-specific inversion event could result in amplification of the plasmid copy number. (Hint: Consider the situation when replication forks have duplicated one recombination site but not the other.)

DATA ANALYSIS PROBLEM

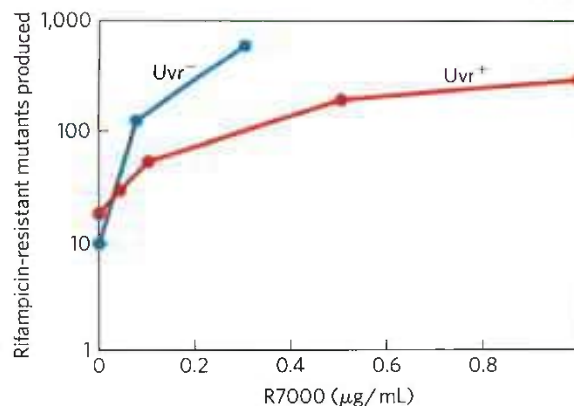
22. Mutagenesis in *Escherichia coli* Many mutagenic compounds act by alkylating the bases in DNA. The alkylating agent R7000 (7-methoxy-2-nitronaphtho[2,1-*b*]furan) is an extremely potent mutagen.



In vivo, R7000 is activated by the enzyme nitroreductase, and this more reactive form covalently attaches to DNA—primarily, but not exclusively, to G=C base pairs.

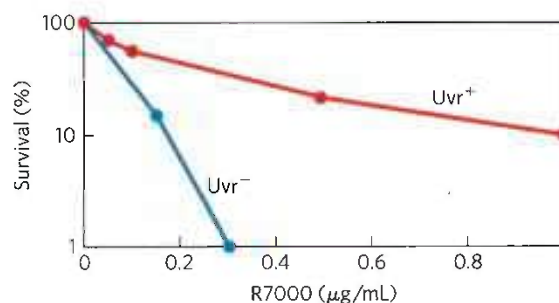
In a 1996 study, Quillardet, Touati, and Hofnung explored the mechanisms by which R7000 causes mutations in *E. coli*. They compared the genotoxic activity of R7000 in two strains of *E. coli*: the wild-type (Uvr^+) and mutants lacking *uvrA* activity (Uvr^-). They first measured rates of mutagenesis. Rifampicin is an inhibitor of RNA polymerase. In its presence, cells will not grow unless certain mutations occur in the gene encoding RNA polymerase; the appearance of rifampicin-resistant colonies thus provides a useful measure of mutagenesis rates.

The investigators determined the effects of different concentrations of R7000. Their results are shown in the following graph:



(a) Why are some mutants produced even when no R7000 is present?

Quillardet and colleagues also measured the survival rate of bacteria treated with different concentrations of R7000, with the following results:



(b) Explain how treatment with R7000 is lethal to cells.

(c) Explain the differences in the mutagenesis curves and in the survival curves for the two types of bacteria, Uvr^+ and Uvr^- , as shown in the graphs.

The researchers went on to measure the amount of R7000 covalently attached to the DNA in Uvr^+ and Uvr^- *E. coli*. They incubated bacteria with [^3H]R7000 for 10 or 70 minutes, extracted the DNA, and measured its ^3H content in counts per minute (cpm) per microgram of DNA.

Time (min)	^3H in DNA (cpm/μg)	
	Uvr^+	Uvr^-
10	76	159
70	69	228

(d) Explain why the amount of ^3H drops over time in the Uvr^+ strain and rises over time in the Uvr^- strain.

Quillardet and colleagues then examined the particular DNA sequence changes caused by R7000 in the Uvr^+ and Uvr^- bacteria. For this, they used six different strains of *E. coli*, each with a different point mutation in the *lacZ* gene, which encodes β -galactosidase. Cells with any of these mutations have a nonfunctional β -galactosidase and are unable to metabolize lactose (i.e., a Lac^- phenotype). Each type of point mutation

required a specific reverse mutation to restore *lacZ* gene function and Lac⁺ phenotype. By plating cells on a medium containing lactose as the sole carbon source, the researchers selected for these reverse-mutated, Lac⁺ cells. And by counting the number of Lac⁺ cells following mutagenesis of a particular strain, they could measure the frequency of each type of mutation.

First, they looked at the mutation spectrum in Uvr⁻ cells. The following table shows the results for the six strains, CC101 through CC106 (with the point mutation required to produce Lac⁺ cells indicated in parentheses).

R7000 ($\mu\text{g/mL}$)	Number of Lac ⁺ cells (average $\pm$ SD)					
	CC101 (A=T to C=G)	CC102 (G=C to A=T)	CC103 (G=C to C=G)	CC104 (G=C to T=A)	CC105 (A=T to T=A)	CC106 (A=T to G=C)
0	6 $\pm$ 3	11 $\pm$ 9	2 $\pm$ 1	5 $\pm$ 3	2 $\pm$ 1	1 $\pm$ 1
0.075	24 $\pm$ 19	34 $\pm$ 3	8 $\pm$ 4	82 $\pm$ 23	40 $\pm$ 14	4 $\pm$ 2
0.15	24 $\pm$ 4	26 $\pm$ 2	9 $\pm$ 5	180 $\pm$ 71	130 $\pm$ 50	3 $\pm$ 2

(e) Which types of mutation show significant increases above the background rate due to treatment with R7000? Provide a plausible explanation for why some have higher frequencies than others.

(f) Can all of the mutations you listed in (e) be explained as resulting from covalent attachment of R7000 to a G=C base pair? Explain your reasoning.

(g) Figure 25-26b shows how methylation of guanine residues can lead to a G=C to A=T mutation. Using a similar pathway, show how an R7000-G adduct could lead to the G=C to A=T or T=A mutations shown above. Which base pairs with the R7000-G adduct?

The results for the Uvr⁺ bacteria are shown in the following table.

R7000 ($\mu\text{g/mL}$)	Number of Lac ⁺ cells (average $\pm$ SD)					
	CC101 (A=T to C=G)	CC102 (G=C to A=T)	CC103 (G=C to C=G)	CC104 (G=C to T=A)	CC105 (A=T to T=A)	CC106 (A=T to G=C)
0	2 $\pm$ 2	10 $\pm$ 9	3 $\pm$ 3	4 $\pm$ 2	6 $\pm$ 1	0.5 $\pm$ 1
1	7 $\pm$ 6	21 $\pm$ 9	8 $\pm$ 3	23 $\pm$ 15	13 $\pm$ 1	1 $\pm$ 1
5	4 $\pm$ 3	15 $\pm$ 7	22 $\pm$ 2	68 $\pm$ 25	67 $\pm$ 14	1 $\pm$ 1

(h) Do these results show that all mutation types are repaired with equal fidelity? Provide a plausible explanation for your answer.

Reference

Quillardet, P., E. Touati, and M. Hofnung. 1996. Influence of the *uvr*-dependent nucleotide-excision repair on DNA adducts formation and mutagenic spectrum of a potent genotoxic agent: 7-methoxy-2-nitronaphtho[2,1-*b*]furan (R7000). *Mutat. Res.* 358:113-122.

RNA METABOLISM

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Expression of the information in a gene generally involves production of an RNA molecule transcribed from a DNA template. Strands of RNA and DNA may seem similar at first glance, differing only in that RNA has a hydroxyl group at the 2' position of the aldopentose, and uracil usually replaces thymine. However, unlike DNA, most RNAs carry out their functions as single strands, strands that fold back on themselves and have the potential for much greater structural diversity than DNA (Chapter 8). RNA is thus suited to a variety of cellular functions.

RNA is the only macromolecule known to have a role both in the storage and transmission of information and in catalysis, which has led to much speculation about its possible role as an essential chemical intermediate in the development of life on this planet. The discovery of catalytic RNAs, or **ribozymes**, has changed the very definition of an enzyme, extending it beyond the domain of proteins. Proteins nevertheless remain essential to RNA and its cellular functions. In the biosphere of today, all nucleic acids, including RNAs, are complexed with proteins. In the case of RNA, these complexes are called **ribonucleoproteins** or **RNPs**. Some of these RNPs are quite elaborate, and RNA can assume both structural and catalytic roles within complicated biochemical machines.

All RNA molecules except the RNA genomes of certain viruses are derived from information permanently stored in DNA. During **transcription**, an enzyme system converts the genetic information in a segment of double-stranded DNA into an RNA strand with a base sequence complementary to one of the DNA strands. Four major kinds of RNA are produced. **Messenger RNAs (mRNAs)** encode the amino acid sequence of one or more polypeptides specified by a gene or set of genes. **Transfer RNAs (tRNAs)** read the information

encoded in the mRNA and transfer the appropriate amino acid to a growing polypeptide chain during protein synthesis. **Ribosomal RNAs (rRNAs)** are constituents of ribosomes, the intricate cellular machines that synthesize proteins. **Noncoding RNAs (ncRNAs)** have a variety of catalytic, structural, and regulatory functions.

During replication the entire chromosome is usually copied, but transcription is more selective. Only particular genes or groups of genes are transcribed at any one time, and some portions of the DNA genome are never transcribed. The cell restricts the expression of genetic information to the formation of gene products needed at any particular moment. The sum of all the RNA molecules produced in a cell under a given set of conditions is called the cellular **transcriptome**. Given the relatively small fraction of the human genome devoted to protein-coding genes (about 2%), we might expect that only a small part of the human genome is transcribed. This is not the case. Transcriptome analyses have revealed that approximately 76% of the human genome is transcribed into RNA. The products are predominantly not mRNAs but rather ncRNAs. Many ncRNAs are involved in regulating gene expression by interaction with other RNAs, genomic DNA, or proteins. However, the rapid pace of their discovery has forced us to realize that we do not yet know the function of the majority of our genomic ncRNA transcripts.

In this chapter we examine the synthesis of RNA on a DNA template and the postsynthetic processing, location, and turnover of RNA molecules. In doing so, we encounter many of the specialized functions of RNA, including catalytic functions. We also describe systems in which RNA is the template and DNA the product, rather than vice versa. The information pathways thus come full circle and reveal that template-dependent nucleic acid synthesis has standard rules, regardless of the nature of template or product (RNA or DNA). This examination of the biological interconversion of DNA and RNA as information carriers leads inevitably to a discussion of the evolutionary origin of biological information and processing. We will be guided by four principles.

P1 RNA is synthesized by RNA polymerases using DNA templates and ribonucleoside 5'-triphosphates. RNA is made in the 5'→3' direction and complementary to the template DNA strand. Transcription is highly regulated

and initiates by recruitment of the transcription machinery to gene promoters. Although bacterial and eukaryotic polymerases share many conserved features, the transcriptional machinery and its regulation are much more complex in eukaryotes.

P2 Many RNAs must be modified and processed to become functional. RNAs can be modified by nucleases, by excision of certain RNA segments, and/or by chemical modification of the RNA nucleotides. In humans, nearly all mRNAs are 5' capped, spliced, and polyadenylated before being exported from the nucleus for translation in the cytoplasm.

P3 RNA can be used as a template for synthesis of DNA by reverse transcriptases. RNA carries genetic information that can be reverse transcribed into DNA. Retroviruses, such as HIV or those responsible for some cancers, must convert their RNA genomes into DNA by reverse transcription. Reverse transcription by telomerase is also responsible for producing the telomeres that protect the DNA found at the ends of eukaryotic chromosomes.

P4 RNA can act as both a catalyst and carrier of genetic information. Ribozymes can catalyze chemical transformations using many of the same strategies as protein-based enzymes. The dual capacity of RNA as both a carrier of genetic information and a catalyst is support for the RNA world hypothesis for the evolution of life on Earth.

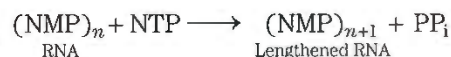
26.1 DNA-Dependent Synthesis of RNA

Our discussion of RNA synthesis begins with a comparison between transcription and DNA replication (Chapter 25). Transcription resembles replication in its fundamental chemical mechanism, its polarity (direction of synthesis), and its use of a template. And like replication, transcription has initiation, elongation, and termination phases. Transcription differs from replication in that it does not require a primer and, generally, involves only limited segments of a DNA molecule. Additionally, only one DNA strand serves as a template for a particular RNA molecule.

RNA Is Synthesized by RNA Polymerases

The discovery of DNA polymerase and its dependence on a DNA template spurred a search for an enzyme that synthesizes RNA complementary to a DNA strand. By 1960, four research groups had independently detected an enzyme in cellular extracts that could form an RNA polymer from ribonucleoside 5'-triphosphates.

Subsequent work on the purified *Escherichia coli* RNA polymerase helped to define the fundamental properties of transcription (Fig. 26-1). **P1 DNA-dependent RNA polymerase** requires, in addition to a DNA template, all four ribonucleoside 5'-triphosphates (ATP, GTP, UTP, and CTP) as precursors of the nucleotide units of RNA, as well as Mg^{2+} . The chemistry and mechanism of RNA synthesis closely resemble those used by DNA polymerases (see Fig. 25-3). RNA polymerase elongates an RNA strand by adding ribonucleotide units to the 3'-hydroxyl end, building RNA in the 5'→3' direction. The 3'-hydroxyl group acts as a nucleophile, attacking the α phosphate of the incoming ribonucleoside triphosphate (Fig. 26-1a) and releasing pyrophosphate. The overall reaction is



RNA polymerase requires DNA for activity and is most active when bound to a double-stranded DNA. As noted above, only one of the two DNA strands serves as a template. The template DNA strand is copied in the 3'→5' direction (antiparallel to the new RNA strand), just as in DNA replication. Each nucleotide in the newly formed RNA is selected by Watson-Crick base-pairing interactions: U residues are inserted in the RNA to pair with A residues in the DNA template, G residues are inserted to pair with C residues, and so on. Base-pair geometry (see Fig. 25-5) may also play a role in base selection.

Unlike DNA polymerase, RNA polymerase does not require a primer to initiate synthesis. Initiation occurs when RNA polymerase binds at specific DNA sequences called promoters (described below). The 5'-triphosphate group of the first residue in a nascent (newly formed) RNA molecule is not cleaved to release PP_i , but instead remains intact and functions in eukaryotes as a substrate for the RNA-capping machinery (see Fig. 26-13). During the elongation phase of transcription, the growing end of the new RNA strand base-pairs temporarily with the DNA template to form a short hybrid RNA-DNA double helix, about 8 bp long (Fig. 26-1b). The RNA in this hybrid duplex "peels off" shortly after its formation, and the DNA duplex re-forms.

To enable RNA polymerase to synthesize an RNA strand complementary to one of the DNA strands, the DNA duplex must unwind over a short distance, forming a transcription "bubble." During transcription, the *E. coli* RNA polymerase generally keeps about 17 bp unwound. The 8 bp RNA-DNA hybrid occurs in this unwound region. Elongation of a transcript by *E. coli* RNA polymerase proceeds at a rate of 50 to 90 nucleotides/s. Because DNA is a helix, movement of a transcription bubble requires considerable strand rotation of the nucleic acid molecules. DNA strand rotation is restricted in most DNAs by DNA-binding proteins and other structural barriers. As a result, a moving RNA polymerase generates

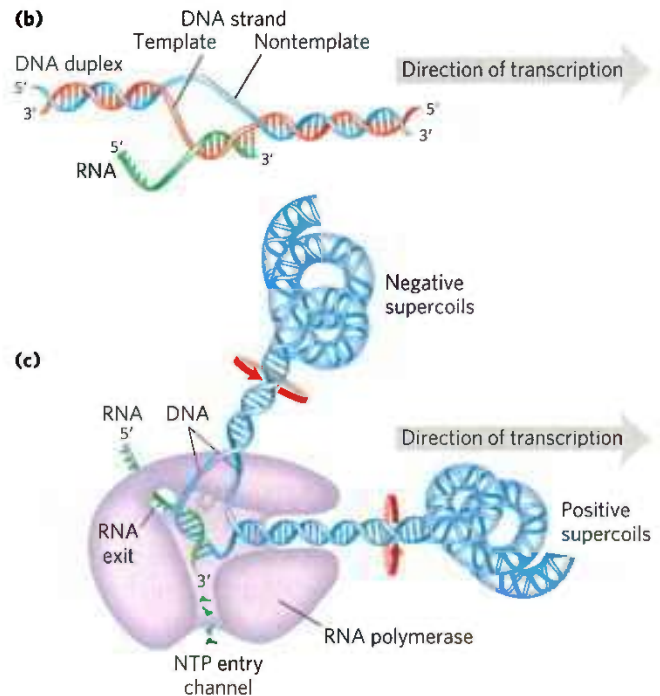
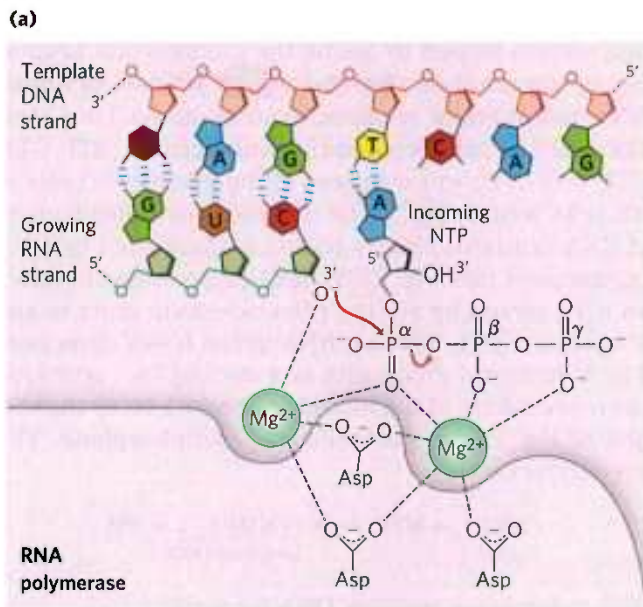


FIGURE 26-1 Transcription by RNA polymerase in *E. coli*. For synthesis of an RNA strand complementary to one of two DNA strands in a double helix, the DNA is transiently unwound. (a) Catalytic mechanism of RNA synthesis by RNA polymerase. Notice that this is essentially the same mechanism used by DNA polymerases. The reaction involves two Mg^{2+} ions coordinated to the phosphate groups of the incoming nucleoside triphosphates (NTPs) and to three Asp residues, which are highly conserved in the RNA polymerases of all species. One Mg^{2+} ion facilitates attack by the 3'-hydroxyl group on the α phosphate of the NTP; the other Mg^{2+} ion facilitates displacement of the pyrophosphate. Both metal ions stabilize the pentacoordinate transition state. (b) About 17 bp of DNA are unwound at any given time. RNA

polymerase and the transcription bubble move from left to right along the DNA as shown, facilitating RNA synthesis. The DNA is unwound ahead and rewound behind as RNA is transcribed. As the DNA is rewound, the RNA-DNA hybrid is displaced and the RNA strand is extruded. (c) Movement of an RNA polymerase along DNA tends to create positive supercoils (overwound DNA) ahead of the transcription bubble and negative supercoils (underwound DNA) behind it. The RNA polymerase is in close contact with the DNA ahead of the transcription bubble as well as with the separated DNA strands and the RNA within and immediately behind the bubble. A channel in the protein funnels new NTPs to the polymerase active site. The polymerase footprint encompasses about 35 bp of DNA during elongation.

waves of positive supercoils ahead of the transcription bubble and negative supercoils behind (Fig. 26-1c). This has been observed both *in vitro* and *in vivo* (in bacteria). In the cell, the topological problems caused by transcription are relieved through the action of topoisomerases (Chapter 24).

KEY CONVENTION The two complementary DNA strands have different roles in transcription. The strand that serves as template for RNA synthesis is called the **template strand**. The DNA strand complementary to the template, the **nontemplate strand**, or **coding strand**, is identical in base sequence to the RNA transcribed from the gene, with U in the RNA in place of T in the DNA (Fig. 26-2). The coding strand for a particular gene may be located in either strand of a given chromosome (as shown in Fig. 26-3 for a virus). By convention, the regulatory sequences that control transcription (described later in this chapter) are designated by the sequences in the coding strand. ■

The DNA-dependent RNA polymerase of *E. coli* is a large, complex enzyme with five core subunits ($\alpha_2\beta\beta'\omega$; M_r 390,000) and a sixth subunit, one of a group designated

(5') CGCTATAGCGTTT(3') DNA nontemplate (coding) strand
(3') GCGATATCGCAA(5') DNA template strand
(5') CGCUAUAGCGUUU(3') RNA transcript

FIGURE 26-2 Template and nontemplate (coding) DNA strands. The two complementary strands of DNA are defined by their function in transcription. The RNA transcript is synthesized on the template strand and is identical in sequence (with U in place of T) to the nontemplate strand, or coding strand.

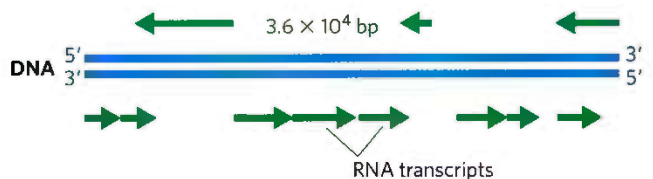


FIGURE 26-3 Organization of coding information in the adenovirus genome. The genetic information of the adenovirus genome is encoded by a double-stranded DNA molecule of 36,000 bp, both strands of which encode proteins. The information for most proteins is encoded by (that is, identical to) the top strand — by convention, the strand is oriented 5' to 3' from left to right. The bottom strand acts as template for these transcripts. However, a few proteins are encoded by the bottom strand, which is transcribed in the opposite direction (and uses the top strand as template).

σ , with variants designated by size (molecular weight). The σ subunit binds transiently to the core and directs the enzyme to specific binding sites on the DNA (described below). These six subunits constitute the RNA polymerase holoenzyme (Fig. 26-4). The RNA polymerase holoenzyme of *E. coli* thus exists in several forms, depending on the type of σ subunit. The most common subunit is σ^{70} (M_r 70,000), and the upcoming discussion focuses on the corresponding RNA polymerase holoenzyme.

RNA polymerases lack a separate proofreading 3'→5' exonuclease active site (such as that of many DNA polymerases), and the error rate for transcription is higher than that for chromosomal DNA replication—approximately one error for every 10^4 to 10^5 ribonucleotides incorporated into RNA. Because many copies of an RNA are generally produced from a single gene, and nearly all RNAs are eventually degraded and replaced, a mistake in an RNA molecule is of less consequence to the cell than a mistake in the permanent information stored in DNA. Many RNA polymerases, including bacterial RNA polymerase and the eukaryotic RNA polymerase II (discussed below),

do pause when a mispaired base is added during transcription, and they can remove mismatched nucleotides from the 3' end of a transcript by direct reversal of the polymerase reaction. But we do not yet know whether this activity is a true proofreading function and to what extent it may contribute to the fidelity of transcription.

RNA Synthesis Begins at Promoters

Initiation of RNA synthesis at random points in a DNA molecule would be an extraordinarily wasteful process.

P1 Instead, an RNA polymerase binds to specific sequences in the DNA called **promoters**, which direct the transcription of adjacent segments of DNA (genes). The sequences where RNA polymerases bind are variable, and much research has focused on identifying the particular sequences that are critical to promoter function.

In *E. coli*, RNA polymerase binding occurs within a region stretching from about 70 bp before the transcription start site to about 30 bp beyond it. By convention, the DNA base pairs that correspond to the beginning of an RNA molecule are given positive numbers, and those preceding the RNA start site are given negative numbers. The promoter region thus extends between positions -70 and +30. Analyses and comparisons of the most common class of bacterial promoters (those recognized by an RNA polymerase holoenzyme containing σ^{70}) have revealed consensus sequences centered about positions -10 and -35 (Fig. 26-5a). Although the sequences are not identical for all bacterial promoters in this class, certain nucleotides that are particularly common at each position form a **consensus sequence**. The consensus sequence at the -10 region is (5')TATAAT(3'); at the -35 region it is (5')TTGACA(3'). A third AT-rich recognition element, called the UP (upstream promoter) element, occurs between positions -40 and -60 in the promoters of certain highly expressed genes. The UP element is bound by the α subunit of RNA polymerase. The efficiency with which an RNA polymerase containing σ^{70} binds to a promoter and initiates transcription is determined in large measure by these sequences, the spacing between them, and their distance from the transcription start site. A change in only one base pair in the promoter can decrease the rate of binding by several orders of magnitude. The promoter sequence thus establishes a basal level of expression that can vary greatly from one *E. coli* gene to the next. The x-ray crystal structure of the σ^{70} RNA polymerase holoenzyme bound to its promoter shows how the σ factor recognizes both the RNA polymerase and the -10 and -35 regions by introducing a large bend in the DNA (Fig. 26-5b). Information about these interactions can also be obtained using the method illustrated in Box 26-1.

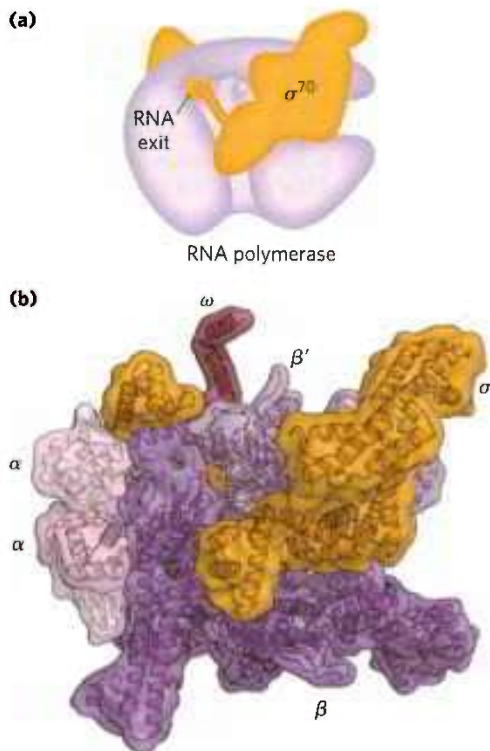


FIGURE 26-4 Structure of the σ^{70} RNA polymerase holoenzyme of *E. coli*. (a) The several subunits of the bacterial RNA polymerase give the enzyme the shape of a crab claw (purple). The σ^{70} subunit rests on top of the crab claw and threads through the RNA exit channel. (b) In this crystal structure of the RNA polymerase holoenzyme, each of the six subunits ($\alpha_2\beta\beta'\omega$ and σ^{70}) can be identified. The pincers of the crab claw are formed by the β and β' subunits. [Data from PDB ID 4MEY, D. Degen et al., *eLife* 3:e02451, 2014.]

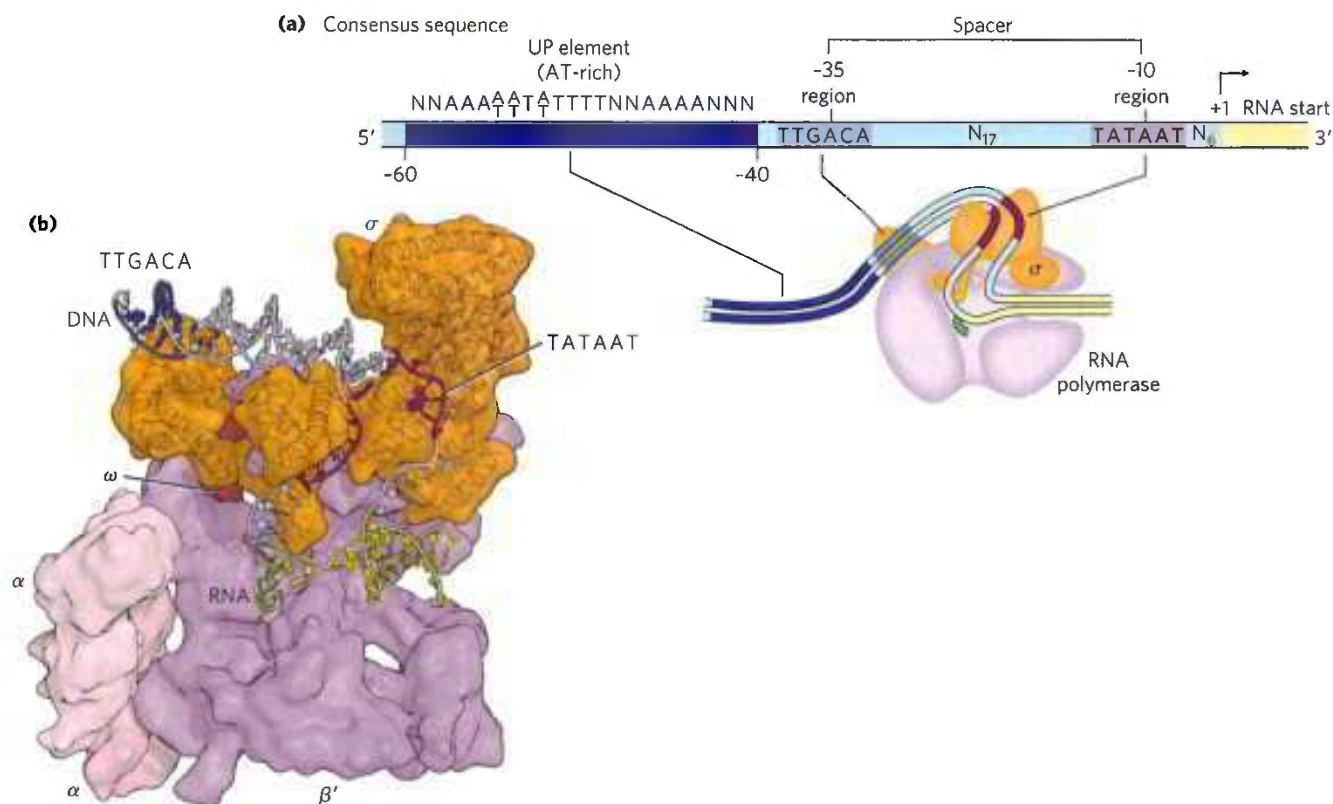


FIGURE 26-5 Promoter recognition by RNA polymerase holoenzymes containing σ^{70} . (a) The nontemplate strand of the consensus sequence for *E. coli* promoters recognized by σ^{70} is shown, read in the 5'→3' direction, as is the convention for representations of this kind. The sequences differ from one promoter to the next, but comparisons of many promoters reveal similarities, particularly in the -10 and -35 regions. The sequence element UP, not present in all *E. coli* promoters, generally occurs in the region between -40 and -60 and strongly stimulates transcription at the promoters that contain them. Spacer regions contain slightly variable numbers of nucleotides (N). Only the first nucleotide coding the RNA transcript

(at position +1) is shown. (b) This x-ray crystallographic structure of the *E. coli* holoenzyme bound to a promoter shows that the σ^{70} subunit introduces a sharp bend in the DNA template, allowing it to simultaneously contact the -35 and -10 regions as well as the RNA polymerase core. In this view the β subunit was omitted to allow the path of the DNA through the polymerase to be more easily seen. Due to the low resolution of the structure (5.5 Å) not all of the DNA backbone could be modeled, and these unmodeled regions are missing from the DNA shown in the figure. A cartoon schematic of the structure is also shown below (a), aligned with consensus promoter elements. [Data from PDB ID 4YLN, Y. Zuo and T. A. Steitz, *Mol. Cell* 58:534, 2015.]

The pathway of transcription initiation and the fate of the σ subunit are illustrated in Figure 26-6. The pathway consists of two major parts, binding and initiation, each with multiple steps. First, the polymerase, directed by its bound σ factor, binds to the promoter. A **closed complex** (in which the bound DNA remains double-stranded) and an **open complex** (in which the bound DNA is partially unwound near the -10 sequence) form in succession. Second, transcription is initiated within the complex, leading to a conformational change that converts the complex to the elongation form, followed by movement of the transcription complex away from the promoter (promoter clearance). Any of these steps can be affected by the specific makeup of the promoter sequences. The σ subunit dissociates at random as the polymerase enters the elongation phase of transcription. The protein NusA (M_r 54,430) binds to the elongating RNA polymerase, competitively with the σ subunit. Once transcription is complete, NusA dissociates from the enzyme, the RNA polymerase dissociates from the

DNA, and a σ factor (σ^{70} or another) can again bind to the enzyme to initiate transcription.

E. coli has other classes of promoters bound by RNA polymerase holoenzymes with different σ subunits, such as the promoters of the heat shock genes. The products of this set of genes are made at higher levels when the cell is exposed to environmental stress, such as a sudden increase in temperature. RNA polymerase binds to the promoters of these genes only when σ^{70} is replaced with the σ^{32} (M_r 32,000) subunit, which is specific for the heat shock promoters (see Fig. 28-3). By using different σ subunits, the cell can coordinate the expression of sets of genes, permitting major changes in cell physiology. Which sets of genes are expressed is determined by the availability of the various σ subunits. This, in turn, is determined by several factors: regulated rates of synthesis and degradation, posttranslational modifications that switch individual σ subunits between active and inactive forms, and a specialized class of anti- σ proteins, each type binding to and sequestering a particular σ subunit to render it unavailable for transcription initiation.